

The Quantum Operating Manual

All of quantum — the physics, the machines, the algorithms, every industry, the money, the history, and the honest frontier — 184 graded node cards on one substrate.

Built 2026-07-08 · Index all, exclude nothing, grade everything · a vendor press release and a peer-reviewed threshold demo do not weigh the same, and this manual always says which is which.

CONTENTS

The Full Map

§01 · Foundations 26

§02 · Hardware 27

§03 · Stack & Algorithms 27

§04 · Adjacent Technologies 22

§05 · Industries 27

§06 · Ecosystem & Geopolitics 22

§07 · History 15

§08 · The Honest Frontier 18

How claims are graded

Conflict register

The Full Map

The Full Map — Atlas of Quantum

The entire territory on one page — from the postulate that a system can be in two states at once, up through the machines that exploit it, and out into every industry that will be rebuilt on top of them.

It's a map, not a front-to-back read. Two things travel with every node: which chapter covers it (§), and how strong the evidence behind it is (graded per evidence/SCHEMA.md). A press release and a peer-reviewed threshold demo do not weigh the same, and the manual always says which is which.

Every node below is a coverage unit. NETWORK-CAPACITY.md counts how many are filled; FRONTIER.md is the live worklist. Node IDs are stable citation anchors.

Layer 0 — Foundations (the physics that makes any of this possible) · §01

Node	What	ID
Superposition & the state vector	A system holds a weighted combination of basis states until measured	F-superpos
Measurement & the Born rule	Probabilities as squared amplitudes; collapse / update	F-measure
Entanglement & nonlocality	Correlations with no classical explanation	F-entangle
Bell inequalities & their experimental violation	The test that ruled out local hidden variables (Aspect → 2022 Nobel)	F-bell
Decoherence & open quantum systems	Why the quantum world looks classical; the enemy of every qubit	F-decoher
The qubit & Bloch sphere	The unit of quantum information	F-qubit
No-cloning theorem	You cannot copy an unknown quantum state — the root of quantum crypto	F-nocloning
Quantum information theory	Von Neumann entropy, channels, fidelity, quantum capacity	F-qinfo

Node	What	ID
Uncertainty & complementarity	Conjugate observables; the Heisenberg bound	F-uncertainty
Interpretations	Copenhagen, many-worlds, Bohmian, QBism, relational, objective-collapse	F-interp
QFT / second-quantization bridge	Where quantum mechanics meets fields and particles	F-qft
Quantum thermodynamics	Work, heat, and information at the quantum scale	F-qthermo
Quantum tunneling	The physics inside every Josephson junction, flux qubit, STM	F-tunneling
Contextuality (Kochen-Specker)	The resource behind magic-state computation; bridges to QEC	F-contextuality
Identical particles & spin-statistics	Exchange symmetry — underlies quantum chemistry, fermionic sim, anyons	F-statistics
Adiabatic theorem	The foundation quantum annealing rests on	F-adiabatic
Teleportation & entanglement swapping	Foundational protocols under the quantum internet & modular QC	F-teleport
Geometric / Berry phase	Geometric gates, topological band theory, links to quantum materials	F-berry
Generalized measurement (POVMs)	Positive-operator measurements, Naimark dilation, weak measurement & instruments — the real measurement model	F-povm
Purification & Stinespring dilation	Every mixed state is a pure state, every channel a unitary, on a larger space — the church of the larger Hilbert space	F-purification
Lieb-Robinson bounds	An emergent finite speed limit / effective light cone in non-relativistic quantum systems	F-liebrobinson
PBR theorem	Pusey-Barrett-Rudolph: the reality status of the quantum state (ψ -ontic vs ψ -epistemic)	F-pbr
		F-wigner

Node	What	ID
Wigner functions & quasiprobability	Phase-space QM; negativity as a nonclassicality / computation resource	
Stabilizer formalism & Gottesman-Knill	Why a large class of quantum circuits is classically simulable; the backbone of QEC	F-stabilizer
Quantum discord	Quantum correlations that survive beyond entanglement	F-discord
Quantum entropy inequalities	Strong subadditivity, monogamy, data-processing — the bookkeeping laws of quantum information	F-entropy-ineq

Layer 1 — Hardware: the qubit modalities & the machines · §02

Node	What / who	ID
Superconducting (transmon)	IBM, Google, Rigetti, IQM, OQC, SEEQC	H-supercon
Trapped ions	IonQ, Quantinuum, Oxford Ionics, Alpine, eleQtron	H-ion
Photonic	PsiQuantum, Xanadu, QuiX, ORCA, TundraSystems	H-photonic
Neutral atoms	QuEra, Pasqal, Atom Computing, Infleqtion, planqc	H-neutral
Spin qubits in silicon	Intel, Diraq, Quantum Motion, SQC, Equal1	H-silicon
Topological / Majorana	Microsoft (Majorana 1), Nokia Bell	H-topo
Bosonic / cat qubits	Alice & Bob, AWS, Nord Quantique	H-bosonic
Quantum annealing	D-Wave	H-anneal
NV centers / diamond	Sensing + compute; Quantum Brilliance	H-nv
Cryogenics & dilution refrigerators	Bluefors, Oxford Instruments, Maybell	H-cryo

Node	What / who	ID
Control electronics & I/O	Qblox, Quantum Machines, Zurich Instruments, Keysight	H-control
Fabrication & supply chain	Foundries, wafers, cabling, rare materials, TLS defects	H-fab
Interconnects & modular QC	Chip-to-chip, photonic links, networked processors	H-intercon
Microwave↔optical transduction	Bridging mK microwave qubits to optical fiber; gates fridge-to-fridge modularity	H-transduce
Single-photon detectors / SNSPDs	Photon counting for photonic QC & QKD; Single Quantum, ID Quantique, Photon Spot	H-detect
Cryogenic control ASICs	Cryo-CMOS control/readout in the fridge; Intel Horse Ridge, SemiQon, Equal1	H-cryocmos
Laser & photonics subsystems	Stabilized lasers/PICs for ion/atom/NV machines; TOPTICA, M Squared, Vescent	H-lasers
Deterministic photon sources	Single-photon / entangled-pair emitters; Quandela quantum dots, SPDC, Sparrow	H-photonsource
Optical frequency combs	Shared ruler-for-light under clocks/timing/atomic-machine lasers; microcombs, NIST	H-frequencycomb
Quantum-limited amplifiers	JPA/TWPA at mK + HEMT at 4 K for single-shot readout; Silent Waves, Low Noise Factory	H-paramp
Packaging & 3D integration	Flip-chip, TSVs, interposers, chiplets; MIT-LL, IBM, Google, QuantWare	H-package
Ion-trap chip fabrication	Microfabricated surface/QCCD traps; Sandia HOA/Enchilada, Honeywell, Infineon	H-iontrap
UHV & atomic-source systems	Ultra-high-vacuum + cryogenic chambers for ion/atom machines; sets atom-loss floor	H-uhv
300 mm foundry ecosystem	Who manufactures qubits at wafer scale; imec, GlobalFoundries, Intel, SkyWater	H-foundry
Quantum-dot vs donor spin fork	The silicon-spin strategic split; SQC (donor) vs Diraq/imec/Intel (dots)	H-spinsplit

Node	What / who	ID
Cryogenic microwave wiring	Passive coax / attenuators / ferrite isolators from 300 K to mK; the line-count scaling wall (split from H-cryo/H-fab)	H-wiring
Spin-photon interfaces	Emitters entangling a matter spin with a flying photon; silicon T-centers, diamond SiV/SnV, erbium — the matter-qubit networking node	H-spinphoton

Layer 2 — The stack: from a noisy qubit to a useful answer · §03

Node	What	ID
Gates & circuits	Universal gate sets, native gates, connectivity	S-gates
Noise, NISQ & error mitigation	Where the field is today; ZNE, PEC	S-nisq
Quantum error correction	Surface code, qLDPC, magic-state distillation, threshold theorem	S-qec
Logical qubits & fault tolerance	The line between demos and utility	S-logical
Algorithms — Shor	Factoring; the reason cryptography must migrate	S-shor
Algorithms — Grover	Unstructured search, quadratic speedup	S-grover
Algorithms — VQE / QAOA	Near-term variational chemistry & optimization	S-variational
Algorithms — HHL & quantum linear algebra	Speedups (and the fine print)	S-hhl
Quantum simulation	Simulating quantum systems — Feynman's original pitch	S-qsim
Quantum machine learning	QML, kernels, hybrid; the hype-heavy corner	S-qml
Compilers, transpilers, middleware	Qiskit, Cirq, PennyLane, tket, Braket SDK	S-software

Node	What	ID
Benchmarks & "advantage" claims	Quantum volume, CLOPS, algorithmic qubits, supremacy/advantage history	S-bench
Cloud QPU access	IBM Quantum, AWS Braket, Azure Quantum, Google	S-cloud
QFT & phase estimation	The primitive under Shor/HHL/simulation; QPE depth is a FT problem	S-qft
Quantum walks & element distinctness	$O(N^{\{2/3\}})$ distinctness, BQP-universal walks, algorithmic framework	S-walk
Quantum Monte Carlo / amplitude estimation	Quadratic speedup for expectation values; feeds finance	S-qmc
Real-time QEC decoding	Decoder as its own discipline — Riverlane, ASIC/FPGA, the throughput wall	S-decoders
Quantum RAM (qRAM)	The load-bearing, largely-unproven data-loading assumption under HHL/QML	S-qram
Certified randomness	First delivered advantage application — JPMorgan/Quantinuum, Nature 2025	S-certrand
QSVT — quantum singular value transformation	The modern unifying lens: HHL/Grover/Hamiltonian-sim as one framework (Gilyén et al. 2019)	S-qsvt
Hamiltonian simulation methods	Trotter / qDRIFT / LCU / qubitization — the how behind digital quantum simulation	S-hamsim
Tensor-network classical simulation	The classical counterfactual that decides every advantage claim (MPS, contraction)	S-tensornet
Classical shadows	Predict many properties from few measurements; size-independent sample complexity	S-shadows
Error mitigation as a discipline	ZNE/PEC/symmetry/virtual-distillation taxonomy; exponential sampling ceiling	S-errmit
Circuit knitting & cutting	Run big circuits on small QPUs via quasiprobability; exponential cut overhead	S-circuitcut

Node	What	ID
Pulse-level & optimal control	GRAPE/CRAB/Krotov/DRAG — where a gate becomes an analog waveform	S-optcontrol
Quantum advantage complexity theory	BQP/BPP, oracle & sampling separations — what "advantage" formally means	S-complexity

Layer 3 — Adjacent quantum technologies (not compute) · §04

Node	What	ID
Quantum key distribution (QKD)	Provably secure key exchange; BB84, E91	A-qkd
Quantum internet & repeaters	Entanglement distribution, memories, the future network	A-qinternet
Satellite QKD	Micius, space-based entanglement	A-satqkd
Post-quantum cryptography (PQC)	NIST standards (Kyber/ML-KEM, Dilithium), migration, HNDL threat	A-pqc
Quantum sensing & metrology	The nearest-term commercial quantum tech	A-sensing
Atomic clocks & timing	Optical clocks, PNT, GPS-free timing	A-clocks
Magnetometry	SQUID, OPM, brain imaging (MEG), mineral exploration	A-magneto
Gravimetry & inertial nav	Cold-atom gravimeters, GPS-free navigation	A-gravimetry
Quantum imaging & radar	Ghost imaging, quantum illumination, LiDAR	A-imaging
Quantum materials	Topological insulators, high-Tc superconductors, 2D materials	A-materials
Quantum RNG	Certified randomness from quantum sources	A-qrng
Quantum memories (component industry)	Rack-mounted/room-temp memories for repeaters; Qunnect, Aliro	A-qmemory-hw

Node	What	ID
Squeezed light (cross-cutting resource)	Sub-shot-noise light; LIGO production use, Xanadu CV/GKP compute	A-squeezed
Quantum/optical time-transfer networks	Comparing distant clocks over fiber/free-space; supports SI-second redefinition	A-timedist
NV-diamond sensing	Room-temp atom-scale sensing; Element Six/ Bosch, SBQuantum, QDTI	A-nvsensing
Assured PNT / GPS-denied navigation	System-level fusion of clock + cold-atom IMU + magnetometry; Q-CTRL, SandboxAQ, DARPA RoQS/PINS	A-pnt
Rydberg RF electrometry / quantum antennas	Atom-based, self-calibrated RF receivers; Rydberg Technologies, Infleqion	A-rydberg
Atom interferometry (platform)	Matter-wave interferometry under gravimetry/ IMU + fundamental physics; MAGIS-100, AION	A-atominterf
Entanglement-based clock networks	GHZ-entangled clocks toward Heisenberg-limited timekeeping (contested advantage)	A-entclock
Quantum-secured / -resistant blockchain	PQC signatures on-chain (Algorand Falcon, QRL) + QKD-secured ledgers (niche)	A-qblockchain
Quantum-enhanced MRI/ NMR & biomagnetism	Hyperpolarization (NVision) + NV/atomic magnet-free NMR + OPM MEG/MCG	A-qmri
Quantum-dot displays (boundary case)	QLED as quantum <i>material</i> not quantum tech; the QD single-photon source is the real thread	A-qdisplay

Layer 4 — Industries: where quantum actually lands · \$05

Industry	Use cases · anchor players	ID
Finance	Portfolio optimization, Monte Carlo pricing, risk, fraud — JPMorgan, Goldman, HSBC, Wells Fargo	I-finance
Pharma & healthcare	Drug discovery, protein folding, molecular sim — Boehringer, Roche, Cleveland Clinic, Moderna	I-pharma

Industry	Use cases · anchor players	ID
Chemicals & materials	Catalysts, batteries, nitrogen fixation, carbon capture — BASF, Dow, Mitsubishi Chemical	I-chem
Energy & utilities	Grid optimization, battery chemistry, fusion, oil & gas — ExxonMobil, E.ON, EDF	I-energy
Logistics & supply chain	Routing, scheduling, traffic — DHL, Volkswagen, Airbus, DB	I-logistics
Automotive	Materials, batteries, aerodynamics, autonomous — BMW, Mercedes, Ford, Hyundai	I-auto
Aerospace & defense	Sensing, nav, sim, codebreaking — Airbus, Boeing, Lockheed, DARPA, DoD	I-aerospace
Telecom & networking	Quantum-secure networks, optimization — BT, Verizon, SK Telecom, NTT	I-telecom
Cybersecurity	PQC migration, "harvest now decrypt later," QKD networks	I-cyber
AI & machine learning	Quantum ML, hybrid pipelines, sampling	I-ai
Climate & sustainability	Carbon capture, fertilizer (Haber-Bosch) chemistry, emissions optimization	I-climate
Insurance & risk	Catastrophe modeling, actuarial optimization	I-insurance
Manufacturing & industrials	Process optimization, digital twins, quality	I-manufacturing
Oil & gas	Seismic inversion, reservoir sim, LNG logistics, materials	I-extractive
Space & Earth observation	Orbit optimization, quantum sensors on satellites	I-space
Government services	Tax/fraud detection, customs, census, public-sector optimization (distinct from defense)	I-gov
Agriculture & food science	Precision ag, fertilizer/nitrogen-fixation chemistry, agri sensing	I-agri

Industry	Use cases · anchor players	ID
Retail	Demand forecasting, pricing, assortment/shelf optimization, recommendation	I-retail
Media & entertainment	Procedural generation, rendering, QRNG content, ad/recommendation optimization	I-media
Construction & built environment	Structural materials sim, project scheduling, urban microclimate	I-construction
Air-traffic management	Traffic-flow optimization, trajectory deconfliction, gate scheduling	I-atm
Semiconductors & EDA	Device-materials sim, computational lithography, quantum-chip EDA; the fab base	I-semiconductor
Weather & climate modeling	NWP / fluid-dynamics PDEs, quantum linear-solvers, QML nowcasting (distinct from I-climate chemistry)	I-weather
Nuclear & fusion	Plasma dynamics + stability (QAOA), nuclear-structure sim, reactor/fuel optimization	I-nuclear
Healthcare imaging & diagnostics	OPM-MEG brain imaging, NV/hyperpolarized MRI contrast, quantum-enhanced imaging (sensing, near-term)	I-healthimaging
Mining & mineral exploration	Diamond-magnetometer + cold-atom gravimeter critical-minerals exploration; mine-plan optimization	I-mining
Intelligence & cryptanalysis	Offensive SIGINT: Shor-driven decryption, harvest-now-decrypt-later, intercept analytics	I-intelligence

Layer 5 — Ecosystem, money & geopolitics · §06

Node	What	ID
National programs — US	National Quantum Initiative, NQI reauthorization, DOE, NSF, NIST	E-us
National programs — China	Reported ~\$15B+ commitment, USTC, Micius, Jiuzhang	E-china

Node	What	ID
National programs — EU	Quantum Flagship (€1B+), EuroQCI, per-member strategies	E-eu
National programs — UK	National Quantum Strategy (£2.5B), NQCC	E-uk
National programs — others	India, Japan, Australia, Canada, Israel, South Korea, Germany	E-others
Private investment & VC	Funding rounds, SPACs, market size, the 2024–26 capital cycle	E-vc
Standards bodies	NIST, ISO/IEC JTC1, ETSI, IEEE, ITU	E-standards
Talent & workforce	The skills gap, university programs, quantum education	E-talent
Market forecasts	BCG, McKinsey, IDC, Gartner hype cycle — with skepticism	E-market
Patents & IP landscape	Who owns what; national patent races	E-patents
Export controls	US 2024 quantum export rule + allied plurilateral coordination (NL, JP, UK, FR)	E-export
Supply-chain chokepoints	Dilution fridges/He-3, quantum-grade lasers, specialty fab (geopolitics vs H-fab technical)	E-supplychain
Open-source ecosystem	Qiskit/Cirq/PennyLane governance as soft power / lock-in; OpenQASM/QIR portability	E-oss
Defense funding channel	DARPA QBI (quantum benchmarking) + defense procurement, distinct from civilian NQI	E-defense
China's parallel software stack	Origin/QPanda, Baidu Paddle Quantum, Huawei HiQ as a fork / soft-power risk vs Qiskit/Cirq	E-china-stack
Consolidation wave (M&A / IPO)	IonQ roll-ups, D-Wave/Xanadu/IQM/Quantinuum exits 2025–26; distinct from primary VC	E-mna
Quantum-safe migration market	PQC migration as its own economic market (HNDL-driven), distinct from A-pqc crypto + E-market	E-pqcmarket

Node	What	ID
Insurance & liability for quantum risk	Cyber/D&O underwriting of the quantum threat; mirror of I-insurance	E-insurance
Sovereign compute & national champions	Gov-hosted machines (EuroHPC six-site fleet) + equity stakes; on-prem sovereignty	E-sovereign
Metrology & SI-redefinition governance	BIPM/CIPM/CGPM, second-redefinition to optical clocks (CGPM 2026), Kibble balance	E-metrology-gov
Workforce immigration & clearance	Visa/deemed-export/clearance filter on who can fill the talent gap; distinct from E-talent	E-immigration
Academic-lab leaderboard	USTC/MIT/Harvard/Delft/Oxford research base; the industry map one generation early	E-labs

Layer 6 — History: the full timeline · §07

Era	Milestones	ID
Old quantum theory (1900–1925)	Planck, Einstein photoelectric, Bohr atom, de Broglie	T-old
The formalism (1925–1935)	Heisenberg matrix mechanics, Schrödinger, Dirac, EPR paradox	T-formalism
Foundations era (1935–1980)	Bell's theorem (1964), Bell tests, Aspect experiments	T-foundations
Birth of quantum computing (1980–1994)	Manin, Feynman (1981), Deutsch, Shor's algorithm (1994)	T-birth
Early experiments (1995–2010)	Cirac-Zoller, first gates, NMR Shor demo, DiVincenzo criteria	T-early
The engineering race (2011–2019)	D-Wave, IBM/Google scale-up, Google "supremacy" (2019)	T-race
The error-correction era (2020–now)	Logical qubits, QuEra 48-logical, Google Willow, utility-scale claims	T-ecera
Loophole-free Bell tests (2015)	Delft/NIST/Vienna close detection + locality together — the foundations capstone	T-belltests

Era	Milestones	ID
The NISQ era (2018)	Preskill's coinage — the vocabulary of the noisy near-term machine	T-nisq
1927 Solvay Conference & Bohr–Einstein debates	The interpretation argument goes public; entanglement's opening act	T-solvay
EPR paradox & Bohr's reply (1935)	The field's most fertile question — locality vs. completeness — as its own milestone	T-epr
Birth of quantum information theory (1970–84)	Wiesner conjugate coding → Holevo bound → no-cloning → BB84	T-qinfobirth
"Second quantum revolution" framing (2003)	Dowling–Milburn coinage — passive understanding → active single-system engineering	T-2ndrev
Nobel Prizes as the quantum spine (1918→2025)	The establishment's ratification timeline; Planck to Clarke/Devoret/Martinis	T-nobel
The "quantum winter" question	AI-winter analogy — the funding-downturn / unmet-promise risk, graded honestly	T-winter

Layer 7 — The honest frontier & open problems · §08

Node	What's unsettled	ID
Fault-tolerant scaling	Millions of physical qubits per useful computation — how far off, really	0-scaling
Error-correction overhead	The brutal physical:logical ratio	0-overhead
Decoherence & materials defects	Two-level systems, the materials science bottleneck	0-materials
Quantum advantage in practice	Which claimed speedups survive classical counterattack	0-advantage
Hype vs reality	Vendor timelines, "quantum-washing," the valuation question	0-hype

Node	What's unsettled	ID
Room-temperature & photonic scaling	The paths that would change the economics	0-roomtemp
The killer app question	Is there a near-term application that pays for itself	0-killerapp
Interpretational & foundational openness	Measurement problem still unsolved	0-foundations
Real-time QEC decoding throughput	The classical co-processor bottleneck (~1 MHz/logical qubit); Riverlane	0-decoder
Chip-to-chip entanglement fidelity	Whether modular interconnects stay below threshold across module boundaries	0-interconnect-loss
Verifying a too-large computation	Certifying a result no classical machine can re-check (Mahadev)	0-verification
Funding-winter & talent attrition	Whether a capital downturn drains the field before roadmaps mature	0-talent-attrition
The classical-simulation frontier	Where the quantum-classical boundary actually sits vs the best classical method (tensor nets, belief propagation)	0-classical-sim
Benchmarking standardization crisis	No agreed cross-vendor yardstick; QV vs #AQ vs "logical qubits"; spoofable metrics	0-benchmark-standard
Business-model & ROI question	Who pays, at what margin — unit economics distinct from technical feasibility	0-roi-business
Energy consumption vs classical	Fixed cryo/control overhead vs joules-per-useful-answer; energetic advantage before computational	0-energy
The "quantum utility" definitional debate	Whether "utility" is a real milestone or a rhetorical hedge (IBM 2023 → classically simulated)	0-utility-definition
Cryptographically-relevant QC timeline	When Shor breaks RSA-2048; HNDL threat; Mosca's inequality (distinct from general FTQC)	0-crqc-timeline

The honest edge of the map

Named out loud so nothing hides: (1) vendor roadmaps are marketing until a peer reviews them — every timeline node carries that caveat; (2) "quantum advantage" is a moving target as classical algorithms improve; (3) QML is the most hype-inflated corner and gets graded hardest; (4) national-program dollar figures are frequently inflated or double-counted; (5) quantum biology (F -adjacent, covered in the biophysics canon) is contested and lives in that branch, cross-linked here rather than duplicated.

Network Capacity — the coverage meter

"Network capacity" = fraction of 00-IDEAL-STATE-MAP.md nodes with a filled, graded research card. Breadth ran to 100%; the loop then flipped to DEPTH, and depth is now complete across every layer.

Meter

```
TOTAL NODES:      184
FILLED (stub+):    184
DEPTH-COMPLETE:    184
NETWORK CAPACITY:  100.0%  ← breadth
DEPTH CAPACITY:    100.0%  ← every card deepened with primary sources + graded c
CONFLICT OBJECTS:  15      (evidence/CONFLICTS.md)
RENDERING:         LaTeX/mhchem math → SVG • WeasyPrint PDF • figures via matplo
```

Per-layer breakdown (counted from disk)

Layer	Nodes	Filled	Depth-complete
L0 Foundations	26	26	26 (100%)
L1 Hardware	27	27	27 (100%)
L2 Stack & algorithms	27	27	27 (100%)
L3 Adjacent tech	22	22	22 (100%)
L4 Industries	27	27	27 (100%)
L5 Ecosystem & geopolitics	22	22	22 (100%)
L6 History	15	15	15 (100%)
L7 Frontier & open	18	18	18 (100%)
Total	184	184	184 (100%)

Trajectory

- **Cycle 1** (2026-07-08): 0→72%. Filled 94 original nodes; random-walk found 36 → map 94→130.
- **Cycle 2** (2026-07-08): 72→100% breadth. Filled the 36 gap nodes (125 cards).
- **Cycle 3 — DEPTH** (2026-07-08): every card deepened (stub → ~450–700 words with primary derivations, exact 2025–26 numbers, per-claim tiers). Random-walk during depth added **52 more nodes** (map 130→182): L0+8, L1+8, L2+8, L3+7, L4+6, L5+8, L6+6, L7+6. Conflict register grew 8→15.
- **Cycle 3b — RENDERING** (2026-07-08): math/chemistry marked up in LaTeX and rendered to SVG (`reports/render_math.py` via `latex+dvisvgm`), figures generated (`reports/gen_figures.py`), manual compiled to HTML (`build_manual.py`) and PDF (`build_pdf.py` , WeasyPrint).

Thin-node flags (honest edge)

`I-gov` , `I-retail` , `I-media` remain T5/T6-heavy (forecasts / classical-AI mislabeled as quantum). Deepened but honestly thin — the manual says so.

Still-open threads (queued, from depth-agent "what I'd chase" notes)

RESOLVED in the polish pass: QuEra 96-logical confirmed peer-reviewed (Nature s41586-025-09848-5 → T2); Duke/IonQ networking primary found (arXiv:2606.17173 → T3); Majorana 2 — no independent replication (kept contested, Legg critique + Microsoft Nature reply); Kalai↔Aaronson Mar-2026 exchange sourced verbatim both sides; Flatiron vs D-Wave — no third-party adjudication (kept contested). Node splits landed: +H-wiring, +H-spinphoton. SKIPPED (no real dated pilot): I-water (desalination/membrane), I-foodbev — only speculative coverage exists, so not added. Remaining candidate: QEC-theory-history (§07).

Last recomputed from disk: polish pass — 184/184 depth, 2026-07-08.

§01 • Foundations

The physics that makes any of this possible — superposition, entanglement, decoherence, quantum information.

Adiabatic Theorem · F-adiabatic

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

If a quantum system starts in an eigenstate of its Hamiltonian and the Hamiltonian is changed slowly enough, the system stays in the corresponding instantaneous eigenstate. "Slowly enough" is set by the spectral gap: the runtime must scale roughly as $1/\text{gap}^2$ (Born & Fock, 1928; rigorous conditions later tightened by Kato and others). If the gap closes or nearly closes, the theorem fails and the system leaks to excited states (Landau-Zener transitions).

CORE IDEA / KEY EQUATION

The whole theorem hangs on one ratio. Call Δ the smallest energy gap between the eigenstate you are riding and its nearest neighbor over the entire schedule, and let the drive matrix element be roughly how fast the Hamiltonian changes, dH/dt . The system stays put as long as the drive is slow compared to the gap squared — the standard adiabatic condition is (rate of change of H)/ $\Delta^2 \ll 1$, so the total runtime scales as $T \sim 1/\Delta^2$. Halve the gap and you must run four times slower. The failure mode is quantitative and named: at an **avoided crossing** the probability of jumping to the excited state is the Landau-Zener formula $P = e^{-2\pi\Gamma}$, with $\Gamma = \Delta^2/(4\hbar v)$, where v is the sweep speed through the crossing. Go slow enough and the jump probability is exponentially suppressed; sweep too fast through a tiny gap and you almost certainly leave the ground state. The exponent Δ^2 is the same one that appears in the runtime — the gap is the single number that governs whether adiabatic evolution works.

WHY IT MATTERS FOR QUANTUM TECH

The adiabatic theorem is the load-bearing foundation under **quantum annealing** and **adiabatic quantum computation** (Farhi et al., 2000): encode a hard optimization problem's solution as the ground state of a final Hamiltonian, start in the easy-to-prepare ground state of an initial one, and interpolate slowly. If you stay adiabatic, you end in the answer. Adiabatic QC is polynomially equivalent to the gate model (Aharonov et al., 2004), which makes it a legitimate computing paradigm rather than a heuristic (see H-anneal). The catch is the gap: for hard

instances it can shrink exponentially, erasing any speedup — the central open question hanging over D-Wave's machines (see C-dwave-advantage). The same $1/\text{gap}^2$ control appears in quantum simulation, where adiabatic state preparation is a standard way to reach a many-body ground state before measuring it (see S-qsim), and the Landau-Zener physics is exactly what STIRAP and other geometric state-transfer protocols exploit (see F-berry). Slow, gap-limited ramps are also how experimentalists load qubits and cold-atom lattices into their target states across every hardware line (see H-ion, H-supercon).

KEY GRADED CLAIMS

- [T1] A system remains in its instantaneous eigenstate under sufficiently slow Hamiltonian change, with error controlled by the inverse spectral gap — Born & Fock, Z. Phys. 51, 165 (1928) (status: established)
- [T2] Adiabatic quantum computation is polynomially equivalent to the standard gate model — Aharonov, van Dam, Kempe, Landau, Lloyd, Regev, SIAM J. Comput. 37, 166 (2007); arXiv:quant-ph/0405098 (status: established)
- [T2] Coherent quantum annealing demonstrated at scale — quantum critical dynamics in a 5,000+-qubit programmable spin glass, faster than the matched classical annealing model — King et al., Nature 617, 61 (2023) (status: demonstrated, classical-advantage claim contested)

CONFLICTS / OPEN QUESTIONS

- Whether the minimum gap for practically important optimization problems stays large enough for a real speedup is unresolved — it is the crux of the annealing-advantage debate.

GO DEEPER

- Farhi et al., "Quantum Computation by Adiabatic Evolution," arXiv:quant-ph/0001106 (2000)
- Albash & Lidar, "Adiabatic quantum computation," RMP 90, 015002 (2018)

SOURCES

- Born & Fock (1928)
- Aharonov et al., arXiv:quant-ph/0405098

- King et al., Nature 617, 61 (2023). doi:10.1038/s41586-023-05867-2

Bell Inequalities & Their Experimental Violation · F-bell

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Bell (1964) proved that any theory of local hidden variables puts a numerical ceiling on correlations between separated measurements; quantum mechanics predicts violations of that ceiling. The CHSH form (1969) made it testable. Six decades of experiments — Freedman–Clauser (1972), Aspect (1982), and the 2015 loophole-free trio — confirm the quantum prediction. Nature violates local realism.

CORE IDEA / KEY EQUATION

CHSH builds one number from four correlation measurements: $S = E(a,b) - E(a,b') + E(a',b) + E(a',b')$, where $E(x,y)$ is the average product of the ± 1 outcomes when Alice measures setting $\mathcal{X}$ and Bob setting $\mathcal{Y}$. Any local hidden-variable theory obeys $|S| \leq 2$ — the outcomes are fixed functions of a shared variable, so the four terms cannot all be large at once. Quantum mechanics, using a Bell state and measurement angles 45° apart, reaches $S = 2\sqrt{2} \approx 2.828$, the Tsirelson bound (the maximum quantum mechanics itself allows). The gap between 2 and 2.828 is the experimental target: measure $S > 2$ with the two stations space-like separated and no detection bias, and no local-realist theory can reproduce it. That gap is what six decades of increasingly airtight experiments have nailed down.

WHY IT MATTERS FOR QUANTUM TECH

Bell violation is the operational certificate that a device holds real entanglement (F-entangle) — the basis of device-independent QKD (A-qkd), where security follows from $S > 2$ alone without trusting the hardware, and of certified quantum randomness (A-sensing, related randomness-expansion protocols), where the violation guarantees the output bits are unpredictable to any adversary. The same self-testing logic lets you verify an untrusted quantum processor. The 2022 Nobel citation names Bell violation as the launch point of quantum information science, and the 2023 superconducting demonstration (below) shows the test now runs on the same chip technology (H-supercon) used to build quantum computers.

KEY GRADED CLAIMS

- [T1] Local hidden-variable theories obey $|S| \leq 2$ (CHSH); quantum mechanics reaches $2\sqrt{2} \approx 2.828$ (Tsirelson bound) — Bell, Physics Physique Fizika 1, 195 (1964); Clauser–Horne–Shimony–Holt, PRL 23, 880 (1969) (status: established)
- [T1] Bell inequalities are violated experimentally with all major loopholes (locality, detection) closed simultaneously — Hensen et al., Nature 526, 682 (2015); Giustina et al., PRL 115, 250401 (2015); Shalm et al., PRL 115, 250402 (2015) (status: established)
- [T1] Loophole-free violation demonstrated with superconducting circuits: two qubits entangled across a 30 m cryogenic link, $S = 2.0747 \pm 0.0033$ over >1 million trials, $p < 10^{-108}$ — Storz et al. (Wallraff group, ETH Zurich), Nature 617, 265 (2023), doi:10.1038/s41586-023-05885-0 (status: established)
- [T1] 2022 Nobel Prize in Physics to Aspect, Clauser, Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science" — nobelprize.org/prizes/physics/2022 (status: established)

CONFLICTS / OPEN QUESTIONS

- Superdeterminism and retrocausality evade Bell's theorem by denying measurement independence; consistent but widely viewed as unfalsifiable. The freedom-of-choice loophole was pushed back by cosmic-photon and Big Bell Test (Nature 557, 212, 2018) experiments, never fully closable.

GO DEEPER

- Aspect, Dalibard, Roger, PRL 49, 1804 (1982); Brunner et al., "Bell nonlocality," RMP 86, 419 (2014)

SOURCES

- Bell (1964) doi:10.1103/PhysicsPhysiqueFizika.1.195 · CHSH (1969) doi:10.1103/PhysRevLett.23.880
- Hensen et al., Nature 526, 682 (2015). doi:10.1038/nature15759 · Giustina et al. doi:10.1103/PhysRevLett.115.250401 · Shalm et al. doi:10.1103/PhysRevLett.115.250402

- Storz et al., "Loophole-free Bell inequality violation with superconducting circuits," Nature 617, 265 (2023). doi:10.1038/s41586-023-05885-0
- Nobel press release: <https://www.nobelprize.org/prizes/physics/2022/press-release/>

Geometric / Berry Phase · F-berry

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

When a quantum system is driven adiabatically around a closed loop in parameter space, its state picks up a phase with two parts: the familiar dynamical phase ($\propto$ energy $\times$ time) and a **geometric phase** that depends only on the *path traced* and stays the same however fast or slow the loop is run. Berry (1984) gave the general formula; the phase is the integral of a "Berry connection," and its curvature is the "Berry curvature." Pancharatnam had found the optical version in 1956, and Aharonov-Bohm is a special case. The geometric phase is gauge-invariant and observable.

CORE IDEA / KEY EQUATION

The geometric phase is a loop integral. As the parameters R traverse a closed path C , the state acquires $\gamma = \oint_C \mathbf{A}(R) \cdot d\mathbf{R}$, where $\mathbf{A}(R) = i\langle n(R) | \nabla_R | n(R) \rangle$ is the **Berry connection** — the same object a vector potential is in electromagnetism. By Stokes' theorem this equals a surface integral of the **Berry curvature** $F = \nabla \times \mathbf{A}$ over any cap bounded by the loop: $\gamma = \iint F \cdot d\mathbf{S}$.

The clean example is a spin- $\frac{1}{2}$ in a magnetic field swept around a cone: the Berry phase equals exactly half the solid angle the field vector sweeps out, $\gamma = -\Omega/2$, independent of how slowly you go — pure geometry. Two consequences follow. Timing errors cancel because γ carries no dependence on the dynamical phase (energy $\times$ time), which is why geometric gates resist certain pulse fluctuations. And when the parameter space is a material's Brillouin zone, the curvature integrated over a filled band is forced to be 2π **times an integer**, the Chern number — a topological invariant that cannot change under smooth deformation, so the response it controls is quantized to extraordinary precision.

WHY IT MATTERS FOR QUANTUM TECH

Because the geometric phase is set by geometry rather than timing, it is intrinsically resistant to certain control errors — the basis of **holonomic / geometric quantum gates**, where a qubit is steered around a loop to enact a

gate that resists fluctuations in pulse duration and amplitude (see H-supercon, H-ion). Berry curvature is also the organizing quantity of modern band theory: it produces the anomalous Hall effect, defines topological invariants (Chern numbers) that classify topological insulators and the quantum Hall effect, and underlies the topological protection sought for qubits (see H-topo, A-materials, F-statistics). Non-abelian generalizations of the Berry phase — matrix-valued holonomies — are the mathematical content of braiding non-abelian anyons, so this node sits directly under topological quantum computing (see H-topo). The same curvature integrated over a Brillouin zone is what makes a material's response quantized and defect-tolerant, which is why classifying it is a target for materials-simulation algorithms (see S-qsim, A-materials).

KEY GRADED CLAIMS

- [T1] Adiabatic cyclic evolution yields a path-dependent geometric phase separate from the dynamical phase — Berry, Proc. R. Soc. Lond. A 392, 45 (1984) (status: established)
- [T1] Berry curvature integrated over a filled band gives a quantized Chern number = the quantum-Hall conductance — TKNN, Thouless, Kohmoto, Nightingale & den Nijs, PRL 49, 405 (1982) (Thouless, Nobel 2016) (status: established)
- [T1] The quantized Hall resistance is reproducible to a few parts in 10^9 , precise enough to define the resistance standard $R_K = 25812.807 \, \Omega$ — von Klitzing, Dorda & Pepper, PRL 45, 494 (1980), Nobel 1985 (status: established)
- [T2] Geometric/holonomic gates realized on superconducting and NV qubits — e.g. Abdumalikov et al., Nature 496, 482 (2013) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Whether geometric gates actually beat optimized dynamical gates in real error budgets is contested — the noise-suppression advantage can be eroded by the longer loop times.

GO DEEPER

- Xiao, Chang & Niu, "Berry phase effects on electronic properties," RMP 82, 1959 (2010)

- Wilczek & Shapere, *Geometric Phases in Physics* (1989)

SOURCES

- Berry, Proc. R. Soc. A 392, 45 (1984). doi:10.1098/rspa.1984.0023
- TKNN, PRL 49, 405 (1982)
- von Klitzing, Dorda & Pepper, PRL 45, 494 (1980). doi:10.1103/PhysRevLett.45.494

Contextuality (Kochen-Specker) · F-contextuality

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Quantum measurement outcomes cannot be assigned pre-existing values independent of which other compatible observables are measured alongside them. The Kochen-Specker theorem (1967) proves this: for a Hilbert space of dimension ≥ 3 , no assignment of definite values (0/1) to all projectors is consistent with the algebraic relations quantum mechanics requires.

Contextuality generalizes Bell nonlocality — Bell inequalities are the special case where the "contexts" are spacelike-separated — and it is the sharper statement of what makes quantum theory non-classical. Spekkens later reformulated it operationally so it applies to noisy, real experiments.

CORE IDEA / KEY EQUATION

The cleanest witness is the **Peres-Mermin square**: a 3×3 grid of nine two-qubit observables, each ± 1 -valued and each commuting with its row-mates and column-mates. Quantum mechanics forces the product of the three observables along every row to be $+1$ and along two columns $+1$ but the third column -1 . Multiply all six line-products together: quantum mechanics gives -1 . Now try to pre-assign a definite value ± 1 to each of the nine boxes independent of context. Each box sits in one row and one column, so it enters the grand product exactly twice, and any ± 1 assigned to it contributes its square, $+1$ — the noncontextual product is forced to be $+1$. The contradiction, $-1 \neq +1$, is state-independent: it holds for every input state, needs no inequality and no entanglement, and no assignment of hidden values escapes it. That algebraic clash is the whole content of Kochen-Specker, compressed to nine boxes.

WHY IT MATTERS FOR QUANTUM TECH

Contextuality is a computational *resource*. Howard, Wallman, Veitch & Emerson (2014) proved that for qudits, contextuality is precisely what "magic states" supply in the leading fault-tolerant model: Clifford operations plus contextuality-free (stabilizer) states are efficiently classically simulable (the Gottesman-Knill

theorem), and the onset of contextuality coincides exactly with the states that magic-state distillation must produce to reach universal quantum computation. This draws a direct line from a 1967 no-go theorem to the resource that error-corrected machines spend most of their overhead manufacturing — magic-state factories are the dominant space and time cost in surface-code architectures (see S-qec, S-logical). The classical-simulability boundary it draws also tells simulator designers exactly which subroutines a classical machine can shadow for free and which demand real quantum hardware (see S-qsim). Contextuality further underlies some certified-randomness and self-testing arguments, where a measured contextuality violation is the certificate that a device is behaving quantumly (see A-qrng, O-verification).

KEY GRADED CLAIMS

- [T1] No non-contextual hidden-variable model reproduces QM in dimension ≥ 3 — Kochen & Specker, J. Math. Mech. 17, 59 (1967) (status: established)
- [T2] Contextuality is the resource enabling universal quantum computation via magic-state distillation (qudit model) — Howard, Wallman, Veitch & Emerson, Nature 510, 351 (2014), arXiv:1401.4174 (status: established)
- [T2] State-independent contextuality demonstrated experimentally, violating a noncontextuality inequality with trapped ions and needing no special input state — Kirchmair et al., Nature 460, 494 (2009), arXiv:0904.1655 (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- The qubit ($d = 2$) case is subtler — Wigner-function negativity and contextuality diverge; the exact resource-theoretic statement for qubits is still refined.

GO DEEPER

- Spekkens, "Contextuality for preparations, transformations, measurements," PRA 71, 052108 (2005)
- Bermejo-Vega et al., PRL 119, 120505 (2017) (contextuality + qubit magic)

SOURCES

- Kochen & Specker (1967)

- Howard et al., Nature 510, 351 (2014). doi:10.1038/nature13460
- Kirchmair et al., Nature 460, 494 (2009). doi:10.1038/nature08172
- Mermin, "Hidden variables and the two theorems of John Bell," RMP 65, 803 (1993) (Peres-Mermin square)

Decoherence & Open Quantum Systems · F-decoher

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

No system is perfectly isolated. When a quantum system couples to its environment, the environment effectively "measures" it: phase relations between superposition components leak into environmental degrees of freedom and become unrecoverable. The reduced state decoheres into an apparent classical mixture in a preferred ("pointer") basis selected by the interaction. Decoherence explains why the everyday world looks classical, and it sets the clock every qubit races against (T1 energy relaxation, T2 dephasing).

CORE IDEA / KEY EQUATION

The state of the system alone is the reduced density matrix $\rho_S = \text{Tr}_E(|\Psi\rangle\langle\Psi|)$, traced over the environment. Coupling entangles system with environment, and the off-diagonal terms of ρ_S (the "coherences" that encode superposition) decay — for a qubit, the $|0\rangle\langle 1|$ element shrinks as $\exp(-t/T_2)$. The continuous-time law for a Markovian environment is the Lindblad (GKS) master equation:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho \} \right)$$

where the jump operators L_k model each channel (relaxation, dephasing). The first term is ordinary unitary evolution; the sum is the irreversible leak. Two rates matter for a qubit: T_1 , the energy-relaxation time ($|1\rangle \rightarrow |0\rangle$), and $T_2 \leq 2 T_1$, the total dephasing time. Decoherence picks out the pointer basis — the states the interaction leaves alone — which for most environments is position/energy, and that is why big warm objects never look superposed.

WHY IT MATTERS FOR QUANTUM TECH

Decoherence is the enemy: it caps circuit depth on NISQ hardware (S-nisq) — you get only $\sim T_2/t_{\text{gate}}$ operations before the state is noise — and dictates the entire architecture of quantum error correction (S-qec), which spreads one logical qubit across many physical ones so syndrome measurements catch errors before they accumulate. Coherence times are the headline spec of every

qubit modality: superconducting transmons and fluxonium (H-supercon) now reach the millisecond range, trapped-ion hyperfine qubits (H-ion) hold coherence for seconds to minutes, and neutral atoms (H-neutral) sit in between. The whole T1/T2 vocabulary comes straight from this section, and pushing those numbers up is the central materials-and-engineering problem of the field.

KEY GRADED CLAIMS

- [T1] Environmental entanglement suppresses interference and selects pointer states — Zeh, Found. Phys. 1, 69 (1970); Zurek, PRD 24, 1516 (1981) and PRD 26, 1862 (1982); review: Zurek, RMP 75, 715 (2003) (status: established)
- [T2] Decoherence of a mesoscopic superposition was watched in real time in cavity QED — Brune et al., PRL 77, 4887 (1996) (Haroche group; 2012 Nobel) (status: demonstrated)
- [T2] Decoherence was turned on as a knob: heating C70 fullerenes to ~3,000 K in a Talbot-Lau interferometer made them emit thermal photons that carry away which-path information, washing out interference fringes in quantitative agreement with theory — Hackermüller, Hornberger, Brezger, Zeilinger, Arndt, Nature 427, 711 (2004), doi:10.1038/nature02276 (status: demonstrated)
- [T2] Coherence-time frontier: a fluxonium qubit reached Ramsey coherence $T_2^* \approx 1.48 \pm 0.13$ ms with single-qubit gate fidelity above 0.9999 — Somoroff et al. (Manucharyan group), PRL 130, 267001 (2023), arXiv:2103.08578 (status: demonstrated)
- [T2] The formal machinery is the theory of open systems: Lindblad/GKS master equations, CPTP maps — Lindblad, Commun. Math. Phys. 48, 119 (1976); Breuer & Petruccione (2002) (status: established)

CONFLICTS / OPEN QUESTIONS

- Decoherence explains the *appearance* of classicality without selecting a single outcome — whether it dissolves or merely reframes the measurement problem is contested (Schlosshauer, Phys. Rep. 831, 1, 2019).

GO DEEPER

- Zurek, "Decoherence, einselection, and the quantum origins of the classical," RMP 75, 715 (2003), arXiv:quant-ph/0105127
- Schlosshauer, *Decoherence and the Quantum-to-Classical Transition* (Springer 2007)

SOURCES

- Zeh (1970) doi:10.1007/BF00708656 · Zurek RMP 75, 715 (2003) doi:10.1103/RevModPhys.75.715
- Brune et al., PRL 77, 4887 (1996). doi:10.1103/PhysRevLett.77.4887
- Hackermüller et al., "Decoherence of matter waves by thermal emission of radiation," Nature 427, 711 (2004). doi:10.1038/nature02276
- Somoroff et al., "Millisecond coherence in a superconducting qubit," PRL 130, 267001 (2023). arXiv:2103.08578

Quantum Discord · F-discord

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Quantum discord measures quantum correlations that survive even when a state has no entanglement. It captures the gap between two expressions for mutual information that are identical in classical probability but diverge for quantum states. Harold Ollivier and Wojciech Zurek introduced it in 2001, with Leah Henderson and Vlatko Vedral independently defining the classical-correlation term the same year. The striking claim: many separable (unentangled) mixed states still carry nonzero discord, so entanglement does not exhaust the ways quantum systems can be correlated.

CORE IDEA / KEY EQUATION

Classically, mutual information can be written two ways — $I(A:B) = H(A) + H(B) - H(A,B)$ and $J(A:B) = H(A) - H(A|B)$ — and they always agree. Quantumly the conditional entropy $H(A|B)$ requires choosing a measurement on B , and the two expressions come apart. Discord is the minimized difference:

$$D(A : B) = I(A : B) - \max_{\{\Pi_k\}} J(A : B)$$

where the maximization runs over local measurements (POVMs) on B and I uses von Neumann entropies. $D \geq 0$ always, and $D = 0$ exactly for states that stay undisturbed by some local measurement on B ("classical-quantum" states). Because the set of zero-discord states is much smaller than the set of separable states, generic separable states have $D > 0$. A concrete example is the two-qubit Werner-like separable mixtures used in the DQC1 "power of one qubit" model, where entanglement is absent yet discord tracks the computational speed-up. Computing discord requires an optimization over measurements, so closed forms exist only for special families (e.g. two-qubit Bell-diagonal states).

WHY IT MATTERS FOR QUANTUM TECH

Discord matters where entanglement is scarce but quantum behavior persists — highly mixed states, room-temperature and noisy platforms. It was proposed as the resource behind the DQC1 model of mixed-state quantum computation,

which shows apparent speed-up with vanishing entanglement (see S-nisq, S-qsim). Discord appears in analyses of quantum metrology, state merging, and remote state preparation, and it connects to the broader map of correlations that includes entanglement and Bell nonlocality (see F-entangle, F-qinfo). For security arguments, the sharper structural constraint comes from entanglement monogamy rather than discord (see A-qkd, F-entropy-ineq).

KEY GRADED CLAIMS

- [T1] Some separable states have strictly positive discord; zero entanglement does not imply classical correlations — Ollivier & Zurek, PRL 88, 017901 (2001); Henderson & Vedral, J. Phys. A 34, 6899 (2001) (status: established)
- [T2] The DQC1 model exhibits computational advantage with negligible entanglement but nonzero discord — Datta, Shaji & Caves, PRL 100, 050502 (2008), arXiv:0709.0548 (status: demonstrated, interpretation debated)
- [T2] Comprehensive framework and measures for discord-type correlations — Modi, Brodutch, Cable, Paterek & Vedral, Rev. Mod. Phys. 84, 1655 (2012), arXiv:1112.6238 (status: established review)

CONFLICTS / OPEN QUESTIONS

- Whether discord is an *operational* resource — one that buys a task no cheaper protocol can — is still contested; several claimed advantages have alternative explanations, and no single agreed resource theory of discord has won out.
- Discord is measurement-optimization-dependent and generally hard to compute (NP-hard in general), so which of the many proposed measures is canonical remains unsettled.

GO DEEPER

- Modi, Brodutch, Cable, Paterek & Vedral, Rev. Mod. Phys. 84, 1655 (2012)
- Ollivier & Zurek, PRL 88, 017901 (2001)

SOURCES

- Ollivier & Zurek, PRL 88, 017901 (2001). doi:10.1103/PhysRevLett.88.017901

- Henderson & Vedral, J. Phys. A 34, 6899 (2001).
doi:10.1088/0305-4470/34/35/315
- Datta, Shaji & Caves, PRL 100, 050502 (2008). arXiv:0709.0548
- Modi et al., Rev. Mod. Phys. 84, 1655 (2012). arXiv:1112.6238

Entanglement & Nonlocality · F-entangle

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Two or more systems are entangled when their joint state cannot be written as a product of individual states — the whole carries information that no part carries alone. Measurement outcomes on separated entangled particles show correlations stronger than any local hidden-variable model allows (quantified by Bell inequalities, see F-bell). Einstein, Podolsky and Rosen flagged it as a paradox in 1935; Schrödinger named it *Verschränkung* the same year and called it *the* characteristic trait of quantum mechanics.

CORE IDEA / KEY EQUATION

A pure two-party state is entangled when $|\Psi\rangle \neq |\varphi\rangle_A \otimes |\chi\rangle_B$ for any single-system states. The canonical example is the singlet Bell state $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$: measure either qubit and you get 0 or 1 at random, but the two results are perfectly anti-correlated no matter how far apart. For mixed states, ρ is separable only if $\rho = \sum_i p_i \rho_A^{(i)} \otimes \rho_B^{(i)}$; anything else is entangled. The amount of entanglement in a pure bipartite state is the entanglement entropy $S = -\text{Tr}(\rho_A \log \rho_A)$, where $\rho_A = \text{Tr}_B(|\Psi\rangle\langle\Psi|)$ is the reduced state of one side — $S = 0$ for a product state and $S = 1$ bit ("one ebit") for a maximally entangled qubit pair. Locally you see only the reduced state, which is why entanglement never lets you signal; the correlations only appear when the two measurement records are later compared.

WHY IT MATTERS FOR QUANTUM TECH

Entanglement is the workhorse resource. A two-qubit gate (S-gates) is the operation that *creates* it — a CZ or CNOT turns a product state into a Bell state, and multipartite entanglement across a register is what a quantum computer *is*. It powers teleportation (F-teleport), entanglement-based QKD (A-qkd, the E91 protocol), and the stabilizer entanglement that error-correcting codes (S-qec) distribute across physical qubits to protect one logical qubit. Every hardware modality is judged partly on how well it entangles: trapped ions (H-ion) via Mølmer-Sørensen gates, superconducting qubits (H-supercon) via tunable couplers, neutral atoms (H-neutral) via Rydberg blockade, photons (H-photonic)

via measurement-induced fusion. Distributing entanglement between distant nodes is the physical layer of the quantum internet.

KEY GRADED CLAIMS

- [T1] Entangled states exist and violate local realism — EPR, Phys. Rev. 47, 777 (1935); Schrödinger, Proc. Camb. Phil. Soc. 31, 555 (1935); experimental record in F-bell (status: established)
- [T1] Entanglement cannot transmit signals — the no-communication theorem keeps quantum correlations consistent with relativity (status: established)
- [T2] Entanglement is a quantifiable, interconvertible resource (entanglement entropy, distillation, monogamy) — Horodecki *et al.*, Rev. Mod. Phys. 81, 865 (2009), arXiv:quant-ph/0702225 (status: established)
- [T1] Entanglement survives distribution over 1,200 km: the Micius satellite delivered polarization-entangled photon pairs (source ~5.9 million pairs/s) to two ground stations 1,203 km apart, violating a Bell inequality by $S = 2.37 \pm 0.09$ under Einstein locality — Yin *et al.*, Science 356, 1140 (2017), doi:10.1126/science.aan3211 (status: established)
- [T2] Many-body entanglement at high fidelity: a 60-atom neutral-atom array ran parallel two-qubit gates at 99.5% fidelity and produced three-qubit GHZ states at 90.9(6)% raw fidelity, above the surface-code threshold — Evered, Bluvstein *et al.* (Lukin group), Nature 622, 268 (2023), doi:10.1038/s41586-023-06481-y (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Deciding whether an arbitrary mixed state is entangled (the separability problem) is NP-hard in general; multipartite entanglement classification is unfinished mathematics.

GO DEEPER

- Horodecki *et al.*, "Quantum entanglement," RMP 81, 865 (2009) — the canonical review
- Nielsen & Chuang §2.6; Preskill lecture notes ch. 4

SOURCES

- Einstein, Podolsky, Rosen, Phys. Rev. 47, 777 (1935). doi:10.1103/PhysRev.47.777
- Schrödinger, Proc. Camb. Phil. Soc. 31, 555 (1935). doi:10.1017/S0305004100013554
- Horodecki, Horodecki, Horodecki, Horodecki, RMP 81, 865 (2009). arXiv:quant-ph/0702225
- Yin et al., "Satellite-based entanglement distribution over 1200 kilometers," Science 356, 1140 (2017). doi:10.1126/science.aan3211
- Evered, Bluvstein et al., "High-fidelity parallel entangling gates on a neutral-atom quantum computer," Nature 622, 268 (2023). doi:10.1038/s41586-023-06481-y

Quantum Entropy Inequalities · F-entropy-ineq

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The structural laws that von Neumann entropy must obey across subsystems. The deepest is strong subadditivity (SSA), proved by Elliott Lieb and Mary Beth Ruskai in 1973 after being conjectured by Robinson, Ruelle, and Lanford in the late 1960s. From SSA follow the monotonicity of relative entropy, the data-processing inequality, the Araki-Lieb triangle bound, and — for entanglement — the Coffman-Kundu-Wootters monogamy inequality. Together they are the arithmetic of how quantum information distributes and degrades, and every capacity theorem and security proof leans on them.

CORE IDEA / KEY EQUATION

Strong subadditivity, for any tripartite state ρ_{ABC} :

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$$

Equivalently, conditioning on more never increases conditional entropy, and the mutual information $I(A : C|B) \geq 0$. This one inequality is the workhorse; several key bounds are corollaries: - **Data processing:** relative entropy is non-increasing under any CPTP channel, $D(\Lambda\rho \parallel \Lambda\sigma) \leq D(\rho \parallel \sigma)$ — you cannot make two states more distinguishable by processing them, so quantum information cannot be increased by local operations. Equivalent in strength to SSA. - **Araki-Lieb (triangle):** $|S(\rho_A) - S(\rho_B)| \leq S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B)$. The lower bound has no classical analog; it forces a global pure state ($S(\rho_{AB}) = 0$) to have $S(\rho_A) = S(\rho_B)$, the entropy of entanglement. - **CKW monogamy:** for three qubits, $C^2(A : B) + C^2(A : C) \leq C^2(A : BC)$ in the concurrence (tangle). Entanglement cannot be freely shared — if A is maximally entangled with B, it has none left for C.

WHY IT MATTERS FOR QUANTUM TECH

Monogamy is the mathematical reason quantum key distribution is secure: an eavesdropper's correlation with the key is bounded by how much the legitimate parties share, so listening in is detectable (see A-qkd). Data processing sets the

ceiling on every quantum channel capacity and underwrites the LSD coherent-information formula and Holevo-type bounds (see F-qinfo). Entropy inequalities gate entanglement distillation and the resource accounting of teleportation and state merging (see F-entangle). In many-body physics and holography, SSA constrains entanglement entropy and area laws (see S-qsim).

KEY GRADED CLAIMS

- [T1] Strong subadditivity holds for von Neumann entropy — Lieb & Ruskai, J. Math. Phys. 14, 1938 (1973) (status: established)
- [T1] Relative entropy is monotone under CPTP maps (data processing), equivalent to SSA — Lindblad, Commun. Math. Phys. 40, 147 (1975); Uhlmann (1977) (status: established)
- [T2] Three-qubit entanglement obeys the CKW monogamy inequality; generalized to $\mathcal{N}$ qubits by Osborne & Verstraete — Coffman, Kundu & Wootters, PRA 61, 052306 (2000), arXiv:quant-ph/9907047; Osborne & Verstraete, PRL 96, 220503 (2006) (status: established)
- [T2] Equality in SSA characterizes quantum Markov chains (recoverable states) — Hayden, Jozsa, Petz & Winter, Commun. Math. Phys. 246, 359 (2004); strengthened by Fawzi & Renner, arXiv:1410.0664 (2015) (status: established)

CONFLICTS / OPEN QUESTIONS

- The remainder-term / approximate-recovery refinements of SSA (how much information a recovery map can restore when SSA is nearly saturated) are still being sharpened.
- Beyond qubits, the exact form of monogamy for general entanglement measures and higher-dimensional systems is not fully settled; monogamy can fail for some correlation measures (e.g. discord is not always monogamous — see F-discord).

GO DEEPER

- Nielsen & Chuang, ch. 11 · Wilde, *Quantum Information Theory* (2013), arXiv:1106.1445
- Lieb & Ruskai, J. Math. Phys. 14, 1938 (1973)

SOURCES

- Lieb & Ruskai, J. Math. Phys. 14, 1938 (1973). doi:10.1063/1.1666274
- Coffman, Kundu & Wootters, PRA 61, 052306 (2000). arXiv:quant-ph/9907047
- Lindblad, Commun. Math. Phys. 40, 147 (1975). doi:10.1007/BF01609396
- Fawzi & Renner, Commun. Math. Phys. 340, 575 (2015). arXiv:1410.0664

Interpretations of Quantum Mechanics · F-interp

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The formalism predicts probabilities perfectly; what it *says about reality* is unsettled. The main families: Copenhagen (Bohr/Heisenberg — the classical measurement context is primitive), many-worlds (Everett 1957 — no collapse, all branches real), de Broglie–Bohm (1952 — particles with definite trajectories guided by the wave), objective collapse (GRW/CSL, Diósi–Penrose — collapse is a physical process, and testable), QBism (states are agents' credences), and relational QM (Rovelli — states are relative to observers). All except objective collapse are empirically identical to standard QM.

CORE IDEA / KEY EQUATION

The whole dispute sits on a mismatch between two rules the standard formalism runs together. Between measurements a state evolves smoothly and deterministically by the Schrödinger equation, $i\hbar \partial_t |\psi\rangle = H|\psi\rangle$, which is linear and never destroys a superposition. At a measurement the Born rule takes over: outcome $\mathcal{A}$ appears with probability $P(a) = |\langle a|\psi\rangle|^2 = \text{Tr}(\rho P_a)$, and the state jumps to the matching eigenstate. Superposition in, one definite outcome out — that discontinuous, probabilistic jump is nowhere in the linear equation, and specifying when and why it happens is the measurement problem. The interpretations are answers to it: many-worlds keeps only the linear equation and denies the jump is real (every outcome happens, on its own branch); de Broglie–Bohm keeps the wave and adds definite particle positions; objective-collapse theories add a small stochastic nonlinear term to the Schrödinger equation so that superpositions of large masses self-destruct on a physical timescale, which makes them the one family with numbers to test. The GRW/CSL collapse rate λ and the Diósi–Penrose gravitational timescale are the parameters experiments try to pin.

WHY IT MATTERS FOR QUANTUM TECH

Mostly none — the no-collapse interpretations agree on every number, so gates, memories, and algorithms behave the same whichever you believe. The exception has teeth: objective-collapse models predict that a superposition of a

large enough mass decoheres on its own, and the platforms that build ever-heavier or ever-more-isolated superpositions are exactly the matter-wave interferometers, levitated nanoparticles, and mechanical oscillators shared with quantum sensing and bosonic hardware (H-bosonic). Each new mass or coherence-time record tightens the bound on the collapse rate. The interpretation debate does not change how you run S-gates or S-qec, but the collapse tests ride on the same isolation-and-readout engineering that H-photonic and sensing programs push.

KEY GRADED CLAIMS

- [T1] All mainstream no-collapse interpretations reproduce identical predictions; no experiment to date distinguishes them (status: established)
- [T2] Objective-collapse models are being squeezed experimentally: an underground test of Diósi–Penrose collapse-induced radiation excluded its natural parameter-free version — Donadi et al., Nat. Phys. 17, 74 (2021), doi:10.1038/s41567-020-1008-4 (status: demonstrated)
- [T2] The superposition principle holds for objects far heavier than atoms: matter-wave interference in a 2-m Talbot–Lau interferometer was shown for oligoporphyrin molecules beyond 25,000 Da made of up to 2,000 atoms — the heaviest objects put into a spatial superposition to date, directly narrowing the window collapse models can occupy — Fein et al., Nat. Phys. 15, 1242 (2019), doi:10.1038/s41567-019-0663-9 (status: demonstrated)
- [T3] Extended Wigner's-friend theorems sharpen what any observer-independent account must give up — Frauchiger & Renner, Nat. Commun. 9, 3711 (2018); Bong et al., Nat. Phys. 16, 1199 (2020) (status: contested)

CONFLICTS / OPEN QUESTIONS

- C-measurement-problem: the field's oldest open question. Physicist polls (e.g. Schlosshauer et al., arXiv:1301.1069) show no consensus — Copenhagen-ish pluralities, growing Everettian and QBist minorities. What would resolve it: a confirmed collapse-model deviation, or nothing ever.

GO DEEPER

- Everett, Rev. Mod. Phys. 29, 454 (1957) · Bohm, Phys. Rev. 85, 166 & 180 (1952) · Ghirardi, Rimini, Weber, PRD 34, 470 (1986)

- Fuchs, Mermin, Schack, Am. J. Phys. 82, 749 (2014) · Rovelli, Int. J. Theor. Phys. 35, 1637 (1996)

SOURCES

- As listed above; Donadi et al. (2021) doi:10.1038/s41567-020-1008-4; Bassi et al., "Models of wave-function collapse," RMP 85, 471 (2013)
- Fein et al., Nat. Phys. 15, 1242 (2019). doi:10.1038/s41567-019-0663-9

Lieb-Robinson Bounds (Emergent Locality) · F-liebrobinson

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Non-relativistic quantum mechanics has no built-in speed limit — the Schrödinger equation is not Lorentz-invariant, and a local perturbation could in principle be felt everywhere at once. Lieb and Robinson (1972) proved that for lattice systems with short-range interactions this does not happen: information propagates at most at a finite emergent velocity, and outside a linear "light cone" correlations are exponentially small. This speed limit is effective rather than fundamental — it emerges from the Hamiltonian's locality — yet it acts like relativity's light cone for spin chains and cold atoms. Hastings and Koma (2006) extended it and used it to prove that a spectral gap forces exponential decay of ground-state correlations, one of the most-used tools in modern many-body physics.

CORE IDEA / KEY EQUATION

For local operators A and B supported on regions separated by distance d , evolving A to time t under a local Hamiltonian gives a commutator bound of the form

$$\|[A(t), B]\| \leq C \|A\| \|B\| \cdot \exp\left(\frac{v|t| - d}{\xi}\right),$$

where v is the Lieb-Robinson velocity, ξ a length scale set by the interaction range, and C depends on the boundary of the supports. The commutator measures how much a probe at B can detect a disturbance at A . It stays exponentially small until $v|t| \approx d$ — the arrival time of the effective light cone — and only then becomes order one.

Plain version: poke one spin, and a spin a distance d away barely notices until a time d/v has passed. A local Hamiltonian — each term touching only nearby sites — is enough to manufacture an approximate causal structure with a group velocity, with no relativity in the theory.

WHY IT MATTERS FOR QUANTUM TECH

Lieb-Robinson velocity sets the clock for how fast entanglement and information spread, bounding how quickly a quantum computer can move data across a chip and how fast entangling gates over distance can be (H-supercon, H-ion, S-gates). It underpins the area law for entanglement entropy and the efficiency of tensor-network (MPS/DMRG) simulation, so it controls what classical machines can and cannot simulate (F-qinfo). The gap-implies-clustering result explains why local error correction and topological order are stable (S-qec, S-logical). It also bounds state-transfer rates in quantum networks and metrology protocols spread over many sites (A-qinternet, A-sensing).

KEY GRADED CLAIMS

- [T1] Short-range lattice Hamiltonians have a finite group velocity; correlations vanish exponentially outside a linear light cone — Lieb & Robinson, Commun. Math. Phys. 28, 251 (1972) (status: established)
- [T1] A nonzero spectral gap implies exponential decay of ground-state correlations, via Lieb-Robinson bounds — Hastings & Koma, Commun. Math. Phys. 265, 781 (2006); Nachtergaele & Sims, Commun. Math. Phys. 265, 119 (2006) (status: established)
- [T2] A light-cone spread of correlations at a finite velocity was observed directly in a quantum gas — Cheneau et al., Nature 481, 484 (2012); trapped-ion light cones, Jurcevic et al. & Richerme et al., Nature 511, 202 & 198 (2014) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- The tight velocity and light-cone shape for long-range (power-law $1/r^\alpha$) interactions are still being pinned down; whether the cone is linear, polynomial, or logarithmic depends on α , and recent work keeps sharpening the boundaries.
- Optimal (tight) Lieb-Robinson velocities for generic interacting models remain open; known bounds are often loose by a constant or more.

GO DEEPER

- Nachtergaele & Sims, "Lieb-Robinson bounds in quantum many-body physics," Contemp. Math. 529, 141 (2010), arXiv:1004.2086

- Chen, Lucas & Yin, "Speed limits and locality in many-body quantum dynamics," Rep. Prog. Phys. 86, 116001 (2023), arXiv:2303.07386

SOURCES

- Lieb & Robinson, Commun. Math. Phys. 28, 251 (1972). doi:10.1007/BF01645779
- Hastings & Koma, Commun. Math. Phys. 265, 781 (2006). doi:10.1007/s00220-006-0030-4
- Cheneau et al., Nature 481, 484 (2012). doi:10.1038/nature10748

Measurement & the Born Rule · F-measure

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Measuring an observable returns one of its eigenvalues, with probability equal to the squared magnitude of the corresponding amplitude — the Born rule. After the measurement the state updates ("collapses") to the outcome's eigenstate (von Neumann's projection postulate), or more generally per the POVM/instrument formalism. The rule is where quantum mechanics touches experiment: everything the theory predicts is a Born-rule probability.

CORE IDEA / KEY EQUATION

$P(a) = |\langle a|\psi\rangle|^2$, the probability of outcome $\mathcal{A}$ when observable A is measured on state $|\psi\rangle$, where $|a\rangle$ is the eigenstate with eigenvalue $\mathcal{A}$. Equivalently the expectation value is $\langle A \rangle = \langle \psi|A|\psi \rangle$, and for a general (mixed) state

$P(a) = \text{Tr}(\rho P_a)$ with P_a the projector onto the $\mathcal{A}$ -eigenspace. The rule is quadratic in the amplitude, which is why it kills the higher-order interference terms a linear or cubic rule would produce — three-path interference must vanish exactly. Gleason's theorem shows this squared-modulus form is forced: on any Hilbert space of dimension ≥ 3 , the only probability measure over projectors that is additive on orthogonal outcomes is $\text{Tr}(\rho P)$. So the "why squared" question has a partial answer — given the Hilbert-space structure and non-contextuality, no other rule is consistent.

WHY IT MATTERS FOR QUANTUM TECH

Every quantum computation ends in measurement — the whole game of algorithm design is arranging interference so the Born-rule statistics concentrate on the answer, which is why Grover (S-grover) and Shor (S-shor) both end with a projective readout that samples the amplified peak. Readout fidelity is a first-class hardware metric: dispersive readout of a transmon (H-supercon) probes the cavity to infer $|0\rangle$ vs $|1\rangle$, and state-dependent fluorescence reads an ion (H-ion) at >99.9%. Error correction (S-qec) runs on repeated *non-destructive* stabilizer measurements — the POVM/instrument formalism, projecting only the syndrome and leaving the logical state intact. Measurement statistics also

underwrite security: QKD (A-qkd) turns an eavesdropper's unavoidable measurement disturbance into a detectable error rate.

KEY GRADED CLAIMS

- [T1] Outcome probabilities are $|\langle a|\psi\rangle|^2$ — Born, Z. Phys. 37, 863 (1926); confirmed in a century of experiments (status: established)
- [T1] The Born rule is the only consistent probability assignment on Hilbert spaces of dimension ≥ 3 given non-contextuality of the measure — Gleason's theorem, J. Math. Mech. 6, 885 (1957) (status: established)
- [T1] Direct test of the rule's quadratic form: a three-slit photon experiment bounds genuine third-order (Sorkin) interference to below $\sim 1\%$ of the pairwise term, $\kappa = 0.0064 \pm 0.0119$, consistent with zero — Sinha, Couteau, Jennewein, Laflamme, Weihs, Science 329, 418 (2010), doi:10.1126/science.1190545 (status: established)
- [T2] Weak measurements trade information gain against disturbance, allowing partial, non-projective readout — Aharonov, Albert & Vaidman, PRL 60, 1351 (1988); single quantum trajectories of a transmon reconstructed under continuous weak monitoring (readout cavity probed with ~ 1 photon, near-quantum-limited parametric amplifier) — Murch, Weber, Macklin, Siddiqi, Nature 502, 211 (2013), doi:10.1038/nature12539 (status: demonstrated)
- [T2] Step-by-step ("progressive") collapse of a cavity field was watched via repeated QND photon-number measurement with non-absorbing atoms — Guerlin et al. (Haroche group), Nature 448, 889 (2007), doi:10.1038/nature06057; 2012 Nobel (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- *Why* the Born rule, and what physically happens during collapse, is the measurement problem — unsolved; interpretations disagree (see F-interp). Derivations (decision-theoretic, envariance) remain contested.

GO DEEPER

- von Neumann, *Mathematical Foundations of Quantum Mechanics* (1932), ch. VI
- Busch, Lahti & Mittelstaedt, *The Quantum Theory of Measurement* (1996)

- Wiseman & Milburn, *Quantum Measurement and Control* (2009)

SOURCES

- Born, "Zur Quantenmechanik der Stoßvorgänge," Z. Phys. 37, 863 (1926). doi:10.1007/BF01397477
- Gleason, J. Math. Mech. 6, 885 (1957)
- Aharonov, Albert, Vaidman, PRL 60, 1351 (1988). doi:10.1103/PhysRevLett.60.1351
- Sinha et al., "Ruling out multi-order interference in quantum mechanics," Science 329, 418 (2010). doi:10.1126/science.1190545
- Murch, Weber, Macklin, Siddiqi, "Observing single quantum trajectories of a superconducting quantum bit," Nature 502, 211 (2013). doi:10.1038/nature12539
- Guerlin et al., "Progressive field-state collapse and quantum non-demolition photon counting," Nature 448, 889 (2007). doi:10.1038/nature06057

No-Cloning Theorem · F-nocloning

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

No physical process can make a perfect copy of an unknown quantum state. The proof is three lines: a universal copier would have to act linearly, and linearity forbids cloning superpositions of states it can clone individually. Proved independently by Wootters & Zurek and by Dieks in 1982 (with an earlier, little-noticed version by Park in 1970), partly in response to a proposed FTL-communication scheme that cloning would have enabled.

CORE IDEA / KEY EQUATION

Suppose a unitary U and a blank register $|b\rangle$ could copy any state:

$U(|\psi\rangle|b\rangle) = |\psi\rangle|\psi\rangle$ for every $|\psi\rangle$. Apply it to two different states,

$U(|\psi\rangle|b\rangle) = |\psi\rangle|\psi\rangle$ and $U(|\varphi\rangle|b\rangle) = |\varphi\rangle|\varphi\rangle$. Unitaries preserve inner products, so taking the overlap of the two input equations against the two output equations gives $\langle\psi|\varphi\rangle = \langle\psi|\varphi\rangle^2$. A number equal to its own square is 0 or 1, which forces the states to be either identical or orthogonal. A machine cannot clone a set that contains any non-orthogonal pair — in particular it cannot clone the two poles and the equator of the Bloch sphere at once. The same one-line argument, run with density matrices, gives no-broadcasting for mixed states. What survives is approximate cloning: the best universal $1 \rightarrow 2$ machine produces two copies each with fidelity $F = \langle\psi|\rho_{\text{out}}|\psi\rangle = 5/6 \approx 0.833$, a ceiling set by the same linearity that forbids the perfect copy.

WHY IT MATTERS FOR QUANTUM TECH

No-cloning is the root of quantum cryptography: an eavesdropper cannot copy qubits in flight without disturbing them, which is what BB84 and E91 detect and is the physical assumption behind the security proofs in S-qkd $\rightarrow$ A-qkd. It also forbids naive backup of quantum data — the reason error correction (S-qec) has to protect information without reading or copying it, spreading one logical qubit across many physical ones instead of duplicating it. It blocks classical-style signal amplification, so long-haul links on photonic hardware (H-photonic) need quantum repeaters and entanglement swapping rather than repeaters that copy-

and-amplify. The $5/6$ optimal-cloner bound is also the yardstick against which any real intercept-resend attack on deployed QKD is measured.

KEY GRADED CLAIMS

- [T1] An unknown pure state cannot be perfectly copied — Wootters & Zurek, Nature 299, 802 (1982); Dieks, Phys. Lett. A 92, 271 (1982) (status: established)
- [T1] Corollaries: no-broadcasting for mixed states (Barnum et al., PRL 76, 2818, 1996) and no-deleting (Pati & Braunstein, Nature 404, 164, 2000) (status: established)
- [T2] Imperfect cloning is possible up to a tight bound — the optimal universal $1 \rightarrow 2$ cloner reaches fidelity $5/6 \approx 0.833$: Bužek & Hillery, PRA 54, 1844 (1996) (status: established)
- [T2] The $5/6$ bound was reached in the lab: single-photon cloning by stimulated parametric down-conversion produced clones at near-optimal fidelity, universal across input states — Lamas-Linares, Simon, Howell & Bouwmeester, Science 296, 712 (2002), arXiv:quant-ph/0205149 (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- None at the theorem level; the live work is in how close practical eavesdropping/cloning attacks get to the optimal-cloner bound in deployed QKD (see A-qkd).

GO DEEPER

- Scarani et al., "Quantum cloning," RMP 77, 1225 (2005)
- Nielsen & Chuang, Box 12.1

SOURCES

- Wootters & Zurek, Nature 299, 802 (1982). doi:10.1038/299802a0
- Dieks, Phys. Lett. A 92, 271 (1982). doi:10.1016/0375-9601(82)90084-6
- Bužek & Hillery, PRA 54, 1844 (1996). doi:10.1103/PhysRevA.54.1844
- Lamas-Linares et al., Science 296, 712 (2002). arXiv:quant-ph/0205149

The PBR Theorem (Reality of the Quantum State) · F-pbr

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The PBR theorem (Pusey, Barrett & Rudolph, 2012) is a no-go result about what the wavefunction is. Foundations splits interpretations into ψ -ontic (the quantum state is a real property of an individual system) and ψ -epistemic (the state is only information about some deeper underlying "ontic" state λ , so two different states could correspond to the same reality). PBR proves that ψ -epistemic models are ruled out, given one natural assumption: systems prepared independently have independent physical states (preparation independence). Under that assumption, distinct pure quantum states must correspond to disjoint sets of ontic states — the state is real. The framework is the ontological-models formalism of Harrigan & Spekkens (2010).

CORE IDEA / KEY EQUATION

Model each preparation of $|\psi\rangle$ as sampling a hidden ontic state λ from a distribution $\mu_\psi(\lambda)$. " ψ -epistemic" means two non-orthogonal states can overlap: there exist $|\psi_0\rangle, |\psi_1\rangle$ with μ_0 and μ_1 sharing support. PBR take $\mathcal{N}$ independent copies, each prepared in $|\psi_0\rangle$ or $|\psi_1\rangle$, so the joint ontic state factorizes (preparation independence). If the single-system distributions overlap with probability $q > 0$, then with probability $q^{\mathcal{N}}$ all $\mathcal{N}$ systems land in the shared region and the model cannot tell which product state was prepared. PBR construct an entangled measurement on the $\mathcal{N}$ copies with an outcome that quantum mechanics assigns probability zero to every prepared product state. A model reproducing those zeros cannot have overlapping μ 's — so distinct pure states are ontologically distinct.

Plain version: if the wavefunction were just our ignorance about a deeper reality, two different wavefunctions could secretly be the same underlying state. PBR show that a joint measurement on independently prepared copies would then sometimes give an outcome quantum theory forbids. Assuming independent preparations are independent, the wavefunction has to be a real feature of the system.

WHY IT MATTERS FOR QUANTUM TECH

PBR sets the terms for how we reason about the state as a resource (F-interp, F-qinfo). It joins Bell nonlocality (F-bell) and Kochen-Specker contextuality (F-contextuality) as the third pillar of quantum no-go theorems, together fencing off classical hidden-variable accounts. The ψ -ontic conclusion supports treating unknown states as carrying real, uncopyable information — the ground under no-cloning and the security proofs of QKD (F-nocloning, A-qkd), and under certified-randomness arguments where the state's reality constrains what an adversary can know (A-qrng, O-verification).

KEY GRADED CLAIMS

- [T2] Under preparation independence, no ψ -epistemic model reproduces quantum statistics; distinct pure states are ontologically distinct — Pusey, Barrett & Rudolph, Nat. Phys. 8, 475 (2012), arXiv:1111.3328 (status: established, given the assumption)
- [T2] Without preparation independence, ψ -epistemic models can still reproduce all of quantum theory in dimension $d \geq 2$ — Lewis, Jennings, Barrett & Rudolph, Phys. Rev. Lett. 109, 150404 (2012) (status: established)
- [T3] Experimental tests bound the extent of ontic overlap, constraining epistemic explanations of indistinguishability — Ringbauer et al., Nat. Phys. 11, 249 (2015), arXiv:1412.6213 (status: demonstrated, model-dependent)

CONFLICTS / OPEN QUESTIONS

- The whole result hinges on preparation independence; drop it and epistemic models survive (Lewis et al. 2012), so critics argue PBR constrains a subclass of models rather than closing the question.
- The ontic/epistemic dichotomy assumes an underlying realist λ exists at all — interpretations that reject that framing (relational QM, QBism, Everett) sit outside PBR's target.

GO DEEPER

- Harrigan & Spekkens, "Einstein, incompleteness, and the epistemic view of quantum states," Found. Phys. 40, 125 (2010), arXiv:0706.2661

- Leifer, "Is the quantum state real? An extended review of ψ -ontology theorems," Quanta 3, 67 (2014), arXiv:1409.1570

SOURCES

- Pusey, Barrett & Rudolph, Nat. Phys. 8, 475 (2012). doi:10.1038/nphys2309, arXiv:1111.3328
- Lewis, Jennings, Barrett & Rudolph, Phys. Rev. Lett. 109, 150404 (2012). doi:10.1103/PhysRevLett.109.150404
- Ringbauer et al., Nat. Phys. 11, 249 (2015). doi:10.1038/nphys3233

Generalized Measurement (POVMs, Naimark, Instruments) · F-povm

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

A POVM (positive operator-valued measure) is the general description of what a quantum measurement can be. Projective (von Neumann) measurement is the special case where the outcome operators are orthogonal projectors. Real detectors — lossy, noisy, coupled to ancillas — are POVMs. Naimark's dilation theorem (1940) proves that every POVM is a projective measurement on a larger space: couple the system to an ancilla, measure sharply there, and the induced statistics on the system reproduce the POVM. A **quantum instrument** upgrades this to also give the post-measurement state, and **weak measurement** (Aharonov, Albert & Vaidman, 1988) is the regime of vanishing coupling where disturbance shrinks and the "weak value" can fall outside the eigenvalue spectrum.

CORE IDEA / KEY EQUATION

A POVM is a set of positive operators $\{E_k\}$, $E_k \geq 0$, with $\sum_k E_k = I$. Outcome k occurs with probability $p(k) = \text{Tr}(\rho E_k)$. Each $E_k = M_k^\dagger M_k$ factors through measurement (Kraus) operators M_k ; the post-measurement state is $\rho \rightarrow M_k \rho M_k^\dagger / p(k)$, and the collection $\{M_k\}$ defines the instrument. Naimark: for any POVM $\{E_k\}$ on $\mathcal{H}$ there is a larger space $\mathcal{H} \otimes \mathcal{H}_{\text{anc}}$, a state of the ancilla, and a projective measurement $\{P_k\}$ such that

$$\text{Tr}(\rho E_k) = \text{Tr}((\rho \otimes |a\rangle\langle a|) P_k).$$

In plain terms: any fuzzy or partial readout you can build in the lab equals a perfect textbook measurement performed on your system plus a helper system you brought along. Nothing exotic happens — the extra outcomes and the softened back-action are bookkeeping for the ancilla you traced out.

WHY IT MATTERS FOR QUANTUM TECH

POVMs are the operational language of every real measurement in the stack. Dispersive readout of a superconducting qubit (H-supercon) and fluorescence readout of a trapped ion (H-ion) are POVMs with finite fidelity, and modeling

them as POVMs is what makes readout-error mitigation and soft-information decoding possible (S-decoders, S-qec). Photon-counting detectors are POVMs (H-photonic). Unambiguous state discrimination, optimal measurements for QKD, and tomography all live in the POVM formalism (A-qkd, O-tomography). Weak measurement and continuous monitoring drive quantum feedback and metrology (A-sensing). Naimark dilation is the measurement-side twin of the Stinespring dilation for channels (F-purification), and both feed the decoherence account of how classical outcomes emerge (F-decoher, F-measure).

KEY GRADED CLAIMS

- [T1] Every POVM is realized by a projective measurement on a system-plus-ancilla space (Naimark dilation) — Naimark 1940; textbook treatment Nielsen & Chuang §2.2.8 (status: established)
- [T2] Weak measurement with pre- and post-selection yields weak values outside the eigenvalue range (proposed value 100 for a spin-1/2) — Aharonov, Albert & Vaidman, Phys. Rev. Lett. 60, 1351 (1988) (status: established, experimentally demonstrated)
- [T2] Weak-value amplification measured a beam deflection of ~560 femtoradians — Hosten & Kwiat, Science 319, 787 (2008) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Whether weak values are a real physical quantity or a statistical artifact of post-selection is still debated; their utility for metrological amplification against technical noise is also contested.
- The instrument (which post-measurement state) is not fixed by the POVM alone — the same $\{E_k\}$ admits many Kraus decompositions, so the physical implementation carries extra content.

GO DEEPER

- Nielsen & Chuang, *Quantum Computation and Quantum Information*, §2.2.6–2.2.8 (POVMs, Naimark)
- Wiseman & Milburn, *Quantum Measurement and Control* (2010), Ch. 1 (instruments, continuous measurement)

SOURCES

- Naimark, Izv. Akad. Nauk SSSR Ser. Mat. 4, 277 (1940)
- Aharonov, Albert & Vaidman, Phys. Rev. Lett. 60, 1351 (1988).
doi:10.1103/PhysRevLett.60.1351
- Hosten & Kwiat, Science 319, 787 (2008). doi:10.1126/science.1152697

Purification & Stinespring Dilation · F-purification

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Purification says every mixed state is the shadow of a pure state living on a larger space: for any density matrix ρ_A there is a pure state $|\psi\rangle_{AB}$ on a joint system such that $\rho_A = \text{Tr}_B |\psi\rangle\langle\psi|$. Stinespring's dilation theorem (1955) is the channel-level version: every physical evolution — every completely positive trace-preserving (CPTP) map — is a unitary (isometry) into a bigger space followed by discarding the added part. Together they are the formal content of what John Smolin nicknamed the "church of the larger Hilbert space": mixedness and irreversibility are never fundamental; they are what remains after you throw away access to a purifying partner. Uhlmann (1976) sharpened purification into a formula for fidelity between mixed states.

CORE IDEA / KEY EQUATION

Purification: given $\rho_A = \sum_i p_i |i\rangle\langle i|$, define $|\psi\rangle_{AB} = \sum_i \sqrt{p_i} |i\rangle_A |i\rangle_B$ on $\mathcal{H}_A \otimes \mathcal{H}_B$ with $\dim \mathcal{H}_B \geq \text{rank } \rho_A$; then $\text{Tr}_B |\psi\rangle\langle\psi| = \rho_A$. The purification is unique up to a unitary on B .

Stinespring: any CPTP map $\mathcal{E}$ on $\mathcal{H}_A$ can be written $\mathcal{E}(\rho) = \text{Tr}_E [V \rho V^\dagger]$, where $V : \mathcal{H}_A \rightarrow \mathcal{H}_A \otimes \mathcal{H}_E$ is an isometry ($V^\dagger V = I$) into a system-plus-environment space. Equivalently $\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$ with $\sum_k K_k^\dagger K_k = I$ (Kraus form); the K_k are the isometry's components along an environment basis. Minimal dilation dimension is the Kraus rank, $\leq (\dim \mathcal{H}_A)^2$.

Plain version: any noisy or lossy process you run in the lab is a clean reversible rotation on your system plus a hidden meter, with the meter then ignored. Noise is entanglement with something you stopped tracking.

WHY IT MATTERS FOR QUANTUM TECH

This is the backbone of quantum error correction and open-system engineering. Modeling decoherence as a CPTP channel (via Stinespring/Kraus) is how amplitude damping, dephasing, and depolarizing noise get simulated and how threshold theorems are proved (S-qec, F-decoher). Purification underlies

entanglement measures, thermofield-double states, and the environment "monitoring" picture of einselection (F-decoher, F-qinfo). It is the channel twin of Naimark dilation for measurements (F-povm) and the engine behind teleportation-based error correction and remote state preparation (F-teleport). Randomized benchmarking and process tomography assume the Kraus/Stinespring structure (O-tomography). In quantum thermodynamics and black-hole information debates, "purify the environment" is the standard move (F-interp).

KEY GRADED CLAIMS

- [T1] Every CPTP map equals an isometry into a larger space followed by partial trace (Stinespring dilation) — Stinespring, Proc. Amer. Math. Soc. 6, 211 (1955) (status: established)
- [T1] Every mixed state has a pure purification on a doubled space, unique up to a unitary on the ancilla — standard corollary; textbook Nielsen & Chuang §2.5 (status: established)
- [T2] Fidelity between mixed states equals the maximal overlap over all purifications — Uhlmann, Rep. Math. Phys. 9, 273 (1976) (status: established)

CONFLICTS / OPEN QUESTIONS

- None mathematically. The interpretive question is whether the purifying partner is a real physical system or a formal device; the "church of the larger Hilbert space" is a modeling stance, and Stinespring's non-uniqueness means many physical environments realize the same channel.

GO DEEPER

- Nielsen & Chuang, §2.5 (purification) and §8.2 (quantum operations, Kraus/Stinespring)
- Paulsen, *Completely Bounded Maps and Operator Algebras* (2002), Ch. 4 (Stinespring, rigorous)

SOURCES

- Stinespring, Proc. Amer. Math. Soc. 6, 211 (1955). doi:10.1090/S0002-9939-1955-0069403-4

- Uhlmann, Rep. Math. Phys. 9, 273 (1976).
doi:10.1016/0034-4877(76)90060-4
- Choi, Linear Algebra Appl. 10, 285 (1975).
doi:10.1016/0024-3795(75)90075-0

QFT / Second-Quantization Bridge · F-qft

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Quantum field theory promotes fields, and the particles are their quanta: second quantization replaces "a wavefunction for N particles" with creation/annihilation operators acting on Fock space, letting particle number change and making quantum mechanics compatible with special relativity. Dirac quantized the electromagnetic field in 1927, giving the photon a rigorous home; QED, and later the Standard Model, followed. QFT is the deepest experimentally confirmed layer of physics.

CORE IDEA / KEY EQUATION

Second quantization writes each field mode as a quantum harmonic oscillator and reads its excitations as particles. A free field expands as

$\varphi(x) = \sum_k [a_k f_k(x) + a_k^\dagger f_k^*(x)]$, where $a_k^\dagger$ creates a quantum in mode k and a_k destroys one, obeying the commutator $[a_k, a_{k'}^\dagger] = \delta_{k,k'}$ for bosons (and

anticommutators for fermions, which forces the Pauli exclusion principle). The number operator $N_k = a_k^\dagger a_k$ counts quanta in mode k , and states live in Fock space built by acting with creation operators on the vacuum $|0\rangle$:

$|n\rangle = (a^\dagger)^n / \sqrt{n!} |0\rangle$. The whole point is that particle number is now a dynamical variable — an interaction term in the Hamiltonian can turn one quantum into three — which is what a fixed- N wavefunction could never describe and what relativistic processes (pair creation, emission, decay) demand. The single-mode oscillator relation $E_n = \hbar\omega(n + \frac{1}{2})$ is the seed of everything, from the photon to the transmon.

WHY IT MATTERS FOR QUANTUM TECH

Photonic quantum computing *is* applied field quantization: the qubits and resources are Fock states, squeezed states, and coherent states of light modes, which is the native language of H-photonic and of the bosonic codes in H-bosonic. Circuit QED treats a microwave resonator as a quantized field mode coupled to an artificial atom, and it is the theoretical backbone of superconducting transmon processors (H-supercon), where the same $a/a^\dagger$

algebra sets qubit frequencies and dispersive readout in S-gates and S-bench. Simulating interacting field theories — scattering amplitudes, lattice gauge dynamics — is a headline target for fault-tolerant machines (S-qec), one of the few applications with a proven super-polynomial speedup argument.

KEY GRADED CLAIMS

- [T1] The quantized EM field: emission/absorption from field quanta — Dirac, Proc. R. Soc. A 114, 243 (1927) (status: established)
- [T1] QED is the most precisely verified physical theory: the electron g-2 measurement (0.13 ppt precision) agrees with QED prediction — Fan et al., PRL 130, 071801 (2023), doi:10.1103/PhysRevLett.130.071801 (status: established)
- [T1] The muon magnetic anomaly is now measured to 127 ppb, the most precise muon g-2 result and one of the sharpest QFT tests: the Fermilab Muon g-2 final result (Runs 1–6) agrees with the revised Standard Model prediction — Muon g-2 Collaboration, final result announced June 2025 (submitted to PRL), improving on the 2021/2023 measurements (status: established)
- [T2] Circuit QED — superconducting circuits realizing quantized field–atom physics — is the theoretical backbone of transmon processors: Blais et al., RMP 93, 025005 (2021), arXiv:2005.12667 (status: established)
- [T3] Efficient quantum algorithms exist for simulating scattering in interacting QFTs — Jordan, Lee & Preskill, Science 336, 1130 (2012), arXiv:1111.3633 (status: demonstrated in theory)

CONFLICTS / OPEN QUESTIONS

- Rigorous mathematical construction of interacting 4D QFT (Yang–Mills mass gap) is an open Clay Millennium Problem; QFT + gravity remains unreconciled.
- The muon g-2 anomaly is unresolved rather than closed: the experimental value is now far more precise than theory, and the tension depends on which Standard Model hadronic-vacuum-polarization input (data-driven vs. lattice QCD) is used.

GO DEEPER

- Peskin & Schroeder, *An Introduction to QFT* · Weinberg, *The Quantum Theory of Fields* vol. 1

SOURCES

- Dirac (1927) doi:10.1098/rspa.1927.0039 · Fan et al. (2023) doi:10.1103/PhysRevLett.130.071801
- Blais et al., RMP 93, 025005 (2021). arXiv:2005.12667
- Fermilab Muon g-2 Collaboration, final result (127 ppb), June 2025 (submitted to PRL); prior: PRL 126, 141801 (2021) and PRL 131, 161802 (2023)

Quantum Information Theory · F-qinfo

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The Shannon theory of quantum states: how much quantum information a state carries (von Neumann entropy $S = -\text{Tr } \rho \log \rho$), how well it can be compressed (Schumacher coding), how much classical information quantum states can convey (Holevo bound), and what noisy quantum channels can transmit (quantum capacity). Fidelity, trace distance, and CPTP channels are the working vocabulary of every hardware benchmark.

CORE IDEA / KEY EQUATION

The central quantity is the von Neumann entropy $S(\rho) = -\text{Tr}(\rho \log \rho)$, which is Shannon's $H = -\sum p \log p$ rewritten with the density matrix's eigenvalues as the probabilities. It counts, in qubits, the irreducible size of a quantum source: Schumacher's theorem says N copies of ρ compress into about $N \cdot S(\rho)$ qubits and no fewer, the quantum echo of Shannon's compression bound. When you want to read *classical* messages out of quantum states, the ceiling is the Holevo χ quantity: for an ensemble $\{p_i, \rho_i\}$ with average state $\rho = \sum_i p_i \rho_i$, the accessible information obeys $I \leq \chi = S(\rho) - \sum_i p_i S(\rho_i)$. A corollary is blunt — $\mathcal{N}$ qubits carry at most $\mathcal{N}$ classical bits of retrievable message. For sending quantum states down a noisy channel $\mathcal{N}$, the analogue of Shannon capacity is the coherent information $I_c(\rho, \mathcal{N}) = S(\mathcal{N}(\rho)) - S((\mathcal{N} \otimes I)(|\psi\rangle\langle\psi|))$, and the quantum capacity is its regularized maximum (the LSD theorem). Regularization is the sting in the tail: unlike the classical case, you generally cannot compute the capacity from a single use of the channel.

WHY IT MATTERS FOR QUANTUM TECH

Channel capacities set the hard limits for quantum communication and the quantum internet, and they are why long links on photonic hardware (H-photonic) need entanglement distillation and repeaters rather than amplification (see F-nocloning). Fidelity and trace distance are how every gate, memory, and teleportation demo is scored across H-supercon, H-ion, and H-neutral, and they are the figures of merit that S-bench randomized benchmarking and S-qec

threshold estimates report. The Holevo bound underwrites the key-rate accounting in QKD security proofs (S-qkd $\rightarrow$ A-qkd). Von Neumann entropy is also the bookkeeping unit for entanglement as a resource, which is the currency spent by teleportation, superdense coding, and the whole S-gates $\rightarrow$ S-qec stack.

KEY GRADED CLAIMS

- [T1] A quantum source is compressible to $S(\rho)$ qubits per signal — Schumacher, PRA 51, 2738 (1995); entropy formalism from von Neumann (1932) (status: established)
- [T1] n qubits cannot convey more than n classical bits — Holevo, Probl. Inf. Transm. 9, 177 (1973) (status: established)
- [T2] Quantum channel capacity is given by regularized coherent information (LSD theorem) — Lloyd, PRA 55, 1613 (1997); Shor (2002); Devetak, IEEE TIT 51, 44 (2005) (status: established)
- [T2] Channel capacities are strange: additivity fails (Hastings, Nat. Phys. 5, 255, 2009) and quantum capacity is superactivatable — two zero-capacity channels can combine to positive capacity (Smith & Yard, Science 321, 1812, 2008) (status: demonstrated)
- [T2] Quantum information survives transmission at continental scale: ground-to-satellite teleportation of single-photon qubits over a 1,400 km uplink to the Micius satellite reached average fidelity 0.80 ± 0.01 across six mutually unbiased input states, above the $2/3$ classical state-estimation limit — Ren et al., Nature 549, 70 (2017), arXiv:1707.00934 (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- No closed-form, single-letter formula for the quantum capacity of a general channel; whether it is even computable in practice remains open.

GO DEEPER

- Nielsen & Chuang chs. 11–12; Wilde, *Quantum Information Theory* (2013), arXiv:1106.1445
- Preskill notes ch. 10

SOURCES

- Holevo (1973); Schumacher, PRA 51, 2738 (1995) doi:10.1103/PhysRevA.51.2738
- Devetak, IEEE Trans. Inf. Theory 51, 44 (2005). arXiv:quant-ph/0304127
- Hastings, Nat. Phys. 5, 255 (2009). arXiv:0809.3972 · Smith & Yard, Science 321, 1812 (2008). arXiv:0807.4935
- Ren et al., Nature 549, 70 (2017). arXiv:1707.00934

Quantum Thermodynamics · F-qthermo

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Thermodynamics rebuilt for single quantum systems, where work and heat are stochastic, information is physical, and coherence itself enters the energy books.

Anchors: Landauer's principle (erasing one bit costs at least $kT \ln 2$),

Bennett's resolution of Maxwell's demon via erasure cost, fluctuation theorems (Jarzynski, Crooks) that hold arbitrarily far from equilibrium, and resource-theoretic "second laws" for quantum states.

CORE IDEA / KEY EQUATION

The founding law is **Landauer's principle**: erasing one bit of information forces the environment to absorb at least $kT \ln 2$ of heat — about 3×10^{-21} J, or 0.018 eV, at room temperature ($T = 300$ K, k = Boltzmann's constant).

Reversible computation dodges this floor; the cost lands only when information is discarded. That is how Bennett exorcised Maxwell's demon: the demon must eventually reset its memory, and that erasure pays back exactly the work it appeared to conjure. Away from equilibrium the accounting sharpens into the **Jarzynski equality**, $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$, with $\beta = 1/kT$. Read it plainly: take the stochastic work W done in any drive of the system — fast or slow, gentle or violent — repeat the drive many times, average $e^{-\beta W}$ over those runs, and the result equals $e^{-\beta \Delta F}$. A free-energy difference, an equilibrium quantity, is recovered from irreversible non-equilibrium pulls. Crooks' theorem refines this into a symmetry between the forward and reverse work distributions.

WHY IT MATTERS FOR QUANTUM TECH

Sets the energetic floor of computation and the physics of cooling: dilution refrigerators, algorithmic cooling, and the (open) question of the true energy cost per logical qubit of a fault-tolerant machine. Landauer's $kT \ln 2$ is the hard floor beneath every erasure in a quantum error-correction cycle — each syndrome reset and each measurement discard has a thermodynamic price that scales with the code's overhead (see S-qec, S-logical). Trapped-ion and

superconducting platforms run their whole logic inside milli-kelvin baths, so the wall-plug cost of pumping that heat out dominates system power budgets (see H-ion, H-supercon). Quantum heat engines and absorption refrigerators are candidate on-chip coolers for keeping qubits near their ground state, and fluctuation theorems give the estimation tools for reading free energies out of the noisy, driven dynamics that quantum simulators produce (see S-qsim).

KEY GRADED CLAIMS

- [T1] Erasure of one bit dissipates $\geq kT \ln 2$ — Landauer, IBM J. Res. Dev. 5, 183 (1961); demon exorcised by erasure cost, Bennett, Int. J. Theor. Phys. 21, 905 (1982) (status: established)
- [T2] Landauer's bound verified experimentally — classically with a colloidal particle (Bérut et al., Nature 483, 187, 2012) and in the quantum regime with a trapped ion (Yan et al., PRL 120, 210601, 2018) (status: demonstrated)
- [T2] Single-atom heat engine operated with one trapped ion — Roßnagel et al., Science 352, 325 (2016), doi:10.1126/science.aad6320 (status: demonstrated)
- [T2] Quantum coherence measurably increases the power output of a microscopic engine in the small-action regime — Klatzow et al., PRL 122, 110601 (2019), NV-center working fluid in diamond (status: demonstrated)
- [T2] Fluctuation theorems: $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$ — Jarzynski, PRL 78, 2690 (1997); quantum versions established (status: established)

CONFLICTS / OPEN QUESTIONS

- Definitions of work and heat for coherent quantum systems remain multiple and inequivalent; whether quantum coherence/entanglement give thermodynamic advantage in engines is case-by-case, still contested.

GO DEEPER

- Goold et al., "The role of quantum information in thermodynamics," J. Phys. A 49, 143001 (2016), arXiv:1505.07835
- Vinjanampathy & Anders, Contemp. Phys. 57, 545 (2016), arXiv:1508.06099

SOURCES

- Landauer (1961) doi:10.1147/rd.53.0183 · Jarzynski (1997) doi:10.1103/PhysRevLett.78.2690
- Roßnagel et al., Science 352, 325 (2016) · Yan et al., PRL 120, 210601 (2018)
- Klatzow et al., PRL 122, 110601 (2019). doi:10.1103/PhysRevLett.122.110601

The Qubit & Bloch Sphere · F-qubit

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The qubit is the unit of quantum information: any two-level quantum system, whose pure states are $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. Up to global phase, every pure qubit state maps to a point on the surface of a unit sphere — the Bloch sphere — with mixed states filling the interior. Single-qubit gates are rotations of this sphere. The term "qubit" was coined by Benjamin Schumacher in 1995; the geometric picture descends from Bloch's treatment of spin- $1/2$ and the Poincaré sphere for polarization.

CORE IDEA / KEY EQUATION

A pure qubit state is $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$, with $0 \leq \theta \leq \pi$ and $0 \leq \varphi < 2\pi$. The two angles (θ, φ) are the polar and azimuthal coordinates of a point on a unit sphere, so every pure state is one point on the surface. The general (possibly mixed) state is the density matrix $\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$, where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices and $\mathbf{r}$ is the Bloch vector; $|\mathbf{r}| = 1$ is the pure-state surface and $|\mathbf{r}| < 1$ is the mixed interior, with $\mathbf{r} = \mathbf{0}$ the maximally mixed center. This works because a 2×2 Hermitian, unit-trace, positive matrix has exactly three real free parameters, the components of $\mathbf{r}$. A unitary gate acts as $U\rho U^\dagger$ and rotates $\mathbf{r}$ rigidly about some axis — the group is $SU(2)$, the double cover of the rotation group $SO(3)$, which is why a 2π turn of the Bloch vector corresponds to a 4π turn in $SU(2)$ and picks up a physical sign. Two real numbers describe the state; one global phase is unobservable and drops out.

WHY IT MATTERS FOR QUANTUM TECH

The qubit is the abstraction that makes hardware interchangeable: transmons (H-supercon), trapped ions (H-ion), photons (H-photonic), neutral atoms (H-neutral), and spins in silicon (H-silicon) all implement the same Bloch-sphere object, so algorithms and error correction are written once against it. Every single-qubit gate in S-gates is a named Bloch rotation (X, Y, Z, Hadamard, T), and a gate's fidelity is literally the accuracy of that rotation, extracted by

randomized benchmarking in S-bench. Error correction (S-qec) is built on the same picture — a logical qubit is a Bloch sphere protected inside many physical ones. Continuous-variable hardware breaks the two-level frame and uses an infinite-dimensional mode instead (H-bosonic).

KEY GRADED CLAIMS

- [T1] Any two-level system's pure states are parameterized by two real angles (θ, φ) on a sphere; unitaries are $SU(2)$ rotations — Bloch, Phys. Rev. 70, 460 (1946); Feynman, Vernon & Hellwarth, J. Appl. Phys. 28, 49 (1957) (status: established)
- [T1] "Qubit" and the qubit as the quantum information unit: Schumacher, "Quantum coding," PRA 51, 2738 (1995) (status: established)
- [T2] What a physical system needs to serve as a qubit register is codified in the DiVincenzo criteria — DiVincenzo, Fortschr. Phys. 48, 771 (2000), arXiv:quant-ph/0002077 (status: established)
- [T2] A single physical qubit can be controlled at the six-nines level: a $^{43}\text{Ca}^+$ trapped-ion hyperfine "clock" qubit reached average single-qubit gate fidelity 99.9999% (error $\approx 1 \times 10^{-6}$), memory coherence $T_2^* = 50$ s, and combined prep+readout 99.93%, all in a room-temperature surface trap — Harty et al., PRL 113, 220501 (2014), arXiv:1403.1524 (status: demonstrated)
- [T2] The record has since improved by an order of magnitude: single-qubit gate error $1.5(4) \times 10^{-7}$ in a trapped-ion qubit — Oxford group, arXiv:2412.04421 (2024) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Qudits (d-level) and continuous-variable/bosonic encodings compete with the two-level abstraction in some hardware (see H-bosonic); which unit wins per modality is an engineering question, still open.

GO DEEPER

- Nielsen & Chuang §1.2, §4.2; Preskill notes ch. 2

SOURCES

- Schumacher, PRA 51, 2738 (1995). doi:10.1103/PhysRevA.51.2738
- DiVincenzo (2000). arXiv:quant-ph/0002077

- Bloch, Phys. Rev. 70, 460 (1946). doi:10.1103/PhysRev.70.460
- Harty et al., PRL 113, 220501 (2014). arXiv:1403.1524 · Oxford single-qubit 10^{-7} gate, arXiv:2412.04421 (2024)

Stabilizer Formalism & Gottesman-Knill · F-stabilizer

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

The stabilizer formalism describes a large, structured class of quantum states not by writing out 2^n amplitudes but by listing the operators that leave them fixed. A stabilizer state on n qubits is the unique common $+1$ eigenstate of a commuting group of n independent Pauli operators. Daniel Gottesman developed the formalism in his 1997 PhD thesis, building on the Heisenberg picture. It is the mathematical backbone of quantum error correction and, through the Gottesman-Knill theorem, marks a sharp boundary between quantum circuits that classical computers can simulate and those they cannot.

CORE IDEA / KEY EQUATION

A state $|\psi\rangle$ is stabilized by a Pauli operator S when $S|\psi\rangle = |\psi\rangle$. Fix a group $\mathcal{S} = \langle S_1, \dots, S_n \rangle$ of n commuting, independent Pauli operators (excluding $-I$); their joint $+1$ eigenspace is a single state — the stabilizer state. This is exponential compression: n generators, each specified by $\sim 2n$ bits, pin down a state that would otherwise need 2^n complex amplitudes.

Gottesman-Knill theorem: a circuit built from state preparations in the computational basis, Clifford gates (Hadamard, phase S , and CNOT), and measurements in the Pauli basis can be simulated efficiently on a classical computer. The reason is that Clifford gates map Pauli operators to Pauli operators under conjugation, so instead of tracking the state you track how a set of $\sim n$ Pauli generators transforms — a polynomial-size bookkeeping problem. Aaronson & Gottesman (2004) sharpened this to an $O(n^2)$ -per-step tableau algorithm (their CHP program handles thousands of qubits) and proved the simulation task is complete for the class $\oplus L$, evidence that stabilizer circuits are not even universal for classical computation, never mind quantum.

WHY IT MATTERS FOR QUANTUM TECH

Every mainstream quantum error-correcting code is a stabilizer code: the surface code, color codes, and the toric code all define the protected logical subspace

as the joint +1 eigenspace of a set of stabilizer generators, and error detection is just measuring those generators to read a syndrome (see S-qec, S-logical). Gottesman-Knill also defines what a quantum computer must add to escape classical simulability: Clifford operations are free, and the non-Clifford resource — magic states, T gates, Wigner negativity, contextuality — is what has to be distilled at great cost (see F-contextuality, F-wigner). Stabilizer simulation is the standard tool for validating error-correction circuits and estimating logical error rates on near-term devices (see S-nisq, S-qsim).

KEY GRADED CLAIMS

- [T1] Clifford circuits on stabilizer states with Pauli measurements are efficiently classically simulable (Gottesman-Knill) — Gottesman, PhD thesis, arXiv:quant-ph/9705052 (1997); Heisenberg-representation form arXiv:quant-ph/9807006 (1998) (status: established)
- [T1] Stabilizer circuits simulate in $O(n^2)$ per gate and the problem is $\oplus L$ -complete — Aaronson & Gottesman, PRA 70, 052328 (2004), arXiv:quant-ph/0406196 (status: established)
- [T2] Adding any single non-Clifford gate (e.g. T) to the Clifford group yields a universal gate set — Boykin et al., 2000; standard result (status: established)

CONFLICTS / OPEN QUESTIONS

- Where exactly the classical-to-quantum boundary sits as non-Clifford resources are added is quantified by stabilizer rank and magic monotones, and tight bounds remain an active research target.
- Efficient simulation degrades gracefully with a few T gates (extended stabilizer / stabilizer-decomposition methods); the precise scaling frontier is still being pushed.

GO DEEPER

- Gottesman, "Stabilizer Codes and Quantum Error Correction," arXiv:quant-ph/9705052 (1997)
- Nielsen & Chuang, §10.5 (stabilizer formalism and codes)
- Aaronson & Gottesman, PRA 70, 052328 (2004)

SOURCES

- Gottesman, arXiv:quant-ph/9705052 (1997) · arXiv:quant-ph/9807006 (1998)
- Aaronson & Gottesman, PRA 70, 052328 (2004). doi:10.1103/PhysRevA.70.052328 · arXiv:quant-ph/0406196

Identical Particles & Spin-Statistics · F-statistics

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Identical quantum particles are indistinguishable: the joint state must be symmetric or antisymmetric under exchange of any two. The spin-statistics theorem (Fierz 1939, Pauli 1940, from relativistic QFT) fixes which: integer-spin particles are **bosons** (symmetric, can share a state) and half-integer-spin particles are **fermions** (antisymmetric, forbidden from sharing — the Pauli exclusion principle). Exchanging two particles multiplies the wavefunction by $+1$ (bosons) or -1 (fermions). In two dimensions the exchange phase can take any value $e^{i\theta}$, giving **anyons**, and non-abelian anyons carry a whole unitary matrix.

CORE IDEA / KEY EQUATION

The rule is a single sign on exchange: $\psi(r_1, r_2) = \pm \psi(r_2, r_1)$, with $+1$ for **bosons** and -1 for **fermions**. The minus sign is the whole Pauli principle — put two fermions in the same single-particle state and $\psi = -\psi$ forces $\psi = 0$, so that state cannot be occupied twice. Written as occupation numbers, the average filling of a level of energy $\mathcal{E}$ follows the two distributions

$n(\varepsilon) = 1/(e^{\beta(\varepsilon-\mu)} \mp 1)$: the minus sign (Bose-Einstein) lets occupation diverge and pile particles into the ground state (condensation), the plus sign (Fermi-Dirac) caps every level at one and stacks fermions up to the Fermi energy. In two dimensions the exchange is no longer a bare sign but a phase $e^{i\theta}$ that can sit anywhere between the boson value $\theta = 0$ and the fermion value $\theta = \pi$ — that is what "any-on" names. For a fractional quantum Hall state at filling $\nu = 1/m$ the quasiparticles carry charge e/m and exchange phase $\theta = \pi/m$. Non-abelian anyons replace the phase with a unitary matrix, so exchanging them rotates the state inside a degenerate subspace.

WHY IT MATTERS FOR QUANTUM TECH

Exchange antisymmetry is why atoms have shell structure and why chemistry exists at all — so any quantum simulation of molecules or materials must enforce fermionic statistics. Mapping fermionic modes onto qubits (Jordan-

Wigner, Bravyi-Kitaev transforms) is a core primitive of quantum chemistry algorithms, and the Jordan-Wigner string overhead is a real cost driver in the circuit depth of those simulations (see S-qsim, S-variational, I-pharma, I-chem). Bosonic statistics underlie photonic quantum computing, where the Hong-Ou-Mandel effect (two indistinguishable photons bunching at a beam splitter) is the basic two-photon gate ingredient, and boson sampling is a near-term quantum-advantage benchmark (see H-photonic, S-bench). Non-abelian anyons are the entire premise of topological quantum computing: braiding them enacts fault-tolerant gates protected by topology, which is what Microsoft's Majorana program pursues (see H-topo, F-berry).

KEY GRADED CLAIMS

- [T1] Integer spin $\rightarrow$ symmetric states (bosons); half-integer spin $\rightarrow$ antisymmetric states (fermions), forced by relativistic locality — Pauli, Phys. Rev. 58, 716 (1940) (status: established)
- [T1] Two-dimensional systems admit anyonic exchange statistics — Leinaas & Myrheim (1977); Wilczek (1982); confirmed via fractional quantum Hall interferometry, Nakamura et al., Nature Physics 16, 931 (2020) (status: established/demonstrated)
- [T2] Anyon collider directly measures fractional exchange statistics at filling $\nu = 1/3$ — anyon-anyon collisions give an exchange phase $\theta \approx \pi/3$, matching the predicted $1/3$ statistics — Bartolomei et al., Science 368, 173 (2020) (status: demonstrated)
- [T2] Bose-Einstein condensation realizes bosons macroscopically sharing one quantum state — ~2000 rubidium-87 atoms condensed at ~170 nK, Anderson et al., Science 269, 198 (1995), Nobel 2001 (status: established)

CONFLICTS / OPEN QUESTIONS

- Non-abelian anyons (Majorana zero modes) as computational objects remain contested at the device level — see C-majorana-existence (evidence/CONFLICTS.md) and H-topo.

GO DEEPER

- Wilczek, "Quantum Mechanics of Fractional-Spin Particles," PRL 49, 957 (1982)

- Nayak et al., "Non-Abelian anyons and topological quantum computation," RMP 80, 1083 (2008)

SOURCES

- Pauli, Phys. Rev. 58, 716 (1940). doi:10.1103/PhysRev.58.716
- Nakamura et al., Nature Physics 16, 931 (2020)
- Bartolomei et al., Science 368, 173 (2020). doi:10.1126/science.aaz5601
- Anderson, Ensher, Matthews, Wieman & Cornell, Science 269, 198 (1995). doi:10.1126/science.269.5221.198

Superposition & the State Vector · F-superpos

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

A quantum system is described by a state vector $|\psi\rangle$ in a Hilbert space, and any normalized linear combination of valid states is itself a valid state. Before measurement the system holds all components of the combination at once, with complex amplitudes; the amplitudes interfere, which is what separates quantum from classical probability. This is the first postulate of quantum mechanics and the raw resource behind everything downstream.

CORE IDEA / KEY EQUATION

$|\psi\rangle = \sum_i c_i |i\rangle$ with complex amplitudes c_i obeying $\sum_i |c_i|^2 = 1$. For a single qubit this is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$ — a point on the Bloch sphere (see F-qubit). The content is Dirac's superposition principle: add any two allowed states with complex weights and you get another allowed state. The weights carry a phase, and phases add and cancel — that is interference. A classical mixture sums probabilities; a quantum superposition sums amplitudes and squares the total (see the Born rule, F-measure), so cross terms $2 \operatorname{Re}(c_i^* c_j)$ appear that have no classical analog. Everything quantum — interference fringes, entanglement, algorithmic speedup — traces back to this one linearity axiom.

WHY IT MATTERS FOR QUANTUM TECH

An n -qubit register in superposition spans 2^n amplitudes simultaneously — the state space quantum algorithms compute in. A physical qubit realizes $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ as two circulating supercurrent directions on a transmon (H-supercon), two hyperfine levels of an ion (H-ion), or two Rydberg/ground states of a neutral atom (H-neutral). Interference between amplitudes is how algorithms extract answers: Shor's algorithm (S-shor) routes the right period into a constructive peak of the quantum Fourier transform (F-qft), and Grover's search (S-grover) amplifies the marked amplitude by repeated reflection. Single-qubit gates (S-gates) are exactly the operations that rotate a superposition on the Bloch sphere. The frontier question — how large a superposition survives — is

set by decoherence (F-decoher), which is why coherence time bounds every architecture.

KEY GRADED CLAIMS

- [T1] Physical states superpose linearly; interference of single particles is observed directly — Schrödinger, *Ann. Phys.* 384, 361 (1926); Dirac, *Principles of Quantum Mechanics* (1930); single-electron double-slit buildup, Tonomura et al., *Am. J. Phys.* 57, 117 (1989) (status: established)
- [T1] Matter-wave interference holds for whole molecules: C60 buckyballs (720 amu, de Broglie wavelength ~ 2.5 pm, $\sim 400\times$ smaller than the molecule) diffracted through a 100 nm-period grating — Arndt, Nairz, Vos-Andreae, Keller, van der Zouw, Zeilinger, *Nature* 401, 680 (1999), doi:10.1038/44348 (status: established)
- [T2] Superposition persists in ever-larger objects: matter-wave interference demonstrated for functionalized oligoporphyrin molecules beyond 25 kDa ($\sim 2,000$ atoms, $\sim 2\times 10^4$ amu) in a Talbot-Lau interferometer — Fein et al., *Nat. Phys.* 15, 1242 (2019), doi:10.1038/s41567-019-0663-9 (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- Where (if anywhere) superposition breaks for macroscopic objects is the empirical frontier — objective-collapse models predict a scale limit; interferometry keeps pushing it upward (see F-interp, F-decoher).

GO DEEPER

- Dirac, *Principles of QM*, ch. 1 (the superposition principle stated as the axiom)
- Nielsen & Chuang, *Quantum Computation and Quantum Information*, §2.1
- Arndt & Hornberger, *Nat. Phys.* 10, 271 (2014) — testing the limits review

SOURCES

- Fein et al., "Quantum superposition of molecules beyond 25 kDa," *Nat. Phys.* 15, 1242–1245 (2019). doi:10.1038/s41567-019-0663-9
- Arndt et al., "Wave–particle duality of C60 molecules," *Nature* 401, 680–682 (1999). doi:10.1038/44348

- Tonomura et al., Am. J. Phys. 57, 117 (1989). doi:10.1119/1.16104

Teleportation & Entanglement Swapping · F-teleport

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Quantum teleportation transfers an unknown state from Alice to Bob using one shared entangled (EPR) pair and two classical bits — no quantum system travels between them, and the original is destroyed (consistent with no-cloning, see F-nocloning). Alice performs a joint Bell-state measurement on her qubit and her half of the pair, sends the two-bit outcome, and Bob applies one of four Pauli corrections to recover the exact state. Bennett, Brassard, Crépeau, Jozsa, Peres & Wootters proposed it in 1993. **Entanglement swapping** (Żukowski, Zeilinger, Horne & Ekert, 1993) is teleportation applied to a member of another entangled pair, so two particles that never interacted end up entangled — the primitive that lets a repeater stitch short entangled links into a long one.

CORE IDEA / KEY EQUATION

The identity that makes it work is a rewriting of three qubits. Alice holds the unknown state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and one half of a shared Bell pair; Bob holds the other half. Expand the joint three-qubit state in the **Bell basis** of Alice's two qubits and it factors exactly into four terms:

$|\psi\rangle|\Phi^+\rangle_{AB} = \frac{1}{2} \sum_i (\text{Alice's Bell state})_i \otimes (U_i|\psi\rangle)_{\text{Bob}}$, where the four U_i are the Pauli operators I, X, Z, XZ . Alice's Bell measurement collapses the sum to one term at random and tells her which i occurred (two classical bits). Bob's qubit is then $U_i|\psi\rangle$, so applying U_i^{-1} — one of I, X, Z , or XZ — recovers $|\psi\rangle$ exactly. The two classical bits are mandatory: before Bob hears them his qubit is the maximally mixed state $I/2$, carrying no information, which is why teleportation transmits nothing faster than light and never clones (α and β are never measured, only moved). Entanglement swapping is the same identity with $|\psi\rangle$ itself half of another Bell pair.

WHY IT MATTERS FOR QUANTUM TECH

Teleportation is the workhorse of quantum networking and modular computing. Entanglement swapping is how quantum repeaters beat photon loss to build a long-range quantum internet (see A-qinternet). Teleporting gates ("gate teleportation") moves logical information between error-correction blocks and is

how magic states are consumed in fault-tolerant circuits — the T-gate that universalizes the Clifford group is applied by teleporting through a magic state (see S-logical, S-qec, F-contextuality). Chip-to-chip and fridge-to-fridge links in modular architectures rely on teleported entanglement (see H-intercon, O-interconnect-loss), and the same photonic Bell-measurement primitive is the fusion operation in measurement-based photonic computing (see H-photonic). Distributed multi-node quantum computing over ion or NV registers stitches processors together by teleporting logical qubits between them (see H-ion). First photonic demonstrations came in 1997 (Innsbruck; Rome).

KEY GRADED CLAIMS

- [T1] An unknown state is teleported with one EPR pair + two classical bits — Bennett et al., PRL 70, 1895 (1993) (status: established)
- [T1] Entanglement swapping entangles independent, never-interacting particles — Żukowski, Zeilinger, Horne & Ekert, PRL 71, 4287 (1993) (status: established)
- [T2] Deterministic teleportation between distant nodes demonstrated — Pfaff et al., Science 345, 532 (2014, NV centers 3 m); ground-to-satellite, Ren et al., Nature 549, 70 (2017) (status: demonstrated)
- [T2] Free-space teleportation over 143 km with active feed-forward, fidelity above the $2/3$ classical bound — Ma et al., Nature 489, 269 (2012), La Palma to Tenerife (status: demonstrated)
- [T2] Teleportation between non-neighboring nodes of a three-node quantum network, using entanglement swapping on the middle node plus a memory qubit — Hermans et al., Nature 605, 663 (2022), NV-center registers in Delft (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- None foundational. Engineering frontier is fidelity and rate over lossy links and quantum memories (see A-qmemory-hw, O-interconnect-loss).

GO DEEPER

- Bennett & Wiesner dense coding, PRL 69, 2881 (1992) (dual protocol)
- Pirandola et al., "Advances in quantum teleportation," Nature Photonics 9, 641 (2015)

SOURCES

- Bennett et al., PRL 70, 1895 (1993). doi:10.1103/PhysRevLett.70.1895
- Żukowski et al., PRL 71, 4287 (1993). doi:10.1103/PhysRevLett.71.4287
- Ma et al., Nature 489, 269 (2012). doi:10.1038/nature11472
- Hermans et al., Nature 605, 663 (2022). doi:10.1038/s41586-022-04697-y

Quantum Tunneling · F-tunneling

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

A particle described by a wavefunction has nonzero amplitude on the far side of an energy barrier it could not classically climb, so it can appear there with finite probability. The transmission amplitude falls off exponentially with barrier width and the square root of barrier height. Tunneling was worked out in 1928 by Gamow, and independently Gurney & Condon, to explain alpha decay — the alpha particle escapes the nucleus by tunneling through the Coulomb barrier — and by Fowler & Nordheim for field emission. It is a direct consequence of the Schrödinger equation, with no classical analogue.

CORE IDEA / KEY EQUATION

Inside a barrier taller than the particle's energy, the Schrödinger equation gives an evanescent (decaying) wave rather than an oscillating one, so the transmission probability through a barrier of height V_0 and width d for a particle of energy E is $T \approx e^{-2\kappa d}$, with the decay constant $\kappa = \sqrt{2m(V_0 - E)}/\hbar$. Two features do the work: the dependence is **exponential in width** (double the barrier and the tunneling rate does not halve — it drops by a huge factor), and κ scales with the **square root of the barrier height** above the particle's energy. This exponential sensitivity is why alpha-decay half-lives span more than twenty orders of magnitude for a narrow range of decay energies (the Geiger-Nuttall law), and why a scanning tunneling microscope resolves single atoms: the tunneling current changes by about a decade for every 1 Å the tip moves. The Josephson junction is the coherent version, carried by the tunneling amplitude of Cooper pairs rather than the incoherent rate of single particles.

WHY IT MATTERS FOR QUANTUM TECH

Tunneling is the physical mechanism inside the Josephson junction: Cooper pairs tunnel coherently across a thin insulating barrier, and the resulting nonlinear inductance is what makes a transmon or flux qubit a usable anharmonic oscillator (see H-supercon). The same $I = I_c \sin \varphi$ relation sets the qubit frequency, the coupling to readout resonators, and the parametric amplifiers that read the qubits out. It is also the readout physics of the scanning

tunneling microscope (Binnig & Rohrer, 1981 Nobel 1986) used to image and place the individual dopants and defects that seed some qubit platforms, the operating principle of flash-memory Fowler-Nordheim programming, and the leakage that limits classical transistor scaling. In quantum annealing, tunneling through (rather than thermal hopping over) energy barriers is the proposed source of any advantage (see H-anneal, F-adiabatic), and tunneling matrix elements govern the reaction and electron-transfer rates that molecular quantum simulators aim to compute (see S-qsim, I-chem, I-pharma).

KEY GRADED CLAIMS

- [T1] A quantum particle transmits through a classically forbidden barrier with exponentially suppressed probability — Gamow (1928); Gurney & Condon, *Nature* 122, 439 (1928) (status: established)
- [T1] Cooper-pair tunneling across a Josephson junction obeys $I = I_c \sin \varphi$, $d\varphi/dt = 2eV/\hbar$ — Josephson, *Phys. Lett.* 1, 251 (1962), Nobel 1973 (status: established)
- [T2] The scanning tunneling microscope resolves single atoms because the tunneling current decays ~ 1 order of magnitude per 1 Å of tip-sample gap, giving ~ 0.1 nm lateral and ~ 0.01 nm vertical resolution — Binnig, Rohrer, Gerber & Weibel, *PRL* 49, 57 (1982), Nobel 1986 (status: established)
- [T3] The tunneling delay time in strong-field ionization of atomic hydrogen is zero within an uncertainty of 12 attoseconds (attoclock) — Sainadh et al., *Nature* 568, 75 (2019) (status: demonstrated, interpretation debated)

CONFLICTS / OPEN QUESTIONS

- Tunneling *time* — how long a particle "spends" in the barrier — remains debated (Hartman effect, attoclock experiments); no impact on device physics.

GO DEEPER

- Razavy, *Quantum Theory of Tunneling* (2003)
- Tinkham, *Introduction to Superconductivity*, ch. 6 (Josephson effects)

SOURCES

- Gamow, *Z. Phys.* 51, 204 (1928)

- Josephson, Phys. Lett. 1, 251 (1962). doi:10.1016/0031-9163(62)91369-0
- Binnig, Rohrer, Gerber & Weibel, PRL 49, 57 (1982). doi:10.1103/PhysRevLett.49.57 · Nobel Prize in Physics 1986
- Sainadh et al., Nature 568, 75 (2019). doi:10.1038/s41586-019-1028-3

Uncertainty & Complementarity · F-uncertainty

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

Conjugate observables (position/momentum, phase/number, orthogonal spin axes) cannot both have sharp values in the same state: $\Delta x \Delta p \geq \hbar/2$.

Heisenberg found the physics in 1927; Kennard and Robertson made it a theorem about state preparation — the bound follows from the operators failing to commute, with no reference to measurement clumsiness. Bohr's complementarity is the companion idea: wave and particle descriptions are mutually exclusive but jointly necessary, and which one shows up depends on the experimental arrangement.

CORE IDEA / KEY EQUATION

The sharp statement is Robertson's: for any two observables A and B in any state, $\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$, where ΔA is the standard deviation and $[A, B] = AB - BA$ is the commutator. For position and momentum $[x, p] = i\hbar$, so the right side is $\hbar/2$ and you recover $\Delta x \Delta p \geq \hbar/2$. The content is that the bound comes from the algebra — non-commuting operators cannot share a full set of eigenstates, so no state makes both spreads zero. This is a fact about how states are prepared, with no clumsy-measurement story attached; the Cauchy–Schwarz inequality applied to the two operators acting on the state is the whole proof. The entropic version replaces variances (which behave badly for periodic or discrete observables) with Shannon entropies of the measurement outcomes: $H(A) + H(B) \geq \log(1/c)$, where c is the largest squared overlap between the two observables' eigenbases. The entropic form is state-independent and is the one QKD proofs actually use.

WHY IT MATTERS FOR QUANTUM TECH

Uncertainty relations set the noise floor of quantum sensing — the standard quantum limit — and they define what squeezing can and cannot buy: pushing noise below the limit in one quadrature by paying it back in the conjugate one. This is the physics LIGO's squeezed-light injection exploits, and the same trade governs bosonic sensing and error correction (H-bosonic). The entropic form underwrites the security proofs behind QKD (S-qkd $\rightarrow$ A-qkd), where an

eavesdropper's information about one basis is bounded by their ignorance of the conjugate basis. Complementarity — the impossibility of reading which-path information without killing the interference — is the operational lever behind interferometric sensors and photonic protocols (H-photonic).

KEY GRADED CLAIMS

- [T1] Preparation uncertainty: $\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$ — Heisenberg, Z. Phys. 43, 172 (1927); Kennard, Z. Phys. 44, 326 (1927); Robertson, Phys. Rev. 34, 163 (1929) (status: established)
- [T1] Entropic form: $H(A) + H(B) \geq \log(1/c)$ — Maassen & Uffink, PRL 60, 1103 (1988); extended with quantum memory, Berta et al., Nat. Phys. 6, 659 (2010) (status: established)
- [T2/T3] Measurement error–disturbance versions of the relation are formalization-dependent: Ozawa's reformulation (PRA 67, 042105, 2003) was supported by neutron/photon experiments, while Busch–Lahti–Werner (PRL 111, 160405, 2013) prove a Heisenberg-type bound under different error definitions (status: contested)
- [T2] Squeezing beats the quantum noise floor in a working instrument: frequency-dependent squeezed vacuum injected into the LIGO detectors during observing run O4 cut quantum noise by 5.8 dB (a factor of 1.9) at Livingston and 4.0 dB at Hanford, surpassing the standard quantum limit by up to 3 dB in the 35–75 Hz band — LIGO O4 Detector Collaboration, "Broadband Quantum Enhancement of the LIGO Detectors with Frequency-Dependent Squeezing," PRX 13, 041021 (2023) (status: demonstrated)

CONFLICTS / OPEN QUESTIONS

- The error–disturbance dispute is about which operational definition of "measurement error" is the right one — both camps' theorems are correct under their own definitions.

GO DEEPER

- Coles et al., "Entropic uncertainty relations and their applications," RMP 89, 015002 (2017)

SOURCES

- Heisenberg (1927) doi:10.1007/BF01397280 · Robertson (1929) doi:10.1103/PhysRev.34.163
- Ozawa, PRA 67, 042105 (2003) · Busch, Lahti, Werner, PRL 111, 160405 (2013). arXiv:1306.1565
- LIGO O4 Detector Collaboration, PRX 13, 041021 (2023). doi:10.1103/PhysRevX.13.041021

Wigner Functions & Quasiprobability · F-wigner

Layer: L0 Foundations · **Chapter:** §01 · **Status:** depth-complete

WHAT IT IS

A Wigner function $W(x,p)$ represents a quantum state as a real distribution over phase space, the closest quantum analog of a classical probability density. Eugene Wigner introduced it in 1932 to compute quantum corrections to classical statistical mechanics. Its defining feature: unlike a genuine probability density, W can take negative values. Those negative regions are the phase-space signature of nonclassicality. The construction generalizes to finite-dimensional (qudit) systems through the discrete Wigner function of Gross and Wootters, where the same negativity plays the same role.

CORE IDEA / KEY EQUATION

$$W(x, p) = \frac{1}{\pi \hbar} \int \langle x + y | \rho | x - y \rangle e^{-2ipy/\hbar} dy$$

a real function whose marginals recover the correct position and momentum distributions, so integrating W reproduces every measurable statistic. What breaks the classical picture is that $W(x,p) < 0$ in some regions for most states.

Hudson's theorem (1974) pins down the boundary: a pure continuous-variable state has a non-negative Wigner function if and only if it is a Gaussian (coherent, squeezed, or thermal-pure) state. Everything with negativity — Fock states, cat states, GKP states — sits outside that Gaussian island. In the odd-prime-dimension discrete case (Gross 2006), the non-negative pure states are exactly the stabilizer states, tying Wigner positivity to the stabilizer formalism (see F-stabilizer). Negativity is then a resource: Wigner-non-negative states, Gaussian/Clifford operations, and positive measurements are efficiently classically simulable, so any quantum speed-up must inject negativity somewhere.

WHY IT MATTERS FOR QUANTUM TECH

Wigner negativity is the continuous-variable currency of quantum advantage. In photonic and bosonic hardware (see H-photonic, H-bosonic), Gaussian states and operations alone are classically simulable; a non-Gaussian, Wigner-negative element (photon subtraction, a cubic phase gate, a GKP state) is what

lifts the platform to universality. In the qudit circuit model, negativity of the discrete Wigner function is the same resource that magic-state distillation must manufacture (see F-contextuality, S-qec, S-logical). Wigner negativity is also a practical witness for benchmarking non-classical state preparation on NISQ and bosonic devices (see S-nisq, S-qsim).

KEY GRADED CLAIMS

- [T1] A pure state has a non-negative Wigner function iff it is Gaussian — Hudson, Rep. Math. Phys. 6, 249 (1974); qudit extension Gross, J. Math. Phys. 47, 122107 (2006), arXiv:quant-ph/0602001 (status: established)
- [T2] Wigner negativity is a necessary resource for super-polynomial quantum speed-up; non-negative states + operations are efficiently classically simulable — Mari & Eisert, PRL 109, 230503 (2012), arXiv:1208.3660; Veitch, Ferrie, Gross & Emerson, New J. Phys. 14, 113011 (2012), arXiv:1201.1256 (status: established)
- [T2] In odd prime dimension, non-negative discrete Wigner functions characterize exactly the stabilizer states — Gross (2006) (status: established)

CONFLICTS / OPEN QUESTIONS

- The qubit ($d = 2$) case is subtle: the standard discrete Wigner construction fails to make stabilizer states non-negative, so the clean "negativity = magic" statement holds cleanly only for odd dimension. Rebit and qubit refinements tie negativity to contextuality instead (see F-contextuality).
- Whether Wigner negativity, contextuality, and magic are the *same* resource or merely coincident measures is still being sharpened.

GO DEEPER

- Veitch, Ferrie, Gross & Emerson, New J. Phys. 14, 113011 (2012)
- Kenfack & Życzkowski, "Negativity of the Wigner function as an indicator of nonclassicality," J. Opt. B 6, 396 (2004)

SOURCES

- Wigner, Phys. Rev. 40, 749 (1932). doi:10.1103/PhysRev.40.749

- Hudson, Rep. Math. Phys. 6, 249 (1974).
doi:10.1016/0034-4877(74)90007-X
- Gross, J. Math. Phys. 47, 122107 (2006). arXiv:quant-ph/0602001
- Veitch et al., New J. Phys. 14, 113011 (2012). arXiv:1201.1256 · Mari & Eisert, PRL 109, 230503 (2012). arXiv:1208.3660

§02 • Hardware

Every qubit modality and the machines around them — superconducting, ions, photonics, neutral atoms, silicon, topological, and the supply chain.

Quantum annealing · H-anneal

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

A special-purpose analog approach that rests directly on the adiabatic theorem (**F-adiabatic**): encode an optimization problem in the couplings of a network of superconducting flux qubits, initialize in the ground state of a simple transverse-field Hamiltonian, then slowly interpolate to the problem Hamiltonian so the system stays near its instantaneous ground state and settles into a (near-)minimum-energy configuration. There are no gates and no universal computation — the machine natively solves Ising/QUBO problems and samples from hard distributions. It is not a gate-model computer wearing a costume; it is a distinct model of computation, and one company defines the commercial category.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **D-Wave:** Advantage2 reached general availability May 2025 — 4,400+ qubits on the Zephyr topology (20-way qubit connectivity), with higher per-qubit coherence and a larger energy scale than Advantage1 (5,000+ qubits, Pegasus, 15-way). Delivered via the Leap cloud; hybrid solvers push problems up to millions of variables by splitting work between the QPU and classical heuristics. Gate-model annealing-adjacent research (coherent annealing) is also on D-Wave's roadmap.
- **Advantage claim:** D-Wave's March 2025 Science paper reported quantum-annealer simulation of quantum magnetic phase-transition dynamics (spin glasses) it argued was beyond reach of classical supercomputers. Within days, groups in Switzerland (EPFL) and the US posted arXiv counter-papers reproducing key results with tensor-network / belief-propagation methods on classical hardware; D-Wave rebutted that the classical methods only cover part of the parameter regime and time scales. Per the schema's default, the advantage stays **contested**.

KEY GRADED CLAIMS

- [T4] Advantage2 GA: 4,400+ qubits, Zephyr, higher coherence — D-Wave release + whitepaper (claimed)

- [T2/contested] Quantum advantage in spin-glass dynamics simulation — Science (Mar 2025); classical tensor-network counterattacks on arXiv (contested)
- [T5] Commercial utility of annealing for production optimization — vendor case studies (claimed; few third-party-audited head-to-head wins)

TRADE-OFFS (VS OTHER MODALITIES)

Thousands of qubits available today and real paying deployments years ahead of any gate-model rival — the modality shipped a usable product first. Against that: it is restricted to optimization/sampling problem classes, has no error correction and no threshold theorem, shows no proven scaling of *solution quality* with qubit count, and every headline speedup so far has been at least partially matched by tuned classical heuristics (simulated/parallel tempering, tensor networks).

CONFLICTS / OPEN QUESTIONS

C-anneal-advantage (also **C-dwave-advantage**): does any annealing workload hold a durable quantum advantage once classical heuristics are tuned against it? Resolution would be a widely reproduced benchmark where the annealer wins on time-to-solution at fixed solution quality against best-known classical methods, audited by third parties — which has not yet happened.

SOURCES

dwavequantum.com GA release + 4,400-qubit whitepaper; Science (Mar 2025); EPFL/US arXiv counter-papers; Physics World + Scientific American + HPCwire contest coverage.

Bosonic / cat qubits · H-bosonic

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Qubits encoded in the many-level state space of a harmonic oscillator (a superconducting cavity or resonator mode) instead of a two-level system, so a single physical mode carries redundancy that a normal qubit does not. **Cat qubits** store information in superpositions of two coherent states $|\alpha\rangle$ and $|\!-\!\alpha\rangle$; a two-photon drive stabilizes them so that bit-flips are suppressed *exponentially in the photon number* $|\alpha|^2$, leaving only phase-flips for a simple 1D repetition code to catch — turning a 2D error-correction problem into a 1D one. GKP (grid) and binomial codes are sibling encodings that protect against small displacements. The goal across all of them: collapse the physical-per-logical overhead that makes the surface code so brutal.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Alice & Bob** (Paris): Boson 4 chip reached >1 hour bit-flip lifetime (Sep 2025, up from 7 minutes in 2024); the "squeezed cat" scheme cut correction cost (2025); the "Elevator Codes" preprint (Jan 2026) claims a ~15:1 physical:logical ratio at ~10,000× lower logical error. Raised €100M (Jan 2025); roadmap: 100 logical qubits by 2030.
- **AWS**: Ocelot chip (Nature, Feb 2025) — first integrated multi-cat-qubit chip combining cat stabilization with a distance-5 repetition code; bit-flip times approaching 1 s; AWS claims up to ~90% QEC-overhead reduction vs surface codes for a target logical error rate.
- **Nord Quantique** (Sherbrooke): multimode bosonic (GKP/binomial) encoding in 3D cavities; demonstrated error correction beyond break-even and pitches high logical density per physical unit — a single mode doing the work of a small code block.
- Yale (the modality's academic origin — Devoret/Schoelkopf lineage) continues to set GKP and cavity-coherence records.

KEY GRADED CLAIMS

- [T2] Ocelot: cat-qubit + repetition-code chip, ~1 s bit-flip times — AWS, Nature (2025) (demonstrated)

- [T4] Boson 4: >1 hr bit-flip lifetime — Alice & Bob release (claimed)
- [T3] Elevator codes: ~15:1 overhead, 10,000× lower error — Alice & Bob preprint, Jan 2026 (claimed, theory + partial demo)
- [T4] 100 logical qubits by 2030 — Alice & Bob roadmap (roadmap)

TRADE-OFFS (VS OTHER MODALITIES)

Reuses the entire transmon fab/cryo stack (H-fab , H-cryo) but promises an order-of-magnitude smaller QEC overhead *if* the noise-bias holds under operation; against that, phase-flips remain and must still be corrected, gate sets on biased-noise qubits are awkward (bias-preserving CNOTs are the hard case), and no bosonic machine has yet run multi-logical-qubit algorithms. Most overhead numbers are extrapolations from single-digit-qubit demos.

CONFLICTS / OPEN QUESTIONS

Does the exponential bit-flip suppression survive once you do *fast gates* on the cat rather than just storing it — i.e., does the noise bias hold during computation, not only in memory? That is the load-bearing assumption behind every overhead claim in this node.

SOURCES

AWS Ocelot (Nature, Feb 2025); alice-bob.com newsroom (squeezed cat, Boson 4, Elevator Codes preprint); postquantum.com cat-qubit + Nord Quantique analyses; The Quantum Insider (Mar 2025).

Control electronics & I/O · H-control

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The classical layer that makes qubits do anything: arbitrary-waveform generators and FPGAs synthesizing shaped microwave pulses (typically upconverted from an IF), fast ADCs digitizing the readout, and real-time feedback engines that must decode errors and branch mid-circuit within qubit coherence times. In the NISQ era this is signal generation; in the fault-tolerant era the controller becomes a real-time computer running a QEC decoder inside a latency budget of hundreds of nanoseconds to a microsecond per cycle (the decoder-throughput wall — see [S-decoders](#) , [0-decoder](#)). The layer is modality-agnostic "picks and shovels" revenue: every architecture in this chapter needs it.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Quantum Machines** (Israel): OPX family with the QUA language for sub-microsecond real-time classical control flow; raised \$170M Series D (Feb 2025); co-built **DGX Quantum** with Nvidia — a Grace-Hopper GPU tightly coupled to the controller for QEC-era feedback and calibration, targeting few-μs round-trip.
- **Zurich Instruments** (Switzerland, Rohde & Schwarz): SHFQC integrated controllers; launched the **ZQCS** platform (Mar 2026) explicitly aimed at operating long-lived logical qubits at large scale, with distributed synchronization across many units.
- **Qblox** (Delft): modular Cluster series; ~\$32M raised; the "open architecture" stack via partnerships with Bluefors (cryo integration), Q-CTRL (autonomous calibration), and QuantWare. **Keysight**: QCS platform plus its test-and-measurement moat. **Zurich/Keysight/QM** dominate room-temperature control.
- Frontier direction: push the controller into the cold as cryo-CMOS ([H-cryocmos](#)) to escape one-coax-per-qubit; couple GPUs for decoding (Nvidia CUDA-Q, [S-software](#)).

KEY GRADED CLAIMS

- [T4] ZQCS targets large-scale logical-qubit operation — Zurich Instruments launch (Mar 2026) (claimed)
- [T4] Sub- μ s real-time feedback + GPU-coupled decoding (DGX Quantum) — QM/Nvidia (demonstrated in deployments; benchmarks vendor-supplied)
- [T5] Quantum funding context: ~\$3.77B raised Jan–Sep 2025, ~3 $\times$ all of 2024 — trade press (reported/forecast)

TRADE-OFFS (VS OTHER APPROACHES)

Room-temperature racks are flexible and vendor-agnostic but cost on the order of ~\$1k+ per qubit-channel and drown in cabling past a few thousand channels; cryo-CMOS cuts the wire count but must run on a milliwatt heat budget with re-characterized transistor models. Real-time decoding forces a hard co-design between the controller, the decoder ASIC/FPGA, and the QEC cycle time — latency, not raw compute, is the binding constraint.

CONFLICTS / OPEN QUESTIONS

Can decoders and controllers keep up with surface-code cycle times ($\sim 1\ \mu$ s) across 1,000+ logical qubits — the decoder-throughput wall? Will hyperscaler/GPU vendors (Nvidia) absorb this layer into their accelerated-computing stack, or does it stay a specialist market?

SOURCES

Zurich Instruments ZQCS (The Quantum Insider, Mar 2026); Quantum Machines / Nvidia DGX Quantum; qblox.com + PitchBook; postquantum.com enabling-technologies analysis.

Cryogenics & dilution refrigerators · H-cryo

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Superconducting, silicon-spin, and bosonic qubits live at 10–20 millikelvin — colder than deep space — reached by dilution refrigerators that exploit the finite solubility of ^3He in ^4He : pumping ^3He across the phase boundary in the mixing chamber absorbs heat continuously. Modern "dry" (cryogen-free) fridges reach the 4 K stage with pulse-tube coolers instead of a liquid-helium bath, which is what made mK an engineering commodity rather than a specialist art. The fridge sets three hard limits on qubit count: **cooling power** at the mixing-chamber stage (a few hundred μW to $\sim 1\text{ mW}$ at 20 mK), **internal volume** for the sample and wiring, and the **number of coax lines** that can be routed from 300 K to the mixing chamber without dumping heat — the passive-wiring bottleneck (coax, attenuators, isolators — see [H-wiring](#)) that cryo-CMOS ([H-cryocmos](#)) exists to break.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Bluefors** (Finland): the market anchor; with Oxford Instruments it holds >70% of the dilution-refrigerator market (ICV 2025). The **KIDE** platform is purpose-built for large-scale QC — 1,000+ qubit support, hexagonal side-loading, and multi-unit clustering; IBM Quantum System Two runs on it. Ultra-Compact LD launched Feb 2025 for space-constrained labs.
- **Oxford Instruments NanoScience** (UK): ProteoxMX/LX line, the other half of the duopoly. **Maybell Quantum** (US): the "Icebox" packing dense wiring (thousands of lines) into one fridge. **Leiden Cryogenics**, **Zero-Point Cryogenics** (Canada), and Chinese entrants (e.g. domestic supply for USTC programs) round out the field.
- **Market:** laboratory dilution-fridge market $\sim \$320\text{M}$ (2025), projected $\sim \$520\text{M}$ by 2034 — quantum programs are the main growth driver.

KEY GRADED CLAIMS

- [T5] Bluefors + Oxford Instruments >70% market share — ICV Global Dilution Refrigerator Report 2025 (claimed)

- [T4] KIDE supports 1,000+ qubit systems — Bluefors (claimed; IBM deployment corroborates at current scales)
- [T5] Market ~\$320M (2025) → ~\$520M (2034) — analyst forecasts (forecast)
- [T1] Dilution refrigeration reaches ~10 mK continuously via $^3\text{He}/^4\text{He}$ mixing — established cryogenics

TRADE-OFFS (VS OTHER MODALITIES/COMPONENTS)

Dry fridges made mK routine, but each ~1 mW of cooling power at 20 mK costs on the order of ~25 kW at the wall — the thermodynamic tax on superconducting QC. Wiring heat-load per qubit forces cryo-CMOS control, photonic I/O, or aggressive multiplexing before a fridge can hold thousands of qubits. Ion, neutral-atom, photonic, and NV modalities skip dilution fridges entirely (photon detectors still want a 2–4 K stage — see [H-detect](#)).

CONFLICTS / OPEN QUESTIONS

Can a single fridge + its wiring scale to million-qubit superconducting machines, or does the endgame require chip-to-chip links across *many* fridges ([H-intercon](#)), which in turn needs microwave-optical transduction ([H-transduce](#)) that does not yet exist? **^3He supply** — a byproduct of tritium decay in nuclear-weapons stockpiles — is a strategic chokepoint rarely priced into roadmaps (see [E-supplychain](#)).

SOURCES

bluefors.com (KIDE, Ultra-Compact LD); The Quantum Insider (Feb 2025); ICV Global Dilution Refrigerator Report 2025; Maybell Icebox coverage; intelmarketresearch/valuates market reports.

Cryogenic control ASICs · H-cryocmos

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Every qubit needs control and readout wires running from room-temperature electronics down into the fridge. At thousands of qubits this **wiring bottleneck** turns fatal: heat load down the coax, cabling bulk, and connector count all explode past what a dilution fridge (H-cryo) can absorb. The fix is **cryo-CMOS** — control/readout ASICs fabricated in standard silicon that run *inside* the cryostat, either at the 4 K stage or, ideally, alongside the qubits at ~10 mK–100 mK — collapsing the wire count by multiplexing many qubits per line and moving the classical/quantum boundary down into the cold. It is a prerequisite for any million-qubit superconducting or silicon-spin machine, and it borrows the entire CMOS supply chain (H-foundry) rather than inventing new fab.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Intel — Horse Ridge I/II:** cryo-CMOS SoCs generating microwave qubit-drive pulses at ~4 K; Horse Ridge II added integrated readout and multi-gate pulsing. Intel's Pando Tree targets scale-up interconnect.
- **SemiQon** (Finland): cryo-CMOS transistors (launched late 2024) claiming control/readout of hundreds of qubits in one cooldown at ~100× lower power than room-temperature electronics; pursues co-locating electronics on the qubit chip.
- **Equal1** (Ireland): UnityQ / Bell-1 rack-mounted silicon-spin system with control integrated in GlobalFoundries 22FDX FD-SOI CMOS; validated the CMOS process April 2025; Bell-1 headed to ESA's Φ -lab (2025).
- **Diraq** (with Emergence Quantum) and **QuTech** (with Intel) also active. Silicon spin qubits (H-silicon) benefit most: same-die integration is native to their CMOS-compatible modality, whereas transmons must place the ASIC at a separate cold stage.

KEY GRADED CLAIMS

- [T3] Cryo-CMOS control of multiple silicon quantum dots in a commercial CMOS process — Equal1, GlobalFoundries 22FDX (demonstrated/validated)

- [T4] SemiQon cryo-CMOS controls hundreds of qubits per cooldown at ~100× lower power — company claim (claimed)
- [T4] Horse Ridge II integrates drive + readout + multi-gate pulsing at 4 K — Intel (demonstrated on Intel's own stack)

TRADE-OFFS

The core tension is **power vs proximity**: dissipation in the ASIC heats the cold stage, so the nearer the qubits (mK, where cooling power is $\sim\mu\text{W}$ – mW) the tighter the power budget. Running at 4 K eases the thermal budget (cooling power $\sim\text{W}$) but reintroduces a wiring hop back down to the qubits. Transistors misbehave at cryo — threshold shifts, flicker noise, and carrier freeze-out all require re-characterized device models. Yield and reliability of cryo-CMOS at scale are unproven, and a failure inside a cold fridge is expensive to reach.

CONFLICTS / OPEN QUESTIONS

Can cryo-CMOS dissipate little enough to sit at the mixing-chamber stage, or is 4 K the practical floor — leaving a residual wiring problem between 4 K and the qubits? Does same-die integration advantage silicon spin qubits enough to offset their smaller lead in qubit quality versus transmons and ions?

SOURCES

intc.com (Horse Ridge II); semiqon.com/technology; quantumzeitgeist.com (SemiQon); thequantuminsider.com 2025/04/17 (Equal1 CMOS validation); design-reuse.com; qutech.nl; arXiv:2604.16216.

Single-photon detectors / SNSPDs · H-detect

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Photonic quantum computing, QKD, and much of quantum networking rest on detecting one photon at a time — fast, with near-unity efficiency and near-zero false counts. The dominant technology is the **superconducting nanowire single-photon detector (SNSPD)**: a current-biased superconducting nanowire (NbN, NbTiN, WSi, MoSi) held just below its critical current, where a single absorbed photon breaks superconductivity locally, forces the current to divert, and produces a measurable voltage spike. As of 2026 the SNSPD is the fastest and most efficient single-photon detector available, outperforming avalanche photodiodes (SPADs) and transition-edge sensors (TES) on efficiency, jitter, and dark counts at once. It is the emitter's counterpart (**H-photon source** makes the photons; SNSPDs catch them) and a quiet load-bearing chokepoint: photonic QC and QKD scale no faster than their detector supply.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Single Quantum** (Netherlands): multi-pixel free-space arrays; a 6×6 NbTiN array built for the NASA/JPL Psyche deep-space optical-comms terminal, >50% system efficiency at 1550 nm, <15 ns per-pixel dead time.
- **ID Quantique** (Switzerland): ships the **ID281** and **ID281 Pro** turnkey SNSPD systems in closed-cycle cryostats; used in QKD and lab research; parallel-nanowire photon-number resolution.
- **Photon Spot** (US): turnkey SNSPD systems and cryostats, common in US labs. Others: **Quantum Opus**, **Scontel**, **Photec**.
- **State of the art**: system detection efficiency demonstrated >99%; dark-count rates ~0.25 counts/hour; timing jitter <3 ps; recovery ~500 ps (giving ~100s of MHz count rates). Large-area devices now reach mid-infrared (to ~7.4 μm). Waveguide-integrated SNSPDs on the same chip as the photonic circuit are the scaling path.

KEY GRADED CLAIMS

- [T2] SNSPD system detection efficiency >99% with picosecond jitter and sub-Hz dark counts — peer-reviewed device papers (demonstrated)

- [T2] 6×6 SNSPD array flown for NASA Psyche deep-space optical comms — SPIE 2025 / Single Quantum (demonstrated)
- [T1] SNSPDs are the fastest/most-efficient single-photon detectors as of 2026 — review literature (established)

TRADE-OFFS

SNSPDs need a cryostat (~1–3 K, a smaller cooler than a dilution fridge but still a system-cost and integration burden) versus room-temperature SPADs, which are cheaper but far noisier and slower. Scaling to the thousands-to-millions of pixels a PsiQuantum/Xanadu-class machine needs strains cryogenic wiring and readout (links to [H-control](#), [H-cryocmos](#), [H-paramp](#)). Photon-number resolution — telling one photon from two — remains hard and is a live research front (parallel nanowires, TES hybrids).

CONFLICTS / OPEN QUESTIONS

Can SNSPD arrays scale to the pixel counts fusion-based photonic FTQC needs without a cryogenic-wiring blowup that recreates the superconducting-QC wiring problem? Does monolithic on-chip integration of detectors with the photonic circuit arrive before detector supply becomes the binding constraint on the whole modality?

SOURCES

Wikipedia SNSPD; Nature Light: Science & Applications s41377-023-01374-1; Science Advances adt0502; idquantique.com (ID281 Pro); ADS 2025SPIE13699E; arXiv:2101.05407.

Fabrication & supply chain · H-fab

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The materials science and manufacturing base under every qubit: Josephson-junction deposition (shadow-evaporated Al/AlO_x, or foundry-compatible variants), substrate choice (sapphire vs high-resistivity silicon), superconducting films (Nb, Al, Ta, TiN), 300 mm foundry processes, and the specialty supply chain (coax, attenuators, isolators/circulators, ³He, isotopically enriched ²⁸Si, single-photon detectors). The central villain is the **two-level system (TLS)** — atomic-scale defects in amorphous oxides and at interfaces that resonantly absorb qubit energy and drift over hours, capping and destabilizing coherence. Getting T₁ up (see the 1.68 ms transmon in **H-supercon**) is mostly a war against TLS.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **TLS science:** 2025 mapping experiments (arXiv:2511.05365) localized most detectable surface TLS to the Josephson-junction leads, clustered where surface roughness and amorphous SiO₂ sit — actionable fab guidance rather than folklore. High-throughput junction studies (arXiv:2602.11469) tie TLS density to structural control; site-specific TLS frequency tuning has been demonstrated (arXiv:2503.04702). Titanium sacrificial layers cut TLS loss in tantalum CPW resonators (arXiv:2601.16369).
- **Materials:** tantalum films (Princeton/Houck lineage) cut microwave loss vs niobium and pushed transmon T₁ past 0.3–0.5 ms; replacing sapphire with high-resistivity silicon then cut bulk-substrate loss further, yielding time-averaged $Q \approx 9.7 \times 10^6$ across 45 qubits and the 1.68 ms record.
- **Foundries:** imec's 300 mm superconducting-qubit CMOS flow (Nature npj 2024); Diraq/imec spin qubits >99% from a standard 300 mm line (2025); SQC's 250k-register patterning (2025); GlobalFoundries manufacturing PsiQuantum's Omega photonic chipset; SkyWater, Intel's 300 mm line, and national fabs (SQMS at Fermilab) round out capacity. Related nodes: **H-**

package (3D/TSV integration), H-iontrap (surface-trap fab),
H-foundry (the 300 mm ecosystem).

KEY GRADED CLAIMS

- [T3] Surface TLS concentrate on junction leads / rough, SiO₂-rich regions — arXiv:2511.05365 (demonstrated, awaiting reproduction)
- [T2] 300 mm CMOS flows yield error-correction-grade spin qubits and high-coherence transmons — Nature/npj papers (demonstrated)
- [T6] TLS problem "engineerable away" by process control alone — no field consensus (speculative)

TRADE-OFFS

Foundry processes buy uniformity and volume but restrict the exotic materials and geometries that still hold coherence records in the lab (shadow-evaporated Al junctions). The specialty supply chain — ³He, dilution fridges (Finnish/UK duopoly, H-cryo), SNSPDs (H-detect), isolators, ²⁸Si — concentrates in few hands and is a geopolitical single point of failure (see E-supplychain).

CONFLICTS / OPEN QUESTIONS

Junction-parameter spread (~1–2% today) vs the <0.1% wanted for fixed-frequency architectures; whether TLS density can be driven down ~10× by process control, or whether architectures must simply tolerate a fixed TLS bath. Yield at wafer scale, not single-device records, is the number that gates manufacturability.

SOURCES

arXiv:2511.05365, 2602.11469, 2503.04702, 2601.16369; PMC11446867 (imec 300 mm); Princeton Nature s41586-025-09687-4; Science Advances ado6240 (phonon engineering); quantumzeitgeist TLS coverage.

The 300 mm / wafer-scale foundry ecosystem · H-foundry

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The distinct-from-materials question of *who can actually manufacture qubits at wafer scale, on industrial tooling, with yield*. **H-fab** covers the materials science (junctions, films, TLS); this node covers the **foundry ecosystem** — the 200/300 mm semiconductor lines being retooled to make superconducting, silicon-spin, and photonic quantum chips, and the strategic fact that quantum is increasingly won or lost on manufacturing uniformity rather than single-device records. The pitch that pulled billions into silicon spin and photonics is precisely this: if a qubit can be built in a CMOS-compatible 300 mm flow, the entire semiconductor industry becomes the quantum fab, and Moore's-law-style volume follows.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **imec** (Belgium): the anchor R&D foundry. Ran a 300 mm superconducting-qubit CMOS flow (Nature npj 2024) and, with **Diraq**, produced gate-defined silicon spin qubits at >99% 2Q fidelity from a standard 300 mm line (Nature, 2025) — the strongest evidence to date that foundry-made qubits can hit error-correction-grade fidelity.
- **GlobalFoundries**: manufactures PsiQuantum's Omega **photonic** chipset on 300 mm silicon photonics, and hosts Equal1's cryo-CMOS control in its 22FDX FD-SOI process (**H-cryocmos**) — one commercial fab serving multiple modalities.
- **Intel**: its own 300 mm line makes Tunnel Falls spin-qubit chips with high-volume test infrastructure. **SkyWater** (US) runs quantum/superconducting programs; **national fabs** (SQMS at Fermilab, LPS/US-government lines) and **TSMC-adjacent** research round out capacity. **SQC** patterned 250k-register donor devices with industrial STM/CMOS tooling (2025).

KEY GRADED CLAIMS

- [T2] Error-correction-grade qubits produced from a standard 300 mm industrial flow (spin) — Diraq/imec, Nature (2025) (demonstrated)

- [T2] Commercial 300 mm silicon-photonics fab makes integrated photonic-QC chips — PsiQuantum/GlobalFoundries (demonstrated, component-level)
- [T4] "Foundry compatibility $\Rightarrow$ Moore's-law-style qubit scaling" — industry thesis (claimed)

TRADE-OFFS

A merchant foundry buys uniformity, volume, metrology, and yield learning that no lab can match — but it constrains you to the materials and geometries the line already qualifies (no exotic substrates, restricted junction processes), which is exactly where lab-scale coherence records still live (**H-fab**). Retooling a fab for quantum (superconducting films, mK-relevant dielectrics, ^{28}Si , single-photon devices) is expensive and slow, and the volume that justifies it does not exist yet — a chicken-and-egg problem.

CONFLICTS / OPEN QUESTIONS

Does foundry uniformity actually hold across thousands of qubits on a wafer, or does device-to-device spread reappear at scale (the historic silicon-spin killer)? Which modality's foundry bet pays off first — silicon spin (native CMOS), photonics (silicon photonics), or superconducting (needs bespoke films)? Whether a true merchant "quantum foundry" market emerges, or each vendor stays captive to one R&D fab (imec, GF, Intel).

SOURCES

PMC11446867 (imec 300 mm superconducting flow); Diraq/imec Nature (2025); PsiQuantum/GlobalFoundries Omega; Intel Tunnel Falls; SQC 250k-register release; postquantum.com silicon-spin + photonic ecosystem analyses.

Optical frequency combs · H-frequencycomb

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

An optical frequency comb is a laser whose spectrum is a set of perfectly evenly spaced sharp lines — a ruler for light. It coherently links optical frequencies (~hundreds of THz) to microwave frequencies (GHz) that electronics can count, which is what lets an optical atomic clock (**A-clocks**) actually output a usable signal, and what lets many stabilized laser tones in an atomic quantum computer share one absolute reference. The 2005 Nobel-winning technology (Hänsch/Hall) is shared infrastructure that sits under three otherwise separate nodes: **clocks** (**A-clocks**), **time/frequency transfer** (**A-timedist**), and **laser subsystems** for ion/atom machines (**H-lasers**). The frontier is shrinking a lab-bench mode-locked comb onto a chip.

KEY PLAYERS & STATE OF THE ART (2025-26)

- **Chip-scale microcombs (Kerr combs):** Kerr-nonlinear microresonators (silicon nitride, on-chip) generate soliton combs at CMOS-compatible scale, replacing rack-sized mode-locked-laser combs. A Feb 2025 Nature Photonics result used a **Vernier dual-microcomb** scheme to divide a stabilized ~871 nm clock laser down to a countable ~235 MHz output — the key step toward a fully integrated optical clock.
- **NIST** and academic groups (Caltech, EPFL, Purdue, Chalmers) drive the physics; **Vescent**, **Menlo Systems**, **TOPTICA**, and **IMRA** supply commercial combs. NIST has demonstrated chip-scale atomic clocks and chip-scale optical frequency synthesizers at $\sim 10^{-16}$ relative uncertainty.
- **Direction:** integrate the dual comb, pump laser, on-chip heaters/tuning, and spectral filters onto a single die — the enabling piece for μ PNT (micro positioning/navigation/timing) and portable optical clocks (**A-pnt**).

KEY GRADED CLAIMS

- [T2] Vernier dual-microcomb optical frequency division of a stabilized clock laser — Nature Photonics s41566-025-01617-0 (2025) (demonstrated)
- [T2] Chip-scale optical frequency synthesizer at 2.7×10^{-16} relative uncertainty — Science Advances (established/demonstrated)

- [T1] Frequency combs coherently link optical and microwave frequencies — established (2005 Nobel)

TRADE-OFFS

Bench mode-locked-laser combs are mature, low-noise, and turnkey but rack-sized and power-hungry. Microcombs promise mass-manufacturable, low-SWaP combs on a chip but currently struggle with octave-spanning bandwidth (needed for self-referencing), spectral flatness, and turnkey soliton initiation. Which you pick trades performance against integration — the same tension running through **H-lasers** .

TRADE-OFFS / WHY IT'S ITS OWN NODE

Because a comb is shared plumbing under clocks, timing networks, and atomic-computer lasers, keeping it as a separate node avoids duplicating the same hardware story three times across **A-clocks** , **A-timedist** , and **H-lasers** . Improvements here propagate to all of them at once.

SOURCES

Nature Photonics s41566-025-01617-0 (Vernier microcombs, 2025); Science Advances (chip-scale synthesizer); NIST optical-frequency-combs program page; Laser Focus World (microcomb clocks, 2025); arXiv:2602.05151, 2512.05005.

Interconnects & modular quantum computing · H-intercon

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

No single chip or trap reaches a million qubits, so every serious roadmap goes modular: multiple processors linked by quantum interconnects that distribute entangled Bell pairs between modules, then consume them for teleported gates (the protocol under **F-teleport**). Three flavors, at three length scales: **short cryogenic microwave/chip-to-chip couplers** inside one fridge (cm-scale, fast, high-fidelity); **optical-photon links** over fiber between fridges/racks/buildings (km-scale, slow, lossy); and **microwave-to-optical transduction** (**H-transduce**) to let a superconducting qubit talk to a fiber at all. Interconnect *quality*, more than raw qubit count, now gates every modality's endgame — a bad link below the QEC threshold poisons the whole logical computation.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Oxford (trapped ions)**: distributed quantum computing across an optical network link — deterministic teleported CZ gates between two photonicallly linked ion-trap modules, executing Grover's algorithm across the boundary (Nature, 2025). The cleanest full-stack demonstration of the modular paradigm to date.
- **Duke Quantum Center + IonQ (Monroe group)**: first *fully-distributed* GHZ (tripartite) state across a **three-node** network of single trapped-ion memories linked by photonic interconnects — three modules ~2 m apart, 3 m single-mode fibers to a central free-space GHZ generator; bounded fidelity $0.841(17) \leq F \leq 0.881(17)$ at a generation rate $0.095(5) \text{ s}^{-1}$, with a Mermin-inequality violation closing the detection loophole for the first time in a distributed multipartite state (Jun 2026). Clears the two-node barrier toward genuine networked/modular QC.
- **IBM**: Nighthawk's 2026–27 plan chains up to three 120-qubit modules (360 qubits) with **I-couplers** (short-range chip-to-chip); **Loon**'s c-couplers add long-range on-chip connectivity for qLDPC codes; **Flamingo** targets ~1 m cryogenic links between modules.
- **Photonic Inc.** (silicon T-centers): distributed entanglement between modules over telecom fiber, using a spin-photon interface native to the T-

center (see **H-spinphoton**); roadmap targets ~200 kHz remote-entanglement rate at 99.8% fidelity.

- **PsiQuantum / Xanadu**: chip-to-chip photonic-qubit interconnects are native to the photonic modality (**H-photonic**) rather than an add-on.
- **Quantinuum / IonQ**: photonic interconnects are the declared scaling path beyond single-trap limits. Silicon-photonic CNOT-gate teleportation between separate chips demonstrated (PRL 2025–26).

KEY GRADED CLAIMS

- [T2] Teleported two-qubit gates between separate ion-trap modules; algorithm run across the link — Nature s41586-024-08404-x (demonstrated)
- [T3] First fully-distributed three-node GHZ state of remote ion memories, $F \approx 0.84\text{--}0.88$, Mermin violation with detection loophole closed — Duke + IonQ, Goetting et al., arXiv:2606.17173 (2026) (demonstrated; preprint)
- [T4] IBM 3×120-qubit multi-module systems in 2026 — IBM roadmap (roadmap)
- [T4] Photonic Inc. 200 kHz / 99.8% distributed entanglement — company target (roadmap)

TRADE-OFFS

Optical links carry entanglement kilometers but at kHz rates and ~90–97% Bell-pair fidelities — orders of magnitude below intra-module gates (99.9%+).

Microwave couplers are fast and high-fidelity but reach only centimeters.

Entanglement distillation buys fidelity back at a further rate cost. The core tension: the faster and cleaner the link, the shorter its reach; bridging fridges cheaply needs a transducer that does not yet exist (**H-transduce**).

CONFLICTS / OPEN QUESTIONS

C-interconnect-loss / O-interconnect-loss: whether modular interconnects can stay below the QEC threshold across module boundaries. Microwave-optical transducers remain far from the efficiency/added-noise needed for fridge-to-fridge superconducting links. Does modularity's overhead (slower inter-module gates) break QEC cycle budgets before it buys scale?

SOURCES

Nature s41586-024-08404-x (Oxford distributed ions); Goetting et al., arXiv:2606.17173 (Duke/IonQ three-node GHZ, 2026); photonic.com networking releases; IBM roadmap/newsroom (Nov 2025); PRL (silicon-photon CNOT teleportation); arXiv:2412.18458.

Trapped-ion qubits · H-ion

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Individual atomic ions (Yb^+ , Ba^+ , Ca^+) held in electromagnetic (Paul) traps in ultra-high vacuum; qubits live in hyperfine or optical levels of identical atoms, so every qubit is perfect by construction and needs no fabrication tuning. Gates are driven with lasers (or on-chip microwaves/electronics) through shared motional modes of the ion chain, giving effectively all-to-all connectivity within a chain. Modern systems use microfabricated **surface traps** (see [H-iontrap](#)) rather than bulk blade traps; Quantinuum's QCCD (quantum charge-coupled device) architecture physically shuttles ions between dedicated storage, gate, and readout zones to sidestep the single-chain length limit.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Quantinuum:** Helios launched Nov 2025 — 98 physical $^{137}\text{Ba}^+$ qubits at 99.921% 2Q / 99.9975% 1Q fidelity, coherence in the seconds-to-minutes range, on a Honeywell-fabricated QCCD trap; results characterized in an arXiv preprint (2511.05465) and independently validated by Sandia in Nature (June 2026). Roadmap: Sol (2027), **Apollo** fault-tolerant system ~2029. Helios already runs dozens of logical qubits with real-time QEC.
- **IonQ:** closed its \$1.075B acquisition of **Oxford Ionics** (Sep 2025); claims a 99.99% 2Q fidelity record (2025); Forte Enterprise and Tempo (~100 algorithmic qubits) systems shipping, Tempo contracted to KISTI (South Korea) and to AFRL. Combined roadmap: 256 physical qubits at 99.99% (2026), ~10,000 (2027), 2M by 2030 — aggressive relative to any demonstrated trap-scaling result.
- **Oxford Ionics** (now IonQ): electronic (laser-free) gate control delivered on chip traps; supplied the QUARTET system to the UK NQCC. **Alpine Quantum Technologies** (Innsbruck): Ca^+ , rack systems. **eleQtron** (Germany): MAGIC microwave-driven gates.

KEY GRADED CLAIMS

- [T2] Helios: 98 qubits, 99.921% 2Q fidelity, seconds-to-minutes coherence — arXiv:2511.05465 + Sandia/Nature validation Jun 2026 (demonstrated)

- [T4] IonQ 99.99% 2Q gate record — company claim 2025 (claimed)
- [T4] IonQ 2M physical qubits by 2030 — post-acquisition roadmap (roadmap)
- [T1] Ion qubits are intrinsically identical (no fab variability) — atomic physics (established)

TRADE-OFFS (VS OTHER MODALITIES)

The best gate fidelities and by far the longest coherence (seconds to minutes) of any modality, all-to-all connectivity, and no dilution fridge (UHV chamber, room-temperature-adjacent — see [H-uhv](#)). Against that: gates are slow (μs – ms , $\sim 1000\times$ slower than transmons), a single trap chain saturates around ~ 50 – 100 ions before motional-mode crowding degrades gates, and scaling then demands ion shuttling (heating, speed cost), multi-zone chips, or photonic interconnects ([H-intercon](#)). The whole machine is a precision laser instrument ([H-lasers](#)), so uptime tracks laser-lock stability.

CONFLICTS / OPEN QUESTIONS

Whether QCCD shuttling or photonic networking wins the scaling race — Quantinuum bets on the former, IonQ (post-Oxford-Ionics) increasingly on the latter. Whether the μs – ms gate-speed penalty erases the fidelity advantage once algorithms reach millions of gates. IonQ's 2030 count is a roadmap number, not a demonstrated one.

CONFLICTS LOGGED

See [C-ion-scaling](#) (shuttling vs photonic networking as the trapped-ion scaling path).

SOURCES

Quantinuum Helios release + [arXiv:2511.05465](#); HPCwire Sandia/Helios validation (Jun 2026); IonQ/Oxford Ionics acquisition filings; DCD KISTI Tempo coverage; [postquantum.com](#) trapped-ion ecosystem + Helios analyses.

Ion-trap chip fabrication (surface traps) · H-iontrap

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The trapped-ion modality (H-ion) has quietly migrated from macroscopic four-rod blade traps to **microfabricated surface-electrode traps**: a planar silicon or glass chip patterned with dozens to hundreds of gold or aluminum electrodes whose shaped RF and DC fields levitate a chain of ions $\sim 50\text{--}100\text{ }\mu\text{m}$ above the surface, in ultra-high vacuum (H-uhv). The chip *is* the processor. Modern designs carve the surface into functional **zones** — storage (long chains wait), gate (2–4 ions isolated for entangling operations), junctions (X/Y intersections to shuttle ions between zones), and readout (fluorescence collection) — the QCCD architecture that lets ion machines grow past a single chain. Integrating optics (waveguides, grating couplers, on-chip photodetectors) and control electronics into the trap chip is the current fabrication frontier.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Sandia National Laboratories**: the reference research foundry. Its **High Optical Access (HOA)** platform (Phoenix, Peregrine traps) gives good laser access skimming the surface; the "Enchilada" trap stores up to ~ 200 ions across a branched layout — among the most complex surface traps ever fabricated. Sandia supplies traps to much of the US government-funded ion community.
- **Honeywell / Quantinuum**: the Helios QCCD trap chip (Nov 2025, 98 qubits at 99.921% 2Q — see H-ion) was fabricated in Honeywell's MEMS/microfabrication facility, a rare vertically integrated trap-fab-to-computer pipeline.
- **Infineon** (with academic partners) fabricates ion-trap chips at scale; **Oxford Ionics** (now IonQ) builds electronic (integrated-electrode) traps for laser-free gates. Sandia's TICTOC program integrates photonics onto trap chips for compact ion clocks.

KEY GRADED CLAIMS

- [T2] Microfabricated surface-electrode traps run multi-zone QCCD processors at record fidelity — Helios/Honeywell + Sandia literature (demonstrated)
- [T2] Branched surface traps store ~200 ions with multiple junctions — Sandia Enchilada (demonstrated)
- [T3] On-chip integrated photonics for laser delivery/readout in trap chips — Sandia TICTOC + academic demos (early)

TRADE-OFFS

Surface traps trade the deep, symmetric confinement of blade traps for lithographic scalability, multi-zone layouts, and integrable optics/electronics. The cost: ions sit close to a surface, so **anomalous heating** from surface electric-field noise is worse (it scales steeply with ion-electrode distance), degrading gate fidelity unless the surface is cryogenically cooled or carefully treated. Junction shuttling adds motional heating and time overhead. Fabrication uniformity and dielectric charging are yield issues.

CONFLICTS / OPEN QUESTIONS

Can integrated photonics and control deliver every laser tone and readout on-chip fast enough to remove the external optics table (**H-Lasers**) that otherwise bounds scaling? Does anomalous surface heating cap how small and dense trap electrodes can go? Whether commercial trap fabrication moves to a merchant-foundry model (like CMOS) or stays captive to Sandia/Honeywell/Infineon.

SOURCES

Sandia HOA / Enchilada trap literature; arXiv:2009.02398 (Phoenix/Peregrine); Quantinuum Helios (Honeywell fab); postquantum.com trapped-ion ecosystem; Sandia TICTOC project page; arXiv:1008.0990 (surface-trap demonstration).

Laser & photonics subsystems · H-lasers

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Trapped-ion, neutral-atom, and NV-diamond machines are, physically, laser instruments. They need many stabilized laser tones — for trapping/cooling, Rydberg excitation, qubit-state manipulation, repumping, and readout — each at the right wavelength with sub-kHz linewidth, mode-hop-free tuning, and long-term frequency stability locked to an optical reference (increasingly an optical frequency comb, see [H-frequencycomb](#)). As atomic machines scale from tens to thousands of atoms, the laser subsystem becomes the dominant cost, footprint, and reliability risk — the atomic-modality analog of superconducting wiring. It is an arms-supplier layer: a handful of photonics vendors sit quietly under most atomic/ionic quantum computers, and the push is toward low-SWaP (size, weight, power) modules and photonic integrated circuits (PICs).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **TOPTICA Photonics** (Germany): the dominant supplier of tunable diode-laser systems and frequency-stabilized references; serves virtually every major neutral-atom and ion lab and commercial system.
- **M Squared Lasers** (UK): high-performance Ti:sapphire and frequency-stabilized systems used for Rydberg excitation (the demanding blue/UV tones).
- **Vescent** (US): precision diode lasers, frequency combs, and servo/lock electronics widespread in atomic physics. Trapping power: **IPG Photonics**, **NKT Photonics**.
- **PIC integration**: IonQ + imec are developing photonic integrated circuits to route and deliver laser light on-chip for ion machines (2025), the same bet neutral-atom vendors are making — replacing free-space beam paths with waveguides to kill drift and shrink the footprint.

KEY GRADED CLAIMS

- [T1] Neutral-atom/ion machines require multiple sub-kHz-linewidth stabilized laser tones — atomic physics literature (established)

- [T4] TOPTICA/M Squared supply the Rydberg-excitation lasers for most neutral-atom systems — supply-chain analyses (claimed/reported)
- [T3] Photonic integrated circuits can shrink ion-machine laser delivery (IonQ × imec) — 2025 collaboration (roadmap/early)

TRADE-OFFS

Free-space bulk-optics laser tables give the best performance but do not scale — large, drift-prone, labor-intensive to align, and sensitive to vibration and temperature. PIC integration promises SWaP and manufacturability but currently trades away optical power and linewidth, especially at the blue/UV wavelengths ion qubits need. Laser reliability (diode lifetime, lock stability) directly bounds machine uptime: a single unlocked tone can halt the whole processor mid-computation.

CONFLICTS / OPEN QUESTIONS

Does laser delivery integrate onto chips fast enough to keep pace with atom-count roadmaps (H-neutral , H-ion), or does the optics table become the scaling wall for atomic modalities the way wiring is for superconducting ones? Blue/UV PIC materials and on-chip high-power delivery are the specific unsolved pieces.

SOURCES

toptica.com (quantum technologies / LASER 2025); globenewswire.com 2025/07/08 (neutral-atom market, Toptica/Hamamatsu); postquantum.com (neutral-atom supply chain); ionq.com/news (IonQ × imec PICs); arXiv:2304.08402.

Neutral-atom qubits · H-neutral

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Neutral atoms (Rb, Sr, Cs, Yb) held in arrays of optical tweezers; qubits live in hyperfine or nuclear-spin states, and two-qubit gates come from exciting atoms to giant **Rydberg** states whose strong interaction blockades neighbors.

Tweezers can be physically rearranged mid-computation, giving reconfigurable, effectively all-to-all connectivity and native support for shuttling logical qubits — the feature that made the modality the fastest-moving of 2024–26. Atoms are laser-cooled but the apparatus itself runs at room temperature inside an ultra-high-vacuum chamber (see [H-uhv](#)); the machine is fundamentally a laser instrument ([H-lasers](#)).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **QuEra / Harvard / MIT:** 2025 Nature results — 48 logical qubits on 280 atoms with below-threshold operation and logical-layer magic-state distillation; a separately reported continuously-operating array kept a ~3,000-atom system running for over two hours by reloading atoms mid-computation, addressing atom-loss as an error channel. **Jan 2026: 96 logical qubits from 448 physical atoms via a high-rate $[[16,6,4]]$ code (~4.7:1 encoding), below-threshold across all 96 — peer-reviewed in Nature (s41586-025-09848-5), the current logical-qubit record.** Roadmap: 100 logical qubits ~2026. Raised \$230M (Feb 2025), Google/SoftBank participating.
- **Pasqal** (France): 1,000+ atom arrays; first neutral-atom demo of logical qubits outperforming physical ones on a differential-equation workload (2026); analog + digital modes; ~3 kW rack systems. Targets 10,000 qubits ~2026–27.
- **Atom Computing** (US): 1,180-atom Sr array (1,225 sites), ~40 s coherence; with Microsoft demonstrated 24 entangled logical qubits (2024); Magne (~50 logical from ~1,200 physical) targeted ~2027.
- **Infleqtion** (US): Sqale platform, UK NQCC deployment. **planqc** (Germany): Sr arrays, DLR/LRZ contracts.

- **Atom-count frontier:** Caltech loaded 6,100 atoms in a single tweezer array (Sep 2025); QuEra/Atom Computing project 100,000 atoms per chamber within a few years.

KEY GRADED CLAIMS

- [T2] 48 logical qubits + logical magic-state distillation — Harvard/QuEra, Nature (2025) (demonstrated)
- [T2] 96 logical qubits via a high-rate $[[16,6,4]]$ code on 448 atoms, below-threshold — QuEra, Nature s41586-025-09848-5 (2026) (demonstrated, peer-reviewed)
- [T2] Continuous >2 hr operation via mid-computation atom reloading (~3,000 atoms) — Harvard/MIT, Nature (2025) (demonstrated)
- [T3] 6,100-atom single-array loading — Caltech (Sep 2025) (demonstrated, static array)
- [T4] Pasqal 10,000 qubits by ~2026–27 — company roadmap (roadmap)

TRADE-OFFS (VS OTHER MODALITIES)

The largest qubit counts of any gate-based modality, reconfigurable connectivity, a room-temperature apparatus, and intrinsically identical atoms; against that, gates are slower than transmons (Rydberg gates ~sub- μ s but with cooling/rearrangement overhead per cycle), atom loss is a distinctive error channel that forces continuous reloading, and 2Q fidelities (~99.5%) still trail trapped ions. The optical-table laser subsystem is the practical scaling wall (**H-Lasers**), the analog of superconducting wiring.

CONFLICTS / OPEN QUESTIONS

Does the logical-qubit lead survive the move from batch-mode demos to continuous, repeatable computation with fast mid-circuit measurement and feed-forward at scale? Rearrangement time and cooling cycles set an effective clock speed that has to fit inside QEC budgets.

SOURCES

quera.com; Nature 2025 logical-qubit + continuous-operation papers; QuEra 96-logical Nature s41586-025-09848-5 (2026); Caltech 6,100-atom (Sep 2025); Pasqal newsroom; IEEE Spectrum "Neutral-Atom QC 2026"; postquantum.com neutral-atom guide.

NV centers / diamond qubits · H-nv

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

A nitrogen-vacancy (NV) center is a point defect in diamond — a substitutional nitrogen atom beside a missing carbon — whose electron spin works as a qubit that can be optically initialized and read out and stays coherent at *room temperature*, because the stiff, low-spin-density diamond lattice isolates it almost as well as a millikelvin vacuum would. Nearby ^{13}C nuclear spins act as long-lived ancilla qubits and quantum memory. The same physics drives quantum sensing, where NV is a commercial reality today (see [A-nvsensing](#) , [A-magneto](#)); for computing, the pitch is a cryogenics-free, rack- or edge-deployable "quantum accelerator."

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Quantum Brilliance** (Australia/Germany): the anchor company. 2025: three QB-QDK2.0 units deployed at Oak Ridge National Laboratory for hybrid quantum-classical work; Europe's first room-temperature NV accelerator went live at Fraunhofer IAF (Jun 2025). Roadmap: 25–100 qubit systems by 2026–27 via incremental qubit-count "drops," moving toward full chip production. Part of a €35M German Cyberagency portable-QC program (with ParityQC) targeting a transportable defense system by 2027.
- **Academic multi-qubit records** (Delft/QuTech, Stuttgart, Ulm, Harvard): ~10-qubit registers on a single NV node with fault-tolerant-grade control and demonstrated small logical operations; NV nodes double as the memory/repeater endpoints in early quantum-network experiments (a bridge to [A-qinternet](#) , [A-qmemory-hw](#)). Coherence: NV electron-spin T2 reaches milliseconds at room temperature and seconds with dynamical decoupling; ^{13}C nuclear memories exceed a minute.

KEY GRADED CLAIMS

- [T4] QB-QDK2.0 accelerators operating at ORNL and Fraunhofer IAF — deployment announcements (demonstrated as installations; compute utility unproven)

- [T4] 25–100 qubits by 2026–27 — Quantum Brilliance roadmap (roadmap)
- [T2] ~10-qubit NV registers with high-fidelity control and small logical ops — academic literature (demonstrated)
- [T1] NV electron/nuclear spins are coherent qubits at room temperature — established physics

TRADE-OFFS (VS OTHER MODALITIES)

Room temperature, no laser-vacuum-cryo stack, car-battery-scale power, and a physically tiny package — the only modality plausibly deployable at the edge or in a vehicle. Against that: qubit counts are the lowest of any modality (single digits per node), entangling *many* NV centers is hard because each defect sits at a random lattice position, and deterministic placement of many high-quality NVs in an array is unsolved fabrication.

CONFLICTS / OPEN QUESTIONS

Is diamond compute a real computing modality, or a superb sensing technology wearing a compute badge? The 25–100-qubit roadmap requires deterministic NV placement and NV-to-NV entangling that nobody has demonstrated at those numbers. The strongest near-term value of the platform is clearly on the sensing side.

SOURCES

quantumbrilliance.com; OLCF/ORNL Q&A (Sep 2025); Fraunhofer IAF release (Jun 2025); The Quantum Insider; IOPscience 2633-4356/ade359 (NV registers); entangledfuture.com NV guide (2026).

Packaging, interposers & 3D integration · H-package

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Once a superconducting chip passes a few hundred qubits, you cannot route every control and readout line in from the edges of a single plane — wiring congestion, microwave crosstalk, and mechanical stress make 2D layout a dead end. The fix is borrowed from classical chip packaging and pushed into the superconducting, millikelvin regime: **flip-chip** bonding (a qubit chip bump-bonded face-to-face to a separate wiring/readout chip), **through-silicon vias (TSVs)** carrying signals vertically through the substrate, and **silicon interposers** hosting superconducting coplanar-waveguide routing between chiplets. This lets a machine partition into a **qubit chiplet + bus/wiring chiplet** stack — the quantum analog of 2.5D/3D heterogeneous integration — so qubit count can grow without the signal fan-out choking it.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **MIT Lincoln Laboratory** pioneered superconducting TSVs and 3D-integrated qubit stacks (high-coherence tileable 3D architectures; silicon hard-stop spacers for controlled flip-chip gaps). **IBM** and **Google** both run flip-chip processes in production (Google's Sycamore/Willow and IBM's Heron/Nighthawk use multi-layer wiring and bump bonds).
- **QuantWare** (Delft): explicitly markets chiplets + 3D packaging ("VIO" vertical I/O) to let smaller labs build high-density modules — packaging-as-a-product.
- **imec**, **Chalmers**, and foundry programs develop superconducting TSV and micro-bump processes; through-sapphire-substrate machining is an alternative route (arXiv:2406.09930). A 2026 review (arXiv:2602.04831) surveys large-scale superconducting integration; microwave-crosstalk characterization in planar/3D devices is an active measurement front (arXiv:2606.02440).

KEY GRADED CLAIMS

- [T2] High-coherence qubits preserved through TSV/flip-chip 3D integration — MIT-LL, arXiv:2107.11140 / 1708.02226 (demonstrated)

- [T4] Chiplet + interposer packaging enables high-density modular qubit stacks — QuantWare (claimed)
- [T3] 3D partition (qubit chiplet + bus chiplet) as the path past 2D wiring limits — IEEE/review literature (demonstrated at small scale)

TRADE-OFFS

3D integration buys signal density and routing headroom but adds lossy interfaces — every bump bond, via, and dielectric layer is a candidate home for TLS defects (H-fab) that can degrade the very coherence the packaging is meant to serve. Flip-chip gaps must be controlled to microns; TSV superconductivity must survive thermal cycling to mK. The engineering is a constant fight between more connectivity and preserved coherence.

CONFLICTS / OPEN QUESTIONS

Can 3D packaging scale line density fast enough to feed thousand-qubit chips without importing enough interface loss to cap coherence? Does the industry converge on a standard interposer/chiplet interface (a "quantum socket"), or stay bespoke per vendor? Packaging is emerging as a distinct chokepoint alongside the fridge and the controller.

SOURCES

arXiv:2107.11140, 1708.02226, 1907.12882 (MIT-LL 3D/TSV); arXiv:2602.04831 (2026 large-scale integration review); 2406.09930 (through-sapphire); 2606.02440 (microwave crosstalk); QuantWare VIO materials; NSF interposer-packaging report (par.nsf.gov/10340124).

Quantum-limited amplifiers (JPA / TWPA / HEMT) · H-paramp

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Reading out a superconducting qubit means measuring a microwave signal carrying, on average, a fraction of a photon — buried in thermal and amplifier noise. A **quantum-limited parametric amplifier** at the mixing-chamber stage (~10 mK) boosts that signal while adding the minimum noise quantum mechanics allows (half a photon for phase-insensitive gain), *before* the signal reaches the noisier high-electron-mobility-transistor (**HEMT**) amplifier at the 4 K stage and the room-temperature electronics (**H-control**). Without this first amplifier, single-shot dispersive readout — telling $|0\rangle$ from $|1\rangle$ in one measurement, fast enough for real-time QEC — is impossible. Two architectures dominate: the resonator-based **Josephson parametric amplifier (JPA)** (high gain, narrow band) and the **Josephson traveling-wave parametric amplifier (TWPA/JTWP)** (near-quantum-limited over a wide band, enabling frequency-multiplexed readout of many qubits on one line).

KEY PLAYERS & STATE OF THE ART (2025–26)

- The canonical readout chain: a near-quantum-limited JPA or TWPA at 10 mK → a HEMT (~2–4 K, adding a few photons) → room-temperature amplifiers. TWPAs deliver high gain over GHz bandwidths at ~1 added photon, letting a single line read out many frequency-multiplexed qubits — the reason they matter for scaling.
- 2025 advances: low-intrinsic-loss coplanar lumped-element JTWPAs (arXiv:2503.07559); inverse-Kerr phase-matching for flatter gain (arXiv:2507.17039); in-operando S-parameter-calibrated characterization (APL 2024); two-mode-squeezing schemes that push readout fidelity past the standard quantum limit (arXiv:2603.15804).
- Suppliers/developers: academic groups (Berkeley/Siddiqi lineage, Chalmers, NIST), plus vendors bundling TWPAs into readout stacks (Silent Waves' "Argo" TWPA, Raytheon/Lincoln Lab, and control-vendor integrations). HEMTs come chiefly from **Low Noise Factory** (Sweden).

KEY GRADED CLAIMS

- [T1] A phase-insensitive linear amplifier must add $\geq \frac{1}{2}$ photon of noise (standard quantum limit) — Caves theorem, established
- [T2] JTWPAs provide near-quantum-limited gain over GHz bandwidth, enabling multiplexed single-shot readout — Science aaa8525 + follow-ons (demonstrated)
- [T3] Two-mode-squeezed readout beats the SQL for qubit measurement — arXiv:2603.15804 (demonstrated, lab-scale)

TRADE-OFFS

JPA's give the cleanest noise but narrow bandwidth and low saturation power (few qubits per amp); TWPAs trade a little noise for wide bandwidth and higher saturation power (many qubits per line) at the cost of a long, fabrication-sensitive Josephson transmission line. The HEMT is unavoidable but sits at 4 K and its power dissipation and noise both scale with channel count — another line-count pressure feeding the **H-cryocmos** and **H-cryo** bottlenecks.

CONFLICTS / OPEN QUESTIONS

Can TWPA fabrication yield and uniformity reach the level needed to read out thousands of qubits without per-device tuning? Do parametric amplifiers get absorbed into cryo-CMOS readout, or stay a distinct superconducting-component layer? Photon-number-resolving and directional (nonreciprocal) amplifiers that drop the bulky isolators are a live front.

SOURCES

Science aaa8525 (near-quantum-limited TWPA); arXiv:2503.07559, 2507.17039, 2603.15804; AIP APL 125/104001 (in-operando calibration); Low Noise Factory HEMT datasheets.

Photonic qubits · H-photonic

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Qubits encoded in light — single photons (dual-rail, path, polarization, time-bin) or continuous-variable squeezed states (see [A-squeezed](#)). Computation interferes photons in integrated waveguide circuits (thin-film lithium niobate, silicon, or silicon-nitride) and measures them; the leading architecture is measurement-based **fusion** computation, where small entangled resource states are stitched into a large cluster state by probabilistic fusion gates. Photons barely decohere and travel telecom fiber, so networking is native and much of the machine runs warm — the catch is that photons do not interact, so gates are probabilistic, and **photon loss** is the dominant error. The modality leans on two other hardware nodes: deterministic photon sources ([H-photonsource](#)) and near-unity single-photon detectors ([H-detect](#)).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **PsiQuantum:** Omega chipset (Feb 2025) manufactured at GlobalFoundries on 300 mm silicon photonics — single-photon sources, fusion-gate circuits, and cryogenic detectors integrated with chip-to-chip qubit interconnects. \$1B Series E at \$7B valuation (Sep 2025); ~\$1B utility-scale datacenter build-outs announced for Brisbane and Chicago (Mar 2026). Bets on million-qubit fault tolerance, skipping NISQ entirely.
- **Xanadu:** Aurora (Jan 2025) — modular networked photonic computer: 12 qubits across 35 photonic chips + 13 km fiber, room-temperature apart from cryogenic detectors; uses GKP/CV encoding. IPO'd on Nasdaq/TSX Mar 2026 (~\$302M raised). Earlier Borealis showed programmable Gaussian-boson-sampling advantage (Nature 2022).
- **QuiX Quantum** (Netherlands): €15M Series A to ship a first-generation ~8-qubit *universal* single-photon computer in 2026, on silicon-nitride chips.
- **ORCA Computing** (UK): PT-series time-bin/boson-sampling systems in national labs, fiber-based, rack-mounted. **TundraSystems** (UK) and **Quandela** (see [H-photonsource](#)) round out the field.

KEY GRADED CLAIMS

- [T3] Omega chipset with integrated fusion + wafer-scale fab at GlobalFoundries — PsiQuantum (2025) (demonstrated, component-level)
- [T4] Aurora: first scalable networked modular photonic QC — Xanadu press (claimed)
- [T2] Gaussian-boson-sampling advantage (Borealis) — Xanadu, Nature (2022) (demonstrated, advantage contested by later classical algorithms)
- [T6] Million-qubit photonic FTQC in datacenter form by late decade — PsiQuantum (roadmap)

TRADE-OFFS (VS OTHER MODALITIES)

Room-temperature circuits, native fiber networking, and reuse of the telecom/CMOS-photonics supply chain; against that, gates are probabilistic (fusion succeeds ~50%, so massive multiplexing and fast switching are required), photon loss compounds along every waveguide and coupler, detectors still need cryogenics (2–4 K), and no photonic machine has run a meaningful gate-based algorithm at scale. The loss budget for million-photon operation is unproven end-to-end.

CONFLICTS / OPEN QUESTIONS

Fusion-based FTQC has no intermediate-scale public benchmark — the approach is close to all-or-nothing, so there is little early signal to grade against a roadmap. Whether deterministic, indistinguishable single-photon sources and low-loss switches arrive fast enough to close the loss budget is the field's binding question (see `H-photonsource`, `H-detect`).

SOURCES

psiquantum.com/news (Omega); xanadu.ai press (Aurora) + Nature 606 (Borealis, 2022); quixquantum.com Series A; Optica OPN photonics roadmap (Jun 2025); The Quantum Insider.

Deterministic single-photon & entangled-pair sources · H-photon source

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The emitter counterpart to the SNSPD (**H-detect**): a photonic quantum computer or QKD link is only as good as its supply of *single, identical, on-demand* photons. Two families compete. **Deterministic** sources use a solid-state quantum emitter — an InGaAs/GaAs semiconductor quantum dot in a micropillar or open cavity, Purcell-enhanced so it emits exactly one photon per trigger — approaching "push a button, get one photon." **Probabilistic** sources use nonlinear optics: spontaneous parametric down-conversion (SPDC) or four-wave mixing that occasionally splits a pump photon into a heralded pair; simple and room-temperature, but the emission time is random, so scaling many sources means heavy multiplexing. The three figures of merit are **brightness** (collection efficiency), **single-photon purity** ($g^2(0) \rightarrow 0$), and **indistinguishability** (Hong-Ou-Mandel visibility $\rightarrow 1$).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Quandela** (France): the anchor deterministic-source company. InGaAs quantum dots in micropillar cavities grown by molecular-beam epitaxy; current devices hit in-fibre brightness $>30\%$, $g^2(0) < 0.05$, and HOM visibility $> 90\%$ from a single dot. Its **Prometheus** is a plug-and-play single-photon-source appliance; Quandela also builds full photonic processors (see **H-photonic**) on this source.
- **2026 milestone:** Quandela + C2N demonstrated $88 \pm 1\%$ indistinguishability between photons from *two independent* quantum-dot sources *without spectral filtering* — a prerequisite for scaling beyond one emitter (arXiv 2602-series). Telecom-band (1550 nm) two-photon-resonant Purcell-enhanced dots also reported (arXiv:2602.06140).
- **SPDC/heralded:** the workhorse of QKD and boson-sampling; bulk-crystal and thin-film-lithium-niobate waveguide pair sources are commercial (e.g. from QKD vendors). **Sparrow Quantum** (Denmark) supplies quantum-dot single-photon chips.

KEY GRADED CLAIMS

- [T2] Deterministic QD source: >30% in-fibre brightness, $g^2(0) < 0.05$, HOM > 90% — Quandela device papers (demonstrated)
- [T3] 88% indistinguishability between two independent QD sources, filter-free — Quandela/C2N, arXiv (2026) (demonstrated)
- [T1] SPDC produces heralded single photons but at random times — established nonlinear optics

TRADE-OFFS

Deterministic dots give high brightness and on-demand timing but need cryogenics (~4 K), suffer chip-to-chip wavelength/efficiency spread (screening and binning, or on-chip tuning, required), and are hard to make *mutually* indistinguishable across many emitters. SPDC is room-temperature and dead-simple but fundamentally probabilistic, forcing multiplexing that reintroduces loss and switching complexity — the very loss budget that gates H-photonics.

CONFLICTS / OPEN QUESTIONS

Which source wins for million-photon FTQC: perfected deterministic dots, or heavily multiplexed SPDC? Cross-source indistinguishability at scale (many identical emitters feeding one interferometer) is the specific unsolved requirement, and 2026's filter-free two-source result is the first real signal it might be reachable.

SOURCES

quandela.com (Prometheus, single-photon-source technology);
arXiv:2602.06140 (telecom QD) + 2026 two-source indistinguishability preprint;
Nano Letters 5c01560 (remote QD-cavity indistinguishability);
quantumzeitgeist.com single-photon-source coverage.

Silicon spin qubits · H-silicon

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Qubits stored in the spin of single electrons (or nuclei) confined in silicon, split into two families that share a substrate but not a philosophy: **gate-defined quantum dots**, where lithographic gates electrostatically trap electrons in a well (prioritizes manufacturability — the devices are CMOS transistor structures), and **donor qubits**, where the spin lives on a single phosphorus atom placed with atomic precision (prioritizes uniformity — every P atom is identical by nature). The pitch for both: qubits are ~100 nm scale (roughly a million times smaller in area than a transmon), so in principle the entire semiconductor industry becomes the quantum fab. Operates around 100 mK–1 K, and — because the devices are silicon — is the natural home for cryo-CMOS co-integration (**H-cryocmos**).

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Diraq + imec** (Sep 2025, Nature): >99% two-qubit fidelity on **gate-defined** dots made in a standard 300 mm CMOS foundry flow — four randomly selected devices from one wafer all under 1% 2Q error, three at 99.9%+ single-shot readout. First foundry-made qubits at error-correction-grade fidelity, and the strongest evidence yet for the "uniformity via lithography" thesis.
- **SQC** (Silicon Quantum Computing, Michelle Simmons): **donor** route — 99.99% 2Q fidelity in atom-precision devices and Grover's algorithm at 98.9% without error correction (Nature, Dec 2025); patterned 250,000-qubit registers in 8 hours with industrial STM/CMOS tooling (Nov 2025).
- **Quantum Motion** (UK): full-stack silicon QC delivered to the UK NQCC (2025), built on 300 mm wafers. **Intel**: Tunnel Falls 12-qubit research chip distributed to labs; 300 mm line with high-volume test. **Equal1** (see **H-cryocmos**): control integrated in GlobalFoundries 22FDX. **SiMOS single-qubit** RB records sit near 99.9%.
- **2026 firsts**: quantum error detection in silicon (Jan 2026); first universal logical operations on silicon spins (Mar 2026).

KEY GRADED CLAIMS

- [T2] >99% 2Q fidelity from a standard 300 mm industrial flow (gate-defined) — Diraq/imec, Nature (2025) (demonstrated)
- [T2] 99.99% 2Q + Grover at 98.9% in donor devices — SQC, Nature (Dec 2025) (demonstrated)
- [T4] 250k-register patterning $\Rightarrow$ manufacturing scalability — SQC release (claimed)
- [T1] Donor P atoms are intrinsically identical; gate-dots buy reproducibility from lithography — device physics (established)

TRADE-OFFS (VS OTHER MODALITIES)

Unmatched density, direct reuse of the CMOS supply chain, and solid-state coherence good enough for foundry-grade fidelity; against that, qubit counts are tiny (tens at most today versus thousands for atoms), historic device-to-device variability, short-range exchange connectivity, and cryogenic control wiring per qubit that is unsolved at scale — cryo-CMOS is the bet that closes it. The quantum-dot vs donor split is a live strategic fork (see [H-spinsplit](#)): manufacturability vs intrinsic uniformity.

CONFLICTS / OPEN QUESTIONS

Foundry-grade fidelity arrived in 2025; the open question is whether uniformity holds across 1,000+ dots on a wafer, and whether anyone stands up a mid-scale (100-qubit) silicon system before other modalities reach fault tolerance. Spin-shuttling fidelity across a chip is the connectivity bottleneck.

SOURCES

Diraq/imec Nature paper + The Quantum Insider (Sep 2025); SQC Nature (Dec 2025); postquantum.com silicon-spin fabrication + ecosystem analyses; PatSnap spin-qubit array review (2026).

Spin-photon interfaces — T-centers, SiV/SnV, rare-earth emitters · H-spinphoton

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The physical component that lets a stationary matter qubit (an electron or nuclear **spin**) emit a **photon** whose state — polarization, time-bin, or frequency — is entangled with the spin. That single-emitter primitive is what turns an isolated qubit into a network node: interfere photons from two remote emitters, herald on a detector click, and the two distant spins are entangled (the heralded-entanglement protocol under **F-teleport**, the building block of modular QC **H-intercon** and the quantum internet **A-qinternet** / repeaters). Unlike microwave↔optical transduction (**H-transduce**), which frequency-converts a photon with no memory attached, and unlike deterministic photon sources for photonic computing (**H-photonsource**), which want indistinguishable photons but no long-lived spin, a spin-photon interface must deliver **both** a coherent optical transition and a spin qubit that outlives many entanglement attempts.

The hard figures of merit: the **zero-phonon-line fraction** (Debye–Waller factor — how much emission lands in the coherent, interference-usable line rather than the phonon sideband), photon **indistinguishability**, spin **coherence** T_2 , emission **wavelength** (telecom O/C-band beats visible for fiber reach), and the resulting remote-entanglement **rate × fidelity**. Nanophotonic cavities rescue weak emitters via the Purcell effect, $F_P = \frac{3}{4\pi^2} \left(\frac{\lambda}{n}\right)^3 \frac{Q}{V}$, funnelling emission into the ZPL and raising cooperativity $C = g^2/(\kappa\gamma)$ toward the deterministic-interface regime.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Silicon T-centers** (Photonic Inc., Canada): spin-photon interface native to silicon, emitting at telecom **1326 nm** — CMOS-foundry-compatible and fiber-ready; distributed entanglement between modules demonstrated, roadmap ~200 kHz remote-entanglement rate at 99.8% fidelity (see **H-intercon**).
- **Group-IV diamond color centers** — **SiV / SnV** (Harvard/Lukin, QuTech, academic + Quantum Brilliance-adjacent): cavity-coupled SiV memory

nodes and **tin-vacancy** centers combine a strong ZPL with better spin coherence than the NV (**H-nv**); high-fidelity spin-photon entanglement in nanophotonic cavities.

- **Rare-earth ions** — **erbium** in silicon or Y_2SiO_5 : telecom **1.5 μm** emission and exceptional coherence, but weak dipoles that require strong cavity Purcell enhancement to emit usefully.
- **Quantum dots**: brightest, most indistinguishable photons, but short spin coherence and spectral diffusion limit them as memory nodes (they excel as sources instead — **H-photonsource**).

KEY GRADED CLAIMS

- [T3] Telecom-band (1326 nm) silicon T-center spin-photon entanglement, foundry-compatible — Photonic Inc. (demonstrated; rate/fidelity targets still roadmap)
- [T2/T3] Cavity-enhanced SiV/SnV spin-photon entanglement at high fidelity in diamond nanophotonics — Harvard/QuTech (demonstrated in lab)
- [T4] 200 kHz / 99.8% remote entanglement over telecom fiber — company target (roadmap)

TRADE-OFFS

Every candidate trades brightness against coherence against wavelength. Diamond centers give clean optics and good spin memory but need exotic fabrication and mostly emit in the visible; T-centers and erbium reach telecom fiber but start dim and lean on cavities; quantum dots are bright but forgetful. No single emitter yet wins all axes at once — which is why this is a distinct node rather than a footnote to **H-silicon** (electrically-controlled spin *compute*) or **H-nv** (NV sensing/compute). The spin-photon interface is the emitter node that makes matter-qubit networking possible at all.

CONFLICTS / OPEN QUESTIONS

Which emitter reaches deterministic, high-rate, high-fidelity, telecom-band operation first — and does cavity integration (**H-photonsource** , **H-package**) hold coherence while boosting brightness? Whether spin-photon links can stay above the entanglement-distillation threshold across real fiber decides if matter-qubit modality networks natively and sidesteps the microwave-transduction wall (**H-transduce**) that superconducting machines face.

SOURCES

Photonic Inc. T-center networking releases; Harvard/Lukin SiV cavity-QED memory-node papers (Nature/Science 2020–25); SnV spin-photon results (Nature Physics/PRX 2024–26); erbium-in-silicon single-photon emission (Nature 2024–26). Cross-links: [H-silicon](#) , [H-nv](#) , [H-photonsource](#) , [H-transduce](#) , [H-intercon](#) , [A-qinternet](#) , [F-teleport](#) , [H-detect](#) .

Quantum-dot vs donor: the silicon-spin fork · H-spinsplit

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Inside the silicon-spin modality (**H-silicon**) there is a strategic fork sharp enough to warrant its own node: two ways to hold a single-electron spin in silicon, optimizing for opposite things. **Gate-defined quantum dots** trap electrons in an electrostatic well shaped by lithographic top-gates (SiMOS or Si/SiGe heterostructures) — the qubit position and count are set by the mask, so the approach inherits CMOS manufacturability and prioritizes *scalability*. **Donor qubits** put the spin on a single dopant atom — usually phosphorus — placed with atomic precision by STM lithography, so every qubit is physically identical by nature, prioritizing *uniformity and raw qubit quality*. Both now clear 99% two-qubit fidelity, so the fork is no longer about which one works; it is about which one scales.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Donor camp — SQC** (Michelle Simmons, Australia): STM-placed phosphorus, 99.99% 2Q fidelity and Grover at 98.9% without error correction (Nature, Dec 2025); patterned 250,000-qubit registers in 8 hours (Nov 2025). The argument: identical atoms remove device-to-device variability at the source.
- **Quantum-dot camp — Diraq + imec**: gate-defined dots at >99% 2Q from a standard 300 mm CMOS flow, with four random wafer devices all under 1% error (Nature, 2025). **Intel** (Tunnel Falls), **Quantum Motion**, and **Equal1** all pursue the dot route because it plugs directly into the foundry (**H-foundry**) and cryo-CMOS (**H-cryocmos**).
- Hybrid ideas (flip-flop qubits, donor-dot hybrids, spin-shuttling buses) blur the line and try to take uniformity from donors and routing from dots.

KEY GRADED CLAIMS

- [T2] Donor P devices: 99.99% 2Q, Grover 98.9% — SQC, Nature (Dec 2025) (demonstrated)
- [T2] Gate-defined dots: >99% 2Q from a 300 mm CMOS flow — Diraq/imec, Nature (2025) (demonstrated)

- [T1] Phosphorus donors are intrinsically identical; dots buy reproducibility from lithography — device physics (established)

TRADE-OFFS

Donors give the best intrinsic uniformity and coherence but atomic-precision STM placement is slow, serial, and hard to interface with industrial CMOS back-ends. Gate-defined dots ride the foundry and cryo-CMOS ecosystems and are naturally reproducible in *layout*, but electrostatic disorder and charge noise make each dot's exact working point drift, so tuning many dots (autotuning, ML calibration) is itself a research problem. Connectivity is short-range exchange in both, pushing spin-shuttling as the bus.

CONFLICTS / OPEN QUESTIONS

Which philosophy reaches a working 100-qubit silicon system first — the donor bet that uniformity beats the variability wall, or the dot bet that foundry volume and autotuning tame it? Does spin-shuttling across a chip preserve fidelity well enough to make either route's short-range connectivity a non-issue? The fork may resolve into a hybrid rather than a winner.

SOURCES

SQC Nature (Dec 2025); Diraq/imec Nature (2025); postquantum.com silicon-spin ecosystem; PatSnap spin-qubit-array review (2026); Intelligent Computing icomputing.0115 (single-electron spin review).

Superconducting (transmon) qubits · H-supercon

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Artificial atoms made from superconducting circuits — a Josephson junction gives a nonlinear inductance, so the two lowest energy levels of the circuit form a qubit. The transmon variant runs the junction shunted by a large capacitor to flatten its sensitivity to charge noise, trading anharmonicity for stability; the fluxonium variant shunts with a large inductance and reaches higher anharmonicity and longer coherence at the cost of a harder control problem. Qubits are driven with shaped microwave pulses at ~4–6 GHz and read out dispersively through a coupled resonator; the whole chip sits at ~10–20 mK in a dilution refrigerator (see [H-cryo](#)), with readout amplified by a near-quantum-limited parametric amplifier (see [H-paramp](#)). This is the most mature gate-based modality by deployed-system count and the one IBM, Google, and Rigetti build on.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **IBM:** Nighthawk (Nov 2025) — 120 transmons on a square lattice targeting 5,000-gate circuits in 2025, rising to 7,500 (2026), 10,000 (2027), 15,000 (2028), chained into three-module 360-qubit configs via I-couplers. Loon (2025) demonstrated c-couplers giving six-way non-nearest-neighbor connectivity for qLDPC codes while holding several-hundred- μ s coherence. Roadmap: Kookaburra (2026, first qLDPC memory + logic-processing unit), Cockatoo (2027), **Starling** fault-tolerant system in 2029 (200 logical qubits, 100M gates), Blue Jay (2033, ~2,000 logical).
- **Google:** Willow, 105 qubits — below-threshold surface-code QEC (Nature 2024): logical error suppressed $\Lambda \approx 2.14\times$ per code-distance step ($d=3\rightarrow5\rightarrow7$), distance-7 logical memory outliving the best physical qubit by $2.4\times$, with a real-time decoder. Roadmap: a long-lived logical qubit, then ~1M physical qubits.
- **Rigetti:** Cepheus-1-108Q GA system (Apr 2026), 99.1% median 2Q fidelity via a modular 4×9 -qubit chiplet tiling; targets 150+ qubits at 99.7% by late 2026, 1,000+ by end-2027.

- **IQM**: Radiance, ~150 qubits, 99.91% 2Q fidelity reported Mar 2026. **OQC** (UK): coaxmon 3D architecture. **SEEQC**: single-flux-quantum (SFQ) digital control on-chip.
- **Coherence records**: a Princeton 2D transmon on tantalum/high-resistivity-silicon reached $T_1 \approx 1.68$ ms ($Q \approx 2.5 \times 10^7$), roughly $3\times$ the prior best (Nature 2025) — direct evidence the TLS/materials bottleneck (see **H-fab**) is still yielding to substrate engineering.

KEY GRADED CLAIMS

- [T2] Below-threshold surface-code error correction on 105-qubit Willow — Nature s41586-024-08449-y (2024) (demonstrated)
- [T2] 2D transmon T_1 up to 1.68 ms on tantalum/Si — Princeton, Nature (2025) (demonstrated)
- [T4] IBM Starling: fault-tolerant, 200 logical qubits by 2029 — IBM roadmap (roadmap)
- [T4] Rigetti 99.1% median 2Q on 108-qubit GA system — company release, Apr 2026 (claimed)
- [T4] IQM 99.91% 2Q fidelity on Radiance — company claim, Mar 2026 (claimed)

TRADE-OFFS (VS OTHER MODALITIES)

Fast gates (tens of ns, $\sim 1000\times$ faster than ions), lithographic fab, and the deepest industrial base; against that, coherence measured in $\sim 0.1\text{--}1.7$ ms, mostly fixed nearest-neighbor connectivity, dilution cryogenics, and TLS defects that cap and drift coherence. Wiring heat-load and fridge volume dominate the scaling problem past $\sim 1,000$ qubits, which is why every roadmap turns to cryo-CMOS (**H-cryocmos**), 3D integration (**H-package**), and modular interconnects (**H-intercon**).

CONFLICTS / OPEN QUESTIONS

Can qLDPC codes (IBM) beat the surface code's overhead in real hardware, or is the surface code's higher threshold worth its worse encoding rate? Does the modality hit a wiring/cryo wall before millions of qubits, forcing fridge-to-fridge links that depend on unsolved microwave-optical transduction (**H-transduce**)?

SOURCES

IBM Quantum blog "large-scale FTQC" + roadmap 2025/2026; Google Quantum AI, Nature s41586-024-08449-y; Princeton Nature s41586-025-09687-4 (millisecond transmon); Rigetti investor release; IQM Radiance release; quantumzeitgeist.com superconducting guide (2026).

Topological / Majorana qubits · H-topo

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Qubits encoded in **Majorana zero modes** — quasiparticles predicted at the ends of a topological superconductor (a semiconductor nanowire proximitized by a superconductor, tuned into a topological phase by magnetic field and gate voltage). A single logical qubit is stored non-locally across a pair (or four) of these modes, so local noise cannot read or corrupt it: the error protection is built into the physics rather than bolted on by a code. Gates would be done by measurement (Microsoft's "measurement-based" scheme) or by braiding modes around each other. The prize is a qubit with a hardware error floor so low that QEC overhead collapses. The problem: after two decades, nobody has an independently verified working topological qubit.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Microsoft:** Majorana 1 (Feb 2025) — an 8-qubit "topoconductor" chip (InAs/Al), claiming measurement-based topological qubits with ~2 ms state lifetimes and an architecture that scales to 1M qubits on a palm-sized chip. The accompanying Nature paper's editors appended a note that the results "do not represent evidence for the presence of Majorana zero modes." Majorana 2 (Build milestone, Jun 2026): aluminum replaced with lead, claimed ~20-second state lifetimes (~1,000× improvement) and an FTQC timeline pulled from 2033 to 2029.
- Microsoft is effectively alone in productizing. **Nokia Bell Labs**, Delft/QuTech, Copenhagen, and other academic groups continue the underlying topological-matter physics; several have published careful null or ambiguous results on the same nanowire signatures.

KEY GRADED CLAIMS

- [T4] Majorana 1 hosts measurable topological qubits — Microsoft (Feb 2025) (contested — Nature editorial note; APS-meeting skepticism; no independent reproduction)
- [T4] Majorana 2: ~20 s lifetimes, 1,000× stability gain — Microsoft (Jun 2026) (contested — critics note the protocol-dependence and the still-

missing independent demonstration of the topological phase). As of mid-2026 there is still **no independent replication**; the sharpest critique remains Henry Legg's formal Nature comment (24 Jun 2026, d41586-026-01788-y) flagging flawed tune-up routines, software errors in the analysis code, and omitted transport measurements — Microsoft filed a formal Nature reply (Nayak: "we stand by our results and our roadmap"). Reproduction is throttled by a de-facto verification vacuum: only a handful of labs can fabricate the InAs/Al (now Pb/Al) devices.

- [T6] Scalable topological FTQC by 2029 — Microsoft roadmap (roadmap)

TRADE-OFFS (VS OTHER MODALITIES)

If real: intrinsic hardware error protection, digital voltage-based control (reusing the semiconductor toolset), a tiny footprint, and a dramatic cut in QEC overhead that would reshape every scaling roadmap in this chapter. Today: zero independently verified qubits, extreme sensitivity to disorder, and a research lineage that already produced one retracted Nature paper (the 2018 Delft/Microsoft-funded ballistic-Majorana quantized-conductance claim, retracted 2021). The modality is graded hardest in the manual for exactly this reason.

CONFLICTS / OPEN QUESTIONS

C-majorana-exists: Microsoft (T4) vs. much of the condensed-matter community, which holds that disorder-induced Andreev bound states can mimic every reported signature. What would resolve it: an independent lab reproducing topological-qubit *operation*, or a braiding/fusion demonstration with published raw data that unambiguously passes the topological-gap protocol.

SOURCES

Nature d41586-026-01788-y (2026 skeptics piece) + Majorana 1 paper editorial note; Science/AAAS coverage (2025, 2026); Physics World expert roundup; postquantum.com Majorana 2 analysis; 2021 Nature retraction of the 2018 quantized-conductance paper.

Microwave↔optical transduction · H-transduce

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Superconducting qubits live at ~10 mK and speak microwave photons (~5–10 GHz). Optical fiber carries quantum information between fridges at ~200 THz with almost no loss. Nothing bridges the ~10,000× frequency gap cheaply, so a **quantum transducer** — a device that coherently converts a single microwave photon into a single optical photon and back — is the missing link for fridge-to-fridge superconducting modularity (**H-intercon**) and for a superconducting-to-photon quantum internet (**A-qinternet**). Three physical routes compete: **electro-optic** (χ^2 crystals, thin-film lithium niobate), **optomechanical** (a vibrating membrane or bulk-acoustic mode couples both fields), and **magneto-optic / atomic** (rare-earth ions like erbium, or magnons in YIG). The figures of merit are conversion **efficiency** η , **added noise** (junk photons per converted photon), and **bandwidth**.

KEY PLAYERS & STATE OF THE ART (2025–26)

- The target is the "quantum-enabled" regime: added noise below 1 photon *and* η above ~50% simultaneously. Theory says $\eta > 1/2$ with low noise is physically allowed, so this is an engineering wall, not a fundamental one.
- 2025 membrane-based **optomechanical** transducers reached input-referred added noise approaching single-photon levels in both directions; one platform reported external $\eta \approx 2.2\%$ at added noise ≈ 0.94 , with an efficiency-bandwidth product ~2 orders of magnitude above prior low-noise demos (arXiv:2509.26349).
- **Electro-optic** thin-film-lithium-niobate devices push bandwidth and integration; **erbium-ion-in-crystal** transducers coupled to planar photonic + superconducting resonators pursue the atomic route (Nature Communications).
- Academic groups (Caltech, JILA/NIST, Delft, Chicago, Yale) dominate; no commercial merchant transducer ships yet. IBM, PsiQuantum, and every superconducting-modularity roadmap treat it as an *open dependency*, not a solved input.

KEY GRADED CLAIMS

- [T3] Membrane optomechanical transduction near single-photon added noise, $\eta \sim 2\%$ — APS SMT 2025 / arXiv:2509.26349 (demonstrated, low efficiency)
- [T1] $\eta > 1/2$ with low added noise is physically allowed — transduction theory (established)
- [T4] Efficiency high enough for fault-tolerant fridge-to-fridge links — no group claims this yet (roadmap)

TRADE-OFFS

The brutal three-way tension is efficiency vs added noise vs bandwidth: pumping harder raises η but heats the device and injects thermal noise; narrow bandwidth mismatches qubit lifetimes and throughput. Percent-level efficiency means most photons are simply lost — unusable for QEC, which needs high-fidelity links. This node, not qubit count, gates whether superconducting machines can ever go truly modular across fridges.

CONFLICTS / OPEN QUESTIONS

Will any of the three routes reach simultaneous high- η , sub-unity-noise, qubit-bandwidth operation — and on what timeline? Or do superconducting machines stay confined to single-fridge scale (**H-cryo** volume limits) while photonic (**H-photonic**) and ion (**H-ion**) modalities network natively over fiber and sidestep the problem entirely? This is one of the quietest but most decisive forks in the whole hardware map.

SOURCES

schedule.aps.org/smt/2025 (MAR-A09); arXiv:2509.26349; Nature Communications s41467-023-36799-0; arXiv:2503.01133; arXiv:2406.02704.

Ultra-high-vacuum & atomic-source systems · H-uhv

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

Trapped-ion (H-ion) and neutral-atom (H-neutral) machines hold their qubits — bare atoms — in an **ultra-high or extreme-high vacuum** chamber (10^{-11} mbar and below). The vacuum *is* the qubit's isolation: a single collision with a stray background gas molecule knocks an atom out of its trap, which is the modality's characteristic error channel (atom/ion loss). The system comprises the chamber and viewports, ion/getter pumps, an atomic source (an oven or dispenser for ions; a 2D-MOT / dispenser cold-atom source for neutral atoms), and — increasingly — a **cryogenic shroud** that both improves the vacuum (cold surfaces cryopump residual gas) and blocks blackbody radiation that would otherwise limit Rydberg-state lifetimes. It is the atomic-modality counterpart of the dilution fridge (H-cryo): unglamorous plumbing that sets a hard ceiling on qubit lifetime and count.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Cryogenic vacuum for atoms:** a high-optical-access cryogenic chamber achieved a **3,000-second single-atom trap lifetime** for Rydberg arrays (arXiv:2412.09780) — using cryogenically cooled metal walls and windows to cryopump the background gas and suppress unwanted transitions. Trap lifetime, historically seconds, is now limited by design rather than by attainable vacuum.
- **Continuous operation:** Harvard/MIT ran a ~3,000-atom array for over two hours by reloading atoms mid-computation (2025), turning atom loss from a hard stop into a managed, replenished error — a vacuum/atom-source engineering result as much as a physics one (see H-neutral).
- **Scale:** Caltech loaded 6,100 atoms in one chamber (Sep 2025); QuEra and Atom Computing project ~100,000 atoms per vacuum chamber within a few years — which pushes chamber size, pumping speed, and source flux hard. Vendors like **ColdQuanta/Infleqtion** and specialist UHV-chamber makers supply the hardware.

KEY GRADED CLAIMS

- [T2] 3,000-second trap lifetime in a cryogenic high-optical-access chamber — arXiv:2412.09780 (demonstrated)
- [T2] Continuous >2 hr operation via mid-computation atom reloading — Harvard/MIT, Nature (2025) (demonstrated)
- [T1] A background-gas collision ejects a trapped atom (loss is vacuum-limited) — established atomic physics

TRADE-OFFS

Room-temperature UHV chambers are simpler and give full optical access but cap trap lifetime at seconds to minutes; cryogenic chambers extend lifetime by orders of magnitude and add blackbody suppression, at the cost of a cooling system and reduced optical access. Bigger chambers hold more atoms but are harder to pump and to keep field-uniform. The atomic source flux trades loading speed against vacuum degradation.

CONFLICTS / OPEN QUESTIONS

Does the path to 100,000-atom machines run through room-temperature chambers with fast reloading, or through cryogenic chambers with long-lived atoms — and can either keep vacuum-limited loss below the QEC threshold at that scale? Optical access vs cryogenic shrouding is a live chamber-design tension for the biggest neutral-atom roadmaps.

SOURCES

arXiv:2412.09780 (3,000 s cryogenic trap); Nature 2025 continuous-operation paper; Caltech 6,100-atom (Sep 2025); APS Physics v18/103 (longer atom trapping); postquantum.com building-a-neutral-atom-QC guide.

Cryogenic microwave wiring — coax, attenuators, isolators · H-wiring

Layer: L1 Hardware · **Chapter:** §02 · **Status:** depth

WHAT IT IS

The passive microwave plumbing that carries signals between room-temperature electronics (**H-control**) and the qubits at the bottom of the fridge (**H-cryo**) — and the single least-glamorous scaling wall in superconducting and spin QC. Every superconducting qubit needs on the order of 2–4 lines (drive, flux, readout in/out), each a coaxial cable running through four temperature stages. Three passive families dominate: **coax** (semi-rigid CuNi or stainless above 4 K for low thermal conductance; superconducting NbTi below 4 K for near-zero loss), **attenuators** anchored and distributed across stages (a canonical drive-line budget is ~20 dB at 4 K, ~10 dB at the still, ~20–30 dB at the mixing chamber), and **isolators** / **circulators** — bulky ferrite, magnetically biased, non-reciprocal parts on the output line that protect the qubit from amplifier back-action (**H-paramp**) but do not shrink or integrate. Filters (IR/low-pass, eccosorb) and thermal anchoring round out the chain.

The attenuators are not there to weaken the signal for its own sake; they thermalize the Johnson–Nyquist noise riding down each line, so the qubit sees a cold photon bath. Residual thermal occupation is $\bar{n} = (e^{\hbar\omega/k_B T} - 1)^{-1}$; at 5 GHz and 20 mK, $\bar{n} \sim 10^{-5}$, but only if every stage is properly attenuated and anchored. Each coax also dumps conducted + radiated heat, so line count is capped by the fridge's cooling power — the bottleneck **H-cryocmos** and photonic I/O exist to break.

KEY PLAYERS & STATE OF THE ART (2025–26)

- **Coax / connectors:** HuberSuhner, Radiall, Maury Microwave; **cryo attenuators:** XMA/Omni-Spectra, Bluefors wiring kits.
- **Delft Circuits** (NL): flexible **Cri/oFlex** ribbon cabling puts dozens of lines in one flex, cutting the per-qubit space and heat-load footprint — the leading answer to the coax-count problem.

- **Low Noise Factory / Silent Waves / Quantum Microwave:** cryo isolators, directional couplers, and the output-chain components feeding the HEMT/TWPA readout.
- **Isolator replacement** is the hot research front: on-chip Josephson circulators, directional/nonreciprocal TWPAs, and multiplexed readout aim to delete the bulky ferrite parts entirely — none is yet a drop-in production replacement.

KEY GRADED CLAIMS

- [T1] Distributed multi-stage attenuation + NbTi superconducting coax deliver $\bar{n} \ll 1$ thermal photons at the qubit — established cryo-microwave practice
- [T3] Flexible ribbon cabling (Cri/oFlex-class) materially raises line density per fridge — Delft Circuits + lab deployments (demonstrated)
- [T4] On-chip nonreciprocity will retire ferrite isolators at scale — research roadmap (not yet productized)

TRADE-OFFS

Every added line buys control fidelity and costs cooling power and volume; the ferrite isolators are the worst offenders on footprint and refuse to integrate. A million-qubit superconducting machine implies millions of coax lines — physically impossible in one fridge, which is exactly why the endgame forks toward cryo-CMOS multiplexing (H-cryocmos), photonic readout, and multi-fridge modularity (H-intercon , H-transduce). This node is the passive-supply-chain twin of the active control problem, split out from H-cryo (the fridge) and H-fab (the chip materials).

CONFLICTS / OPEN QUESTIONS

Does the wiring bottleneck get solved by better passive engineering (denser ribbons, integrated isolators) or does it force an architectural escape (cryo-CMOS in-fridge control, optical I/O) before superconducting qubit counts reach the tens of thousands? The ferrite-isolator replacement is the pivotal unsolved passive component. Supply concentrates in few vendors — a chokepoint mirrored in E-supplychain .

SOURCES

Krinner et al., "Engineering cryogenic setups for 100-qubit scale systems," EPJ Quantum Technology 6, 2 (2019); Delft Circuits Cri/oFlex documentation; Bluefors wiring guides; Low Noise Factory / Silent Waves component notes.

Cross-links: [H-cryo](#) , [H-cryocmos](#) , [H-control](#) , [H-paramp](#) , [H-fab](#) , [H-intercon](#) , [E-supplychain](#) .

§03 · Stack & Algorithms

From a noisy qubit to a useful answer — error correction, compilers, Shor/Grover/VQE, benchmarks, and the honest speedup accounting.

Benchmarks & "advantage" claims · S-bench

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

How the field measures machines and adjudicates beyond-classical claims.

Holistic metrics: Quantum Volume (IBM 2019 — the largest square random circuit a device passes, 2^n ; conflates width, depth, fidelity into one number), CLOPS (circuit layers per second, a speed metric), Algorithmic Qubits (IonQ, application-weighted — disputed methodology). **Component metrics:** gate fidelity via randomized benchmarking / cross-entropy benchmarking / cycle benchmarking (S-gates), plus IBM's "layer fidelity." **Advantage claims** rest on sampling tasks — random circuit sampling (RCS, scored by cross-entropy benchmarking / XEB) and boson sampling — where classical simulation cost is *conjectured* exponential, tying the claim to a complexity assumption rather than a proof (S-complexity , S-tensornet).

WHERE IT STANDS (2025–26)

Every advantage claim has drawn a classical counterattack; several fell. **Google 2019** (Sycamore, 53-qubit RCS, "10,000 years"): IBM answered 2.5 days immediately; Pan–Chen–Zhang tensor-network methods (2021–22) cut it to hours-then-minutes on GPU clusters — effectively matched (status: fell). **USTC Jiuzhang** boson sampling (2020): substantially eroded by spoofing and tensor methods. **Willow** (Dec 2024): 105-qubit RCS claimed at $\sim 10^{25}$ classical-years; no classical match to date, tensor-network teams still narrowing the gap — contested-by-default. **Willow "quantum echoes"** (Oct 2025): 13,000× on an OTOC measurement (S-qsim), marketed as *verifiable* advantage — under scrutiny, unresolved. Quantinuum holds the QV record ($\sim 2^{25}$ region, self-reported, 2025); IBM de-emphasized QV in favor of gate-count / layer-fidelity metrics; the resulting **benchmark fragmentation is itself a finding**. IBM staked "verified quantum advantage by end of 2026" on Nighthawk. The reference implementation models the honest small-scale version of this discipline: it reports kernel RMSE and diagonal-deviation as per-run noise gauges rather than a headline advantage number (reference-impl/MATH.md §6–7).

KEY GRADED CLAIMS

- [T2] Sycamore 2019 RCS beyond-classical — Arute et al., Nature 574, 505 (2019), then matched classically: Pan et al., PRL 129, 090502 (2022) (status: fell)
- [T3/T4] Willow 2024 RCS unmatched classically so far — Google blog + Nature 638 (2024) (claimed; contested by default, standing)
- [T4] QV $\approx 2^{25}$ record — Quantinuum blog (vendor self-reported)
- [T4] Algorithmic Qubits metric — IonQ; publicly disputed by Quantinuum ("Debunking algorithmic qubits") (contested)

SPEEDUP / CAVEAT

Sampling tasks have **no known application**; their hardness is conjectural and erodes as classical algorithms improve (`S-tensornet` , `0-advantage`). A benchmark controlled by the vendor reporting it is **T4 by rule** (`evidence/SCHEMA.md`). "Verifiable" advantage matters because RCS results are themselves hard to check — Google's OTOC claim leans on this (`0-verification`).

CONFLICTS / OPEN QUESTIONS

C-advantage: does Willow-class RCS stay unmatched? What replaces QV as the cross-platform standard — DARPA QBI / QED-C application benchmarks are the leading candidates for an application-grounded, vendor-neutral metric.

SOURCES

Nature 574, 505 (2019); PRL 129, 090502 (2022); arXiv:2408.13687; [blog.google quantum-echoes](https://blog.google/quantum-echoes/); quantinuum.com/blog. Cross-links: `S-tensornet` , `S-complexity` , `S-qsim` , `S-gates` , `0-advantage` , `0-verification` , `reference-impl/` .

Certified randomness · S-certrand

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Certified randomness is a protocol that produces bits *provably* random — not merely unpredictable to an observer, but backed by a mathematical certificate that no adversary (not even the device manufacturer) could have known or biased them. The construction (Aaronson, 2018–20) piggybacks on random circuit sampling, the same task used for quantum-advantage claims: a classical server sends hard random-circuit challenges to a remote quantum computer, times the responses, and later spot-checks them on a classical supercomputer via cross-entropy benchmarking. Because a genuine quantum device is the only thing that could answer fast enough, passing responses certify that fresh entropy was generated. It is the first delivered *application* of quantum advantage — a working product built on top of the sampling experiments that were previously only benchmarks. It bridges to **A-qrng** (physical randomness sources): certified randomness adds the remote, adversary-proof *certificate* that a local quantum RNG cannot give.

WHERE IT STANDS (2025–26)

The landmark is JPMorgan × Quantinuum × Argonne/Oak Ridge × UT Austin, published in *Nature* (26 Mar 2025). Running on Quantinuum's 56-qubit trapped-ion H2, the team generated 71,313 certified-random bits; verification consumed ~1.1 exaFLOPS across supercomputers — the classical check itself is what makes the quantum step meaningful (only a real quantum device answers in time, only a classical HPC cluster can verify). Provenance matters: JPMorgan (a customer with a cryptography use case) and Quantinuum (the hardware vendor) co-authored, but this cleared peer review in *Nature*, so it grades above a vendor PR. The honest caveat is that certified randomness is a niche, if real, first application — its commercial pull (lotteries, cryptographic key material, auditable fairness) is genuine but narrow, and the verification cost is enormous.

KEY GRADED CLAIMS

- [T2] Protocol: certified randomness expansion from random circuit sampling — Aaronson–Hung, STOC 2023 (arXiv:2303.01625) (established theory)
- [T2] First experimental certified randomness: 71,313 bits on 56-qubit H2, verified at ~1.1 exaFLOPS — Nature 640 (2025), s41586-025-08737-1 (peer-reviewed)
- [T4→T2] "First commercial application of quantum computing" framing is vendor-sourced (Quantinuum/JPMorgan), but the underlying result is peer-reviewed (claimed marketing / demonstrated result)

SPEEDUP / CAVEAT

Advantage as a *product*, not a raw speedup. It exploits the fact that random circuit sampling is classically intractable to run but checkable, turning a supremacy benchmark into a service. Caveat: verification requires exaFLOPS-class classical compute per batch, and the application space (adversary-proof entropy) is narrow versus the general-purpose promise of quantum computing.

CONFLICTS / OPEN QUESTIONS

Whether the RCS foundation stays classically hard (spoofing/classical-simulation counterattacks target the same experiments — see [0-advantage](#) , [S-bench](#)), and whether the verification cost can drop enough to make this a routine service rather than a showcase.

SOURCES

Nature 640, s41586-025-08737-1 (2025); arXiv:2303.01625; JPMorganChase / UT Austin CNS / ORNL press (2025). Cross-links: [A-qrng](#) , [S-bench](#) , [0-advantage](#) , [I-finance](#) .

Circuit knitting & cutting · S-circuitcut

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Circuit knitting runs a circuit that is too large for one QPU by decomposing it into smaller subcircuits, executing them separately, and reconstructing the full result classically. Two cut types: **gate cuts** replace a non-local two-qubit gate with a quasiprobability sum of local operations, and **wire cuts** break a qubit's timeline into a measure-and-reprepare step. The reconstruction uses a **quasiprobability decomposition** — write the non-local operation as a signed combination of local channels, sample from it, and combine with the right signs in post-processing. This trades quantum width for classical sampling: it lets a 100-qubit problem run on smaller devices, or partitions a circuit across multiple QPUs (relevant to modular/interconnect hardware, **H-intercon**). IBM's Circuit Knitting Toolbox / qiskit-addon-cutting includes an automated cut-finder that searches for the cuts minimizing overhead.

WHERE IT STANDS (2025-26)

The technique is mature as software but bounded by a hard scaling law: each cut multiplies the sampling cost, and the **total overhead grows exponentially in the number of cuts** (γ^2 per gate cut, worse for wire cuts). So knitting is practical for a handful of cuts — separating weakly-entangled blocks, or shaving a few qubits past a device limit — and hopeless for cutting a densely entangled circuit into many pieces. The 2024–25 research pushed on the exponent: classical side-information from intermediate measurements improves post-processing (Phys. Rev. Research 2025), and a Dec 2025 result reduces the measurement overhead from exponential to **polynomial** in special cases via a quantum-tomography reformulation (arXiv:2512.19623) — a meaningful but still-narrow escape from the exponential wall. It shares a mathematical core with error mitigation (**S-errmit**): both are quasiprobability sampling, both pay exponential sampling cost, both estimate expectation values rather than states.

KEY GRADED CLAIMS

- [T2] Gate/wire cutting via quasiprobability; overhead exponential in cut count — Bravyi–Smith–Smolin, PRX 6, 021043 (2016); Peng et al., PRL 125, 150504 (2020) (established)
- [T2] Automated cut-finding minimizing sampling overhead — IBM Circuit Knitting Toolbox, IBM Research QCE 2024 (established, software)
- [T2] Classical side-information improves reconstruction — Phys. Rev. Research (2025) (peer-reviewed)
- [T3] Exponential→polynomial overhead in special cases via tomography reformulation — arXiv:2512.19623 (2025) (preprint)

SPEEDUP / CAVEAT

Not a speedup — a **space-for-time trade** (fewer qubits, exponentially more shots). It is a bridge technology for the pre-fault-tolerant / modular era, useful only when cuts are few and cross-cut entanglement is low; for a highly entangled cut it costs as much as the classical simulation it was trying to avoid (**S-tensornet**). Overlaps error mitigation's cost structure, so the same "shallow enough to knit $\approx$ shallow enough to simulate classically" caution applies.

CONFLICTS / OPEN QUESTIONS

How far the polynomial-overhead special cases generalize, and whether knitting across real chip-to-chip links (**H-intercon** , **0-interconnect-loss**) beats simply building a bigger monolithic device. Whether it earns a durable role once large fault-tolerant machines exist.

SOURCES

PRX 6, 021043 (2016); PRL 125, 150504 (2020); IBM Research QCE 2024; arXiv:2411.17756; arXiv:2512.19623. Cross-links: **S-errmit** , **S-tensornet** , **S-shadows** , **H-intercon** , **0-interconnect-loss** .

Cloud QPU access · S-cloud

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

How anyone actually touches a quantum computer: cloud platforms queueing circuits to remote QPUs. Three models dominate. **IBM Quantum** — own fleet, per-QPU-minute runtime, tiered Open/Pay-as-you-go/Premium. **AWS Braket** — an aggregator fronting IonQ, Rigetti, IQM, QuEra, and others, priced per-task plus per-shot. **Azure Quantum** — aggregator fronting IonQ and Quantinuum, plus the Azure resource estimator, priced per-program or by subscription. Google Quantum AI stays research-partner-gated. The compilation and submission all run through the software layer (`S-software`). The reference implementation is written to hit exactly these backends — Qiskit Runtime to IBM's least-busy device, Braket to IonQ — with only the backend flag changing (`reference-impl/README.md`).

WHERE IT STANDS (2025-26)

Access is a solved problem; the cost structure shapes research. Representative 2026 numbers: IBM PAYG $\approx$ **\$96/QPU-minute** (\$48–72 with Flex/Premium tiers); Braket $\approx$ **\$0.30/task** plus per-shot fees that vary $\sim 25\times$ by hardware (a 100-shot circuit runs $\sim$ \$8.30 on IonQ Forte versus $\sim$ \$0.34 on Rigetti); a 200k-shot VQE run spans **\$85–\$6,000** depending on backend. Quantinuum sells H-series/Helios access directly and via Azure. NVIDIA's CUDA-Q and "quantum-centric supercomputing" partnerships (IBM–RIKEN, various HPC centers) are pulling QPUs into HPC job schedulers as accelerators alongside GPUs. On-prem purchases (IQM, Rigetti, IBM systems at national labs) grew as sovereignty concerns rose (`E-supplychain` , `E-us`). Free tiers shrank across the board as demand from error-correction experiments soaked up device time.

KEY GRADED CLAIMS

- [T2] Multi-vendor cloud access operational since 2016 (IBM) / 2019 (Braket, Azure) — platform documentation (established)
- [T4] Published pricing: IBM $\sim$ \$96/min PAYG; Braket \$0.30/task + per-shot — aws.amazon.com/braket/pricing, IBM docs (vendor-published, verifiable)

- [T5] "Quantum cloud market" TAM projections — analyst reports (forecast; double-counting risk, E-market)

SPEEDUP / CAVEAT

Cloud access sets the **economics** of experimentation. At ~\$100/minute, algorithm development happens on simulators and only validation touches hardware — exactly the workflow the reference impl documents (develop on Aer, validate on IBM/IonQ). Queue times and calibration drift make published fidelity numbers optimistic relative to what a random user session sees; a device benchmarked at 99.9% on a good day may deliver noticeably worse mid-drift.

CONFLICTS / OPEN QUESTIONS

Aggregator versus vertical model — do Braket/Azure margins survive if the winning hardware vendors (IBM, Quantinuum) keep their best machines first-party? How fast does HPC-integrated access displace pure cloud submission as QEC workloads demand tight classical-quantum coupling (S-decoders)?

SOURCES

aws.amazon.com/braket/pricing; learn.microsoft.com/azure/quantum/pricing; IBM Quantum docs. Cross-links: S-software , S-decoders , E-market , E-supplychain , reference-impl/ .

Quantum advantage complexity theory — BQP, BPP & the separations · S-complexity

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The complexity-theory scaffolding that tells you *what* quantum computers can provably do faster, and what "quantum advantage" formally means. **BQP** (bounded-error quantum polynomial time) is the class of problems a quantum computer solves efficiently; **BPP** is its classical randomized counterpart. The central facts: $\text{BPP} \subseteq \text{BQP} \subseteq \text{PSPACE}$, so quantum computers are no more powerful than classical ones up to polynomial space, and a proof that $\text{BQP} \neq \text{BPP}$ would imply $\text{P} \neq \text{PSPACE}$ — beyond current mathematics. This is why **every** quantum speedup is either (a) relative to the *best known* classical algorithm, not a proven separation (Shor, S-shor), or (b) proven only in a restricted **oracle / query model** (Grover, S-grover ; Simon's problem, the first exponential query separation), or (c) proven under a **complexity-theoretic assumption** (sampling advantage rests on the non-collapse of the polynomial hierarchy). The map's 0-advantage node is the empirical face of this node's theory.

WHERE IT STANDS (2025-26)

The strongest unconditional evidence that quantum beats classical is relativized. **Raz–Tal (STOC 2019)** built an oracle under which $\text{BQP} \not\subseteq \text{PH}$ (the polynomial hierarchy) — refuting the natural conjecture that quantum power stays inside the classical hierarchy, using the Forrelation problem and a deep bound against constant-depth (AC^0) circuits. **Shallow-circuit separations** (Bravyi–Gosset–König, Science 2018, and the noise-robust follow-ups) prove a *provable, unconditional* advantage of constant-depth quantum over constant-depth classical circuits — one of the few speedups needing no complexity assumption at all. **Sampling advantage** (random circuit sampling, boson sampling; S-bench) is proven hard for classical computers *only if* the polynomial hierarchy does not collapse — a widely believed but unproven assumption, and one that erodes empirically as tensor-network methods improve (S-tensornet). **Interactive verification** (Mahadev 2018) lets a classical verifier check a quantum computation it cannot itself run, underwriting the "verifiable advantage"

framing (0-verification). Dequantization (Tang, S-hh1 / S-qm1) is the counter-current: several claimed exponential separations collapsed when the classical input model was matched.

KEY GRADED CLAIMS

- [T1] $BPP \subseteq BQP \subseteq PSPACE$; $BQP \subseteq P^{\#P}$ — Bernstein–Vazirani, SIAM J. Comput. 26 (1997) (established)
- [T1] Oracle separation $BQP \not\subseteq PH$ — Raz–Tal, STOC 2019 (established, relativized)
- [T1] Unconditional constant-depth quantum > constant-depth classical — Bravyi–Gosset–König, Science 362, 308 (2018) (established)
- [T2] Sampling advantage hard unless PH collapses — Aaronson–Arkhipov (boson sampling, 2011); Bouland et al., Nat. Phys. 15 (2019) (established, conditional)
- [T2] Classical verification of quantum computation — Mahadev, FOCS 2018 (established)

SPEEDUP / CAVEAT

This node is the *definition* of the caveat. No exponential quantum speedup over classical is unconditionally proven in the standard (non-oracle) model — proving one would resolve open problems in classical complexity. So the honest hierarchy of claims is: unconditional-but-shallow (Bravyi–Gosset–König) > relativized (Raz–Tal, Simon) > assumption-based (sampling) > best-known-classical (Shor, and the algorithms most people mean by "quantum advantage"). Grading every speedup by which rung it sits on is exactly the manual's job (evidence/SCHEMA.md).

CONFLICTS / OPEN QUESTIONS

Whether $BQP \neq BPP$ at all (unknown, and likely as hard as P vs PSPACE).

Whether any *practical* problem sits in the proven-hard regime with an efficient input model (the running S-hh1 / S-qram question). Whether sampling hardness assumptions survive continued classical progress (S-tensornet , S-bench).

SOURCES

SIAM J. Comput. 26, 1411 (1997); Raz–Tal, STOC 2019; Science 362, 308 (2018); Nat. Phys. 15, 159 (2019); Mahadev, FOCS 2018. Cross-links: [S-shor](#) , [S-grover](#) , [S-bench](#) , [S-tensor](#) , [0-advantage](#) , [0-verification](#) .

Real-time QEC decoding · S-decoders

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Error correction (*S-qec*) is only half the job — something must read the stream of syndrome measurements and infer, in real time, which correction to apply before the next round of gates. That something is the *decoder*, and it has become its own engineering discipline. The throughput wall is brutal: a surface-code logical qubit emits a syndrome every QEC cycle ($\sim 1\ \mu\text{s}$ on superconducting hardware), and the decoder must keep up or the error backlog grows without bound (the "decoding backlog problem," Terhal). At scale this is a firehose — Riverlane estimates a large machine produces on the order of $\sim 100\ \text{TB/s}$ of syndrome data. General software decoders (minimum-weight perfect matching, union-find, belief-propagation) are accurate but too slow, so the field has moved to dedicated FPGA and ASIC decoders co-located with the control electronics (*H-control*).

WHERE IT STANDS (2025–26)

Riverlane is the pure-play here (Deltaflow QEC system; Deltaflow 2 shipped 2025, Deltaflow 3 targeted late 2026 with continuous "streaming" decoding). In Dec 2025 they published a hardware decoder in Nature Electronics: a Local Clustering Decoder on FPGA that decodes a round in under $1\ \mu\text{s}$ while adaptively modeling noise, cutting the physical-qubit count per logical qubit $\sim 4\times$ under a leakage-dominated model. Google's Willow result (2024) used a real-time-capable decoder to show below-threshold scaling, and its AlphaQubit work applied ML decoding for higher accuracy at the cost of speed — the central tension of the field: accuracy vs latency vs power. This is a named open problem in its own right (*0-decoder*): the classical co-processor bandwidth may bottleneck fault-tolerant scaling as hard as the qubits do.

KEY GRADED CLAIMS

- [T2] Decoding must run inside the QEC cycle or the backlog diverges — Terhal, Rev. Mod. Phys. 87, 307 (2015) (established)
- [T2] Sub- μs adaptive hardware (FPGA) decoder, $\sim 4\times$ qubit reduction under leakage noise — Riverlane, Nature Electronics (Dec 2025) (peer-reviewed)

- [T2] Real-time decoding enabled Willow below-threshold demo; ML decoders (AlphaQubit) trade latency for accuracy — Google, Nature 638 (2024/25) (peer-reviewed)

SPEEDUP / CAVEAT

Not an algorithm speedup — an enabling constraint. Fault tolerance is impossible without a decoder that sustains syndrome throughput at scale; a slow decoder silently caps how large a computation can run regardless of qubit quality.

CONFLICTS / OPEN QUESTIONS

Whether ASIC/FPGA decoders scale to millions of qubits (100 TB/s-class data movement, cryo-adjacent power budgets) or become the true bottleneck — see `0-decoder`. Accuracy-vs-latency-vs-power has no settled optimum.

SOURCES

Rev. Mod. Phys. 87, 307 (2015); Riverlane Nature Electronics 2025 (HPCwire coverage); Google Willow, Nature 638 (2025); AlphaQubit, Nature 2024. Cross-links: `S-qec`, `S-logical`, `H-control`, `0-decoder`.

Error mitigation as a discipline · S-errmit

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The technique taxonomy that S-nisq names in passing, treated as its own layer because it now sits *underneath* early fault tolerance, not only on bare NISQ. Error mitigation reduces bias in an estimated **expectation value** by spending extra circuit runs; it does not protect a quantum state (that is correction, S-qec). The main families (Cai et al., Rev. Mod. Phys. 95, 045005, 2023): - **Zero-Noise Extrapolation (ZNE)** — run at amplified noise, extrapolate to zero. - **Probabilistic Error Cancellation (PEC)** — learn the noise, invert it via a quasiprobability sum (unbiased, exponential sampling cost). - **Symmetry verification / post-selection** — discard runs that violate a known conserved quantity. - **Virtual distillation / ESD** — prepare M copies and measure to suppress errors as the state's purity, cancelling incoherent noise to $\mathcal{O}(1/M)$. - **Probabilistic Error Amplification (PEA)** — the learned-noise-model backbone of IBM's 2023 utility experiment. - **Learning-based / Clifford-data regression** — train the correction on classically simulable Clifford circuits.

WHERE IT STANDS (2025-26)

The defining theoretical result is a hard ceiling: the sampling overhead of *any* mitigation grows **exponentially** with circuit depth and fault rate (Takagi et al.; Quek et al., "exponentially tighter bounds," Nat. Phys. 2024). So mitigation is a constant-to-polynomial extension of reach, never an asymptotic fix — this is what separates it from correction as a matter of principle. The 2025 shift is that mitigation is being layered **on top of logical qubits**: demonstrations of error mitigation applied to already-error-corrected circuits (Nat. Commun. 2025) and "mitigation for early fault tolerance" show the discipline surviving into the FTQC transition rather than being abandoned with NISQ. PEC and PEA also power the standing (and contested) utility-scale expectation-value claims that tensor networks (S-tensornet) keep answering.

KEY GRADED CLAIMS

- [T2] Comprehensive discipline review; ZNE/PEC/symmetry/virtual-distillation taxonomy — Cai et al., Rev. Mod. Phys. 95, 045005 (2023) (established)
- [T2] Exponential sampling-overhead lower bound for generic mitigation — Quek et al., Nat. Phys. 20 (2024); Takagi et al., npj QI (2022) (established)
- [T2] Virtual distillation suppresses errors as $\mathcal{O}(1/M)$ in copy number — Huggins et al., PRX 11, 041036 (2021); Koczor, PRX 11, 031057 (2021) (established)
- [T2] Mitigation demonstrated on logical qubits — Nat. Commun. (2025), s41467-025-67768-4 (peer-reviewed)

SPEEDUP / CAVEAT

Never a speedup — a bias-reduction technique bought with exponentially many shots in the worst case. The rule follows directly ([evidence/SCHEMA.md](#)): any advantage claim resting on mitigation alone is exposed to classical counterattack, because a circuit shallow enough for mitigation to converge is usually shallow enough to simulate ([S-tensornet](#)). Mitigation's honest role is to buy accuracy on small logical devices during the early-FT window.

CONFLICTS / OPEN QUESTIONS

Where the crossover sits — at what device quality does adding a logical layer beat spending the same resources on more mitigation shots? Whether learning-based mitigation generalizes beyond the Clifford-regression regime it is trained on.

SOURCES

Rev. Mod. Phys. 95, 045005 (2023); Nat. Phys. 20 (2024); PRX 11, 041036 (2021); PRX 11, 031057 (2021); Nat. Commun. (2025). Cross-links: [S-nisq](#) , [S-qec](#) , [S-tensornet](#) , [S-bench](#) , [0-advantage](#) .

Gates, circuits & universal sets · S-gates

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Quantum computation is expressed as circuits: sequences of unitary gates on qubits, followed by measurement. A finite gate set is **universal** if it approximates any unitary to arbitrary precision; the canonical example is Clifford (H, S, CNOT) plus the T gate. The Solovay–Kitaev theorem guarantees efficient synthesis — approximating any single-qubit unitary to precision ϵ costs $O(\log^c(1/\epsilon))$ gates ($c \approx 2-4$ depending on the variant), i.e. polylogarithmic overhead. The Gottesman–Knill theorem draws the dividing line: **Clifford-only** circuits are classically simulable in polynomial time, so the T gate is the source of quantum hardness and the currency of fault-tolerant cost. Every hardware platform exposes a small **native** set — CZ or iSWAP-family on superconducting chips, Mølmer–Sørensen on trapped ions, Rydberg-blockade CZ on neutral atoms — into which the compiler (S-software) rewrites everything. The Hadamard gate that powers the interference in the manual's swap/Hadamard-test reference implementation (reference-impl/MATH.md §1) is the simplest such primitive.

WHERE IT STANDS (2025-26)

Two-qubit gate fidelity is the binding constraint on everything upstream (S-qec thresholds, S-logical counts). Best reported: **Quantinuum Helios** at 99.921% two-qubit / 99.9975% single-qubit across all qubit pairs (Nov 2025, trapped ions, all-to-all connectivity); superconducting devices sit around 99.5–99.9%; neutral atoms are climbing past 99.5%. Connectivity matters as much as raw fidelity: heavy-hex lattices (IBM) and 2D grids (Google) pay SWAP-routing overhead to move qubits together, while trapped ions and reconfigurable atom arrays offer all-to-all connectivity that removes routing entirely. Dynamic circuits — mid-circuit measurement plus classical feed-forward — became standard on major platforms in 2024–25 and are prerequisite for real-time QEC (S-decoders).

KEY GRADED CLAIMS

- [T1] Clifford+T is universal; Solovay–Kitaev gives polylog-overhead synthesis — Nielsen & Chuang; Dawson–Nielsen, quant-ph/0505030 (established)
- [T1] Clifford-only circuits classically simulable in poly time — Gottesman–Knill, quant-ph/9807006 (established)
- [T2] Helios 99.921% two-qubit fidelity, all-to-all, 98 qubits — Quantinuum (Nov 2025); postquantum.com analysis (demonstrated; vendor-built device, third-party analyzed)
- [T4] Vendor "gate count" roadmaps (IBM Nighthawk 5,000→15,000 two-qubit gates by 2028) — IBM Quantum blog (roadmap)

SPEEDUP / CAVEAT

Universality is a **compilation** statement, not a speed statement — it says any unitary is reachable, not that it is cheap. T gates dominate fault-tolerant cost (each needs a distilled or cultivated magic state, S-qec), so **T-count** and **T-depth** are the real currency of algorithm resource estimates (S-shor , S-qsim). A circuit that looks short in physical gates can be enormous in T-count.

CONFLICTS / OPEN QUESTIONS

Cross-platform fidelity numbers are measured with different protocols — randomized benchmarking (RB), cross-entropy benchmarking (XEB), component/cycle benchmarking — and are only loosely comparable (S-bench). A single "99.9%" can mean different things on two vendors' datasheets.

SOURCES

Nielsen & Chuang (2000); quant-ph/0505030; quant-ph/9807006; Quantinuum Helios (2025); ibm.com/quantum/blog. Cross-links: S-qec , S-software , S-bench , S-shor , reference-impl/ .

Grover's algorithm · S-grover

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Grover (1996) finds a marked item among N unstructured possibilities in $O(\sqrt{N})$ oracle queries, versus $O(N)$ classically. The mechanism is **amplitude amplification**: start in a uniform superposition, then repeat a two-reflection step (oracle phase-flip of the marked state, then reflection about the mean) $\sim (\pi/4)\sqrt{N}$ times, rotating amplitude toward the solution in the 2D subspace spanned by marked and unmarked states. The quadratic speedup is provably optimal for black-box search — the BBBV theorem (Bennett–Bernstein–Brassard–Vazirani, 1997) shows no quantum algorithm beats $\Omega(\sqrt{N})$ queries. Amplitude amplification generalizes Grover to boost the success probability of any subroutine that outputs a "good" flag with amplitude a from $O(1/a^2)$ to $O(1/a)$ repetitions, making it a workhorse inside counting, quantum Monte Carlo / amplitude estimation (S-qmc), and optimization heuristics.

WHERE IT STANDS (2025–26)

Grover is textbook-established and has been demonstrated on a handful of qubits on every major platform. The 2020s reassessment is sobering. Careful architecture-level accounting (Babbush et al. 2021; Hoefler–Häner–Troyer, CACM 2023) shows quadratic speedups almost certainly deliver **no practical advantage** on foreseeable fault-tolerant hardware: the QEC overhead, the $\sim$ kHz–MHz logical clock speed, and the cost of implementing the oracle as a reversible circuit eat the $\sqrt{N}$ gain unless the classical runtime is already astronomically large (roughly, the crossover sits beyond problem sizes any real workload reaches). The reason is structural — Grover needs $\sim \sqrt{N}$ *sequential* coherent iterations and admits no parallel speedup (Zalka 1999), so throwing more qubits at it does not help wall-clock time. The field now treats Grover as a building block and a lower-bound landmark; nobody credible pitches bare Grover search as a near-term product.

KEY GRADED CLAIMS

- [T1] $O(\sqrt{N})$ unstructured search, provably optimal — Grover, quant-ph/9605043; Bennett–Bernstein–Brassard–Vazirani, SIAM J. Comput. 26, 1510 (1997) (established)
- [T2] Quadratic speedups washed out by fault-tolerance overhead for practical sizes — Babbush et al., PRX Quantum 2, 010103 (2021); Hoefler et al., CACM 66(5), 82 (2023) (established analysis, some parameter dependence)
- [T1] No parallel advantage; $\sim \sqrt{N}$ iterations must be sequential — Zalka, PRA 60, 2746 (1999) (established)

SPEEDUP / CAVEAT

Proven quadratic query speedup, optimal in the query model. Fine print: (1) query complexity only — a cheap classical oracle call versus an expensive reversible quantum circuit is an unfair comparison; (2) the sequential-iteration requirement kills wall-clock advantage at realistic N and logical clock rates; (3) the symmetric-key-crypto consequence is modest — simply double key lengths (AES-128 $\rightarrow$ AES-256), which is why NIST treats Grover as a minor concern next to Shor.

CONFLICTS / OPEN QUESTIONS

Whether amplitude-amplification subroutines embedded in larger algorithms (Monte Carlo pricing in `S-qmc`, optimization) survive the same overhead accounting. Current evidence leans no for quadratic-only gains; near-quadratic amplitude estimation is the most likely survivor. The two-reflection structure is a special case of QSVT (`S-qsvt`), which is the modern lens for reasoning about its cost.

SOURCES

quant-ph/9605043; SIAM J. Comput. 26, 1510 (1997); PRX Quantum 2, 010103 (2021); CACM 66(5), 82 (2023); PRA 60, 2746 (1999). Cross-links: `S-qmc`, `S-qsvt`, `S-walk`, `0-advantage`.

Hamiltonian simulation methods — Trotter / qDRIFT / LCU / qubitization · S-hamsim

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The *how* behind digital quantum simulation (`S-qsim`): concrete algorithms that implement the evolution operator e^{-iHt} on a gate-based machine for a Hamiltonian $H = \sum_k h_k P_k$. Four method families, each with a different complexity profile. - **Trotter–Suzuki product formulas.** Approximate e^{-iHt} by interleaving $e^{-ih_k P_k t/r}$ short steps. Simple, ancilla-free, and it enjoys **commutator scaling** — error depends on how much the terms fail to commute, so it is often far better than worst-case bounds suggest (Childs–Su–Tran–Wiebe–Zhu 2021). A p -th order formula gives error $O((t/r)^{p+1})$ per step. - **qDRIFT.** Randomly compose terms sampled with probability $\propto |h_k|$, so cost scales with the L1 norm $\lambda = \sum |h_k|$ and is **independent of the number of terms** — good for molecular Hamiltonians with many small terms. Trades a random-sampling error for term-count independence (Campbell 2019). - **LCU (linear combination of unitaries).** Write the propagator (e.g. its Taylor series) as a weighted sum of unitaries and apply it with an ancilla "select/prepare" pair. Achieves $O(\log(1/\varepsilon))$ precision scaling — exponentially better in ε than Trotter (Berry–Childs–Cleve–Kothari–Somma 2015). - **Qubitization / quantum walk.** Block-encode H and walk its eigenphases; **optimal** query complexity $O(\lambda t + \log(1/\varepsilon))$, the provable lower bound (Low–Chuang 2017/2019). This is the QSVT (`S-qsvt`) instance for time evolution.

WHERE IT STANDS (2025–26)

The methods are complementary, and the 2024–25 literature is about hybridizing them. For a fixed accuracy, **second-order Trotter** is often competitive with qubitization/LCU/qDRIFT on real molecular systems because commutator scaling beats the λ -scaling those methods pay, especially for geometrically local Hamiltonians; but LCU/qubitization win at high precision. Recent work compensates Trotter error with an LCU correction using one ancilla, getting both good system-size scaling and high accuracy (PRX Quantum 6, 010359, 2025), and Richardson-extrapolated qDRIFT (Watson 2024) reduces the standard

qDRIFT step count. Fault-tolerant chemistry resource estimates (the FeMoco line in `S-qsim`) now overwhelmingly use qubitization for its optimal T-count. Everything here is a **fault-tolerant** primitive: the deep coherent evolution requires `S-qec`.

KEY GRADED CLAIMS

- [T1] Trotter with commutator error scaling — Childs–Su–Tran–Wiebe–Zhu, PRX 11, 011020 (2021); Lloyd, Science 273 (1996) (established)
- [T1] qDRIFT: cost $\propto \lambda$, independent of term count — Campbell, PRL 123, 070503 (2019) (established)
- [T1] LCU / Taylor-series simulation, $O(\log 1/\varepsilon)$ precision — Berry et al., PRL 114, 090502 (2015) (established)
- [T1] Qubitization: optimal $O(\lambda t + \log 1/\varepsilon)$ query complexity — Low–Chuang, Quantum 3, 163 (2019) (established, matches lower bound)
- [T2] Hybrid Trotter+LCU error compensation, one ancilla — PRX Quantum 6, 010359 (2025) (peer-reviewed)

SPEEDUP / CAVEAT

The *enabling* engine for the best-founded quantum advantage (quantum simulation, `S-qsim`) — exponential over classical for generic dynamics. Caveats: (1) all four assume you can already load / block-encode H and prepare a useful initial state; (2) the constant factors and T-counts are large, keeping useful runs in the fault-tolerant era; (3) classical tensor-network methods (`S-tensornet`) absorb the low-entanglement / short-time instances, so the advantage lives in long-time, high-entanglement regimes.

CONFLICTS / OPEN QUESTIONS

Which method wins for a given target is instance-dependent and actively contested (Trotter's commutator advantage versus qubitization's optimal asymptotics). Whether early-fault-tolerant machines favor the low-ancilla randomized methods (qDRIFT, single-ancilla LCU) over qubitization's block-encoding overhead.

SOURCES

PRX 11, 011020 (2021); PRL 123, 070503 (2019); PRL 114, 090502 (2015); Quantum 3, 163 (2019); PRX Quantum 6, 010359 (2025). Cross-links: S-qsim, S-qsvt, S-qft, S-tensornet, S-qec, I-chem, I-pharma.

HHL & quantum linear algebra · S-hhl

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

HHL (Harrow–Hassidim–Lloyd, 2009) solves the linear system $Ax=b$ by preparing a quantum state $|x\rangle \propto A^{-1}|b\rangle$ in time $O(\log(N) \cdot s^2 \cdot \kappa^2/\varepsilon)$ — polylogarithmic in the dimension N , where S is sparsity, K the condition number, ε the target accuracy. Mechanism: phase-estimate (S-qft) the eigenvalues of A on $|b\rangle$, rotate an ancilla by $1/\lambda$ conditioned on each eigenvalue, then uncompute — inverting A eigenvalue-by-eigenvalue. Against classical Gaussian elimination's $O(N^3)$, or conjugate gradient's $O(Ns\kappa)$ for sparse systems, the $\log(N)$ scaling is an **exponential** improvement in dimension. HHL seeded a whole family: quantum linear algebra, quantum singular value transformation (S-qsvt), quantum recommendation systems, and the ML applications that fueled the 2015-era hype. The inner-product primitive at its core — estimating $\langle u|v\rangle$ between amplitude-encoded states — is exactly what the manual's **reference implementation** builds and validates on real hardware via the swap and Hadamard tests (reference-impl/MATH.md §3, §5, §8).

WHERE IT STANDS (2025–26)

HHL is the canonical "read the fine print" algorithm (Aaronson 2015). The exponential speedup survives only if **four assumptions hold together**: (1) A is sparse and well-conditioned — runtime is quadratic in K , disqualifying most engineering-grade (ill-conditioned) systems; (2) $|b\rangle$ can be prepared efficiently — a qRAM assumption (S-qram) that may itself hide exponential cost; (3) A can be Hamiltonian-simulated efficiently (S-qsim); (4) you want a scalar *property* of $\mathcal{X}$ ($\langle x|M|x\rangle$) rather than the vector itself — reading out all N amplitudes takes $O(N)$ tomography and kills the speedup. Ewin Tang's 2018 dequantization of quantum recommendation systems, and the quantum-inspired classical algorithms that followed (Chia et al., Gilyén et al.), removed the exponential gap entirely for **low-rank** instances. Matrix inversion is BQP-complete, so a hard regime provably exists — sparse, high-rank, well-

conditioned, with efficient state prep and property readout — but no practical problem has been demonstrated to live there fifteen years on.

KEY GRADED CLAIMS

- [T1] HHL $O(\log N \cdot s^2 \kappa^2 / \varepsilon)$ runtime under stated assumptions; matrix inversion BQP-complete — PRL 103, 150502 (2009) (established)
- [T2] Caveat catalogue (state prep, κ , sparsity, output access) — Aaronson, Nat. Phys. 11, 291 (2015) (established)
- [T2] Low-rank dequantization removes the exponential gap — Tang, STOC 2019, arXiv:1807.04271; Chia et al., STOC 2020 (established)
- [T1/T2] QSVT subsumes HHL with improved $\mathcal{K}$, ε dependence — Gilyén–Su–Low–Wiebe, STOC 2019, arXiv:1806.01838 (established)

SPEEDUP / CAVEAT

Exponential in dimension, conditional on four assumptions that rarely hold together in applications. Where they fail, classical or quantum-inspired methods match. QSVT (`S-qsvt`) achieves linear rather than quadratic $\mathcal{K}$ -scaling — the modern way to state the resource cost.

CONFLICTS / OPEN QUESTIONS

Does any end-to-end application (differential-equation solvers, finance PDEs, `I-finance`) satisfy all four assumptions with real data and beat classical? None demonstrated. The state-preparation bottleneck (`S-qram`) is the load-bearing unknown.

SOURCES

PRL 103, 150502 (2009); Nat. Phys. 11, 291 (2015); arXiv:1807.04271; arXiv:1806.01838. Cross-links: `S-qsvt` , `S-qft` , `S-qram` , `S-qsim` , `S-qml` , `reference-impl/` .

Logical qubits & fault tolerance · S-logical

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

A logical qubit is an error-corrected qubit (S-qec) — the unit that algorithms in S-shor / S-qsim actually consume. Fault tolerance means every operation (gates, measurement, and the correction step itself) is performed so that a single physical fault cannot cascade into a logical error. This node marks the line between demos and utility. Useful chemistry or cryptanalysis needs **hundreds to thousands of logical qubits** at logical error rates of 10^{-9} or better, executing **millions to billions** of logical gates — an entirely different regime from the shallow error-detected circuits shown so far. The key subtleties that make bare counts meaningless: code distance d , measured logical error rate per cycle, whether a *universal* (non-Clifford, S-gates) gate set was actually exercised, and whether the device did true correction or only post-selected error *detection*.

WHERE IT STANDS (2025-26)

The logical-qubit count became the field's headline metric, which invites cherry-picking. Milestones: Harvard/QuEra ran circuits on **48 logical qubits** of neutral atoms (Bluvstein et al., Nature 2024 — error-detected, shallow).

Microsoft+Quantinuum demonstrated **12 logical qubits** with 22× error suppression (2024). **Quantinuum Helios** (Nov 2025) produced **48 error-corrected logical qubits at a 2:1 physical:logical ratio** — thought impossible a few years ago — plus a **94-logical-qubit GHZ state**, both with better-than-break-even fidelity (the logical qubits outperform physical qubits running the same task). QuEra demonstrated **96 logical from 448 physical** (a high-rate $[[16,6,4]]$ code, ~4.7:1 encoding) with below-threshold error suppression across all 96 logical qubits — peer-reviewed in Nature (Jan 2026), which makes it the strongest-graded logical-qubit record to date. Caution: a distance-2 detection-only "logical qubit" and Willow's distance-7 corrected memory are different animals, and none of these systems yet runs deep universal computation across many logical qubits simultaneously.

KEY GRADED CLAIMS

- [T2] 48-logical-qubit circuits on neutral atoms — Bluvstein et al., Nature 626, 58 (2024) (demonstrated; error-detected, shallow)
- [T2] 12 logical qubits, 22× error suppression — Microsoft/Quantinuum, arXiv:2404.02280 (demonstrated)
- [T3/T4] Helios: 48 error-corrected logical at 2:1 encoding + 94-qubit GHZ, break-even fidelity — Quantinuum (Nov 2025) + Iceberg-code paper (claimed; partially peer-reviewed)
- [T2] 96 logical qubits via a high-rate $[[16,6,4]]$ code on 448 neutral atoms (~4.7:1), below-threshold across all logical qubits — QuEra, Nature s41586-025-09848-5 (doi:10.1038/s41586-025-09848-5, Jan 2026) (demonstrated, peer-reviewed)
- [T4] IBM Starling: 200 logical qubits / 100M gates by 2029 — IBM roadmap (roadmap)

SPEEDUP / CAVEAT

Logical-qubit counts are **incomparable** across vendors without stating code distance, logical error rate, and gate-set completeness. Treat any bare "N logical qubits" headline as marketing until those three numbers appear (`S-bench`). Break-even (logical beats physical) is the meaningful 2025 threshold; deep universal computation on many logical qubits is the next.

CONFLICTS / OPEN QUESTIONS

"FTQC by 2029" (IBM roadmap, Google-adjacent optimism) versus "not before the mid-2030s" (much of academia) is the field's central unresolved timeline conflict (`C-ftqc-timeline`). Whether the physical:logical ratio keeps falling — Helios's 2:1 is architecture-specific and does not include the deep-circuit distillation overhead.

SOURCES

Nature 626, 58 (2024); arXiv:2404.02280; Quantinuum Helios (2025); QuEra 96-logical — Nature s41586-025-09848-5 (2026); ibm.com/quantum/blog. Cross-links: `S-qec` , `S-decoders` , `S-gates` , `S-bench` , `O-scaling` , `O-overhead` .

Noise, NISQ & error mitigation · S-nisq

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

NISQ — Noisy Intermediate-Scale Quantum, coined by Preskill (2018) — names the current era: 50–1,000+ physical qubits at error rates around 10^{-3} , too noisy for deep circuits and too few for full error correction (S-qec). Error **mitigation** trades extra circuit runs for accuracy without encoding logical qubits. The two workhorses (Temme et al. 2017): **Zero-Noise Extrapolation (ZNE)** runs the circuit at deliberately amplified noise (gate folding or pulse stretching), then extrapolates the observable back to the zero-noise limit; **Probabilistic Error Cancellation (PEC)** learns a noise model and inverts it by sampling from a quasiprobability decomposition, which is unbiased but carries a sampling overhead that grows exponentially in circuit size. Mitigation is not correction — it reduces bias in an *estimated expectation value*, it does not protect a quantum state. It sits distinct enough from correction to deserve its own discipline card (S-errmit).

WHERE IT STANDS (2025-26)

IBM's 2023 "utility" experiment — a 127-qubit kicked-Ising circuit with ZNE (Nature) — was the high-water mark of the mitigation era, and was matched within weeks by classical tensor-network methods running on a laptop (S-tensornet). That exchange set the field's expectations: mitigation extends the reach of noisy hardware by a constant factor, while theory shows generic mitigation costs grow **exponentially** with circuit depth and fault rate (Takagi et al.; Quek et al., Nat. Phys. 2024, giving exponentially tighter lower bounds). The field's center of gravity moved to early fault tolerance (S-qec , S-logical); mitigation persists as a complement layered on top of small logical devices, and 2025 work demonstrates mitigation *on logical qubits* (Nat. Commun. 2025). The reference implementation names ZNE and readout calibration as its own natural next lever against the hardware-noise floor (reference-impl/MATH.md §7).

KEY GRADED CLAIMS

- [T2] Preskill's NISQ framing — arXiv:1801.00862, Quantum 2, 79 (2018) (established as terminology)

- [T2] 127-qubit kicked-Ising beyond brute-force simulation — Kim et al., Nature 618, 500 (2023) (demonstrated)
- [T2] Same observables reproduced classically with tensor networks — Tindall et al., PRX Quantum 5, 010308 (2024) (contested→resolved: classical matched it)
- [T2] Error mitigation has exponential sampling overhead in general — Quek et al., Nat. Phys. 20 (2024); Takagi et al. (established)
- [T2] ZNE/PEC origins — Temme–Bravyi–Gambetta, PRL 119, 180509 (2017) (established)

SPEEDUP / CAVEAT

Mitigation buys accuracy at sampling cost and **never** creates asymptotic speedup. Any "utility" claim demonstrated with mitigation alone is exposed to classical counterattack by construction — if the hardware is shallow enough for mitigation to work, it is usually shallow enough for tensor networks to simulate (`S-tensornet` , `0-advantage`).

CONFLICTS / OPEN QUESTIONS

Whether any mitigated-NISQ computation will ever beat the best classical method on a problem anyone cares about. The 2023–25 record is uniformly negative once a competent classical team responds (conflict `C-nisq-utility`).

SOURCES

arXiv:1801.00862; Nature 618, 500 (2023); PRX Quantum 5, 010308 (2024); Nat. Phys. 20 (2024); PRL 119, 180509 (2017). Cross-links: `S-errmit` , `S-qec` , `S-tensornet` , `reference-impl/` , `0-advantage` .

Pulse-level & quantum optimal control · S-optcontrol

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The bottom of the stack, where an abstract gate becomes an actual analog waveform driving a qubit. Quantum optimal control (QOC) shapes the microwave, laser, or flux pulse that steers the system's Hamiltonian to realize a target unitary with maximum fidelity in minimum time, respecting hardware constraints (bandwidth, power, leakage to non-computational levels). It sits below the gate/compiler layer (`S-gates` , `S-software`) and directly on the control electronics (`H-control`). Core methods: - **GRAPE** (gradient ascent pulse engineering) — piecewise-constant controls, all updated together via gradient/L-BFGS. - **Krotov** — monotonically-convergent gradient method, updates controls sequentially within an iteration. - **CRAB** — expand the pulse in a truncated random basis and optimize the few coefficients (gradient-free, good with a physical simulator in the loop). - **GOAT / analytic-gradient** methods, and **DRAG** — an analytic correction that removes leakage for weakly-anharmonic transmons, the workhorse for single-qubit gates. - **Reinforcement-learning control** — increasingly used where a model is expensive or unknown.

WHERE IT STANDS (2025-26)

QOC is standard practice for squeezing the last fidelity out of a device and is how record two-qubit gates are calibrated. The honest 2025–26 finding is a note of restraint: a systematic transmon study concluded that **properly calibrated DRAG already operates near the decoherence floor** for single-qubit gates, so heavy numerical optimization (GRAPE) buys little there — the win from fancy control is real mainly for two-qubit gates, leakage-heavy or strongly-coupled systems, and robustness against calibration drift (arXiv:2511.12799, 2025). Reinforcement-learning control matured for robust perfect-entangling gates (npj QI 2025). The practical tension is the same everywhere: model-based optimal pulses are only as good as the device model, so closed-loop calibration against the live hardware (and periodic recalibration against drift) matters as much as the optimizer. This layer is also where analog quantum simulation (`S-qsim`) and pulse-efficient compilation of `S-hamsim` blocks live.

KEY GRADED CLAIMS

- [T1] GRAPE gradient pulse optimization — Khaneja et al., J. Magn. Reson. 172, 296 (2005) (established)
- [T1] DRAG leakage suppression for weakly-anharmonic qubits — Motzoi et al., PRL 103, 110501 (2009) (established)
- [T1] Krotov / CRAB optimal-control frameworks — Krotov (1996); Caneva–Murphy–Calarco, PRA 84, 022326 (2011) (established)
- [T3] Calibrated DRAG near the decoherence floor; numerical optimization helps mainly for hard cases — arXiv:2511.12799 (2025) (preprint)
- [T2] Reinforcement-learning robust entangling gates — npj Quantum Information (2025), s41534-025-01065-2 (peer-reviewed)

SPEEDUP / CAVEAT

Not an algorithmic speedup — a **fidelity and time-optimality** layer. It raises gate quality (feeding the thresholds in `S-qec`) and can shorten gate duration to beat decoherence, but it is bounded by the quantum speed limit and by hardware (bandwidth, coherence T_1/T_2 , control-line crosstalk). Over-optimizing to a stale device model can *reduce* real fidelity, so the practical caveat is that calibration and drift-tracking often dominate the theoretical pulse shape.

CONFLICTS / OPEN QUESTIONS

When numerical optimal control actually beats simple analytic pulses (DRAG/Gaussian) is instance-specific and debated. How to make optimal pulses robust to drift without constant recalibration; how well RL control transfers across devices.

SOURCES

J. Magn. Reson. 172, 296 (2005); PRL 103, 110501 (2009); PRA 84, 022326 (2011); arXiv:2511.12799 (2025); npj QI s41534-025-01065-2 (2025). Cross-links: `S-gates`, `S-software`, `H-control`, `S-qsim`, `S-qec`.

Quantum error correction · S-qec

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

QEC encodes one *logical* qubit into many physical qubits so errors are detected and corrected faster than they accumulate. The **threshold theorem** (Aharonov–Ben-Or; Kitaev; Knill–Laflamme–Zurek, 1996–98) proves that if the physical error rate p sits below a code-dependent threshold p_{th} , arbitrarily long computation is possible with only $\text{polylog}(1/\epsilon)$ overhead per logical operation. The **surface code** (Fowler et al. 2012; threshold $\approx 1\%$, nearest-neighbor 2D layout, distance d protects against $\lfloor (d-1)/2 \rfloor$ errors using d^2 data + measure qubits) has been the default. **qLDPC codes** promise far better encoding rate — more logical qubits per physical qubit — at the price of long-range connectivity. Non-Clifford gates need **magic-state distillation** (or newer **cultivation**, **S-gates**), historically the dominant cost of fault-tolerant computing. Syndrome extraction runs every cycle and must be decoded in real time (**S-decoders**).

WHERE IT STANDS (2025–26)

Google's **Willow** result (Nature 2024) is the era-defining demo: a distance-7 surface-code logical qubit whose error rate **halves** with each increase in code distance (suppression factor $\Lambda \approx 2.14$) — the first clear below-threshold memory, meaning bigger codes now help rather than hurt. IBM pivoted to qLDPC: the $[[144,12,12]]$ "gross code" protects 12 logical qubits with 288 physical qubits, versus roughly 3,000 physical for equivalent surface-code protection — about a 10× encoding-rate win; the **Loon** chip (2025) demonstrated the required long-range couplers. Gidney's magic-state **cultivation** (2024) cut projected T-gate costs by roughly 10×, feeding directly into the falling Shor estimates (**S-shor**). Quantinuum, QuEra, and Harvard demonstrated logical operations on trapped-ion and neutral-atom platforms (**S-logical**).

KEY GRADED CLAIMS

- [T1] Threshold theorem — Aharonov–Ben-Or, STOC 1997; Knill–Laflamme–Zurek (established)

- [T2] Below-threshold surface-code memory, $\Lambda = 2.14$, $d = 7$ — Google, Nature 638 (2024), arXiv:2408.13687 (demonstrated)
- [T2] Gross code $[[144,12,12]]$ $\sim 10\times$ better encoding rate — Bravyi et al., Nature 627, 778 (2024) (demonstrated in theory + partial hardware)
- [T3] Magic-state cultivation reduces distillation overhead $\sim 10\times$ — Gidney et al., arXiv:2409.17595 (claimed)
- [T4] IBM Loon validates qLDPC architectural components — IBM blog, Nov 2025 (claimed, awaiting peer review)

SPEEDUP / CAVEAT

Overhead is the whole game (0-overhead): $\sim 100\text{--}1,000$ physical qubits per logical qubit at useful error rates, and QEC **slows** the logical clock because each logical gate spans many syndrome cycles plus decode latency. Encoding rate (surface) versus connectivity demand (qLDPC) is the central engineering tension.

CONFLICTS / OPEN QUESTIONS

Surface code versus qLDPC is a live architectural fork (Google and most of the field versus IBM); whether qLDPC's long-range connectivity is buildable at scale, and stays below threshold across the couplers, is unproven ($0\text{-interconnect-loss}$). Real-time decoding throughput at scale (0-decoder) is a separate open bottleneck.

SOURCES

arXiv:2408.13687; Nature 627, 778 (2024); arXiv:2409.17595; ibm.com/quantum/blog; PRA 86, 032324 (2012). Cross-links: $S\text{-logical}$, $S\text{-decoders}$, $S\text{-gates}$, $S\text{-shor}$, 0-overhead , 0-decoder .

Quantum Fourier Transform & Phase Estimation · S-qft

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The Quantum Fourier Transform (QFT) is the change of basis that maps amplitudes to their discrete-Fourier coefficients using only $O(n^2)$ gates on n qubits — exponentially fewer than the $O(n \cdot 2^n)$ of the classical FFT on 2^n points. On its own the QFT is not a speedup (you cannot read the coefficients out), but it is the engine of Quantum Phase Estimation (QPE): given a unitary U and an eigenstate $|\psi\rangle$ with $U|\psi\rangle = e^{2\pi i\varphi}|\psi\rangle$, QPE estimates the phase φ to n bits by applying controlled powers of U into a register and running an inverse QFT. QPE is the primitive that sits *underneath* the famous algorithms — Shor's period-finding (S-shor), HHL eigenvalue inversion (S-hhl), quantum-chemistry energy estimation (S-qsim), and amplitude estimation (S-qmc) are all QPE in different clothing.

WHERE IT STANDS (2025-26)

QFT (Coppersmith, 1994) and QPE (Kitaev, 1995) are textbook-established. The live engineering story is depth: textbook QPE demands $\sim O(1/\varepsilon)$ coherent applications of U plus a deep controlled-rotation ladder, which is hopeless on NISQ hardware. The response is a family of low-depth variants — iterative/Kitaev-style QPE (one ancilla, classical feedback), Bayesian and statistical phase estimation, Lin–Tong Heisenberg-limited ground-state energy estimation for *early* fault tolerance (PRX Quantum 3, 010318, 2022, shorter maximal circuit depth via a step-function filter), Wan–Berta–Campbell randomized statistical phase estimation (PRL 129, 030503, 2022), and the optimal-precision refinements (arXiv:2403.18927) — that trade circuit depth for repetitions. Approximate QFT (dropping small-angle rotations) cuts gate count with negligible error and is standard in fault-tolerant compilation. QPE is also a special case of QSVT (S-qsvt) — the eigenvalue-transform lens. It remains a fault-tolerant algorithm: the canonical demonstration of why long coherent circuits need S-qec .

KEY GRADED CLAIMS

- [T1] QFT in $O(n^2)$ gates; QPE estimates eigenphase to $\mathcal{N}$ bits — Coppersmith 1994 (quant-ph/0201067); Kitaev 1995 (quant-ph/9511026) (established)
- [T1] QPE is the shared subroutine under Shor, HHL, quantum chemistry, amplitude estimation — Nielsen & Chuang Ch. 5 (established)
- [T2] Iterative/low-depth, statistical, and optimal-precision QPE variants for near-term and early-FT use — Lin–Tong, PRX Quantum 3, 010318 (2022); Wan–Berta–Campbell, PRL 129, 030503 (2022); arXiv:2305.04908; arXiv:2403.18927 (established/demonstrated)

SPEEDUP / CAVEAT

QPE delivers exponential precision in the phase ($\mathcal{N}$ bits from $O(2^n)$ total controlled- U applications) — the engine behind Shor's exponential factoring speedup. Caveat: it needs a good eigenstate to start from and deep coherent evolution of U , so it is a fault-tolerant, not NISQ, primitive; the reference implementation at `reference-impl/` estimates inner products via the shallower swap/Hadamard test precisely to sidestep full QPE depth.

CONFLICTS / OPEN QUESTIONS

How much of the eigenstate-preparation cost (the same qRAM-style assumption that haunts `S-hhl` / `S-qram`) is hidden inside "just run QPE" for real chemistry and finance instances.

SOURCES

quant-ph/0201067; quant-ph/9511026; Nielsen & Chuang §5.2; arXiv:2305.04908; arXiv:2403.18927. Cross-links: `S-shor`, `S-hhl`, `S-qmc`, `S-qsim`, `reference-impl/`.

Quantum Monte Carlo / amplitude estimation · S-qmc

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Quantum Amplitude Estimation (QAE) estimates the amplitude a in a state $\sqrt{1-a} |0\rangle |\psi_0\rangle + \sqrt{a} |1\rangle |\psi_1\rangle$ to additive error ε using $O(1/\varepsilon)$ calls to the state-preparation oracle — versus the $O(1/\varepsilon^2)$ samples any classical Monte Carlo needs, since classical error shrinks only as $O(1/\sqrt{M})$. That is a quadratic speedup for estimating expectation values, the core operation in Monte Carlo integration. Montanaro (2015) generalized it to arbitrary-variance mean estimation. Because expected values price options, aggregate risk (VaR/CVaR), and compute Greeks, QAE is the headline near-to-mid-term algorithm for **I-finance** (JPMorgan, Goldman Sachs, HSBC all have published implementations). Canonically QAE = amplitude amplification (**S-grover**) + phase estimation (**S-qft**); "Grover-free" iterative and maximum-likelihood variants remove the deep QPE register.

WHERE IT STANDS (2025-26)

The quadratic speedup is proven and demonstrated at small scale on trapped-ion and superconducting hardware (arXiv:2109.09685, arXiv:2201.06987). Two caveats dominate the honest picture. (1) Depth: canonical QAE needs very deep circuits — the reason iterative QAE (Grinko et al.) and maximum-likelihood QAE were invented, trading depth for repetitions to reach NISQ. (2) The oracle problem: the speedup assumes an efficient circuit that loads the target probability distribution into amplitudes without approximation error — a state-preparation / **S-qram** assumption that, as with **S-hhl**, can silently cost as much as it saves. Resource estimates (Chakrabarti et al., 2021) put useful financial QAE firmly in the fault-tolerant era, needing many logical qubits and long coherent runs.

KEY GRADED CLAIMS

- [T1] QAE gives $O(1/\varepsilon)$ vs classical $O(1/\varepsilon^2)$ — quadratic Monte Carlo speedup — Brassard–Høyer–Mosca–Tapp 2002 (quant-ph/0005055); Montanaro, Proc. R. Soc. A 471 (2015) (established)

- [T2] Grover-free / iterative / MLE amplitude estimation for shallower circuits — Grinko et al., npj QI 7 (2021); Suzuki et al. 2020 (established)
- [T3/T4] Financial QAE (option pricing, risk) demonstrated small-scale; useful scale needs FTQC — arXiv:1905.02666 (JPMorgan); Chakrabarti et al., Quantum 5, 463 (2021) (demonstrated / roadmap)

SPEEDUP / CAVEAT

Proven quadratic speedup over classical Monte Carlo. Caveats: (1) only quadratic, so fault-tolerance overhead can wash it out for realistic runtimes (S-grover , 0-advantage); (2) requires deep circuits and an efficient, exact distribution-loading oracle — the loading cost is the crux and links directly to S-gram .

CONFLICTS / OPEN QUESTIONS

Does the distribution-loading oracle for real market data admit an efficient circuit, and does the quadratic gain survive full FT resource accounting? Unresolved — the same fine print that dogs Grover.

SOURCES

quant-ph/0005055; Proc. R. Soc. A 471, 20150301 (2015); npj QI 7, 52 (2021); arXiv:1905.02666; Quantum 5, 463 (2021); arXiv:2109.09685. Cross-links: S-grover , S-qft , S-gram , I-finance , 0-advantage .

Quantum machine learning · S-qml

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

QML covers quantum kernels, variational quantum classifiers, quantum neural networks, and hybrid pipelines — the proposal that quantum feature maps or quantum linear algebra could accelerate learning. The clearest instance is the

quantum kernel: encode each datum $\mathcal{X}$ into a state $|\phi(x)\rangle$ and use

$k(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2$ as the kernel a classical SVM consumes. The manual's

reference implementation is exactly this object built honestly end to end —

amplitude-encode vectors, estimate the overlap with a swap/Hadamard test,

assemble the Gram matrix, feed it to a classical SVM (100% on Iris, kernel

RMSE 0.007), with a full shots-versus-noise error budget (`reference-impl/`

`MATH.md` §6–8). It is a correct, hardware-validatable quantum kernel that also

names, in the same breath, why it is **not** a speedup (classical cosine similarity is

$\mathcal{O}(d)$). QML is the most hype-inflated corner of the stack and this manual

grades it accordingly.

WHERE IT STANDS (2025–26)

The record on **classical** data is negative. Tang's 2018 dequantization killed the

flagship (quantum recommendation systems, `S-hhl`); random-Fourier-feature

methods reproduce quantum-kernel performance classically; and a Dec 2025

peer-reviewed benchmark found plain logistic regression beating quantum

SVMs, quantum k-NN, and variational classifiers on 4 of 5 realistic tabular

datasets — one variational classifier scored below random guessing. Barren

plateaus (`S-variational`) apply with full force to trainable models. What

survives is narrow and real: **provable** kernel separations exist for contrived,

cryptographically structured data (discrete-log — Liu–Arunachalam–Temme

2021), and an **exponential** advantage exists for learning from *quantum* data /

experiments when a quantum memory is available (Huang et al., Science 2022).

The defensible position: QML on classical data has no evidence of practical

advantage; QML on quantum data is a real but niche research direction awaiting

hardware.

KEY GRADED CLAIMS

- [T2] Low-rank QML dequantized — Tang, STOC 2019, arXiv:1807.04271 (established)
- [T2] Rigorous quantum-kernel advantage on a crypto-structured task — Liu–Arunachalam–Temme, Nat. Phys. 17, 1013 (2021) (established; artificial problem)
- [T2] Exponential advantage learning from quantum experiments with quantum memory — Huang et al., Science 376, 1182 (2022) (demonstrated; quantum data only)
- [T2] Classical baselines beat QML across realistic tabular benchmarks — Scientific Reports (Dec 2025) (demonstrated)
- [T5] "18% of quantum-algorithm revenue from AI by 2026" — Hyperion Research (forecast; graded skeptically)

SPEEDUP / CAVEAT

No proven or empirical speedup on classical data. Provable advantages exist only where the problem is built from cryptographic hardness or the data is itself quantum. Even when a quantum kernel is *expressive*, the state-loading cost (`S-gram`) and the $O(m^2)$ pairwise-overlap cost of building the Gram matrix (as the reference impl makes concrete) dominate. Assume any commercial "quantum AI" pitch is unsubstantiated until it names its separation.

CONFLICTS / OPEN QUESTIONS

Whether "quantum data" applications (sensor streams `A-sensing` , simulation outputs) become a real market, and whether generative "GenQAI" vendor framings (Quantinuum) ever acquire evidence. Classical shadows (`S-shadows`) are the leading rigorous tool for turning quantum data into predictions with few measurements — a more defensible route than variational QML.

SOURCES

arXiv:1807.04271; Nat. Phys. 17, 1013 (2021); Science 376, 1182 (2022); Sci. Rep. (Dec 2025). Cross-links: `S-hhl` , `S-variational` , `S-gram` , `S-shadows` , `reference-impl/` , `0-hype` .

Quantum RAM (qRAM) · S-qram

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Quantum RAM is the assumed device that loads classical data into quantum superposition: given an address register in a superposition of indices, qRAM returns the corresponding data entries entangled with those addresses, ideally in time polylogarithmic in the number of entries N . It is the load-bearing, largely-unproven assumption under the exponential-speedup algorithms — HHL (S-hhl), quantum recommendation systems, and most quantum machine learning (S-qml). If you cannot get N classical numbers into the machine faster than $\mathcal{O}(N)$, the exponential advantage of algorithms that then process them in polylog time evaporates. The canonical design is the bucket-brigade architecture (Giovannetti–Lloyd–Maccone, 2008), which routes an address through a tree of quantum switches so that only $\mathcal{O}(\log N)$ of the $\sim N$ nodes are actively entangled per query, making it far more error-tolerant than a naive fanout.

WHERE IT STANDS (2025–26)

qRAM remains the field's most honest weak link. The cost debate: bucket-brigade buys gentle (polylogarithmic) query-infidelity scaling but at the price of $\mathcal{O}(N)$ physical components / exponentially many ancillae, and if each must be error-corrected the fault-tolerant overhead can swallow the algorithmic gain (Arunachalam et al. analysis; the recurring theme of S-hhl). 2025 brought the first experimental demonstrations — a Zhejiang University superconducting team realized bucket-brigade qRAM addressing 4-bit and 8-bit data (query fidelities ~ 0.81 and ~ 0.60 , Nature Physics 2026) — proof of principle at tiny scale, far from the millions of addressable cells the speedup algorithms assume. Dequantization results (Tang et al.) further showed that *if* you grant qRAM-style access, many "quantum" ML speedups can be matched classically, sharpening the question of what qRAM actually buys.

KEY GRADED CLAIMS

- [T1/T2] Bucket-brigade qRAM: $O(\log N)$ active switches per query, more noise-tolerant than naive fanout — Giovannetti–Lloyd–Maccone, PRL 100, 160501 (2008) (established)
- [T2] Fault-tolerant qRAM overhead can negate the algorithms it feeds; polylog infidelity vs $O(N)$ components tradeoff — arXiv analyses (Arunachalam et al. 2015; redundancy-repair, Sci. Rep. 2025) (established/contested)
- [T3] First experimental bucket-brigade qRAM (4-/8-bit, fidelity 0.81/0.60) — Zhejiang, Nature Physics (2026), arXiv:2506.16682 (demonstrated, small scale)

SPEEDUP / CAVEAT

Enabling assumption, not a speedup. When qRAM is granted for free, HHL/QML claim exponential gains; when its real cost ($O(N)$ hardware, error correction, state-prep fidelity) is charged, most of those gains shrink or vanish. No practical, scalable qRAM exists — this is the single biggest asterisk on data-heavy quantum advantage.

CONFLICTS / OPEN QUESTIONS

Is scalable, fault-tolerant qRAM physically buildable at the sizes (millions–billions of cells) the speedup algorithms need, and does any application clear the bar once loading is charged honestly? Open — parallels the `S-hhl` fine print directly.

SOURCES

PRL 100, 160501 (2008); arXiv:2506.16682; Nat. Phys. (2026); Sci. Rep. 15 (2025), arXiv:2312.17483; Tang, STOC 2019. Cross-links: `S-hhl`, `S-qml`, `S-qmc`, `reference-impl/` (amplitude encoding = the manual data-loading it abstracts).

Quantum simulation · S-qsim

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Using a controllable quantum system to simulate another quantum system — Feynman's original 1981–82 pitch and still the most credible application class. Two flavors. **Digital:** implement the evolution operator e^{-iHt} on a gate-based machine by Trotter–Suzuki product formulas, linear combination of unitaries (LCU), qubitization, or QSVT (the methods are unpacked in S-hamsim and unified by S-qsvt), then extract observables — often via quantum phase estimation (S-qft) for ground-state chemistry. **Analog:** build a purpose-made quantum system whose native Hamiltonian *is* the target (e.g. Rydberg-atom arrays realizing spin models). Lloyd (1996) proved that local-Hamiltonian dynamics is simulable in polynomial time on a quantum computer — the theorem that makes the whole class rigorous.

WHERE IT STANDS (2025–26)

Analog simulators lead on scale: QuEra/Harvard 256+ atom arrays probing quantum phase transitions, spin liquids, and quench dynamics that strain classical methods (H-neutral). Digital simulation is the target of the current advantage push. IBM's 2023 kicked-Ising "utility" claim on 127 qubits was matched within weeks by classical tensor-network methods on a laptop (see S-nisq , S-tensornet) — the canonical example of the classical counterattack. Google's Oct 2025 "quantum echoes" experiment on Willow measures an out-of-time-order correlator (OTOC) and claims a **verifiable 13,000× speedup** over the best classical estimate, with a molecular-structure (NMR) application hook — under active classical-counterattack scrutiny, unresolved. For fault-tolerant chemistry the showcase target is the FeMoco cofactor (biological nitrogen fixation, I-chem / I-agri): resource estimates sit around ~4M physical qubits and days of runtime, after a decade of refinement that cut earlier figures by orders of magnitude.

KEY GRADED CLAIMS

- [T1] Local-Hamiltonian dynamics simulable in poly time on a quantum computer — Lloyd, Science 273, 1073 (1996) (established)

- [T1] Feynman's simulation proposal — Int. J. Theor. Phys. 21, 467 (1982) (established, foundational)
- [T2] 256-atom analog simulation of quantum phases — Ebadi et al., Nature 595, 227 (2021) (demonstrated)
- [T3/T4] "Quantum echoes" OTOC $13,000\times$ beyond-classical, verifiable — Google, Nature (Oct 2025) + blog.google (claimed; contested-by-default, classical response pending)
- [T3] FeMoco ground-state estimate $\sim 4\text{M}$ qubits / $\sim \text{days}$ — Lee et al., PRX Quantum 2, 030305 (2021) lineage (claimed resource estimate)

SPEEDUP / CAVEAT

Exponential over known classical methods for **generic** quantum dynamics — the best-founded advantage in the portfolio (`O-advantage` , `O-killerapp`). Fine print: classical tensor-network and neural-quantum-state methods keep absorbing "hard" instances, especially anything with limited entanglement or short evolution time (`S-tensornet`); static ground-state estimation (chemistry's real prize) is QMA-hard in general, so the machine helps with *specific* instances without a blanket guarantee. Digital simulation also inherits the state-preparation problem — you must load a good approximate ground state before phase estimation refines it.

CONFLICTS / OPEN QUESTIONS

Is quantum simulation the first paying application (field consensus: most likely) and when — 2027 (vendor optimism) versus mid-2030s (conservative academia)? See conflict `C-killerapp-timing` .

SOURCES

Int. J. Theor. Phys. 21, 467 (1982); Science 273, 1073 (1996); Nature 595, 227 (2021); blog.google quantum-echoes (2025); PRX Quantum 2, 030305 (2021). Cross-links: `S-hamsim` , `S-qsvt` , `S-qft` , `S-tensornet` , `S-nisq` , `I-pharma` , `I-chem` , `O-killerapp` .

Quantum singular value transformation (QSVT) · S-qsvt

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

QSVT is the modern unifying lens for quantum algorithm design. Given a matrix A block-encoded inside a larger unitary (A sits in a corner of U , accessed by ancilla-controlled calls), QSVT applies a **polynomial transformation to the singular values of A** using $O(\deg(p))$ calls to U and its inverse, plus one ancilla and a sequence of single-qubit rotations whose angles encode the polynomial p . It generalizes **quantum signal processing** (Low–Chuang 2017), which does the same for the eigenvalues of a single qubit's rotation, up to arbitrary block-encoded operators (Gilyén–Su–Low–Wiebe, STOC 2019). The payoff is conceptual and practical: pick the right polynomial and you recover most known quantum algorithms as special cases. $p \approx \text{sign function} \rightarrow$ **amplitude amplification / Grover** (S-grover); $p \approx 1/x$ on the spectrum $\rightarrow$ **matrix inversion / HHL** (S-hhl); $p \approx e^{-ixt} \rightarrow$ **Hamiltonian simulation** (S-hamsim , S-qsim); a rectangular window $\rightarrow$ **phase estimation** (S-qft) and eigenstate filtering; a threshold $\rightarrow$ quantum walks (S-walk). Martyn–Rossi–Tan–Chuang (2021) laid this out as an explicit "grand unification of quantum algorithms."

WHERE IT STANDS (2025–26)

QSVT is now the standard framework in which fault-tolerant algorithms are designed and their costs are stated, because it gives near-optimal query complexity with transparent resource accounting: the cost is the polynomial degree, and Chebyshev-approximation theory tells you the degree needed for a target accuracy. For matrix inversion it achieves **linear** dependence on the condition number κ (versus HHL's κ^2) and additive $\log(1/\varepsilon)$ precision — the tightest statement of the linear-systems cost. It also delivers optimal-scaling Hamiltonian simulation matching the qubitization lower bound (S-hamsim). The practical friction is **phase-angle computation**: finding the rotation angles that realize a target polynomial was numerically unstable for high-degree polynomials; 2020–24 work (Haah; Dong–Meng–Whaley–Lin; Motlagh–Wiebe) produced stable $O(\deg)$ angle-finding, largely closing that gap. QSVT is a fault-

tolerant tool — it assumes a block-encoding, which itself needs coherent access to A (a qRAM-style or sparse-oracle assumption, $S\text{-qram}$), so it inherits the same input-model caveats as the algorithms it unifies.

KEY GRADED CLAIMS

- [T1] QSVT: polynomial transform of singular values of a block-encoded operator, $O(\deg)$ queries — Gilyén–Su–Low–Wiebe, STOC 2019, arXiv:1806.01838 (established)
- [T1] Quantum signal processing (the single-qubit precursor) — Low–Chuang, PRL 118, 010501 (2017) + PRX 6, 041067 (2016) (established)
- [T2] Grand unification: amplitude amplification, HHL, Hamiltonian simulation as QSVT instances — Martyn–Rossi–Tan–Chuang, PRX Quantum 2, 040203 (2021) (established, tutorial)
- [T2] Stable phase-angle finding for high-degree polynomials — Dong et al., PRA 103, 042419 (2021); Motlagh–Wiebe, PRX Quantum (2024) (established)
- [T2] QSVT is dequantizable in the low-rank / sampling-access regime — Gharibian–Le Gall, STOC 2022, arXiv:2111.09079 (established; bounds where it does NOT help)

SPEEDUP / CAVEAT

Not itself a speedup — a **design framework** that yields near-optimal versions of the speedups already in the map. Its honest limit is the same one that runs through $S\text{-hhl}$ / $S\text{-qram}$: the whole edifice sits on top of an efficient block-encoding of the input, and where that block-encoding is only "sampling access" to a low-rank matrix, Gharibian–Le Gall showed QSVT is classically dequantizable. So QSVT sharpens *what* is provably fast and *where* the classical counterattack bites, in one language.

CONFLICTS / OPEN QUESTIONS

Which real applications admit an efficient block-encoding that is not itself the bottleneck (the recurring $S\text{-hhl}$ question). Whether angle-finding and block-encoding constant factors are small enough for early-fault-tolerant machines, or whether QSVT stays a large-scale-only tool.

SOURCES

arXiv:1806.01838; PRL 118, 010501 (2017); PRX Quantum 2, 040203 (2021); PRA 103, 042419 (2021); arXiv:2111.09079. Cross-links: [S-hhl](#) , [S-grover](#) , [S-hamsim](#) , [S-qsim](#) , [S-qft](#) , [S-walk](#) , [S-qram](#) , [reference-impl/](#) .

Classical shadows · S-shadows

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Classical shadows (Huang–Kueng–Preskill, Nat. Phys. 2020) is a measurement protocol that builds a compact classical sketch of an unknown quantum state ρ from very few measurements, then predicts many properties from that one sketch. Recipe: apply a random unitary from a fixed ensemble (random global Cliffords, or — more practically — random single-qubit Pauli rotations), measure in the computational basis, and invert the measurement channel to get a classical "snapshot." Average $O(\log M/\epsilon^2)$ snapshots and you can estimate M **different observables** to accuracy ϵ with high probability. The headline result: for local observables the number of measurements is **independent of system size** and saturates information-theoretic lower bounds — a fundamentally different scaling from full quantum state tomography, which needs exponentially many measurements. Crucially the target properties can be chosen *after* the data is collected.

WHERE IT STANDS (2025–26)

Classical shadows became one of the most-used primitives in near-term quantum computing and a rigorous alternative to variational readout. Applications: estimating energies and forces for chemistry, entanglement entropies, fidelities, two-point correlators, and — the deepest connection — as the efficient tool that makes "learning from quantum data" provable (the exponential advantage in `S-qm1`, Huang et al. Science 2022, uses shadow-style measurements). The active research frontier is robustness and specialization: noise-robust shadows under gate-dependent noise, **fermionic** shadows with mode-independent sample complexity for chemistry (2026), qudit and magic-gate variants, and shadows combined with circuit cutting (`S-circuitcut`). The Pauli-measurement version is directly runnable on today's hardware, which is why it displaced heavier tomography for benchmarking and for the manual's kind of kernel/overlap estimation (`reference-impl/` — shadows are the sample-efficient cousin of the swap/Hadamard-test readout, predicting many overlaps from one measurement budget).

KEY GRADED CLAIMS

- [T2] Predict M observables from $O(\log M/\varepsilon^2)$ measurements, size-independent for local observables — Huang–Kueng–Preskill, Nat. Phys. 16, 1050 (2020), arXiv:2002.08953 (established)
- [T2] Sample complexity saturates information-theoretic lower bounds — same, with matching lower bounds (established)
- [T2] Shadows enable provable exponential advantage learning from quantum experiments — Huang et al., Science 376, 1182 (2022) (demonstrated; quantum data)
- [T3] Noise-robust and fermionic shadows extend applicability — Chen et al., PRX Quantum (2021); fermionic-shadow work arXiv:2606.27254 (2026) (established/preprint)

SPEEDUP / CAVEAT

A **measurement-efficiency** result, not a computational speedup — it reduces how many times you must run and measure a circuit, saturating a fundamental lower bound. Caveats: the estimator variance can be large for global or high-weight observables (random-Pauli shadows scale as 3^k for k -local operators), so it shines for **low-weight, many-observable** workloads and degrades for global ones; and it presupposes you can prepare p repeatedly, so it inherits the state-preparation costs of whatever produced p .

CONFLICTS / OPEN QUESTIONS

Which measurement ensemble is optimal for a given task (global Clifford = strong but hard circuits; local Pauli = easy but weaker for global observables). How much shadow variance survives realistic hardware noise at scale — an active robustness question.

SOURCES

Nat. Phys. 16, 1050 (2020), arXiv:2002.08953; Science 376, 1182 (2022); arXiv:2011.09636 (robust shadows); arXiv:2606.27254. Cross-links: `S-qm1`, `S-variational`, `S-circuitcut`, `S-bench`, `reference-impl/`.

Shor's algorithm · S-shor

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Shor (1994) showed a quantum computer can factor integers and compute discrete logarithms in polynomial time. The engine is **period-finding**: factoring N reduces to finding the period $\mathcal{P}$ of $f(x) = a^x \bmod N$, and the period is read off by applying the quantum Fourier transform (QFT, see S-qft) to a superposition of function evaluations, then measuring and post-processing with continued fractions. Gate cost is $O((\log N)^2 \cdot \log \log N)$ with fast integer multiplication, i.e. polynomial in the bit-length. The best classical factoring algorithm, the general number field sieve, runs in sub-exponential $\exp(O((\log N)^{1/3}(\log \log N)^{2/3}))$ time — so Shor is a **superpolynomial** separation and breaks RSA, finite-field Diffie–Hellman, and elliptic-curve cryptography. That is the entire reason the world is migrating to post-quantum cryptography (A-pqc) and the source of the harvest-now-decrypt-later threat (I-cyber).

WHERE IT STANDS (2025–26)

No cryptographically relevant number has ever been factored quantumly. Honest records are tiny: 15 (NMR, Vandersypen 2001) and 21; larger published "factorings" used variational or annealing shortcuts that presuppose structure in the answer and do not scale (survey arXiv:2410.14397). The live action is **resource estimation**. Gidney (May 2025) cut the projected machine for RSA-2048 from his own 2019 estimate of ~20M noisy qubits over 8 hours to **<1M noisy qubits running under a week**, via approximate residue arithmetic, yoked surface codes, and magic-state cultivation (see S-qec). That ~20× drop in six years is why NIST/NSA migration deadlines (2030–2035) keep tightening rather than relaxing. The cost is dominated by modular exponentiation depth and by the T-gate / magic-state budget (S-gates), not by the QFT.

KEY GRADED CLAIMS

- [T1] Polynomial-time quantum factoring and discrete log — Shor, SIAM J. Comput. 26, 1484 (1997) (established)

- [T2] Largest honest quantum factorization remains trivial (15, 21) — Vandersypen et al., Nature 414, 883 (2001); survey arXiv:2410.14397 (established)
- [T3] RSA-2048 with <1M noisy qubits in <1 week — Gidney, arXiv:2505.15917 (claimed; widely cited, unrefuted)
- [T3] 2019 baseline: 20M qubits / 8 hours — Gidney–Ekerå, arXiv:1905.09749 (established as the prior estimate)
- [T5/T6] "Q-Day" dates (2030–2035) — NIST IR 8547 migration timeline + analyst punditry (roadmap/speculative)

SPEEDUP / CAVEAT

Superpolynomial over the best *known* classical algorithm. Factoring has no proven classical lower bound, so a classical breakthrough is possible in principle and Shor is not a complexity-theoretic separation the way sampling tasks aim to be (`S-bench` , `0-advantage`). Practical fine print: it needs full fault tolerance — millions of physical qubits, deep coherent circuits — so Shor is a 2030s+ threat under every credible hardware roadmap. HNDL makes it a present-day data-security problem even though the machine does not exist yet.

CONFLICTS / OPEN QUESTIONS

Resource estimates keep falling — is <100k qubits reachable with better codes and arithmetic? (See conflict `C-shor-timeline` .) Whether any state actor is ahead of public estimates is unknowable from open sources.

SOURCES

SIAM J. Comput. 26, 1484 (1997); arXiv:2505.15917; arXiv:1905.09749; arXiv:2410.14397; NIST IR 8547. Cross-links: `S-qft` , `S-qec` , `S-gates` , `A-pqc` , `I-cyber` , `0-advantage` .

Compilers, transpilers & middleware · S-software

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The software layer between an abstract circuit and pulses on hardware: SDKs for authoring circuits, **transpilers** that rewrite them into native gates (`S-gates`) while routing around limited connectivity (inserting SWAPs), optimizers that cut depth and two-qubit count, and runtime/middleware for hybrid execution.

Compilation quality changes what a chip can do — a 2× depth reduction is worth roughly a hardware generation, because circuit fidelity decays exponentially in two-qubit-gate count. The problems it solves are hard in their own right: qubit routing on a fixed coupling graph is NP-hard, so transpilers use heuristics (SABRE-style routing, template matching, ZX-calculus rewrites) with no guarantee of optimality.

WHERE IT STANDS (2025-26)

Qiskit (IBM, Apache-2.0) is the de-facto standard: v2.x rebuilt the core in Rust, added a C API, and claims best-in-class transpilation on the Benchpress suite.

Cirq (Google) serves the Google-stack research community. **PennyLane** (Xanadu) owns the differentiable-programming/QML niche (`S-qml`) with autodiff through circuits. **tket/pytket** (Quantinuum) is the leading retargetable third-party compiler. **Braket SDK** (AWS) and **Q#/QIR** (Microsoft, with the Azure resource estimator) round out the cloud layer (`S-cloud`); NVIDIA **CUDA-Q** pushes GPU-hybrid workflows and drives the "quantum-centric supercomputing" framing.

Interoperability runs through **OpenQASM 3** and **QIR** (an LLVM-based intermediate representation). The frontier: dynamic-circuit compilation (mid-circuit measurement / feed-forward), error-aware routing using live calibration data, and the first **fault-tolerant compilers** targeting a *logical* instruction set (lattice-surgery operations) rather than physical gates. The reference implementation leans on exactly this layer — Qiskit's exact state-preparation compiler builds the amplitude-encoding circuits it runs (`reference-impl/` `MATH.md` §2).

KEY GRADED CLAIMS

- [T2] All major SDKs actively maintained; OpenQASM 3 / QIR as interchange — GitHub release records (established)
- [T4] Qiskit "most performant transpiler" — IBM Benchpress benchmarking, arXiv:2409.08844 (vendor-run benchmark; methodology public)
- [T2] Compilation reduces two-qubit counts 2–50% versus naive lowering across suites — tket paper, Quantum Sci. Technol. 6, 014003 (2021) + Benchpress (demonstrated)
- [T4] Vendor "quantum-centric supercomputing" middleware roadmaps — IBM/NVIDIA announcements (roadmap)

SPEEDUP / CAVEAT

Software **multiplies** hardware; it never creates asymptotic advantage. Between-SDK benchmark claims are usually vendor-run — check who compiled the competitor's baseline, since a poorly configured rival transpiler is an easy strawman (`S-bench`).

CONFLICTS / OPEN QUESTIONS

Whether the field consolidates on Qiskit the way ML consolidated on PyTorch, or QIR-level interop keeps the layer plural (open-source governance is itself soft power, `E-oss`). How fault-tolerant ISAs (lattice-surgery instruction sets) get standardized across vendors.

SOURCES

github.com/Qiskit/qiskit; quantumai.google/cirq; pennylane.ai; github.com/CQCL/tket; arXiv:2409.08844; Quantum Sci. Technol. 6, 014003 (2021). Cross-links: `S-gates` , `S-cloud` , `S-qml` , `E-oss` , `reference-impl/` .

Tensor-network classical simulation — the counterfactual · S-tensornet

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

The classical method that decides most quantum-advantage arguments (S-bench , 0-advantage). A tensor network represents an n -qubit state or a whole circuit as a graph of small tensors; contracting the graph computes amplitudes or expectation values. The key resource is **entanglement**: a 1D state with bounded entanglement is captured exactly by a **matrix product state (MPS)** of small bond dimension χ , and cost grows only polynomially in n and χ . States with more entanglement need 2D networks (PEPS) or full amplitude-tensor contraction, whose cost is set by the network's treewidth. The upshot is a sharp dividing line: circuits that stay **low-entanglement** — shallow, geometrically local, noisy, or short-time — are classically simulable, sometimes on a laptop; circuits that generate volume-law entanglement across a hard-to-contract geometry are where a quantum device might win. This is the honest counterfactual the manual holds every advantage claim against.

WHERE IT STANDS (2025-26)

Tensor networks have repeatedly erased "beyond-classical" claims. **Sycamore 2019** RCS was cut from "10,000 years" to hours-then-minutes by hyper-optimized contraction on GPU clusters (Pan–Chen–Zhang; and 2024 work ran 1,432 GPUs $\sim 7\times$ faster than Sycamore). **IBM's 2023 kicked-Ising "utility"** experiment (S-nisq) was matched within weeks by an MPS/belief-propagation tensor-network simulation on a laptop (Tindall–Fishman–Stoudenmire–Sels 2024) — the cleanest recent example, because the circuit's limited entanglement made it easy. Gaussian and lossy boson sampling fell to MPS methods too. The 2025 Nature Reviews Physics survey frames this as a productive arms race: each advantage claim forces better contraction heuristics, and each classical advance raises the bar for what counts as advantage. Google's Dec 2024 Willow RCS and Oct 2025 "quantum echoes" (S-qsim) are the current standing claims that tensor networks have **not** yet matched — the live test.

KEY GRADED CLAIMS

- [T2] Sycamore RCS classically matched via tensor contraction — Pan–Zhang, PRL 129, 090502 (2022); "Leapfrogging Sycamore" arXiv:2406.18889 (2024) (established)
- [T2] IBM kicked-Ising utility reproduced by laptop MPS — Tindall et al., PRX Quantum 5, 010308 (2024) (established)
- [T2] MPS captures bounded-entanglement states in $\text{poly}(n, \chi)$; DMRG foundation — Vidal, PRL 91, 147902 (2003); White DMRG (1992) (established)
- [T2] Tensor networks as verification + counterattack tool for advantage — Nature Reviews Physics (2025), arXiv:2503.08626 (established survey)

SPEEDUP / CAVEAT

This is the classical baseline, so its "caveat" runs the other way: tensor networks are efficient **only** for low-entanglement or low-treewidth instances, and cost blows up exponentially with entanglement across the contraction cut. A quantum advantage survives precisely where no efficient contraction exists — which is why long-time, high-entanglement dynamics (`S-hamsim`) is the most defensible advantage target, and shallow/noisy circuits are the most vulnerable.

CONFLICTS / OPEN QUESTIONS

Whether Willow-class RCS and the OTOC "quantum echoes" experiment stay uncontracted, or fall like their predecessors (`C-advantage`). How far belief-propagation-augmented tensor networks push the simulable frontier for structured circuits.

SOURCES

PRL 129, 090502 (2022); PRX Quantum 5, 010308 (2024); PRL 91, 147902 (2003); Nature Rev. Phys. (2025), arXiv:2503.08626; arXiv:2406.18889. Cross-links: `S-bench` , `S-nisq` , `S-qsim` , `S-hamsim` , `O-advantage` , `O-verification` .

Variational algorithms — VQE & QAOA · S-variational

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Variational quantum algorithms (VQAs) pair a shallow parameterized circuit $U(\theta)$ with a classical optimizer in a feedback loop. **VQE** (Peruzzo et al., 2014) minimizes $\langle \psi(\theta) | H | \psi(\theta) \rangle$ for a molecular Hamiltonian H to estimate ground-state energy — the variational principle guarantees the estimate is an upper bound. **QAOA** (Farhi–Goldstone–Gutmann, 2014) encodes a combinatorial problem (MaxCut, etc.) in p alternating layers of problem-phase and mixer unitaries $e^{-i\gamma C} e^{-i\beta B}$, interpolating toward the optimum as p grows. They were designed for NISQ hardware (S-nisq): short circuits, hybrid by construction, partly noise-tolerant. Gradients are computed by the parameter-shift rule; the objective is estimated by repeated measurement (shots), so cost scales with the number of observables and the desired precision.

WHERE IT STANDS (2025-26)

The theory turned against the program on three fronts. **Barren plateaus** (McClean et al., 2018): for expressive random ansätze the gradient variance vanishes exponentially in qubit count, so training needs exponentially many shots to see a signal. **Soft dequantization** (Cerezo et al., Nat. Commun. 2025, arXiv:2312.09121): the known families of circuits that provably *avoid* barren plateaus turn out to be classically simulable in polynomial time — if a landscape is trainable it is likely also classically tractable, squeezing the useful regime from both sides. **Classical optimization competition:** fixed-depth QAOA is provably beaten by classical algorithms on broad instance classes (Bravyi et al. 2020; Farhi–Gamarnik–Gutmann obstruction bounds), and no VQE/QAOA run has beaten best-in-class classical chemistry (DMRG, coupled cluster CCSD(T)) or optimization (Gurobi-class solvers) on any real instance. Add measurement/optimization overhead (thousands of shots per energy evaluation, noisy gradients) and the near-term value proposition is thin. Community response: use variational circuits as *state-preparation subroutines* feeding fault-tolerant phase estimation, or move on.

KEY GRADED CLAIMS

- [T2] VQE original demonstration — Peruzzo et al., Nat. Commun. 5, 4213 (2014) (demonstrated, toy scale)
- [T2] Barren plateaus in random parameterized circuits — McClean et al., Nat. Commun. 9, 4812 (2018) (established)
- [T2] Absence of barren plateaus implies classical simulability for known families — Cerezo et al., Nat. Commun. 16 (2025), arXiv:2312.09121 (established, with stated caveats)
- [T2] Fixed-depth QAOA obstructed / classically matched on many instances — Bravyi et al., PRL 125, 260505 (2020) (established)

SPEEDUP / CAVEAT

No proven speedup exists for VQE or QAOA — both are heuristics. Every empirical claim to date has been matched or beaten classically once a competent classical team responds. The training cost (barren plateaus) and measurement cost (shot budget) compound. The escape hatches — quantum-data inputs, initial-state prep for phase estimation, problem-tailored non-random ansätze — remain unproven at advantage scale.

CONFLICTS / OPEN QUESTIONS

Whether *any* ansatz family is simultaneously trainable, classically hard, and noise-tolerant is the open question. Cerezo et al. conjecture essentially no for barren-plateau-free landscapes; parts of the field dispute how far the "simulable" argument generalizes to structured (non-random) circuits.

SOURCES

Nat. Commun. 5, 4213 (2014); arXiv:1411.4028; Nat. Commun. 9, 4812 (2018); arXiv:2312.09121; PRL 125, 260505 (2020). Cross-links: [S-nisq](#) , [S-qsim](#) , [S-qml](#) , [S-qft](#) , [I-pharma](#) , [0-advantage](#) .

Quantum walks & element distinctness · S-walk

Layer: L2 Stack & algorithms · **Chapter:** §03 · **Status:** depth

WHAT IT IS

Quantum walks are the quantum analog of the classical random walk — a superposition spreading over a graph, in discrete-time (coined) or continuous-time (Hamiltonian) form. They are both an algorithmic framework and a universality result. Ambainis's element-distinctness algorithm (2004) walks on the Johnson graph of colliding subsets to find two equal items among N in $O(N^{2/3})$ queries — provably optimal, and a genuinely super-quadratic improvement over the $O(\sqrt{N})$ a naive Grover approach would give. The framework generalizes to subgraph/triangle finding, spatial search, and matrix-verification, and underpins the MNRS (Magniez–Nayak–Roland–Santha) quantum-walk search meta-algorithm.

WHERE IT STANDS (2025–26)

Two structural results anchor the node. First, universality: a multi-particle continuous-time quantum walk on an unweighted graph is BQP-complete (Childs–Gosset–Webb, Science 2013) — quantum walks alone can run any quantum computation, so they are a full model, not a trick. Second, unification: Childs (2010) showed discrete- and continuous-time walks are interconvertible, and recent "multidimensional quantum walk" work (Jeffery–Zur, STOC 2023, arXiv:2208.13492) finally gave a continuous-time $O(N^{2/3})$ element-distinctness algorithm and a time-efficient k-distinctness walk, closing a long-open gap. Walks also power the fastest known algorithms for several graph problems. Like Grover, the honest caveat is that most speedups are query-model and polynomial, so fault-tolerant overhead accounting (S-grover , 0-advantage) applies.

KEY GRADED CLAIMS

- [T1] Element distinctness in $O(N^{2/3})$ queries, optimal — Ambainis, SIAM J. Comput. 37 (2007); FOCS 2004 (established)
- [T1] Multi-particle continuous-time quantum walk is universal for BQP — Childs–Gosset–Webb, Science 339, 791 (2013) (established)

- [T2] Discrete/continuous-time walk equivalence; multidimensional walks give CT element distinctness + time-efficient k-distinctness — Childs 2010; Jeffery–Zur, arXiv:2208.13492 (established/preprint)

SPEEDUP / CAVEAT

Provable polynomial speedups (up to $O(N^{2/3})$ for distinctness, quadratic for many searches), often optimal in the query model. Caveat: these are query-complexity results — the same fault-tolerance overhead that erodes Grover's practical advantage applies here, and the walk oracle must be cheap to implement reversibly for any wall-clock win.

CONFLICTS / OPEN QUESTIONS

Whether any quantum-walk speedup on a real problem (not a black-box oracle) survives end-to-end resource estimation. As with `S-grover`, current evidence leans no for the polynomial-only cases.

SOURCES

SIAM J. Comput. 37, 210 (2007); Science 339, 791 (2013); arXiv:2208.13492; arXiv:1302.7316; Childs arXiv:0810.0312. Cross-links: `S-grover`, `S-qft`, `0-advantage`.

§04 · Adjacent Technologies

Quantum that isn't computing — QKD, the quantum internet, post-quantum crypto, sensing, metrology, imaging.

Atom interferometry (as its own platform) · A-atominterf

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Atom interferometry is the *technique* underneath several sensing cards, promoted here to its own node because it is a distinct platform, not an application. Laser-cool a cloud of atoms, then use light pulses to coherently split, redirect, and recombine each atom's matter-wave along two paths; the phase difference at recombination encodes whatever acted differently on the two arms — gravitational acceleration, rotation, or a fundamental constant. Because the de Broglie wavelength is tiny and the atom is a perfect, drift-free test mass referenced to atomic structure, atom interferometers are **absolute** instruments. The same hardware measures gravity (A-gravimetry), rotation (gyroscopes), acceleration (the IMU under A-pnt), and probes physics: the fine-structure constant, tests of the equivalence principle, and searches for dark matter and gravitational waves in a new band.

MATURITY & REAL DEPLOYMENTS (2025–26)

A mature laboratory platform; early-commercial as gravimeters; a physics tool at the frontier. - **Commercial instruments:** Exail's AQG absolute gravimeter (A-gravimetry) and AOSense inertial/gravity sensors are atom interferometers you can buy. - **Fundamental physics:** atom interferometers have measured the fine-structure constant α to ~ 0.2 parts per billion and run the tightest lab tests of the weak equivalence principle (Stanford 10 m tower, and the MICROSCOPE-successor concepts). **AION** (UK) and **MAGIS-100** (Fermilab, ~ 100 m baseline) are building long-baseline atom interferometers to search for ultralight dark matter and mid-band gravitational waves (~ 0.1 – 10 Hz, between LIGO and LISA). Space concepts (STE-QUEST, and Bose-Einstein-condensate interferometry on the ISS Cold Atom Lab) extend the free-fall time that sets sensitivity. - **The sensitivity lever:** interferometer phase sensitivity scales with the interrogation time squared and the momentum splitting — which is why the field pushes toward longer drop towers, microgravity, and large-momentum-transfer beam-splitters.

KEY GRADED CLAIMS

- [T1] Light-pulse atom interferometry gives absolute, atomic-referenced measurement of acceleration/rotation — Kasevich–Chu lineage (established)
- [T2] Fine-structure constant measured to sub-ppb by atom interferometry — Parker et al./Morel et al. (established)
- [T2] Long-baseline atom interferometers (MAGIS-100, AION) under construction for dark-matter / GW search — collaboration reports (demonstrated construction, roadmap science)
- [T3] BEC atom interferometry in microgravity (Cold Atom Lab, ISS) — NASA/JPL results (demonstrated)

CONFLICTS / OPEN QUESTIONS

- **The moving-platform problem** (shared with A-pnt / A-gravimetry): a launched cloud dislikes vibration and rotation, so turning a beautiful lab gravimeter into a strapdown navigation-grade 6-DoF sensor on a maneuvering vehicle is unsolved engineering.
- **Physics upside is speculative**: whether long-baseline interferometers actually *detect* dark matter or mid-band gravitational waves (rather than just set limits) is unknown — that is the science bet, not a guaranteed return.

THE HONEST CALL

A mature, foundational quantum-sensing platform — Nobel-recognized physics (1997/2005 laser cooling and precision measurement lineage), buyable as gravimeters, and the engine of the field's most precise fundamental-constant measurements. Its two frontiers are opposite in character: *down* into rugged, small navigation sensors (engineering-hard, A-pnt), and *up* into kilometre-scale physics detectors (science-speculative). Grade the instrument as established; grade each application on its own card.

SOURCES

- <https://arxiv.org/abs/...> (MAGIS-100 detector, Fermilab)
- <https://aion-project.web.cern.ch/> (AION long-baseline atom interferometer)
- Parker et al., "Measurement of the fine-structure constant as a test of the Standard Model," *Science* 360 (2018)

- NASA JPL Cold Atom Lab BEC interferometry results

Atomic & Optical Clocks · A-clocks

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Clocks locked to atomic transition frequencies — the most precise instruments humans have built. Cesium fountains define the SI second today at $\sim 10^{-16}$ fractional uncertainty; optical clocks (trapped ions, optical lattices) run their reference transition $\sim 10^5$ times faster and reach systematic uncertainties near 10^{-18} – 10^{-19} , good to better than a second over the age of the universe. Two quality metrics matter: **accuracy** (systematic uncertainty — how close to the true transition frequency) and **stability** (Allan deviation — how well it averages down over time). Downstream: GPS itself, telecom/finance timestamping, geodesy (a 10^{-18} clock senses a 1 cm change in height via gravitational redshift), and GPS-free PNT (A-pnt).

MATURITY & REAL DEPLOYMENTS (2025–26)

Layered maturity — from mass-market chips to lab instruments defining the SI second. - **Chip-scale atomic clocks (CSACs)** — e.g. Microchip's CSAC (rubidium, ~ 120 mW, $\sim 10^{-10}$ stability) — are long-standing commercial products in defense, telecom, and undersea gear. - **Lab optical clocks** are established metrology. **NIST's Al^+ quantum-logic ion clock** set the accuracy record in July 2025 at $\sim 8.1 \times 10^{-19}$ **systematic uncertainty (~19 decimal places), 41% better than the prior best**. A **six-country coordinated optical-clock comparison** (June 2025) is the largest cross-border agreement test yet — direct groundwork for redefining the second. - **Portable/deployed optical clocks** are commercializing: **QuantX Labs** (rubidium two-photon optical clock, operated at sea), Adelaide's warm-ytterbium-vapor clock, and **Infleqtion's Ticker** — which was deployed on the **UK Royal Navy's XV Excalibur** uncrewed submarine in October 2025, a first-of-kind at-sea optical-clock deployment.

THE SI-SECOND REDEFINITION (THE BIG INSTITUTIONAL EVENT)

BIPM/CCTF's roadmap conditions redefinition on optical standards beating cesium by $\geq 100\times$, ≥ 5 **independent systems** running continuously in different labs, and > 1 **year** of traceable agreeing data. The CGPM (2026) is

expected to endorse the roadmap, with the actual redefinition targeted around **2030**. This is why time-transfer (*A-timedist*) is load-bearing: the clocks are now better than the links that compare them.

KEY GRADED CLAIMS

- [T2] Al^+ optical clock at $\sim 8.1 \times 10^{-19}$ systematic uncertainty, record accuracy — NIST, July 2025 (established)
- [T2] Six-country coordinated optical-clock comparison as redefinition groundwork — Optica/NMI collaboration, June 2025 (demonstrated)
- [T3] Infleqtion Tiqker optical clock deployed on Royal Navy XV Excalibur submarine — Oct 2025 (demonstrated, vendor+navy)
- [T5/T6] SI second redefined on an optical transition ~2030 — CIPM/BIPM roadmap, arXiv:2307.14141 (roadmap)

CONFLICTS / OPEN QUESTIONS

- Which transition wins the redefinition — Sr lattice (most-run), Yb , Yb^+ E3, Al^+ (most accurate), or a weighted ensemble of several — is unsettled; a single-transition definition versus an ensemble is an open governance choice.
- Portable optical clocks trade 2–4 orders of accuracy for size, weight and power. Whether field PNT actually needs optical performance or just a better vapor-cell/CSAC holdover clock is an open market question — for many jam-resistant timing use cases a nanosecond-per-day holdover is enough.

THE HONEST CALL

Split maturity, all real. CSACs are a decades-old commercial commodity; lab optical clocks are the most accurate machines ever built and are about to redefine the second; portable optical clocks are just now leaving the lab (Tiqker at sea). The near-term commercial pull is defense/PNT holdover timing and geodesy, not the 19-digit lab record.

SOURCES

- <https://www.nist.gov/news-events/news> (Al^+ clock record, July 2025)
- https://www.optica.org/about/newsroom/news_releases/2025/unprecedented_optical_clock_network_lays_groundwork_for_redefining_the_second/

- <https://spaceinsider.tech/2026/04/10/quantum-sensing-for-pnt-nears-deployment/> (Tiqker on XV Excalibur)
- <https://arxiv.org/pdf/2307.14141> (redefinition roadmap)

Entanglement-based clock networks · A-entclock

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Distinct from both the clocks (A-clocks) and the classical time-transfer plumbing (A-timedist): this node is the proposal to **entangle distant clocks** so a network of them behaves as one collective quantum sensor. The theory (Kómár et al., Nat. Phys. 2014) is clean: with K clocks of N atoms each linked by entanglement into a shared GHZ state, the frequency uncertainty scales toward the **Heisenberg limit** $\propto 1/(K \cdot N)$ instead of the standard quantum limit $\propto 1/\sqrt{(K \cdot N)}$ — a network of 10 optical-lattice clocks of 1,000 atoms each could in principle reach an Allan deviation of $\sim 2 \times 10^{-18}$ at 1 s, beating any single clock. It would also give a *distributed* reference immune to a single node being spoofed or lost, and enable clock comparisons whose precision beats the link noise.

MATURITY & REAL DEPLOYMENTS (2025–26)

Research — theoretically compelling, experimentally embryonic, and its practical advantage is contested. - **First realization (2022):** two Sr^{88+} ion clocks separated by ~2 m were entangled via a photonic link, and entanglement reduced the number of measurements needed toward the Heisenberg-limit scaling — a proof of principle at benchtop distance, not a network. -

Metropolitan groundwork (2025): an entanglement-based clock **syntonization** demonstration synchronized two rubidium clocks to **<12 ps** at all times over **48–50 km** of deployed fiber across the Métropole Côte d'Azur, riding an entanglement-based QKD link (~7 kbit/s key). This shows entanglement infrastructure and clock timing can share a real fiber network — a step toward, not a demonstration of, Heisenberg-limited networked clocks. - **On paper:** multiple 2025–26 surveys/roadmaps (arXiv:2604.04437, 2606.15421) map the architectures; long-baseline entanglement distribution (A-qinternet) is the missing enabler.

KEY GRADED CLAIMS

- [T1] GHZ-entangled clock networks scale to Heisenberg-limited frequency precision $\propto 1/(KN)$ — Kómár et al., Nat. Phys. 10 (2014) (established theory)

- [T2] Two remote ion clocks entangled via photonic link, sub-Heisenberg measurement reduction — 2022 experiment (demonstrated, ~2 m)
- [T2] Entanglement-based clock syntonization, <12 ps over 48 km deployed fiber — Appl. Phys. Lett. 126, 174003 (2025) (demonstrated)
- [T6] Continental Heisenberg-limited entangled clock network — roadmap only (speculative)

CONFLICTS / OPEN QUESTIONS

- **Does entanglement actually help in practice?** A 2026 critical assessment (arXiv:2604.10243) argues that for real time-synchronization tasks, classical comb-stabilized optical transfer already saturates the useful precision, and GHZ states are fragile (one lost photon collapses the whole network state), so the theoretical $1/N$ gain may not survive loss and decoherence over network distances. This is the same skepticism flagged in `A-timedist`.
- **Enabler dependency:** it cannot exist without long-distance entanglement distribution and quantum memories (`A-qinternet` , `A-qmemory-hw`), which themselves have not reached the repeater crossover.

THE HONEST CALL

A beautiful idea that is nowhere near a network. The Heisenberg-scaling theory is solid and the benchtop/metropolitan building blocks are real, but there is no fielded entangled clock network, and credible physicists question whether entanglement beats good classical optical transfer for actual timekeeping once loss is included. Grade as research/speculative — the near-term "quantum clock network" is classically-linked quantum clocks (`A-timedist`), not entangled ones.

SOURCES

- <https://pubs.aip.org/aip/apl/article/126/17/174003/3345796/> (entanglement-based clock syntonization, 48 km, 2025)
- Kómár et al., "A quantum network of clocks," Nature Physics 10 (2014)
- <https://arxiv.org/pdf/2604.04437> (Quantum Clock Synchronization Networks: A Survey)
- <https://arxiv.org/pdf/2604.10243> (critical assessment of quantum time-sync)

Gravimetry & Inertial Navigation · A-gravimetry

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Cold-atom interferometry: drop or launch a cloud of laser-cooled atoms, split each atom's wavefunction with a sequence of light pulses (a $\pi/2$ – π – $\pi/2$ Mach–Zehnder), and read gravity (or acceleration/rotation) from the accumulated interference phase. Because the atom is the test mass and the laser wavelength is the ruler, the measurement is **absolute and drift-free** — classical spring gravimeters and mechanical/FOG inertial units drift and need recalibration against a reference. Applications: subsurface surveying (voids, tunnels, aquifers, ore, volcano/CO₂-storage monitoring), and drift-free inertial navigation for the GPS-denied problem (A-pnt).

MATURITY & REAL DEPLOYMENTS (2025–26)

Early commercial for gravimetry; pre-deployment for navigation. - **Absolute gravimeters:** **Exail** (ex-Muquans) has sold its **AQG** (Absolute Quantum Gravimeter) for years — including continuous monitoring on Mt Etna. Independent evaluation (Journal of Geodesy 2025) of the AQG-A02/B10 against a reference measured $\sim 500 \text{ nm}\cdot\text{s}^{-2}/\sqrt{\text{Hz}}$ **sensitivity**, reaching $\sim 10 \text{ nm}\cdot\text{s}^{-2}$ ($\approx 1 \mu\text{Gal}$) **after 1 h integration** with drift-free sub- μGal long-term stability — and notably **no active vibration isolation** (it measures microseismic noise with an accelerometer and compensates on the laser phase). - **Gradiometry:** the University of Birmingham's cold-atom gravity **gradiometer detected a buried structure in an open field** (Stray et al., Nature 602, 2022) — the landmark proof that differential (gradiometer) measurement cancels vibration noise outdoors, where a single gravimeter is swamped. - **Navigation:** the hot line. **Q-CTRL's Ironstone Opal** fuses quantum magnetometry/gravimetry with map-matching; airborne trials claim up to **111× better GPS-denied accuracy** than the best conventional alternative and **~4 m over 700 km flights**; a maritime trial logged **144+ hours** unattended at sea (2025–26). **AOSense** cold-atom IMUs target **~5 m/hour** navigation drift without external signals (DARPA PINS lineage). DARPA's **RoQS** funds ruggedization (\$24.4M to Q-CTRL); **Lockheed Martin + Q-CTRL** won a March 2025 DoD prototype quantum-INS contract. France (Exail/ONERA) and the UK (Infleqtion, ship/aircraft trials) run parallel programs.

KEY GRADED CLAIMS

- [T2] Exail AQG: $\sim 500 \text{ nm}\cdot\text{s}^{-2}/\sqrt{\text{Hz}}$, $\sim 1 \text{ }\mu\text{Gal}$ after 1 h, drift-free — independent eval, J. Geodesy 99 (2025); Ménoret et al., Sci. Rep. 2018 (demonstrated)
- [T2] Cold-atom gradiometer located a buried structure in open-field conditions — Stray et al., Nature 602 (2022) (demonstrated)
- [T3] Cold-atom IMU $\sim 5 \text{ m/hour}$ navigation drift, unaided — AOSense / DARPA PINS (demonstrated, program-level)
- [T4] Ironstone Opal: GPS-denied navigation up to $111\times$ better than conventional, $\sim 4 \text{ m}$ over 700 km — Q-CTRL, 2025 (claimed, vendor field trials)

CONFLICTS / OPEN QUESTIONS

- **No independent benchmark for the nav claims:** quantum-INS vendor numbers are self-reported field trials; there is no peer-reviewed head-to-head against classified military-grade ring-laser-gyro/FOG INS.
- **SWaP and dynamics** are the wall between trials and fleet use: size/weight/power, and robustness to vibration and hard maneuvers (a launched atom cloud dislikes a jinking aircraft). Gravimeters mostly measure stationary or slow platforms; strapdown atom-interferometer accelerometers/gyros for full 6-DoF navigation on a moving platform are the harder, less-mature problem.

THE HONEST CALL

Gravimetry is a real, shipping instrument (Exail sells them; independent labs verify the μGal numbers). **Quantum inertial navigation is not yet fielded** — it is in well-funded DoD prototyping with impressive but unverified trial claims, gated by SWaP and dynamic robustness. Split the two: survey gravimeters = commercial; GPS-denied quantum nav = advanced R&D nearing deployment (see A-pnt).

SOURCES

- <https://link.springer.com/article/10.1007/s00190-025-01995-x> (independent Exail AQG evaluation, 2025)
- <https://q-ctrl.com/blog/q-ctrls-new-maritime-quantum-navigation-solution-successfully-undergoes-first-defense-trials-at-sea>

- Stray et al., "Quantum sensing for gravity cartography," Nature 602 (2022)
- <https://aosense.com/> (cold-atom IMU / gradiometer)

Quantum Imaging & Radar · A-imaging

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Imaging and ranging schemes that exploit photon correlations. **Ghost imaging** reconstructs an object from a beam that never touched it, via correlations with a reference beam that did. **Quantum illumination (QI)** (Lloyd 2008; Tan et al. 2008) sends one half of an entangled signal–idler pair at a target and keeps the idler; joint correlation detection gives a theoretical **6 dB advantage in the error exponent** in the high-noise, low-reflectivity, low-signal regime — the basis of "quantum radar" claims. Related: sub-shot-noise microscopy, NOON-state phase imaging, and single-photon LiDAR. A crucial taxonomic split runs through this node — **truly quantum** (entangled/squeezed light) vs **quantum-inspired** (classical correlations, quantum detectors) — because almost all the commercial wins are the latter.

THE QUANTUM VS QUANTUM-INSPIRED LIDAR SPLIT (THE LOAD-BEARING DISTINCTION)

- **Single-photon / quantum-inspired LiDAR is genuinely deployed.** SPAD arrays and photon-counting time-of-flight sensors ship in automotive and remote sensing for photon-starved conditions — but they use **quantum detection with classical light**, not entanglement. A compact all-fiber **quantum-inspired LiDAR** demonstrated **>100 dB rejection** of in-band noise with single-photon sensitivity using classical time-frequency correlations (Nature Communications 2023); a 2025 **photon-number-resolving** single-photon LiDAR approached the **standard quantum limit** in ranging precision (Light: Science & Applications 2025). These keep the noise-rejection benefit without needing entanglement or cryogenics — which is exactly why they, not QI radar, reach the field.
- **Entanglement-based QI** stays in the lab. Microwave QI was demonstrated in principle inside dilution refrigerators, showing **~4 dB** measured enhancement over a classical-noise-radar benchmark (below the 6 dB theoretical ceiling), but only over a narrow parameter range and requiring cryogenic idler storage.

MATURITY & REAL DEPLOYMENTS

Mostly research; the commercial wins are quantum-inspired. **Quantum radar** is the graded-hardest corner: fielded long-range quantum radar is implausible near-term because idler storage over realistic ranges is impossible (you'd need a lossless quantum memory for the round-trip light-time), entanglement generation rates are tiny, and the advantage evaporates the moment you can just transmit more classical power. Defense-lab and academic assessments are broadly skeptical despite recurring Chinese vendor claims. Ghost imaging and sub-shot-noise microscopy (Padgett, Genovese groups) are established lab techniques with niche microscopy use.

KEY GRADED CLAIMS

- [T2] QI's 6 dB error-exponent advantage in the noise-limited regime — Tan et al., PRL 101, 253601 (2008); microwave lab demos ~4 dB (established theory, demonstrated small-scale)
- [T2] Ghost imaging and sub-shot-noise imaging demonstrated repeatedly — Padgett group, Genovese reviews (established)
- [T2] Noise-tolerant / photon-number-resolving LiDAR near the standard quantum limit — Light Sci. Appl. 2025; Nat. Commun. 2023 (demonstrated, quantum-inspired)
- [T6] Fielded long-range quantum (entanglement) radar — vendor/press claims, no verified system (speculative; contested)

CONFLICTS / OPEN QUESTIONS

- **C-qradar**: proponents cite the QI advantage; skeptics (RAND-adjacent and academic reviews) note it assumes conditions — pre-shared entanglement surviving a lossy round trip, known target range, extreme background noise, no option to just raise transmit power — that no fielded radar meets. Resolution: any independently verified over-the-air quantum (not quantum-inspired) radar demo. None exists.
- Where entanglement-based quantum imaging beats cheap computational classical imaging is still case-by-case and mostly loses on cost.

THE HONEST CALL

**Quantum-inspired imaging/LiDAR is commercial and shipping;
entanglement-based quantum radar is not, and probably won't be soon.**

The single most useful thing this node does is police the label: a "quantum LiDAR" in a product is almost always a photon-counting classical system, and "quantum radar" as a fielded weapon is, as of 2026, vaporware.

SOURCES

- <https://www.nature.com/articles/s41377-025-01880-4> (photon-number-resolving single-photon LiDAR, 2025)
- <https://www.nature.com/articles/s41467-023-40914-6> (quantum-inspired LiDAR, >100 dB rejection)
- <https://arxiv.org/abs/2211.05684> (quantum advantage in microwave quantum radar, ~lab)
- <https://arxiv.org/pdf/2103.12548> (Quantum Technology for Military Applications — skeptical review)

Magnetometry · A-magneto

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Measuring magnetic fields with quantum systems, down to femtotesla (fT) — the brain's magnetic signals are ~50–500 fT, roughly a hundred-million-times weaker than Earth's ~50 μ T field. Three platforms with different trade-offs: - **SQUIDs** — superconducting loops, the cryogenic gold standard since the 1960s; ~**1–5 fT/ $\sqrt{\text{Hz}}$** , but need liquid helium and fixed dewars. - **OPMs (optically pumped magnetometers)** — alkali (Rb/Cs) or ^4He vapor cells, near-room-temperature, small and wearable; commercial zero-field OPMs (QuSpin QZFM Gen-2) reach ~**7–15 fT/ $\sqrt{\text{Hz}}$** (1–100 Hz), approaching SQUID sensitivity without cryogenics. - **NV-diamond** — nanoscale, ambient, vector; best NV-MEG-oriented sensors reach ~**9.4 pT/ $\sqrt{\text{Hz}}$** (5–100 Hz) — three orders worse than OPMs, but at atom-scale spatial resolution (see [A-nvsensing](#) , [H-nv](#)).

MATURITY & REAL DEPLOYMENTS (2025–26)

Commercial today across several niches, with neuroimaging the standout disruption. - **MEG (magnetoencephalography)**: SQUID-MEG has imaged brain fields in hospitals for decades inside **\$2–3M shielded rooms** with fixed helmets. The disruption is **OPM-MEG** — wearable helmets the patient can move in. **Cerca Magnetics** (Nottingham spin-out) sells OPM-MEG and ran epilepsy/Parkinson's clinical work through 2024–25; **FieldLine Medical** installed 64-channel wearable OPM-MEG at two US academic medical centres in 2024 for pediatric epilepsy pre-surgical mapping without sedation; **MAG4Health** (France) sells a ^4He -OPM system; an 80-sensor OPM-MEG study (NeuroImage 2025) marks research-grade adoption. - **Cardiac**: SandboxAQ's **CardiAQ** (with Mayo Clinic) applies OPMs to magnetocardiography. - **Navigation**: magnetic-anomaly map-matching (Q-CTRL Ironstone Opal, SandboxAQ AQNav — see [A-sensing](#) , [A-pnt](#)). - **Geophysics/defense**: SQUID transient-EM sensors (CSIRO LANDTEM) have found ore bodies; OPM/NV airborne surveys are emerging; magnetic-anomaly submarine detection (MAD) is a classified but active area.

KEY GRADED CLAIMS

- [T1] SQUIDs ($\sim 1\text{--}5\text{ fT}/\sqrt{\text{Hz}}$) and zero-field OPMs ($\sim 7\text{--}15\text{ fT}/\sqrt{\text{Hz}}$) reach femtotesla sensitivity — decades of metrology + QuSpin datasheets (established)
- [T2] Wearable OPM-MEG matches cryogenic MEG signal quality while allowing head motion — Boto et al., Nature 555 (2018); 80-sensor system, Neurolmage 2025 (demonstrated)
- [T2] NV-diamond magnetometer at $\sim 9.4\text{ pT}/\sqrt{\text{Hz}}$ enabling ambient-condition MEG-scale sensing — Phys.org/2024 diamond magnetometer result (demonstrated)
- [T4] CardiAQ OPM cardiac diagnostics in clinical evaluation with Mayo Clinic; Cerca OPM-MEG clinical work 2024–25 — company announcements (claimed)

CONFLICTS / OPEN QUESTIONS

- **Reimbursement, not physics, gates OPM-MEG.** The sensitivity is there; whether it displaces SQUID suites this decade turns on FDA/CE clearance and insurance reimbursement pace, plus the still-needed magnetically shielded room (OPMs cut cost but don't fully eliminate shielding).
- **NV fit is unsettled:** NV's ambient operation and spatial resolution are unmatched, but it lags OPM/SQUID by 2–3 orders in raw sensitivity — so its commercial niche (current mapping inside chips, single-cell biology, scanning probe) is real but narrow.

THE HONEST CALL

Commercial and the most clinically consequential quantum-sensing modality. SQUID magnetometry is a mature medical/industrial product; OPM-MEG is a live, well-funded disruption already installed in hospitals and limited by regulatory/reimbursement timelines rather than performance; NV magnetometry is real but a specialist tool, not an OPM/SQUID replacement.

SOURCES

- <https://www.sciencedirect.com/science/article/pii/S1053811925001843> (80-sensor OPM-MEG, 2025)
- <https://entangledfuture.com/quantum-sensors/magnetometers/> (OPM/SQUID/NV sensitivity comparison)

- <https://phys.org/news/2024-06-highly-sensitive-diamond-quantum-magnetometer.html> (9.4 pT/ $\sqrt{\text{Hz}}$ NV)
- Boto et al., "Moving magnetoencephalography towards real-world applications," Nature 555 (2018)

Quantum Materials · A-materials

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Materials whose macroscopic behavior is set by quantum effects — **topological insulators** (insulating bulk, symmetry-protected conducting surface states), **high-Tc superconductors** (cuprates since 1986, iron-based since 2008), and **2D materials** (graphene, transition-metal dichalcogenides, and twisted bilayers where a "magic angle" $\sim 1.1^\circ$ produces superconductivity). They sit in two roles at once: a physics frontier in their own right, and the **substrate layer under qubits** — Majorana platforms (H-topo), Josephson-junction materials, and the two-level-system (TLS) defects that cap coherence (H-fab , 0-materials). This node treats quantum materials as an adjacent *technology*, so the grading question is: which of these ship in a product, and which are still lab physics?

MATURITY & REAL DEPLOYMENTS (2025–26)

Split — a spectrum from industrial commodity to contested lab claim. -

Low-Tc superconductors (NbTi , Nb_3Sn) are fully industrial: MRI magnets,

SQUIDs (A-magneto), and the transmon circuits of H-supercon all run on them.

- **High-Tc REBCO tape** crossed from lab to product: it is load-bearing in **Commonwealth Fusion's SPARC** high-field magnets (the $\sim 20\text{ T}$ magnets

that make compact tokamaks viable) — arguably the biggest commercial

deployment of a high-Tc material to date. - **Topological insulators and 2D-material devices remain research-stage** — no volume product ships *on*

topological protection yet, which is precisely the risk in Microsoft's Majorana bet

(H-topo). Recent motion: UChicago/WVU showed **topological**

superconductivity in $\text{FeTe}_{1-x}\text{Se}_x$ can be tuned in/out by composition

(Nature Communications, 2025/26); 2D magnetic/topological heterostructures

are engineered for ultralow-power spintronics; topological phase transitions were

used to raise thermoelectric performance in Cr-doped PbSe .

KEY GRADED CLAIMS

- [T1] Topological insulators and cuprate high-T_c superconductivity are established, Nobel-lineage physics — Kane–Mele / Bednorz–Müller (established)
- [T2] Magic-angle twisted bilayer graphene superconducts at $\sim 1.1^\circ$ — Cao et al., Nature 556 (2018) (established)
- [T2] REBCO high-T_c tape deployed in fusion magnets (SPARC) — Commonwealth Fusion / MIT, 2021+ (demonstrated, commercial)
- [T2/T3] Composition-tuned topological superconductivity in FeTeSe — Nat. Commun. 2025/26 (demonstrated)
- [T6] Room-temperature ambient-pressure superconductor — none verified; **LK-99 (2023)** and the **Ranga Dias retractions** are the cautionary tales (contested/refuted)

CONFLICTS / OPEN QUESTIONS

- **C-highT_c-mechanism**: after ~40 years there is still no accepted microscopic theory of cuprate superconductivity — the field's standing embarrassment and its biggest open prize. Without it, high-T_c discovery stays empirical.
- **Can topological protection be shown cleanly in a device?** The Majorana debate (H-topo) remains contested after multiple retracted/withdrawn zero-bias-peak claims; a decisive, reproducible topological-qubit demonstration would resolve it.
- **Room-temperature superconductivity** is the recurring hype magnet — every few years a claim (LK-99, Dias's hydrides) collapses under replication. Grade any new one T6-contested until independent labs confirm.

THE HONEST CALL

Commercial at the boring end, research at the exciting end. Low-T_c and now high-T_c REBCO are real industrial materials doing real work (magnets, qubits, fusion). Topological materials — the ones that would rewrite computing — remain lab physics with a credibility overhang from retractions. The valuable discipline this node enforces: separate "superconductor that ships" from

"topological phase we hope powers a qubit," and treat every room-temperature-superconductor headline as guilty until replicated.

SOURCES

- <https://phys.org/news/2026-02-tuning-topological-superconductors-adjusting-ratio.html>
- <https://arxiv.org/pdf/2102.02644> (2021 Quantum Materials Roadmap)
- Cao et al., "Unconventional superconductivity in magic-angle graphene superlattices," Nature 556 (2018)
- Commonwealth Fusion Systems / MIT PSFC SPARC magnet reports (REBCO 20 T magnet)

NV-diamond sensing · A-nvsensing

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

The nitrogen-vacancy (NV) center — a nitrogen atom next to a missing carbon in diamond — is an atom-sized quantum sensor whose spin can be initialized, manipulated, and read out optically at **room temperature and ambient pressure**. That combination (no cryostat, nanoscale spatial resolution, vector sensitivity) makes NV-diamond its own sensing platform, spanning magnetometry, electrometry, thermometry, and pressure. This node consolidates NV *sensing* as a discipline; it deliberately cross-links **A-magneto** (where NV competes with SQUID/OPM), **H-nv** (NV as a compute/qubit modality), and **A-sensing** (the broader quantum-sensing market). Its niche is where you need a sensor *at* the sample — scanning-probe imaging, single-cell biology, current mapping inside a chip — not just the highest raw sensitivity.

MATURITY & REAL DEPLOYMENTS (2025–26)

- **Element Six** (De Beers) supplies the quantum-grade CVD diamond (DNV series) underneath most NV sensors. **Bosch + Element Six** formed a JV, "Bosch Quantum Sensing," in 2025 to scale NV sensors toward mass-market (mobility, medical); Element Six holds ~25%.
- **SBQuantum** (Canada) — field-proven diamond vector magnetometers for defense, navigation, and mineral exploration; a handheld unit at ~400 pT/√Hz vector sensitivity, heading errors <5 nT; flying an NGA MagQuest payload with Spire Global to map Earth's field for GPS-free navigation.
- **QDTI** (Quantum Diamond Technologies, Harvard spin-out) — NV "magnetic microscope" for biomarker diagnostics: <1 pg/mL detection from a 5 μL sample in <1 hour.
- Academic: commercial scanning-NV magnetometers in closed-cycle cryostats; fully integrated nanotesla-sensitivity NV chips (arXiv 2508.03237); a 2025 Quantum Diamond Workshop findings report (arXiv 2511.11791) marks the field's coalescence.

KEY GRADED CLAIMS

- [T2] NV magnetometry operates at room temperature with nanoscale resolution — established NV literature / NIST (established)
- [T4] SBQuantum handheld $\sim 400 \text{ pT}/\sqrt{\text{Hz}}$ vector sensitivity, $<5 \text{ nT}$ heading error — company figures (claimed)
- [T4] QDTI $<1 \text{ pg/mL}$ biomarker detection from $5 \text{ }\mu\text{L}$ in $<1 \text{ hr}$ — company claim (claimed)
- [T3] Bosch×Element Six JV to industrialize NV sensing (2025) — press + corporate filings (roadmap)

CONFLICTS / OPEN QUESTIONS

NV's ambient operation and spatial resolution are unmatched, but its raw magnetic sensitivity still lags SQUID ($\sim 1\text{--}5 \text{ fT}/\sqrt{\text{Hz}}$) and OPM ($\sim 7\text{--}15 \text{ fT}/\sqrt{\text{Hz}}$) by 2–3 orders — the best NV ensembles reach $\sim 9.4 \text{ pT}/\sqrt{\text{Hz}}$ for MEG-scale work (see [A-magneto](#)). So its commercial fit vs the [A-magneto](#) incumbents is unsettled: NV wins wherever you must put the sensor *at* the sample (current mapping inside a live chip, single-cell biology, high-pressure diamond-anvil metrology) and loses on any application that just wants the lowest field noise in a shielded room.

THE HONEST CALL

A real, shipping specialist instrument — with a mass-market bet still unproven. NV magnetometers and NV "magnetic microscopes" sell today into defense navigation, materials/failure analysis, and biomarker diagnostics; Element Six supplies the diamond and independent groups reproduce the room-temperature physics. What is *claimed* rather than established is the Bosch×Element Six thesis that NV goes to automotive/consumer volume — that is a roadmap, not a deployment. Grade NV as commercial-in-niches, mass-market-aspirational.

SOURCES

e6.com/about/news (Bosch JV; SBQuantum MagQuest); thequantuminsider.com 2025/04/07 (Bosch×Element Six); sbquantum.com; spacenews.com (SBQuantum + Spire); qdti.com/instrument; arXiv:2508.03237; arXiv:2511.11791 (2025 Quantum Diamond Workshop); nist.gov/noac (NV magnetometry).

Assured PNT / GPS-denied navigation · A-pnt

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Assured Positioning, Navigation, and Timing is the *system-level* problem of knowing where and when you are without GPS/GNSS — because GNSS is jammable, spoofable, and unavailable underwater, underground, indoors, and in contested airspace. This node exists because the quantum answer to that problem is not one sensor but a **fusion**: a drift-free timing source (optical/atomic clock, **A-clocks**), a drift-free inertial reference (cold-atom accelerometers/gyros, **A-gravimetry**), and a map-matching field reference (quantum magnetometry, **A-magneto**; NV vector magnetometry, **A-nvsensing**). Previously this was smeared across the gravimetry, magnetometry, and clock cards; A-pnt is the integration layer where they combine into a navigation solution. The core physics payoff: a quantum clock plus a quantum IMU lets you *dead-reckon* accurately for hours, and a magnetic/gravity map lets you *fix* your absolute position against Earth's own signatures — neither of which an adversary can jam.

MATURITY & REAL DEPLOYMENTS (2025–26)

Advanced R&D nearing deployment — well-funded, defense-led, not yet fielded at fleet scale. - **Fusion navigators:** **Q-CTRL's Ironstone Opal** is the most public — magnetic + gravity map-matching with ML, claiming up to **111× better GPS-denied accuracy** than the best conventional alternative, **~4 m over 700 km** airborne, and **144+ hours** unattended at sea (2025–26); TIME Best Inventions 2025. **SandboxAQ's AQNav** does AI + magnetic-anomaly navigation and has flown on multiple aircraft types. - **Quantum IMUs:** **AOSense** cold-atom inertial units target **~5 m/hour** drift unaided (the DARPA PINS lineage) — vs a conventional tactical IMU that can drift kilometers in the same time. - **Timing anchor:** **Infleqtion's Ticker** optical clock deployed on the **Royal Navy XV Excalibur** uncrewed submarine (Oct 2025) — a holdover timing source for a platform that cannot surface for GPS. - **Programs:** DARPA **Robust Quantum Sensors (RoQS)** (ruggedization; \$24.4M to Q-CTRL), DARPA **PINS** (precision inertial nav), the DIU, and a March 2025 **Lockheed Martin + Q-CTRL** prototype quantum-INS contract. A 2036 PNT-stack forecast keeps clocks dominant

(~76%) with quantum IMUs reaching ~18% as they progress to platform integration (helicopter, maritime, space).

KEY GRADED CLAIMS

- [T4] Ironstone Opal: up to 111× better GPS-denied accuracy, ~4 m over 700 km, 144+ h at sea — Q-CTRL field trials, 2025 (claimed)
- [T3] Cold-atom IMU ~5 m/hour unaided navigation drift — AOSense / DARPA PINS (demonstrated, program-level)
- [T3] Optical clock (Tiqker) deployed as holdover timing on an uncrewed submarine — Inflection / Royal Navy, Oct 2025 (demonstrated)
- [T5] Quantum PNT market shifts toward clocks-dominant + ~18% quantum-IMU stack by 2036 — analyst forecast (forecast)

CONFLICTS / OPEN QUESTIONS

- **No independent benchmark against black-program INS.** Every headline number is a vendor field trial against a specified (often unnamed) baseline; classified military ring-laser/FOG INS performance is the real comparator and it is not public.
- **Fusion is the hard part:** individual quantum sensors work, but robustly combining a jittery atom-interferometer IMU, a map-matching magnetometer (which needs a good, current, classified magnetic map), and a clock into one resilient solution on a maneuvering platform is a systems problem, not a physics one.
- **SWaP + dynamics:** launched atom clouds dislike vibration and hard turns; shrinking the whole stack to fit a drone or missile, not just a ship, is unsolved.

THE HONEST CALL

The most strategically important integration story in quantum sensing, and the reason the defense money is flowing — but it is prototyping, not deployment. The pieces (clock, IMU, magnetometer) are individually real; the fused, ruggedized, independently-benchmarked GPS-denied navigator on a fleet of moving platforms does not yet exist in the open literature. Grade A-pnt as "advanced R&D nearing first operational deployments," with the flashiest accuracy multipliers unverified.

SOURCES

- <https://spaceinsider.tech/2026/04/10/quantum-sensing-for-pnt-nears-deployment/> (Tiqker on XV Excalibur; market split)
- <https://q-ctrl.com/blog/2025-year-in-review-realizing-true-commercial-quantum-advantage-in-the-international-year-of-quantum> (Ironstone Opal)
- <https://www.darpa.mil/news/features/quantum-sensing-computing> (RoQS / PINS)
- <https://link.springer.com/article/10.1007/s10291-026-02030-y> (quantum sensors for PNT, review)

Post-Quantum Cryptography · A-pqc

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Classical cryptography built on problems believed hard even for quantum computers (structured lattices, hash functions, error-correcting codes, isogenies), replacing the RSA and elliptic-curve schemes that Shor's algorithm (S-shor) breaks in polynomial time. The driver is **"harvest now, decrypt later"** (HNDL): adversaries record encrypted traffic today to decrypt once a cryptographically-relevant quantum computer (CRQC) exists — so any secret with a shelf life past the CRQC date (state secrets, health records, genomic data, long-term IP) is *already* at risk, regardless of when the machine actually arrives. PQC runs on today's classical hardware; it is the pragmatic answer to the quantum threat, and the one all four Five-Eyes-adjacent signals agencies endorse over QKD (A-qkd).

MATURITY & REAL DEPLOYMENTS (2025–26)

Commercial and deploying at internet scale now — the most consequential quantum-adjacent technology of the decade. - **Standards:** NIST finalized **FIPS 203 ML-KEM** (Kyber), **FIPS 204 ML-DSA** (Dilithium), and **FIPS 205 SLH-DSA** (SPHINCS+) in **August 2024**. **HQC** (code-based) was selected March 2025 as a backup KEM on a different math assumption, finalization ~2027. - **Live deployment:** hybrid **X25519 + ML-KEM-768** key exchange is default in **Chrome, Firefox, Cloudflare** and major CDNs — a large fraction of TLS 1.3 handshakes on the open web are already post-quantum. **Signal (PQXDH)** and **Apple iMessage (PQ3)** shipped PQC messaging; PQ3 does continuous rekeying for forward secrecy against HNDL. - **Blockchain adoption:** Algorand executed quantum-resistant **Falcon**-signed transactions and enabled Falcon-based accounts on mainnet (Nov 2025) — an early production PQC signature in public-ledger infrastructure. - **Mandated deadlines:** NSA **CNSA 2.0** requires new national-security-system acquisitions to be PQC-capable by **Jan 1 2027**, full NSS migration by **2030–2035**; NIST IR 8547 deprecates quantum-vulnerable algorithms by **2030**, disallows after **2035**.

KEY GRADED CLAIMS

- [T2] ML-KEM/ML-DSA/SLH-DSA standardized after multi-year public cryptanalysis — NIST FIPS 203/204/205, Aug 2024 (established)
- [T2] Hybrid PQC key exchange live across major browsers/CDNs at internet scale — Cloudflare/Google deployment telemetry (demonstrated)
- [T2] Falcon-signed accounts live on Algorand mainnet — Nov 2025 (demonstrated)
- [T1/T2] Shor breaks RSA/ECC given an FTQC — Shor 1994 (established); the *date* such a machine exists is T6
- [T5] HNDL is economically rational for state actors today — CSA / arXiv analyses (claimed)

CONFLICTS / OPEN QUESTIONS

- **C-pqc-confidence**: lattice hardness has *no* unconditional security proof — a classical or quantum cryptanalytic surprise (cf. **SIKE's 2022 one-afternoon collapse** on a classical laptop) would force re-migration. Hence HQC as a math-diverse hedge, and **hybrid modes** (classical + PQC together) as the default so a break in either half is not catastrophic.
- **Migration inventory is the real bottleneck**: the algorithms are ready, but most organizations cannot enumerate where cryptography lives in their stacks (embedded keys, protocols, certificates, HSMs), so "crypto-agility" and discovery tooling — not new math — is the gating work.

THE HONEST CALL

The single most real, most deployed, most economically-important item in the whole adjacent-tech chapter — and it isn't even quantum hardware.

PQC is classical software, shipping now, on a legally-mandated clock. If any node in §04 pays for itself this decade, it is this one. The open risk is cryptanalytic (an assumption breaks), not deployment.

SOURCES

- <https://csrc.nist.gov/projects/post-quantum-cryptography>
- <https://pages.nist.gov/nccoe-migration-post-quantum-cryptography/>
- <https://algorand.co/technology/post-quantum> (Falcon on mainnet, 2025)
- NSA CNSA 2.0 FAQ / timeline; NIST IR 8547

Quantum-secured & quantum-resistant blockchain · A-qblockchain

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Blockchains are unusually exposed to quantum attack: their security rests almost entirely on the elliptic-curve signatures (ECDSA/EdDSA) that Shor's algorithm (S-shor) breaks, and every public key ever exposed on-chain is *permanently* harvestable (A-pqc 's HNDL threat, but worse — the ciphertext-equivalent is the immutable ledger itself). Two responses, often conflated: 1. **Quantum-resistant blockchain** — swap ECDSA for a PQC signature (lattice-based Dilithium/ML-DSA, or hash/NTRU-based **Falcon**, or stateful hash schemes). This is classical software; the dominant, practical track. 2. **Quantum-secured blockchain** — use QKD (A-qkd) or QRNG (A-qrng) to protect the *network* links or seed keys. Mostly experimental and niche.

MATURITY & REAL DEPLOYMENTS (2025–26)

Quantum-resistant is deploying now; quantum-secured is a research sideshow. - **Algorand** executed quantum-resistant **Falcon**-signed transactions and, in **November 2025**, enabled **Falcon-based accounts on mainnet** — an early production PQC signature scheme on a mainstream public ledger (via state proofs already using Falcon since 2022). - **QRL (Quantum Resistant Ledger)** has run a hash-based-signature (XMSS) chain since 2018 — a purpose-built PQC blockchain. - **Bitcoin/Ethereum** face the hardest migration: proposals exist (e.g. quantum-resistant address types, a "P2QRH" BIP for Bitcoin; account-abstraction paths for Ethereum) but require contentious hard/soft forks and a plan for the millions of coins in exposed-public-key addresses (including dormant early-Bitcoin wallets that cannot be moved by their owners). Research consolidations (Hyperledger Fabric + Kyber/Dilithium/Falcon; SoK arXiv:2512.13333) map the design space. - **Threat timing:** consensus estimates put a CRQC able to break ECC at **~5–15 years** out, which is *inside* the lifetime of long-lived on-chain keys — the reason migration is being taken seriously now.

KEY GRADED CLAIMS

- [T2] Falcon-signed accounts live on Algorand mainnet (Nov 2025); QRL XMSS chain live since 2018 — project releases (demonstrated)
- [T2] PQC signatures (ML-DSA/Falcon/SLH-DSA) integratable into blockchains without new math — NIST FIPS 204/205; Hyperledger integrations (established)
- [T5] A CRQC threatening ECDSA is ~5–15 years out — analyst/CSA consensus (claimed/forecast)
- [T4] QKD/QRNG "quantum-secured" ledgers as a needed product — vendor/academic pilots (claimed, niche)

CONFLICTS / OPEN QUESTIONS

- **Signature bloat:** PQC signatures are large (Dilithium ~2.4 kB, Falcon ~0.7 kB, SPHINCS+ ~8–50 kB) vs ECDSA's ~64 bytes — inflating block size, fees, and validation cost. This is the real engineering tax, and it is why chains pick Falcon (small but complex to implement) over Dilithium where they can.
- **The unmovable-coins problem:** exposed public keys on existing UTXOs (and lost-key wallets) can't be retroactively protected — a fork can require new PQC addresses going forward but cannot rescue the old exposed ones from a future quantum attacker.
- **Quantum-secured (QKD) blockchain** inherits all of QKD's limits (distance, trusted relays, `C-qkd-vs-pqc`) and adds little a PQC signature doesn't — mostly a solution in search of a problem.

THE HONEST CALL

Quantum-resistant blockchain is real, live, and mostly a PQC-migration story (so it's `A-pqc` applied to ledgers) — the interesting nuance is signature-size economics and the unmovable-coins overhang. "Quantum-secured" (QKD-based) blockchain is a research curiosity with weak motivation. The genuine near-term risk is not exotic: it's that Bitcoin/Ethereum are slow to fork and hold trillions in ECDSA-exposed value on a ~5–15-year clock.

SOURCES

- <https://algorand.co/technology/post-quantum> (Falcon on mainnet, 2025)

- <https://www.theqrl.org/the-definitive-guide-to-post-quantum-blockchain-security/>
- <https://arxiv.org/pdf/2512.13333> (SoK: post-quantum attackers, blockchain security & performance)
- <https://www.chainalysis.com/blog/quantum-computing-crypto-security/> (threat timing)

Quantum-dot displays (boundary case) · A-qdisplay

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node — boundary case)

WHAT IT IS

A deliberately-included **boundary case**, here to police the word "quantum" in the largest-revenue product that carries it. A **quantum dot** is a semiconductor nanocrystal (~2–10 nm) whose emission wavelength is set by quantum confinement — make the dot smaller and the bandgap widens, so size *tunes color* with near-perfect purity. This is real quantum mechanics (particle-in-a-box confinement). But whether a QD *display* belongs in a quantum-technology atlas depends entirely on which of three things you mean: 1. **Photoluminescent QLED (today's "QLED" TVs)** — QDs sit in a film and simply down-convert blue LED backlight into pure red/green. Confinement sets the color, but the device is an ordinary LCD. **Quantum physics, not quantum technology** in the manual's sense — no superposition, entanglement, or single-quantum control is exploited. 2. **Electroluminescent QD (EL-QD / "NanoLED" / true QLED)** — QDs that conduct electricity and emit their own light, no backlight, no OLED. A real display breakthrough, still QD-as-a-material. 3. **QD single-photon / entangled-photon sources** — the one that *is* quantum technology: individual dots emit one photon at a time on demand (Quandela, [H-photonsource](#)), a building block for photonic QC and QKD. That belongs to the hardware chapter, cross-linked here.

MATURITY & REAL DEPLOYMENTS (2025–26)

- **Photoluminescent QLED / QD-OLED:** fully commercial, mass-market — Samsung, TCL, Sony ship tens of millions of QD TVs; **Nanosys** (acquired by Shoen Chemical) is the dominant QD-material supplier. This is a multi-billion-dollar consumer market and the reason "quantum" is a household display word.
- **Electroluminescent NanoLED / EL-QD:** prototype stage. Samsung Display showed brighter EL-QD prototypes (2025–26); Nanosys/Shoen repeatedly pushed the "ready ~2025–26" target — and, as of 2026, slipped it to ~**2029**. Not yet a product.

- **QD single-photon sources:** research/early-commercial for quantum-photonics labs (Quandela's Prometheus), covered under **H-photonsource** .

KEY GRADED CLAIMS

- [T1] Quantum confinement sets QD emission color (size-tunable bandgap) — established solid-state physics (established)
- [T2] Photoluminescent QD displays are a mass-market commercial product — Samsung/TCL/Nanosys shipments (established, commercial)
- [T4/T6] Electroluminescent EL-QD ("true QLED") displays for consumers ~2029 — Samsung/Nanosys roadmap, repeatedly slipped (roadmap)
- [T2] On-demand single-photon emission from individual quantum dots — established QD-photonics literature (established; see **H-photonsource**)

CONFLICTS / OPEN QUESTIONS

- **The definitional one, which is the point of this card:** is a QLED TV "quantum technology"? By the manual's standard (exploiting a controlled quantum state — superposition/entanglement/single-quantum readout), **no** for photoluminescent QLED — it uses a quantum *material* the way a laser or an LED does, not a quantum *information* resource. **Yes** for QD single-photon sources.
- EL-QD's engineering blockers (blue-dot lifetime/efficiency, cadmium-free formulations, electrical stability) keep slipping the timeline.

THE HONEST CALL

Mostly not a quantum technology in this atlas's sense — included to say so out loud. Today's QD displays are a huge, real consumer market built on a quantum *material*, and calling them "quantum tech" alongside qubits and QKD is a category error the marketing invites. The genuinely quantum-technology thread is the QD *single-photon source* (**H-photonsource**), not the TV. Grade the TV as commercial-but-boundary; grade the photon source in hardware.

SOURCES

- <https://www.nanosys.com/blog-newsroom/nanosys-quantum-dot-technology-at-display-week-2025>

- <https://www.flatpanelshd.com/news.php?subaction=showfull&id=1777964273> (Samsung EL-QD/NanoLED prototypes)
- https://en.wikipedia.org/wiki/Quantum_dot_display (PL vs EL QLED distinction)
- Cross-link: [H-photonsource](#) (QD single-photon sources — the actual quantum-tech thread)

Quantum Internet & Repeaters · A-qinternet

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

A network that distributes *entanglement* between distant nodes — enabling QKD without trusted relays, blind/distributed quantum computing, and networked quantum sensors (**A-sensing** , entanglement-based clock networks). Photon loss in fiber is exponential (~ 0.2 dB/km) and optical amplification is forbidden by no-cloning (**F-nocloning**), so long links need **quantum repeaters**: quantum memories (**A-qmemory-hw**) that store entanglement in short segments, then fuse segments by entanglement swapping (**F-teleport**) and purification. The figure of merit is delivered **entanglement rate vs distance**, and the field's watershed is the moment a repeater beats direct transmission at that.

MATURITY & REAL DEPLOYMENTS (2025–26)

Research — no repeater has yet beaten direct fiber in delivered entanglement rate. That crossover has not happened; everything below is a building block toward it. - **Metropolitan entanglement network:** QuTech (Delft) linked **Delft–The Hague over ~25 km of deployed fiber** with heralded entanglement between processor nodes (Stolk et al., Sci. Adv. 2024) — the first multi-node network on installed fiber. - **Long-distance memory entanglement:** a preprint reports entangling quantum memories over **420 km** of fiber (arXiv:2504.05660); USTC reported **memory–memory entanglement + device-independent QKD across 100 km** (Feb 2026); long-lived **remote ion–ion entanglement** suited to repeaters appeared in Nature (2026). ICFO Barcelona demonstrated **multiplexed** memory-based distribution (2025), attacking the rate problem. - **Programs & vendors:** Germany's **TD.QR** repeater project (started Jan 2026) targets viable repeater demos by **2028**. Startups sell early network gear — **Qunnect** (room-temperature memories, **A-qmemory-hw**) and **Aliro** (network orchestration/SDN). Testbeds run in Boston, Chicago (the Chicago Quantum Exchange), Amsterdam, and Hefei.

KEY GRADED CLAIMS

- [T2] Multi-node heralded entanglement between two cities over ~25 km deployed fiber — Stolk et al./QuTech, Sci. Adv. 2024 (demonstrated)

- [T3] Quantum memories entangled over 420 km fiber — arXiv:2504.05660 (claimed, preprint)
- [T2/T3] Memory–memory entanglement + DI-QKD across 100 km — USTC, Feb 2026 (demonstrated; verify final journal ref)
- [T6] Useful multi-node quantum internet by the early 2030s — various national roadmaps (roadmap)

CONFLICTS / OPEN QUESTIONS

- **Which memory platform wins** is unsettled: atomic ensembles (highly multiplexable, good rate) vs single ions/NV/atoms (processable, can do logic at the node) vs rare-earth crystals (long storage). The network market may fork by use case.
- **Repeater-vs-direct crossover date is unproven**, and some argue **satellite links** ([A-satqkd](#)) will reach continental scale before fiber repeaters do — making the "quantum internet" partly a space story.
- **The demand question underneath it all**: beyond relay-free QKD (itself contested vs PQC, see [C-qkd-vs-pqc](#)), the killer app for a quantum internet — distributed QC, blind computing, sensor networks — is real in principle but has no paying customer yet.

THE HONEST CALL

Genuinely research-stage, and honest about it. The milestones are real and accelerating (city links, 100–420 km memory entanglement, a repeater program aiming at 2028), but the defining achievement — a repeater that beats direct transmission — has not been reached, and the economic case rests on applications that don't yet have buyers. This is a decade-plus infrastructure bet, not a near-term product.

SOURCES

- <https://www.science.org/doi/10.1126/sciadv.ad....> (QuTech metropolitan entanglement network, 2024)
- https://qt.eu/news/2025/2025-10-14_interconnecting-quantum-memories-for-the-quantum-internet
- <https://arxiv.org/pdf/2504.05660> (420 km memory entanglement)
- <https://www.innovationnewsnetwork.com/german-quantum-repeater-project-advances-future-quantum-internet/68370/> (TD.QR)

Quantum Key Distribution · A-qkd

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

QKD lets two parties grow a shared secret key whose security rests on physics — the no-cloning theorem (F-nocloning) and measurement disturbance — rather than computational hardness. BB84 (Bennett–Brassard 1984) encodes bits in non-orthogonal photon states; E91 (Ekert 1991) derives keys from entangled pairs and certifies security via Bell violation. An eavesdropper unavoidably raises the quantum bit error rate (QBER) and is detected. The security proof is information-theoretic in the ideal model — but every real deployment inherits two hard constraints: fiber loss is exponential ($\approx 0.16\text{--}0.2\text{ dB/km}$) so unamplified reach caps near a few hundred km, and QKD only distributes *keys*, so it still needs classically-authenticated channels and symmetric encryption to do anything useful.

MATURITY & REAL DEPLOYMENTS (2025–26)

Commercial today, in niches — a two-decade-old product line with government and finance as the buyers. **Measured performance envelope:** at inter-city distances ($\sim 200\text{ km}$) twin-field QKD reaches **111.74 kbit/s** secure key rate; the same protocol has been pushed to **1002 km** of ultralow-loss fiber (USTC/Pan, PRL 2023) but at only $\sim 0.0034\text{ bit/s}$ — a headline distance, not a usable rate. Practical metro QKD (Toshiba, ID Quantique) runs tens-of-kbit/s to Mbit/s over **10–100 km**. - **Toshiba** runs commercial metro QKD, launched a Paris service in 2025, and (with Quantum Corridor, Dec 2025) demonstrated cross-state QKD over live commercial fiber (Chicago→Indiana), including a **21.8 km** production segment carrying ordinary traffic. - **ID Quantique** has sold QKD boxes since the early 2000s; anchors Geneva/Korea (SK Telecom) networks. - **China** operates the $\sim 2000\text{ km}$ Beijing–Shanghai backbone (trusted-relay), the largest fielded network, plus the space segment (A-satqkd). - **EU EuroQCI:** a continent-wide quantum-secure network is being built — Madrid's MadQCI testbed (IDQ, Toshiba, AIT), Hungary's first multi-node net with Magyar Telekom, national arms across most member states.

KEY GRADED CLAIMS

- [T1] BB84/E91 offer information-theoretic key security in the ideal model — Bennett & Brassard 1984; Ekert PRL 1991 (established)
- [T2] Real devices admit side-channel attacks (detector blinding, Trojan-horse); implementation $\neq$ ideal proof — Lydersen et al., Nat. Photonics 2010 (established)
- [T2] Twin-field QKD over 1002 km fiber at ~ 0.0034 bit/s; 111.74 kbit/s at 202 km — Liu et al., PRL 130, 210801 (2023) (demonstrated)
- [T2] Beijing–Shanghai 2000 km QKD backbone operational via trusted relays — Chen et al., Nature 589 (2021) (demonstrated)
- [T4] Toshiba/Quantum Corridor cross-state QKD over live commercial metro fiber, 21.8 km segment — Toshiba PR, Dec 2025 (claimed)

CONFLICTS / OPEN QUESTIONS

- **C-qkd-vs-pqc:** NSA, UK NCSC, French ANSSI, and German BSI advise *against* QKD for national-security systems — citing cost, the trusted-relay hole, no protection against store-and-forward at relays, and the fact that authentication still needs classical (ideally PQC) crypto — and recommend PQC (A-pqc) instead. QKD vendors and China's program argue physics-based security is worth the cost for the highest-value links. Resolution: field success/failure of large deployments + whether PQC survives cryptanalysis over the next decade.
- Trusted relay nodes break end-to-end security — every backbone today depends on them until repeaters (A-qinternet) arrive. This is the single biggest gap between the marketing ("unhackable") and the deployed reality.

THE HONEST CALL

Commercial but structurally niche. QKD sells real boxes and secures real links today, but the addressable market is capped by three things it cannot escape without a quantum internet: distance (repeaterless $\sim$ a few hundred km), the trusted-relay compromise, and the four-agency government advice steering critical infrastructure toward PQC. It is a genuine product in a small, contested market — not the coming default for internet security.

SOURCES

- <https://thequantuminsider.com/2025/12/09/quantum-corridor-toshiba-demonstrate-cross-state-quantum-key-distribution-over-live-commercial-metro-fiber-network/>
- <https://link.aps.org/doi/10.1103/PhysRevLett.130.210801> (1002 km twin-field QKD)
- <https://www.toshiba.eu/quantum/>
- NSA "Quantum Key Distribution (QKD) and Quantum Cryptography" position page
- Chen et al., "An integrated space-to-ground quantum communication network," Nature 589 (2021)

Quantum memories as a component industry · A-qmemory-hw

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

A quantum memory stores a photonic qubit and releases it on demand with its quantum state intact. It is the missing hardware component that turns point-to-point QKD into a real **quantum internet**: repeaters need memories to hold entanglement while neighboring links succeed, defeating fiber loss over distance (see [A-qinternet](#), [F-teleport](#)). Historically memories were cryogenic lab curiosities. The shift being tracked here is memories becoming a **buyable component** — rack-mounted, room-temperature, sold to network builders — the same maturation single-photon detectors and sources went through. Platforms: warm atomic vapor (rubidium/cesium), rare-earth-doped crystals, cold-atom ensembles, and defect centers.

MATURITY & REAL DEPLOYMENTS (2025–26)

- **Qunnect** (Brooklyn, US) — the standout commercializer. Its rubidium-**vapor** approach runs at room temperature with no cryogenics or frequency conversion. Demonstrated entanglement between telecom-wavelength photons and a room-temperature memory at 90.2% fidelity, generating ~1,200 entangled photon-memory pairs/second. Sells the "GothamQ" testbed hardware; closed a \$10M Series A extension (Airbus Ventures, Cisco Investments, Quantonation) in June 2025; ~\$60M+ raised total.
- **Aliro Quantum** (US) — not a memory maker but the orchestration/SDN layer that manages multi-vendor quantum networks; its platform interoperates with Qunnect, Single Quantum, Cisco, IonQ, Thorlabs, Keysight, and others. Passed 50 supported network devices (Sept 2025).
- Academic memories (rare-earth crystals, cold atoms) still lead on storage time × efficiency but do not ship as products.

KEY GRADED CLAIMS

- [T3] Room-temperature telecom-band photon↔memory entanglement at 90.2% fidelity, ~1,200 pairs/s — Qunnect, 2025 (demonstrated)

- [T4] Qunnect memory hardware is commercially deployable outside the lab — company positioning (claimed)
- [T2] Repeater-based long-distance entanglement distribution requires quantum memories — network theory (established)

CONFLICTS / OPEN QUESTIONS

Room-temperature vapor memories trade storage time and efficiency for practicality; cryogenic rare-earth-doped crystals win on the storage-time × efficiency product (the metric that actually sets repeater performance) but not deployability. Which wins the network market is unsettled. The deeper problem: a repeater needs memory *and* the delivered entanglement rate to beat direct transmission, and that crossover has not been reached anywhere ([A-qinternet](#)).

THE HONEST CALL

A component industry being built ahead of its own market. Qunnect's room-temperature memories are a real, demonstrated product with fidelity/rate numbers you can cite, and Aliro's orchestration layer treats quantum networks as a systems-integration problem — both are early-commercial. But the whole industry is only worth building if a quantum internet has a paying use case beyond QKD (itself contested vs PQC, [C-qkd-vs-pqc](#)). This is picks-and-shovels for a gold rush that hasn't been confirmed — well-funded, technically real, demand-unproven.

SOURCES

qunnect.inc/press-release-2025-06-24; thequantuminsider.com 2025/03/19 (room-temp memory entanglement); thequantuminsider.com 2025/09/22 (Aliro 50 devices); postquantum.com/quantum-networks/quantum-memories.

Quantum-enhanced MRI / NMR & biomagnetism · A-qmri

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

Two distinct quantum leverages on magnetic-resonance and biomagnetic imaging, grouped because both aim at the same clinical/biological target from different physics: 1. **Hyperpolarization** — force nuclear spins far from thermal equilibrium (parahydrogen-induced polarization/SABRE, dynamic nuclear polarization, or brute-force) so the MR signal jumps by **10,000–100,000×**, turning slow *metabolic* MRI (watching, e.g., pyruvate→lactate in a tumor in real time) from impossible to feasible. 2. **Quantum-sensor detection** — read magnetic-resonance signals with **NV-diamond** (A-nvsensing) or atomic magnetometers (A-magneto) instead of pickup coils, enabling **nano-/micro-scale NMR**, **zero-to-low-field NMR** (no giant superconducting magnet), and biomagnetism (MEG/MCG) — see the OPM-MEG story in A-magneto .

MATURITY & REAL DEPLOYMENTS (2025–26)

Hyperpolarization is entering the clinic; quantum-sensor NMR is lab/preclinical. - **NVision Quantum Technologies** (Germany) built the **POLARIS** parahydrogen hyperpolarizer, boosting sugar/metabolite MRI signal **>10,000×** (up to ~100,000× claimed), and in 2025 ran feasibility work with **Memorial Sloan Kettering** toward standardized high-throughput metabolic imaging; its earlier collaboration seeded a Cambridge quantum-MRI hub. Hyperpolarized ¹³C-pyruvate MRI is already in human oncology/cardiology trials at multiple academic centers (the field predates NVision, but hyperpolarizer commercialization is the 2025 story). - **Diamond/NV NMR:** MCQST (Munich) combined optical microscopy with NV-detected NMR to push MRI toward the microscopic scale, converting MR signals to optical readout on a camera (2025). **Zero-/low-field J-spectroscopy with a diamond magnetometer** (arXiv:2512.05776) and NV nuclear-spin-locking for microscale high-field NMR (arXiv:2504.00887) demonstrate NMR in regimes conventional coils can't reach — no superconducting magnet, no large shield.

KEY GRADED CLAIMS

- [T2] Parahydrogen/DNP hyperpolarization raises MR signal 10^4 – $10^5\times$ enabling real-time metabolic MRI — established MR physics; hyperpolarized ^{13}C human trials (established/demonstrated)
- [T3] NVision POLARIS hyperpolarizer feasibility for standardized metabolic imaging with MSK — company + 2025 collaboration (demonstrated, preclinical/translational)
- [T3] NV-diamond micro-scale and zero-field NMR without superconducting magnets — MCQST 2025; arXiv:2512.05776, 2504.00887 (demonstrated, lab)
- [T2] OPM/atomic-magnetometer biomagnetism (MEG/MCG) — see **A-magneto** (demonstrated, clinical trials)

CONFLICTS / OPEN QUESTIONS

- **Hyperpolarization's clock problem:** hyperpolarized states decay in seconds to a couple of minutes (T_1 -limited), so the polarizer must sit beside the scanner and imaging must be fast — a workflow/logistics barrier, not a physics one, and the main thing standing between trials and routine clinical use.
- **NV/atomic NMR's sensitivity–volume tradeoff:** quantum sensors excel at *tiny* samples (nano/micro) but do not replace whole-body clinical MRI; their clinical path is benchtop diagnostics, point-of-care low-field NMR, and research microscopy, not the hospital 3 T magnet.

THE HONEST CALL

The most clinically plausible near-term quantum-biomedical story — but "quantum-enhanced MRI" bundles two very different maturities.

Hyperpolarized metabolic MRI is in real human trials and hyperpolarizers are being commercialized (NVision) — this is close. Quantum-sensor (NV/atomic) NMR is exciting lab physics enabling magnet-free micro-NMR, years from clinical routine. Biomagnetism via OPMs is already clinical (**A-magneto**). Grade each strand separately and resist the umbrella hype.

SOURCES

- <https://www.nvision-quantum.com/quantum-enhanced-mri> (POLARIS hyperpolarization)

- <https://analyticalscience.wiley.com/content/news-do/new-quantum-sensor-elevates-magnetic-resonance-imaging-microscopic-level> (MCQST NV-NMR, 2025)
- <https://arxiv.org/pdf/2512.05776> (zero-to-low-field J-spectroscopy with a diamond magnetometer)
- <https://arxiv.org/pdf/2504.00887> (NV nuclear-spin-locking, microscale high-field NMR)

Quantum Random Number Generation · A-qrng

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Randomness drawn from quantum measurement, where outcomes are unpredictable by physics (**F-measure**) rather than by algorithmic complexity. Three rungs of trust: 1. **Hardware QRNG** — trust the device; measure vacuum-field fluctuations, photon arrival times, or beam-splitter path. Fast and cheap. 2. **Device-independent (DI) randomness** — trust only physics; a Bell-inequality violation certifies genuine unpredictability even with untrusted hardware. Slow, needs loophole-free Bell setups. 3. **Certified randomness via random circuit sampling (RCS)** — a quantum computer produces bits a remote user can verify are fresh and random even while distrusting the server (Aaronson's protocol).

MATURITY & REAL DEPLOYMENTS (2025-26)

The oldest commercial quantum product line — and the fastest-moving on raw throughput. - **Hardware:** ID Quantique's **Quantis** chips have sold for two decades and shipped inside Samsung Galaxy "Quantum" phones; **Quside, Toshiba, Quantum Dice** sell QRNG hardware. Buyers are lotteries, casinos, HSMs, and key generation. - **Speed race:** a **photonic-integrated QRNG hit 18.8 Gbit/s** real-time output (USTC, 2021) — still a benchmark; **Quantum Dice's VERTEX** PCIe card does 2.66 Gbit/s post-processed; and a 2025 **Toshiba chip-scale QRNG** reached **~3 Gbit/s** while shrinking to a device that could embed in everyday hardware (Optica, 2025). Throughput and integration, not sensitivity, are the competitive axes. - **Certified randomness (the 2025 headline):** JPMorgan Chase, Quantinuum, Argonne, Oak Ridge, and UT Austin demonstrated certified randomness on Quantinuum's **56-qubit H2** (Nature 640, 2025) — the first claimed commercial-relevant application of a quantum computer beyond classical capability, using Aaronson's RCS protocol with supercomputer verification at **Frontier**. **Quantinuum's Quantum Origin** became the first software QRNG to earn **NIST ESV validation** (April 2025), and a certified-randomness product line followed.

KEY GRADED CLAIMS

- [T1] Measurement outcomes on superposed states are irreducibly random — quantum theory + loophole-free Bell tests 2015 (established)
- [T2] Chip/photonic hardware QRNG at multi-Gbit/s (18.8 Gbit/s record; ~3 Gbit/s integrated, 2025) — USTC 2021; Toshiba/Optica 2025 (demonstrated)
- [T2] Certified randomness expansion on a 56-qubit trapped-ion processor, verified at exascale — Liu et al., Nature 640 (2025) (demonstrated)
- [T2] Quantum Origin first software QRNG with NIST ESV validation — Quantinuum/NIST, April 2025 (demonstrated)
- [T4] Certified randomness as a commercial product with near-term customer demand — Quantinuum/JPMC (claimed)

CONFLICTS / OPEN QUESTIONS

- **Aaronson himself flags the certified-randomness demo's limits:** certification needed exascale classical verification (only a handful of machines on Earth can do it), and the guarantee is *computational* — it assumes the server can't out-compute Frontier at spoofing, so it weakens as classical simulation improves. It is a beautiful proof-of-principle, not yet a product anyone must buy.
- **Does anyone need it?** Most applications are fine with a \$50 hardware chip or a well-seeded CSPRNG. The market question for both certified and device-independent RNG is whether the extra trust guarantee is worth the price — outside high-assurance government/finance niches, the answer today is usually no.

THE HONEST CALL

Hardware QRNG is a mature, real, commodity product (two decades of sales, in your phone). **Certified randomness is a landmark 2025 demonstration** — arguably the first genuinely-useful thing a beyond-classical quantum computer has done — **but not yet a business**; its guarantee is computational and its verification cost is exascale. The gap between "shipping chip" and "verifiable quantum-advantage service" is the whole story of this node.

SOURCES

- <https://www.nature.com/articles/s41586-025-08737-1> (certified randomness, Nature 640, 2025)
- https://www.optica.org/about/newsroom/news_releases/2025/quantum_random_number_generator_combines_small_size_and_high_speed/ (Toshiba ~3 Gbit/s chip)
- <https://scottaaronson.blog/?p=8746> (Aaronson on the demo's limits)
- <https://www.jpmorgan.com/technology/news/certified-randomness>

Rydberg RF electrometry / quantum antennas · A-rydberg

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3, new node)

WHAT IT IS

A **Rydberg atom** has an electron excited to a very high principal quantum number ($n \sim 50\text{--}100$), giving it a huge electric dipole moment — which makes it exquisitely sensitive to RF electric fields. Shine two lasers through a room-temperature alkali (rubidium/cesium) vapor cell to create **electromagnetically induced transparency (EIT)**; an incident RF field splits the EIT line (Autler–Townes splitting), and the split is read *optically*. The result is an **RF receiver whose sensing element is atoms, not metal** — self-calibrated (referenced to atomic constants and Planck's constant), physically tiny, and broadband from MHz to sub-THz in one cell. This is quantum sensing pointed at the electromagnetic spectrum instead of at magnetic or gravitational fields, and it is distinct enough from **A-magneto / A-nvsensing** to warrant its own node.

MATURITY & REAL DEPLOYMENTS (2025–26)

Early commercial for niche RF sensing; research for general communications. - **Rydberg Technologies** (US, Michigan spin-out) is the furthest along — SI-traceable Rydberg field sensors and a demonstrated over-the-air Rydberg RF communication receiver for defense; DARPA/Army-funded. - **Infleqtion** markets a **quantum spectrum sensing** ("SqyWire") product line using Rydberg atomic sensing to detect, locate, and classify RF signals across a broad band, positioned to replace conventional antenna+receiver front-ends for spectrum awareness. - **ESA** commissioned Rydberg sensor commercialization work beginning 2025 (satellite RF/comms). - **2025 research surge:** an all-optical Rydberg radio antenna (Oct 2025); a metamaterial GRIN lens boosting receiver gain/bandwidth (arXiv:2512.04298); a scalable Rydberg vapor-cell array with a "Stark comb" for arbitrary instantaneous bandwidth (arXiv:2509.26026); quantum-limited microwave electrometry in a Rydberg **atom array** (arXiv:2512.05413); and demonstrations of angle-of-arrival, digital beamforming, and moving-target detection with atomic receivers.

KEY GRADED CLAIMS

- [T2] Rydberg EIT gives SI-traceable, self-calibrated RF electric-field measurement — established atomic-physics literature (established)
- [T3] Over-the-air Rydberg atomic communication receiver demonstrated — Rydberg Technologies / arXiv (demonstrated)
- [T3] Rydberg receiver angle-of-arrival, beamforming, moving-target detection — 2025 arXiv results (demonstrated, lab/field)
- [T4] Inflection quantum spectrum sensing as a deployable product replacing antenna front-ends — company positioning (claimed)

CONFLICTS / OPEN QUESTIONS

- **Sensitivity vs the best classical receiver:** standard Rydberg receivers have historically *not* beaten a good cryogenic low-noise-amplifier classical receiver on raw sensitivity; the advantage is self-calibration, bandwidth coverage in one device, small size, and immunity to some jamming/damage (no metal to fry). Whether quantum-enhanced (atom-array, spin-squeezed) Rydberg sensing decisively beats classical on sensitivity is the open research question.
- **Instantaneous bandwidth** and dynamic range have been real limits; the 2025 array/Stark-comb work is aimed squarely at them but is not yet productized.

THE HONEST CALL

Genuinely novel, early-commercial for specialized RF/spectrum sensing — not a general antenna replacement yet. Rydberg receivers ship into defense spectrum-awareness and metrology (self-calibrated field standards), which is a real if narrow market. The bigger claims — replacing communications antennas, beating classical receivers on sensitivity — are research with fast 2025 momentum. Grade as "real product in a niche, contested advantage in the mainstream RF use case."

SOURCES

- <https://phys.org/news/2025-10-quantum-radio-antenna-rydberg-states.html>
- <https://inflection.com/quantum-spectrum/> (Inflection quantum spectrum sensing)

- <https://arxiv.org/pdf/2507.13111> (Perspective: Practical Atom-Based Quantum Sensors)
- <https://arxiv.org/pdf/2512.05413> (quantum-limited Rydberg electrometry in an atom array)

Satellite QKD · A-satqkd

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Distributing quantum keys (and entanglement) via satellite, where photon loss through vacuum beats fiber's exponential attenuation over continental distances. A satellite either acts as a **trusted relay** between two ground stations (it holds both keys and XORs them — secure only if you trust the spacecraft) or **beams entangled pairs** to two stations at once (end-to-end secure, no trust in the satellite, but far lower rate). This is the near-term route to intercontinental quantum-secure links while fiber repeaters ([A-qinternet](#)) mature.

MATURITY & REAL DEPLOYMENTS (2025–26)

Demonstrated at scale by China; entering an early commercial phase globally. - **Micius** (launched 2016, USTC/CAS) is the flight-heritage benchmark: entanglement distribution over 1,200 km (Yin et al., Science 2017), decoy-state QKD downlink, and the 2017 Beijing–Vienna intercontinental video call secured by satellite-relayed keys. It is a ~600 kg spacecraft needing large, expensive ground stations. - **Jinan-1** (2022, operational) is the practicality milestone. It shrank the payload to **~22.7 kg** and the satellite to **~95.9 kg**, cutting cost roughly **45×** while raising key rates **2–3 orders of magnitude** over Micius. Using an 850 nm source at 625 MHz rep rate (~250 million photons/s transmitted), it generated **up to 1.07 million secure bits in a single pass** and ran **real-time** QKD with portable (~100 kg) ground stations, enabling one-time-pad-encrypted image transfer between China and South Africa over 12,900 km (Nature, March 2025). Real-time on-board key handling — not post-pass processing — is the operational jump. - **Europe:** EAGLE-1 (ESA/SES) is Europe's first end-to-end satellite QKD system, launch slated 2026, feeding EuroQCI's space segment. **Toshiba** announced a satellite QKD transmitter–receiver interoperable with terrestrial fiber QKD (Jan 2026). **SealSQ** began launching a six-satellite QKD/PQC constellation (Jan 2025).

KEY GRADED CLAIMS

- [T2] Satellite entanglement distribution over 1,200 km — Yin et al., Science 356 (2017) (established)

- [T2] Real-time microsatellite QKD, up to 1.07 Mbit secure key/pass, 12,900 km China–South Africa link, ~45× cheaper spacecraft — Li et al., Nature (2025) (demonstrated)
- [T4] EAGLE-1 launch and end-to-end European satellite QKD service by ~2026–27 — ESA/SES, arXiv:2505.20838 (roadmap)
- [T4/T6] Chinese multi-satellite quantum constellation / space quantum internet by ~2030 — CAS/USTC statements (roadmap)

CONFLICTS / OPEN QUESTIONS

- **Trust model:** downlink QKD via a single satellite is trusted-node — the spacecraft (and whoever controls it) knows the key — unless it distributes *entanglement* end-to-end, which costs orders of magnitude in rate. Most fielded/announced systems are the trusted-relay kind.
- **Operations:** daytime operation (sky background swamps single photons), weather/cloud outage, and pointing over fast LEO passes limit availability. Whether the key-rate-per-dollar beats just deploying PQC (A-pqc) on the same links is the live commercial question.

THE HONEST CALL

Demonstrated and near-commercial, led decisively by China. Jinan-1 turned satellite QKD from a hero experiment into something that looks like a deployable service, and Europe is a launch or two behind. But it inherits QKD's core limitation — it distributes keys, still needs classical authentication, and the cheap version trusts the satellite — so it is a strategic-sovereignty capability for states and critical infrastructure, not a mass-market technology.

SOURCES

- <https://www.nature.com/articles/s41586-025-...> (Jinan-1 real-time microsatellite QKD)
- <https://postquantum.com/industry-news/microsatellite-qkd-record/>
- https://en.wikipedia.org/wiki/Quantum_Experiments_at_Space_Scale
- <https://arxiv.org/pdf/2505.20838> (EAGLE-1 / European satellite QKD)
- <https://spaceinsider.tech/2025/12/12/top-10-qkd-players-and-the-road-to-commercial-qkd-in-space-based-secure-communications/>

Quantum Sensing & Metrology · A-sensing

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Using quantum systems — atomic transitions, spin states, entanglement, squeezing — as measurement instruments. The same fragility that plagues qubits (F-decoher) becomes the product: a state that decoheres at the slightest field perturbation is an exquisite detector of that field. This card is the **overview node** for the sensing cluster: clocks (A-clocks), magnetometry (A-magneto), gravimetry/inertial (A-gravimetry), imaging (A-imaging), NV-diamond (A-nvsensing), Rydberg RF (A-rydberg), and the assured-PNT integration (A-pnt). The unifying physics: interferometric phase readout at or below the standard quantum limit, where signal scales with coherence time and entanglement can push sensitivity toward the Heisenberg limit ($\propto 1/N$ rather than $1/\sqrt{N}$).

MATURITY & REAL DEPLOYMENTS (2025–26)

The nearest-term commercial quantum technology — revenue exists today, unlike compute. Chip-scale atomic clocks have shipped for over a decade; LIGO has run squeezed light in production since 2019 (A-squeezed); OPM magnetometers sell into neuroimaging (A-magneto). The 2024–26 wave is **quantum navigation** for the GPS-denied problem: - **Q-CTRL's Ironstone Opal** (magnetic/gravity map-matching + AI) ran air, land, and maritime defense trials, claiming up to **~100–111× better GPS-denied accuracy** than the best conventional INS and **~4 m positioning over 700 km flights**; a maritime trial logged **144+ hours** unattended at sea (2025–26). It made TIME's Best Inventions 2025. - **SandboxAQ's AQNav** does AI + magnetic-anomaly navigation; both companies hold US DoD contracts (DARPA RoQS — \$24.4M to Q-CTRL; Lockheed Martin/Q-CTRL prototype quantum-INS award, March 2025). - **Medical:** OPM-MEG helmets in clinical trials, SandboxAQ/Mayo CardiaQ cardiac sensing (A-magneto).

Market: analysts put quantum sensors at roughly **\$0.4–0.9B in 2025** growing **~15–23% CAGR** to ~\$1.5B by the mid-2030s (Fortune Business Insights and peers). Small but real and mostly defense-led — a 2036 forecast has clocks still dominant (~76% of the PNT-sensing stack) with quantum IMUs reaching ~18%.

KEY GRADED CLAIMS

- [T1] Quantum-limited metrology (squeezing, entanglement-enhanced sensitivity) beats classical shot-noise limits — LIGO squeezed-light operation, PRL 2019+ (established)
- [T4] Ironstone Opal delivers ~4 m accuracy over 700 km flights, up to 111× better than best conventional alternative — Q-CTRL announcements 2025 (claimed, awaiting independent verification)
- [T5] Quantum sensors market ~\$435M (2025) → ~\$1.5B (2034), ~15% CAGR — Fortune Business Insights (forecast)

CONFLICTS / OPEN QUESTIONS

- **Vendor navigation numbers are self-reported field trials.** No peer-reviewed head-to-head against classified military-grade ring-laser/fiber-optic INS is public — the "111×" is against a specified baseline, not the state of the art in a black program.
- **The label problem:** where is the line between "quantum sensor" and mature atomic physics (MRI, cesium clocks, SQUIDs all predate the "quantum tech" branding)? Marketing stretches "quantum" to anything using atoms — grade accordingly, and see the per-modality cards for the honest split.

THE HONEST CALL

The revenue-bearing corner of quantum technology, and the one most likely to matter this decade — but the fielded wins are concentrated in defense PNT and neuroimaging, and the flashiest navigation claims still lack independent benchmarks. Treat sensing as "real and shipping in niches," with the per-modality cards carrying the exact numbers and the honest maturity call.

SOURCES

- <https://q-ctrl.com/blog/2025-year-in-review-realizing-true-commercial-quantum-advantage-in-the-international-year-of-quantum>
- <https://spaceinsider.tech/2026/04/10/quantum-sensing-for-pnt-nears-deployment/>
- <https://thequantuminsider.com/2026/04/10/overview-15-plus-key-quantum-companies-2026/>
- <https://www.fortunebusinessinsights.com/quantum-sensors-market-110331>

Squeezed light as a cross-cutting resource · A-squeezed

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Squeezed light is a quantum state of an optical field where the uncertainty in one quadrature is pushed below the vacuum (shot-noise) limit, at the cost of more uncertainty in the conjugate one — Heisenberg's bound rearranged, not beaten (see **F-uncertainty**). It is a genuine cross-cutting quantum resource: it sharpens metrology below the standard quantum limit, and it is the *substrate* for continuous-variable photonic quantum computing. Squeezing strength is quoted in decibels below shot noise; more dB means a better resource, bounded in practice by optical loss.

MATURITY & REAL DEPLOYMENTS (2025–26)

- **LIGO / Virgo — the flagship production use.** Squeezed vacuum is injected into the interferometer dark port to cut quantum shot noise every observing run. In O3, Hanford measured ~2.0 dB and Livingston ~2.7 dB of shot-noise reduction, raising the detection rate ~40–50%. O4 uses **frequency-dependent squeezing** (a filter cavity) to squeeze the right quadrature across the band — the first at-scale deployment of a technique that reduces both shot noise and radiation-pressure noise. This is the clearest case of a quantum resource doing paid work daily.
- **Xanadu — squeezing as compute.** Its photonic machines encode GKP bosonic qubits in squeezed states via optical parametric oscillators; the Aurora system (2025) networks 35 photonic chips over 13 km of fiber, and Xanadu demonstrated on-chip GKP-state generation on 300 mm silicon-nitride wafers (June 2025).
- **Integrated squeezing** is advancing fast: 18 dB squeezing / 20 dB anti-squeezing at 1570 nm demonstrated in a thin-film lithium-niobate waveguide — the highest for any integrated photonic platform.

KEY GRADED CLAIMS

- [T2] LIGO uses squeezed light in production to boost sensitivity ~40–50% via shot-noise reduction — Phys. Rev. X 13, 041021 (established/demonstrated)

- [T3] 18 dB on-chip squeezing in TFLN waveguide — arXiv 2025 (demonstrated)
- [T3] On-chip GKP states from squeezed light on 300 mm wafers — Xanadu, June 2025 (demonstrated)

CONFLICTS / OPEN QUESTIONS

Squeezing is loss-limited: every lost photon degrades it toward vacuum, so end-to-end optical efficiency caps usable dB — the same wall that gates continuous-variable photonic QC. The gap between the ~18–20 dB generated on-chip and the ~2–3 dB LIGO actually uses in production is *all loss* (mirrors, mode-matching, the km-scale interferometer). Whether squeezing-based (CV/GKP) architectures reach fault tolerance before discrete-photon ones is open (H-photonic , 0-roomtemp).

THE HONEST CALL

The clearest case in the whole manual of a quantum resource doing paid work every day — LIGO/Virgo inject squeezed vacuum on every observing run and it demonstrably raises the detection rate ~40–50%, which is why more gravitational-wave events get caught. That is established, deployed, and reproduced. The compute use (CV/GKP photonic QC, Xanadu) is real hardware progress but still research: on-chip GKP generation and a 35-chip networked machine are demonstrations, not fault tolerance. So: sensing use = production; compute use = frontier.

SOURCES

Phys. Rev. X 13, 041021 (frequency-dependent squeezing at LIGO); arXiv:2202.00847 (Advanced LIGO O4 review); arXiv:2603.02744 (12–18 dB waveguide squeezing); thequantuminsider.com 2025/06/05 (Xanadu Aurora / on-chip GKP).

Quantum/optical time-transfer networks · A-timedist

Layer: L3 Adjacent tech · **Chapter:** §04 · **Status:** deepened (cycle 3)

WHAT IT IS

Distinct from the clocks themselves (**A-clocks**): this node is the **network problem** of comparing and distributing ultra-stable time and frequency between distant clocks without degrading their accuracy. Optical clocks are now so good (~19 decimal places) that ordinary GPS/satellite time links are the limiting error, not the clocks. So a parallel infrastructure is being built — stabilized optical fiber links, free-space laser links, and frequency-comb transfer — that carries an optical clock's stability across cities and countries. This is the plumbing without which the **redefinition of the SI second** cannot happen: BIPM's roadmap requires multiple optical standards in different countries to agree, which means comparing them, which means time transfer.

MATURITY & REAL DEPLOYMENTS (2025-26)

- **International optical-clock comparison across six countries** reported June 2025 — a coordinated cross-border measurement, the largest step yet toward a global optical timescale.
- **Deployed multicore-fiber transfer:** ultrastable optical signals sent through installed multicore fiber alongside live telecom traffic, reaching fractional instability $\sim 3 \times 10^{-19}$ over nearly 3 hours (Optica, 2025) — showing time transfer can share commercial fiber.
- **QTF-Backbone** (Germany): proposal for a nationwide optical-fiber backbone for both quantum tech and time/frequency metrology (arXiv 2506.03998).
- **BIPM/CCTF** updated the second-redefinition roadmap in early 2025: optical standards must beat cesium by 100×, ≥ 5 independent systems running continuously, >1 year of traceable data — every threshold implies robust time-transfer.
- Underpins GPS-independent PNT, financial-transaction timestamping, telecom sync, and VLBI/geodesy.

KEY GRADED CLAIMS

- [T2] Optical-frequency transfer over deployed multicore fiber at $\sim 3 \times 10^{-19}$ instability alongside telecom traffic — Optica, 2025 (demonstrated)
- [T2] Six-country optical-clock comparison completed — Optica newsroom, June 2025 (demonstrated)
- [T4] SI-second redefinition on track for late-2020s/ ~ 2030 — BIPM/CCTF roadmap (roadmap)

CONFLICTS / OPEN QUESTIONS

Most record-setting transfer uses purpose-run fiber; scaling to a routine, always-on continental network sharing commercial infrastructure is unproven. **The naming trap:** "quantum" time transfer (entanglement-based clock synchronization — see the entanglement-based clock-network node `A-entcLock`) is proposed and theoretically elegant, but a 2026 critical assessment (arXiv 2604.10243) questions whether it beats classical optical/comb-stabilized methods in practice, since the record-holding links are all classically stabilized. So the near-term network is optical/frequency-comb-stabilized, not entanglement-based — the word "quantum" here mostly refers to the *clocks*, not the *transfer*.

THE HONEST CALL

Real, demonstrated metrology infrastructure — and mostly classical despite the "quantum" framing. Six-country optical-clock comparison and $\sim 3 \times 10^{-19}$ transfer over shared commercial multicore fiber are done and published; this plumbing is a hard requirement for the SI-second redefinition, so it will get built regardless of market pull. What is *not* established is that entanglement adds anything over comb-stabilized optical links — so treat the transfer layer as advanced classical photonics serving quantum clocks.

SOURCES

optica.org newsroom 2025 (six-country clock network; multicore-fiber transfer); arXiv:2506.03998 (QTF-Backbone); arXiv:2503.13278 (SI-second from multiple optical transitions); arXiv:2604.10243 (quantum time-sync assessment); bipm.org/faq-redefinition-second.

§05 • Industries

Where quantum actually lands — 21 industries from finance and pharma to air-traffic management, with proven results separated from promise.

Quantum in Aerospace & Defense · I-aerospace

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Defense is where quantum tech is *closest to fielded* — and the payoff comes from **sensing and navigation**, not computing: GPS-free inertial navigation, quantum magnetometry/gravimetry for detection, secure comms, plus longer-horizon codebreaking (Shor, **S-shor**) and simulation. The strategic driver is GPS-denied/contested environments and the "harvest now, decrypt later" threat to military secrets. (Pure SIGINT/cryptanalysis is split into **I-intelligence**.)

REAL ACTIVITY (NAMED, DATED)

- **DARPA + Q-CTRL** (Aug 2025) — two awards under the **Robust Quantum Sensors (RoQS)** program (~\$24.4M) for navigation sensors on moving platforms without shielding; **Lockheed Martin** subcontractor. Q-CTRL's April 2025 flight/maritime trials (incl. Royal Australian Navy MV *Sycamore*, ~180 W package) validated quantum inertial nav against military-adequate thresholds.
- **Boeing + US Space Force** — X-37B spaceplane (launched Aug 2025) carried "the highest-performing quantum inertial sensor to fly in space." Boeing also completed ground testing of its **Q4S** quantum-networking satellite payload (entanglement swapping).
- **PsiQuantum + Lockheed Martin** (Nov 2025) — MoU on quantum computing for national security/aerospace.
- **DARPA Quantum Benchmarking Initiative (QBI)** — vetting whether any vendor can reach industrial-scale FTQC by ~2033; the defense channel's reality check (see **E-defense**).
- **Airbus** — magnetic-navigation research; broad quantum program.

KEY GRADED CLAIMS

- [T3] Airborne/maritime quantum inertial navigation field trials — Q-CTRL, April 2025 (demonstrated, military-adequate performance claimed)
- [T3/T4] Highest-performing quantum inertial sensor flown in space — Boeing/Space Force X-37B, 2025 (claimed)

- [T4] Quantum computing for defense applications — PsiQuantum/ Lockheed MoU, Nov 2025 (roadmap)
- [T5] Defense quantum-market forecasts — analyst; grade hard

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** quantum inertial navigation has cleared airborne, maritime, and space field trials — the nearest-term real quantum deployment in the manual.
- **Promise:** quantum magnetometry for submarine/mine detection, space quantum networking — advancing, not fielded.
- **Hype:** "quantum computer breaks enemy crypto now," near-term battlefield quantum computing.

HONEST ASSESSMENT

Aerospace/defense has the **nearest-term real quantum payoff**, and it comes from **A-sensing / A-gravimetry / A-magneto** rather than computing. Quantum inertial navigation has cleared field and space trials and is on a credible path to deployment this decade. Quantum *computing* for defense (codebreaking, sim) is the same fault-tolerance-gated future as everyone else, plus the classified overhang that adversary capability is unknown. Split the story: sensing/nav is arriving; compute is a bet. This is the strongest "quantum is real now" case in the manual, and the thing that is real is a sensor.

SOURCES

- DARPA/Q-CTRL RoQS: <https://thequantuminsider.com/2025/08/27/darpa-selects-q-ctrl-to-develop-next-generation-quantum-sensors-for-navigation-on-advanced-defense-platforms/>
- Q-CTRL/Lockheed (GPS World): <https://www.gpsworld.com/q-ctrl-lockheed-to-develop-quantum-navigation-for-darpa/>
- Aerospace America "A year of acceleration and the rise of quantum sensing"
- Lockheed Martin quantum technology page; War on the Rocks "Genesis Mission"

Quantum in Agriculture & Food Science · I-agri

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Split out of **I-climate**, this node covers two threads. (1) **Food/fertilizer chemistry** — the FeMoco/nitrogenase and Haber-Bosch-replacement story that is really quantum chemistry (**I-chem** , **S-qsim**), plus catalyst and soil-chemistry modeling. (2) **Precision-agriculture optimization** — variational/QUBO methods for crop-yield prediction, irrigation and fertilizer scheduling, plus **quantum sensing** (**A-sensing**) for soil and canopy measurement, which is a nearer-term, more credible thread than compute. (Food & beverage *manufacturing/supply chain* rides **I-manufacturing** / **I-logistics** .)

REAL ACTIVITY (NAMED, DATED)

- **Alice & Bob** (Oct 2025) — FeMoco resource estimate showing ~99,000 physical cat qubits (27x cut vs 2021 Google), explicitly framed for fertilizer/agriculture. An estimate, not a run. (Shared with **I-climate** / **I-chem** .)
- **qHPC-GREEN** (CASUS, Dr. Werner Dobrautz, launched Jan 2025 → 2029) — quantum-HPC hybrid research into energy-efficient nitrogen-fixation simulation. Funded research program, not a deployed tool.
- **Precision-ag method papers** — VQC crop-yield/resource algorithms claiming ~30% better yield prediction and ~25% less water/fertilizer, incl. a "5-hectare wheat" pilot writeup (EPJ Web of Conferences 2025). Small academic trials; treat the percentages skeptically.
- **Quantum-sensing review** — *Potential applications of quantum sensors in agriculture* (Computers and Electronics in Agriculture, 2025) surveys magnetometry/gravimetry/NV soil-and-nutrient sensing as the credible near-term hook.

KEY GRADED CLAIMS

- [T3] FeMoco needs ~99k physical qubits with cat qubits — Alice & Bob (demonstrated *estimate*, not execution)

- [T3] VQC crop-yield/resource algorithms show ~30%/25% gains on small trials — EPJ 2025 (contested; toy-scale, headline numbers unverified, classical ML matches)
- [T5/T6] Quantum precision agriculture at farm scale — analyst/vendor scenarios (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** quantum sensors for soil/nutrient/water measurement are real lab-grade instruments (need no fault-tolerant computer).
- **Promise:** nitrogen-fixation chemistry ($\text{N}_2 + 3 \text{H}_2 \longrightarrow 2 \text{NH}_3$) — the textbook killer app, ~99k physical qubits away.
- **Hype:** "quantum boosts crop yields 30%," farm-scale quantum precision agriculture.

HONEST ASSESSMENT

The fertilizer-chemistry payoff is real as a target and distant as a deliverable — the same ~99k-qubit / 2030s+ verdict as `I-climate`, gated entirely on fault tolerance. The precision-ag optimization papers are the hype-heavy corner: small, lightly-refereed trials with headline percentages that classical ML matches. The one thread with a plausible short horizon is quantum sensing for soil/nutrient/water measurement. Present-day deployed agricultural value from quantum computing is zero.

SOURCES

- <https://alice-bob.com/newsroom/alice-bob-quantum-computing-applications-health-agriculture/>
- <https://www.hpcwire.com/off-the-wire/quantum-hpc-hybrid-research-targets-energy-efficient-fertilizer-production/> (qHPC-GREEN)
- https://www.epj-conferences.org/articles/epjconf/abs/2025/10/epjconf_iemphys2025_01004/epjconf_iemphys2025_01004.html (VQC precision-ag)
- <https://dl.acm.org/doi/10.1016/j.compag.2025.110420> (quantum sensors in agriculture review)

Quantum in AI & Machine Learning · I-ai

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Quantum machine learning (QML, `S-qmL`) promises quantum kernels, quantum neural networks, faster sampling, and quantum-enhanced optimization inside classical ML pipelines — plus, aspirationally, help training or running large models. It's pitched as the fusion of the two hottest technologies. It is also the **most hype-inflated corner of the entire field** and gets graded hardest (per the map's honest edge).

REAL ACTIVITY (NAMED, DATED)

- Enterprise pilots exist in finance, pharma, logistics, and aerospace, and enterprises test QML as a specialized hybrid-pipeline component rather than replacing classical AI stacks.
- **Quantum-kernel methods** for credit scoring, classification, and generative sampling appear in academic + vendor studies (e.g. credit-scoring arXiv 2308.03575), typically on tiny datasets with no advantage shown.
- Vendors (IBM, IonQ, Quantinuum, Xanadu/PennyLane, Multiverse) ship QML SDKs and demos; most "results" prove *feasibility*, not advantage.
- **The dequantization record** — Tang et al. (2019, quantum recommendation systems) and successors produced classical algorithms inspired by the quantum method that matched the claimed speedups, retroactively erasing several QML "advantages."
- **NVIDIA CUDA-Q / cuQuantum** — the dominant tooling is *classical GPU simulation* of quantum circuits, a tell about where the compute actually runs.

KEY GRADED CLAIMS

- [T3/T4] Quantum-kernel classifiers competitive on small datasets — assorted arXiv + vendor studies (no advantage at scale)
- [T2] Certified randomness (JPMC/Quantinuum, *Nature* 2025) has downstream ML/privacy uses — established primitive, not "QML" per se
- [T6] Quantum acceleration of large-model training/inference — aspirational (speculative)

- [T5] "Quantum AI" market-size projections — forecast; among the most inflated in the field

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** QML circuits run and classify toy datasets — feasibility, not advantage. Certified randomness (adjacent) is real.
- **Promise:** quantum-native data (simulating quantum systems for ML) may be the one durable niche.
- **Hype:** "quantum AI accelerates LLMs," quantum-kernel "advantage" on real data — repeatedly dequantized.

HONEST ASSESSMENT

QML is where skepticism should be maximal. Multiple provisional quantum-ML "advantages" have been dequantized — classical algorithms inspired by the quantum method matched them — so the speedup evaporated. Today's hardware (tens–hundreds of noisy qubits, shallow circuits) can't run QML at data scales where it would matter, and loading classical data into quantum states (the I/O bottleneck, `S-gram`) often erases any theoretical gain. Business leaders who bet on "quantum AI" in 2025 largely walked it back. The science is worth following; the near-term product is not. Realistic advantage: unproven, possibly never for mainstream ML, plausibly niche for quantum-native data.

SOURCES

- postquantum.com "Quantum Machine Learning in 2026: State of the Field": <https://postquantum.com/quantum-ai/quantum-machine-learning-reality/>
- techrevolt.news "What 500 Business Leaders Got Wrong About Quantum AI in 2025"
- Quantum ML for credit scoring: <https://arxiv.org/pdf/2308.03575>
- Tang, "A quantum-inspired classical algorithm for recommendation systems" (dequantization)

Quantum in Air-Traffic Management · I-atm

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Air-traffic management (ATM) is split from logistics (I-logistics) and aerospace (I-aerospace) because it is a distinct optimization problem: traffic-flow management, multi-aircraft trajectory deconfliction, arrival-slot/gate scheduling, and airspace-density maximization under weather and congestion constraints. These map cleanly to QUBO/QAOA formulations, and the sector is a favorite for quantum-optimization demos because the combinatorics are brutal and the fuel/delay payoff is quantifiable.

REAL ACTIVITY (NAMED, DATED)

- **Airbus + QC Ware** — Airbus-run quantum optimization challenges report large speedups in airspace/traffic analysis (widely cited "~400% faster analysis vs classical-only" and "up to ~70% higher airspace utilization"). Vendor/challenge results — treat percentages as marketing until refereed.
- **Airbus + BMW Quantum Mobility Quest** (2024) — a broader transport-optimization challenge including traffic-flow problems. Competition, not deployment.
- **Method literature** — *Mini-scale traffic flow optimization: an iterative QUBOs approach* (Nature Scientific Reports s41598-025-04568-2, 2025); *Quantum Computing Applications for Flight Trajectory Optimization* (arXiv 2304.14445); *Exploring Airline Gate-Scheduling Optimization Using Quantum Computers* (arXiv 2111.09472). Small/hybrid-solver studies.
- **SESAR/Eurocontrol & NASA** — exploratory quantum-optimization studies for trajectory/flow management; framing and feasibility, no operational use.

KEY GRADED CLAIMS

- [T3] QUBO traffic-flow / trajectory / gate-scheduling solved on quantum + hybrid solvers at small scale — Nature Sci Rep 2025, arXiv 2304.14445 / 2111.09472 (demonstrated on mini instances)
- [T4] Airbus/QC Ware "~400% faster" / "~70% higher airspace utilization" — challenge/vendor claims (not independently reproduced)

- [T4] Quantum-inspired conflict-resolution speedups on existing HPC/GPU — vendor framing (quantum-inspired = classical; grade accordingly)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** small QUBO deconfliction/scheduling demos run on quantum + hybrid solvers — classically matchable.
- **Promise:** operational traffic-flow optimization — gated on fault tolerance *and* safety certification (a high, slow bar for aviation).
- **Hype:** the Airbus challenge percentages read as operational gains; many are quantum-inspired (classical) or constrained-problem results.

HONEST ASSESSMENT

No air-navigation service provider (FAA, Eurocontrol/SESAR, NATS) runs a quantum computer in operational ATM. The credible near-term wins are frequently quantum-inspired — classical algorithms borrowing quantum structure — honest optimization progress that is not quantum hardware advantage, and the two get blurred in the press. The Airbus challenge percentages are competition results on constrained problems, not certified operational gains. The QUBO demos are real but small and classically matchable. Realistic operational value: gated on fault tolerance and safety certification, late 2020s at the earliest; today it is exploratory.

SOURCES

- <https://www.nature.com/articles/s41598-025-04568-2> (mini-scale traffic-flow QUBO)
- <https://arxiv.org/html/2304.14445> (flight trajectory optimization)
- <https://arxiv.org/pdf/2111.09472> (airline gate scheduling)
- <https://www.airbus.com/en/innovation/digital-transformation/quantum-technologies/airbus-and-bmw-quantum-computing-challenge>

Quantum in Automotive · I-auto

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Automotive spans several quantum threads: battery/electrolyte and fuel-cell-catalyst simulation (materials), aerodynamics and crash/metal-forming simulation, production-line scheduling and paint-shop optimization, and autonomous-driving ML. German OEMs are the most active enterprise cohort outside finance and pharma, and automotive holds one of the field's most defensible "quantum in production" claims.

REAL ACTIVITY (NAMED, DATED)

- **Ford Otosan + D-Wave** (2024) — **deployed a hybrid-quantum app in production** on the Ford Transit line, cutting vehicle sequencing/scheduling time ~50% (roughly 30 min → under 5 min per ~1,000-vehicle run). One of the few real production deployments in any industry (annealing-based).
- **BMW** — quantum-enhanced solvers for sheet-metal forming and crash-safety (2024); co-ran the 2024 Airbus/BMW Quantum Challenge; earlier fuel-cell catalyst work.
- **BMW + Airbus + Quantinuum** (2023) — hybrid quantum-classical **fuel-cell catalyst** simulation (oxygen reduction on platinum).
- **Hyundai + IonQ** — battery chemistry (electrolytes, cathode catalysts) for next-gen batteries.
- **Ford + Quantinuum** — lithium-ion battery reaction modeling. **Volkswagen + IQM** — hybrid battery-simulation study (2024–25).
- **Mercedes-Benz + PsiQuantum / IBM** — battery-materials feasibility work.
- **Volkswagen + D-Wave** (earlier program) — paint-shop color-switching optimization, a QUBO scheduling problem framed to cut solvent/flush waste; the archetypal automotive annealing demo classical solvers also handle.
- **Toyota / Toyota Tsusho** — participation in Japanese quantum consortia (Q-STAR) on materials and mobility optimization; exploratory.

KEY GRADED CLAIMS

- [T3/T4] Ford Otosan production sequencing, ~50% scheduling-time cut — D-Wave/Ford Otosan, 2024 (demonstrated, in production; annealing not gate-model)
- [T4] Fuel-cell catalyst simulation — BMW/Airbus/Quantinuum (exploratory)
- [T4] Battery-chemistry pilots — Hyundai/IonQ, Ford/Quantinuum, VW/IQM (claimed)
- [T5] Automotive slice of quantum economic-value forecasts — analyst; grade hard

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** Ford Otosan runs an annealing-based scheduling app in a live plant — a legitimate deployed workflow.
- **Promise:** battery/electrolyte and fuel-cell catalyst design — small-molecule exploratory studies short of design-grade accuracy.
- **Hype:** "quantum computer beats classical on scheduling" (contested), quantum-accelerated autonomous driving.

HONEST ASSESSMENT

The Ford Otosan case is one of the most defensible "quantum in production" claims anywhere, and it is *quantum annealing* solving a scheduling problem where D-Wave competes head-to-head with classical solvers (advantage contested — see [H-anneal / 0-advantage](#)). Battery and catalyst simulations are small-molecule exploratory studies far from design-grade accuracy. Autonomous-driving "quantum ML" is mostly aspirational. Honest read: optimization/scheduling has real *today* value via annealing and quantum-inspired methods (advantage debated); materials simulation is a **2030s** bet.

SOURCES

- D-Wave/Ford Otosan production: <https://www.dwavequantum.com/company/newsroom/press-release/d-wave-highlights-quantum-optimization-customer-growth-and-introduces-expanded-offering-to-accelerate-adoption-and-deployment/>
- [entangledfuture.com](#) automotive sector; [fuld.com](#) "Quantum Computing Is Entering the Automotive Workshop"

- postquantum.com aerospace & automotive use cases

Quantum in Chemicals & Materials · I-chem

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Like pharma, chemicals is a native quantum-simulation target (`S-qsim`): designing catalysts, battery electrolytes, photochromic and semiconductor materials, and reaction pathways by computing electronic structure directly instead of trial-and-error lab work. McKinsey frames it as compressing decades of chemistry. This is the domain with the clearest *theoretical* quantum advantage — the algorithms (VQE, quantum phase estimation) map onto exactly what quantum computers do naturally.

REAL ACTIVITY (NAMED, DATED)

- **BASF** — collaboration with **SEEQC** on quantum for homogeneous catalysis; with **Kipu Quantum** (2024) on optimization algorithms; broader hybrid catalyst-design program via the German QUTAC consortium.
- **Mitsubishi Chemical + PsiQuantum** — simulating **excited states of photochromic molecules** (smart windows, solar storage, data storage). Also **Mitsubishi + Xanadu** on semiconductor materials; long-running IBM case study.
- **Dow** — quantum chemistry exploration (historical IBM Quantum Network member).
- **Microsoft Azure Quantum Elements** — AI+quantum chemistry/materials platform pitched to chemical companies. **NVIDIA cuEST** (GPU electronic-structure library) — TSMC reports ~50x faster chemistry simulations; a *classical*-acceleration reminder that the near-term competition to quantum chemistry is GPU HPC, not other quantum machines.
- **Merck, Covestro, Evonik** — QUTAC members probing polymer/formulation chemistry.

KEY GRADED CLAIMS

- [T1] Molecular electronic-structure simulation is a provably natural quantum-computer task — textbook (established in principle; hardware not yet capable at industrial scale)

- [T3/T4] Photochromic excited-state simulation — Mitsubishi Chemical/PsiQuantum (exploratory)
- [T4] Homogeneous-catalysis quantum modeling — BASF/SEEQC (research collaboration)
- [T5] Chemicals/materials as a leading slice of McKinsey's ~\$2.7T-by-2035 value estimate and BCG's \$450–850B-by-2040 — analyst forecasts; grade hard, inflation-unadjusted and double-counted with pharma/energy

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** VQE/QPE run on real hardware for tiny molecules (H_2 , LiH , small actives) — results classical DFT/coupled-cluster already match or beat.
- **Promise:** catalyst and photochromic design at useful accuracy — the strongest theoretical case in any industry, gated on thousands of logical qubits.
- **Hype:** "quantum is designing industrial catalysts now," trillion-dollar near-term chemicals TAM.

HONEST ASSESSMENT

The theoretical case is the strongest in any industry — simulating quantum chemistry is what quantum computers are *for*. But FeMoco-class catalysts and industrial reaction networks need error-corrected machines with thousands of logical qubits; today's devices simulate only tiny molecules classical methods already handle, and GPU-accelerated classical chemistry keeps raising the bar quantum must clear. Every named "pilot" is a research collaboration probing algorithms, not a deployed design tool. Honest timeline: fault-tolerant chemistry advantage is a **2030s** proposition. Hold the distinction — strong theory, essentially zero present-day production advantage.

SOURCES

- McKinsey "Solving chemistry's toughest problems": <https://www.mckinsey.com/capabilities/mckinsey-technology/our-insights/solving-chemistrys-toughest-problems-the-quantum-computing-advantage>
- BASF/QUTAC catalysts: <https://www.qutac.de/en/basf-how-quantum-computing-can-help-develop-chemical-catalysts/>

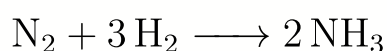
- Mitsubishi Chemical/IBM: <https://www.ibm.com/case-studies/mitsubishi-chemical>
- NVIDIA cuEST / TSMC 50x chemistry: nvidianews.nvidia.com (GTC 2025)
- C&EN "Will quantum computing be chemistry's next AI?" (Nov 2025)

Quantum in Climate & Sustainability · I-climate

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Climate is largely a *downstream application* of quantum chemistry (**I-chem** , **S-qsim**): better catalysts for CO₂ capture, more efficient fertilizer synthesis (cracking the FeMoco nitrogenase problem to replace energy-hungry Haber-Bosch), plus grid/emissions optimization. The prize is enormous — ammonia production alone is up to ~1.8% of global CO₂ and ~1–2% of world energy. (Precision agriculture is split into **I-agri** ; numerical weather/climate *modeling* is split into **I-weather** .)



REAL ACTIVITY (NAMED, DATED)

- **Alice & Bob** (Oct 2025) — published a quantum resource estimate for simulating FeMoco (biological nitrogen fixation) showing a **~27x reduction in physical qubits** vs a 2021 Google study (2.7M → ~99,000 qubits) using cat qubits. A resource estimate, not a run — a notable efficiency milestone that still points at hardware that does not exist.
- **Microsoft / Riverlane / McKinsey** — frame carbon-capture sorbent (MOF) design and nitrogen-cycle modeling as flagship "quantum for the planet" targets.
- **qHPC-GREEN** (CASUS, launched Jan 2025 → 2029) — quantum-HPC hybrid research into energy-efficient fertilizer/nitrogen-fixation simulation (shared with **I-agri**).

KEY GRADED CLAIMS

- [T3] FeMoco simulation needs ~99k physical qubits (27x cut) with cat qubits — Alice & Bob, Oct 2025 (demonstrated *estimate*, not execution)
- [T1] Haber-Bosch's carbon/energy footprint — established chemistry
- [T5] "Quantum computing just might save the planet" TAM framing — McKinsey; grade hard, downstream of the same chemicals value pool, inflation-unadjusted

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** resource *estimates* have improved (27x fewer qubits for FeMoco) — an engineering datapoint, not a computation.
- **Promise:** carbon-capture and fertilizer catalyst design — the textbook killer app, ~99k physical qubits away, i.e. thousands of logical qubits.
- **Hype:** "quantum will decarbonize industry this decade," present-day climate impact from quantum computers (zero).

HONEST ASSESSMENT

The nitrogen-fixation story is the honest headline and the honest caution at once. FeMoco is the textbook "killer app" for quantum chemistry, and the best current estimate still says ~99,000 physical qubits — many thousands of logical qubits with error correction, hardware that does not exist and is years to a decade+ away. The 27x improvement is worth reporting and a reminder of the distance to payoff. Carbon-capture sorbent design shares the same fault-tolerance wall. Realistic climate impact from quantum computing: **2030s at the earliest**, entirely gated on fault tolerance. Present-day value is essentially zero.

SOURCES

- Alice & Bob FeMoco/fertilizer: <https://alice-bob.com/newsroom/alice-bob-quantum-computing-applications-health-agriculture/>
- McKinsey "Quantum computing just might save the planet": <https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/quantum-computing-just-might-save-the-planet>
- Riverlane "How quantum computing can help tackle climate change"
- Microsoft Research "What problems will we solve with a quantum computer?"

Quantum in Construction & Built Environment · I-construction

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Construction/built-environment cuts across three quantum threads: (1) **structural materials simulation** — quantum chemistry (`I-chem` , `S-qsim`) for greener cement/concrete and durable materials; (2) **project scheduling and supply-chain optimization** — QUBO/QAOA for construction schedules, resource conflicts, and prefab supply chains, sharing the `I-logistics` template; (3) **urban microclimate and energy modeling** — CFD-style built-environment and HVAC/EV-charging optimization for greener cities.

REAL ACTIVITY (NAMED, DATED)

- **Concordia / UCF / Syracuse / Tongji** — Liangzhu Leon Wang et al., "*Take a BITE for Built Environment and Urban Microclimate Research*" (arXiv 2604.18407) maps quantum methods onto urban microclimate and building-energy problems. Academic survey/framing.
- **Civil-engineering reviews** — *Quantum and quantum-inspired computing in civil engineering* (ScienceDirect 2025) and *Quantum computing in civil engineering: Potentials and Limitations* (arXiv 2402.14556) document QUBO scheduling and structural-analysis demos on small instances.
- **Prefab supply chain** — *Unlocking the potential of quantum computing in prefabricated construction supply chains* (ScienceDirect 2025) — trends/challenges review, not deployment.
- **The honest datapoint** from these reviews: ~42% of "quantum in construction" research is actually **quantum materials** (`A-materials`), and compute/QML applications are "predominantly theoretical and confined to small-scale simulations."

KEY GRADED CLAIMS

- [T3] QUBO project scheduling finds near-optimal construction schedules faster than classical on small cases — civil-eng reviews (demonstrated on toy instances; no advantage at scale)

- [T3] Quantum methods framed for urban microclimate / building energy — Wang et al. arXiv 2604.18407 (framing/survey)
- [T5/T6] Quantum-designed sustainable construction materials — review scenarios (speculative, FTQC-gated)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** small QUBO scheduling and structural demos exist in the literature — classical MILP still beats them.
- **Promise:** greener cement/concrete via quantum materials chemistry — inherits I-chem 's fault-tolerance wall.
- **Hype:** "quantum optimizes construction projects" as a near-term product — no industrial pilot runs quantum hardware in the loop.

HONEST ASSESSMENT

Construction is an early, academic node with no named industrial pilot running quantum hardware in the loop. Its own literature admits most activity is either materials science (belongs in A-materials) or small-scale QUBO demos that classical MILP still beats. The materials-simulation payoff is real and inherits the fault-tolerance timeline of I-chem . Scheduling and microclimate are optimization pitches without demonstrated advantage. Present-day value is zero; realistic horizon is 2030s and gated on FTQC.

SOURCES

- <https://arxiv.org/abs/2604.18407> (BITE — built environment & urban microclimate)
- <https://www.sciencedirect.com/science/article/pii/S1474034625008535> (quantum in civil engineering)
- <https://arxiv.org/html/2402.14556v2> (civil engineering potentials & limitations)
- <https://www.sciencedirect.com/science/article/abs/pii/S1566253525001162> (prefab supply chains)

Quantum in Cybersecurity · I-cyber

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Cybersecurity is the industry with the most *certain* quantum impact — and it lands as a **threat before it lands as a tool**. A cryptographically relevant quantum computer (CRQC) running Shor's algorithm (**S-shor**) breaks RSA and elliptic-curve crypto, so the internet's encryption must migrate to post-quantum cryptography (**A-pqc**). "Harvest now, decrypt later" (HNDL) means adversaries are already storing encrypted traffic to decrypt once a CRQC exists. This is a real, funded, deadline-driven market today.

REAL ACTIVITY (NAMED, DATED)

- **NIST** — finalized first PQC standards **FIPS 203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA)** in Aug 2024, triggering the largest mandated crypto migration in history. NIST IR 8547 sets deprecation timelines (2030 deprecate, 2035 disallow).
- **NSA CNSA 2.0** — quantum-safe algorithms required for new national-security systems by **Jan 2027**, full app migration by 2030, infrastructure by 2035.
- **Resource-estimate shift** — work published May 2025–March 2026 cut the estimated qubits to break RSA-2048 from ~20M (Gidney/Ekerå 2019) toward **<1M, potentially ~100k** on newer architectures — sharpening the HNDL urgency without a CRQC yet existing.
- **Enterprise reality** — a May 2025 survey of 1,000+ security managers found only **~5% had quantum-safe encryption deployed**; 81% said libraries/HSMs weren't PQC-ready.

KEY GRADED CLAIMS

- [T2] NIST PQC standards final (FIPS 203/204/205), migration mandated — NIST, Aug 2024 (established)
- [T1] Shor breaks RSA/ECC given a CRQC — established theory (**S-shor**); the CRQC itself is [T6]
- [T3] RSA-2048 resource estimate reduced toward <1M qubits — 2025–26 preprints (revised estimate, not a demonstration)

- [T5] ">\$15B migration market by 2030" / 5% deployment rate — surveys + analyst projections (forecast; inflation-unadjusted)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** the threat model and the standards. Shor's math is established; FIPS 203/204/205 are final; deadlines are legally binding for regulated/defense sectors.
- **Promise:** actual enterprise migration — mandated but barely started (~5% deployed).
- **Hype:** "Q-Day is next year," QKD as the answer (PQC software is), and CRQC timelines stated with false precision.

HONEST ASSESSMENT

This is the one industry where quantum's impact is **not** overstated — the threat is mathematically certain, the standards are final, the deadlines are binding. The uncertainty is *timing*: CRQC estimates range 2030 to "not before 2040+" (a first-class conflict, 0-scaling), and the recent qubit-count reductions tighten but do not resolve it. Because HNDL makes the risk retroactive, migration must start now regardless. The honest caveat here runs opposite to the rest of L4 — the danger is *under-reacting*. QKD is over-marketed; classical-software PQC is the real answer.

SOURCES

- NIST IR 8547 draft: <https://nvlpubs.nist.gov/nistpubs/ir/2024/NIST.IR.8547.ipd.pdf>
- PQShield on NIST timelines: <https://pqshield.com/nist-recommends-timelines-for-transitioning-cryptographic-algorithms/>
- "Harvest now, decrypt later" — The Quantum Insider (May 2026), CSA Q-Day clock
- PRNewswire "The \$15 Billion Post-Quantum Migration"

Quantum in Energy & Utilities · I-energy

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Two threads: (1) optimization — grid balancing, unit commitment, EV-charging scheduling, demand forecasting, LNG shipping routes; and (2) simulation — battery chemistries, fusion plasma, carbon-capture sorbents. Utilities face combinatorial grid problems classical solvers approximate; energy majors want better materials. (Fusion plasma modeling and nuclear are deep enough to split into **I-nuclear** ; oil/gas reservoir work lives in **I-extractive** .)

REAL ACTIVITY (NAMED, DATED)

- **EDF + Pasqal + GENCI** (Jan 2025) — neutral-atom quantum computing demonstrated on **EV smart-charging and energy-demand forecasting** at 100+ scale; explicitly a pilot, not commercially viable.
- **EDF + Alice & Bob + Quandela + CNRS** (July 2024) — collaboration on optimizing the *energy consumption of quantum computing itself* (the fridge/laser power budget).
- **E.ON + IBM** — quantum algorithms for electricity-distribution complexity, energy pricing, risk management.
- **ExxonMobil** — first energy company to join the IBM Quantum Network (2019); quantum algorithms for LNG shipping-route optimization and carbon capture.
- **Iberdrola, Enel, Vattenfall** — European utilities running grid-optimization feasibility studies with vendors.
- **Aramco** — targeting local QCaaS access ~2025; **Shell** exploring reservoir optimization (see **I-extractive**).

KEY GRADED CLAIMS

- [T3/T4] EV-charging & demand forecasting on neutral atoms — Pasqal/GENCI/EDF, Jan 2025 (demonstrated pilot, not commercially viable)
- [T4] LNG shipping-route optimization — ExxonMobil/IBM (exploratory)
- [T4] Grid/pricing quantum algorithms — E.ON/IBM (claimed)

- [T5] "Energy transition" TAM lines in analyst reports — forecast; grade hard, entangled with the climate/chemicals figures

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** dated feasibility pilots (EDF/Pasqal EV-charging) run and return results — pre-commercial by the participants' own framing.
- **Promise:** grid dispatch, unit commitment, battery-chemistry simulation — real hard problems, no demonstrated advantage over classical/quantum-inspired solvers yet.
- **Hype:** "quantum will run the grid this decade," trillion-dollar energy-transition quantum TAM.

HONEST ASSESSMENT

Grid optimization is a real, valuable, hard problem, and classical plus quantum-inspired solvers (and D-Wave annealing) currently match or beat gate-model quantum on the instances tried. The EDF/Pasqal work is a credible, dated pilot stated openly as pre-commercial. Battery and carbon-capture simulation share pharma/chem's fault-tolerance wall. No utility runs quantum in dispatch. The nearer-term energy payoff more plausibly comes from quantum *sensing* (grid fault detection, magnetometry) than compute. Realistic optimization payoff: **late 2020s**; simulation payoff: **2030s**.

SOURCES

- Pasqal/GENCI/EDF: <https://thequantuminsider.com/2025/01/21/pasqal-genci-and-edf-use-neutral-atom-quantum-computing-to-advance-ev-smart-charging-and-energy-forecasting/>
- EDF/Alice&Bob/Quandela/CNRS: <https://alice-bob.com/newsroom/edf-alice-bob-quandela-cnrs-quantum-computing/>
- Hello Tomorrow "Quantum Computing in the Energy Sector"
- postquantum.com energy & utilities use cases

Quantum in Oil & Gas · I-extractive

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Oil & gas pitches quantum for **seismic data processing and subsurface imaging** (faster inversion, pattern-finding in seismic/well-log big data), **reservoir simulation** (multi-week optimization compressed to hours), materials discovery (catalysts, carbon capture), and LNG logistics. Big analyst TAM numbers attach here. (**Mining and mineral exploration** — a distinct, sensing-led, nearer-term story — is split into I-mining .)

REAL ACTIVITY (NAMED, DATED)

- **ExxonMobil, bp** — joined the **IBM Quantum Network** (ExxonMobil 2019, first energy major) to explore reservoir simulation, LNG shipping-route optimization, and advanced materials.
- **Shell + D-Wave** — quantum annealing applied to **North Sea reservoir mapping** optimization (reported multi-week → hours framing). Shell has a multi-year quantum research program (also with IBM, Zapata historically).
- **Aramco** — targeting **local QCaaS** access ~2025 for Saudi research; invested in Pasqal (2022) toward an on-premise neutral-atom machine.
- **Seismic method work** — quantum travel-time seismic inversion (arXiv 2208.05794) and QUBO formulations of inversion sub-problems; preprints on small instances.

KEY GRADED CLAIMS

- [T4] Shell/D-Wave reservoir-mapping speedup (weeks→hours) — vendor+operator framing (annealing, advantage contested)
- [T4] Reservoir sim & materials exploration — ExxonMobil/bp/IBM (exploratory)
- [T3] Quantum seismic-inversion formulations — arXiv 2208.05794 (demonstrated on toy instances)
- [T5] "\$2.6T oil-and-gas quantum impact by 2035 / ~30% CAGR" — market-research projections (forecast; among the most inflated in L4, inflation-unadjusted)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** operators run feasibility studies and publish sub-problem demos — no operator runs quantum in its production geophysics pipeline.
- **Promise:** reservoir optimization and seismic inversion at field scale — gated on fault tolerance; classical HPC dominates today.
- **Hype:** trillion-dollar "impact by 2035" TAM lines — grade hardest in the manual.

HONEST ASSESSMENT

Oil & gas is heavy on analyst TAM and light on reproduced results. The seismic/reservoir "advantage" claims are annealing or quantum-inspired optimization on subproblems; classical HPC still dominates production geophysics, and no operator runs quantum in its exploration pipeline. The genuinely nearer-term extractive thread is *quantum sensing* (cold-atom gravimeters, NV magnetometers) — covered under **I-mining** and **A-magneto / A-gravimetry**. Quantum *computing* for reservoirs/seismic is a fault-tolerance-gated **2030s** bet dressed in trillion-dollar forecasts. Grade the TAM numbers hard.

SOURCES

- JPT (SPE) "Quantum Computing: A Beacon of Transformation for Oil and Gas": <https://jpt.spe.org/guest-editorial-quantum-computing-a-beacon-of-transformation-for-the-oil-and-gas-industry>
- oilandgasiq.com "Quantum Computing's Impact on Oil & Gas"
- energyconnects.com "The oil and gas industry aims to take a leap with quantum computing" (June 2024)
- Quantum travel-time seismic inversion: <https://arxiv.org/pdf/2208.05794>

Quantum in Finance · I-finance

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Finance is the flagship "quantum use case": portfolio optimization, Monte Carlo derivative pricing, risk (VaR/CVA), and fraud/anomaly detection all map to problems where classical compute is expensive. Vendors promise quadratic speedups on Monte Carlo (amplitude estimation, see `S-qmc`) and better solutions to NP-hard portfolio selection (QAOA, `S-variational`). Two sub-verticals behave differently: **retail/commercial banking** chases fraud graphs and credit scoring; **capital markets** (trading desks, derivatives, reinsurance-adjacent risk) chases pricing and optimization. Banks were the earliest and heaviest enterprise experimenters, with 15+ global banks running programs by March 2026 (JPMorgan, Goldman, HSBC, BBVA, Barclays, BNP Paribas, Wells Fargo).

REAL ACTIVITY (NAMED, DATED)

- **JPMorganChase** — the most substantive results. With Quantinuum, Argonne, ORNL, UT Austin, published **certified randomness** on the 56-qubit Quantinuum H2-1 (*Nature*, 26 Mar 2025) — a genuine "beyond classical" task (see `S-certrand`). Separately showed a **theoretical QAOA speedup** for constrained optimization (*Science Advances*, 29 May 2025). A JPMorgan+AWS hybrid decomposition (2024) reported a modest **~12% runtime reduction** on a constrained optimization by partitioning. "Up to 100x speed" figures appear in controlled-test PRs and are T4.
- **HSBC + IBM** (Sept 2025) — claimed **up to 34% improvement** predicting whether a corporate bond trade fills at the quoted price vs classical, on real algorithmic bond-trading data using an IBM Heron processor in a hybrid workflow. Framed as "first empirical evidence quantum adds value in production-representative trading." Bank+vendor co-announcement, not independently reproduced.
- **Goldman Sachs** — early leader on Monte Carlo option pricing (with IonQ/QC Ware), then reportedly **scaled back** after concluding practical advantage remains far off — the honest counter-signal to the bullish PRs.

- **Multiverse Computing** — Spanish vendor selling "quantum-inspired" (tensor-network) finance tooling to banks; revenue is classical software, useful context on where the money actually flows.

KEY GRADED CLAIMS

- [T2] Certified randomness on H2-1, classically unachievable — JPMC/Quantinuum, *Nature* s41586-025-08737-1 (demonstrated; a cryptography primitive, not trading P&L)
- [T3] QAOA constrained-optimization speedup (theoretical) — JPMC, *Science Advances* 2025
- [T4] HSBC/IBM 34% bond-fill improvement — co-announcement Sept 2025 (controlled POC)
- [T4] Portfolio-optimization / Monte-Carlo "100x" speedups — bank+vendor PRs (no production deployment)
- [T5] Finance as a headline share of McKinsey's ~\$2.7T-by-2035 economic-value estimate — analyst forecast; grade hard, double-counted across verticals and inflation-unadjusted (2026 dollars)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** certified randomness (peer-reviewed, beyond-classical) — a cryptographic primitive with privacy/audit uses, not alpha.
- **Promise:** portfolio optimization, Monte Carlo pricing, bond-fill prediction — real POCs on tiny instances, awaiting fault tolerance to beat Gurobi/CPLEX at scale.
- **Hype:** "quantum trading edge today," "100x in production," multi-trillion-dollar finance TAM lines.

HONEST ASSESSMENT

No bank runs a quantum computer in live operations as of mid-2026. The certified-randomness result is real and refereed and is a cryptography primitive. The optimization and pricing "advantages" are controlled POCs on small instances that classical methods still beat at scale, and the amplitude-estimation speedup that finance has chased for years needs fault-tolerant depth that erases the near-term gain. Goldman's pullback and JPMorgan's careful language are the honest signals inside the sector. Realistic operational value: late 2020s at the earliest, gated on fault tolerance.

SOURCES

- JPMorgan certified randomness: <https://www.jpmorgan.com/technology/news/certified-randomness> · Nature: <https://www.nature.com/articles/s41586-025-08737-1>
- HSBC/IBM bond trading (Sept 2025): reported via The Quantum Insider / FStech
- "15+ global banks probing quantum" — The Quantum Insider, 27 Mar 2026
- "Goldman Sachs, JPMorgan sharply diverge on quantum" — TheStreet
- McKinsey Quantum Technology Monitor 2026 (economic-value figure)

Quantum in Government Services · I-gov

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Government is pitched as a quantum customer distinct from defense (I-aerospace) and intelligence (I-intelligence): tax administration (real-time audits, anomaly/fraud detection), customs and border flow, census and statistical disclosure, benefits fraud, and general public-sector optimization. The problems map to the same QUBO optimization and anomaly-detection templates sold to finance — combinatorial matching, graph community detection over transaction networks, pattern-finding in large administrative datasets.

REAL ACTIVITY (NAMED, DATED)

- **Academic / preprint** — the substantive work is QUBO-formulated fraud detection via community detection in transaction graphs (*Quantum Computing in Community Detection for Anti-Fraud Applications*, PMC 2025) and unsupervised quantum ML for fraud (arXiv 2208.01203). Method papers on small instances, no agency deployment.
- **Advisory framing** — EY and the Inter-American Center of Tax Administrations (CIAT) publish "how quantum will change tax administration" guides. Consulting scenarios, not pilots.
- **The mislabel trap** — HMRC's £175M 10-year Quantexa contract (2026) to hunt tax fraud is **classical AI/decision-intelligence**, despite the "Quant-" branding. Much "quantum government" press is this pattern: AI relabeled, or PQC/export-control work (covered in A-pqc , E-export).
- **Real government quantum spend** is on the *defensive* side — national PQC-migration mandates (US OMB M-23-02, NSA CNSA 2.0) — which is I-cyber , not offensive optimization.

KEY GRADED CLAIMS

- [T3] QUBO community detection improves anti-fraud risk scoring on transaction graphs — PMC/arXiv method papers (demonstrated on toy data)
- [T4] Vendor "quantum for public sector" optimization offerings — press/marketing (claimed)

- [T5] Tax authorities should prepare for real-time quantum-enabled audits — EY/CIAT advisory (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** nothing offensive is deployed. The genuine government quantum program that exists is defensive PQC migration.
- **Promise:** anti-fraud graph analytics and public-sector optimization — method-paper stage, inheriting finance's caveats.
- **Hype:** headline "quantum government" wins that are actually classical AI (the Quantexa/HMRC pattern).

HONEST ASSESSMENT

There is no known quantum computer running in production inside any tax, customs, or census agency. The genuine near-term government exposure to quantum is defensive — PQC migration and the harvest-now-decrypt-later threat to citizen data — which lives in **A-pqc / I-cyber**. Offensive/optimization use is at the method-paper stage and inherits every caveat of **I-finance** optimization (tiny instances, classical methods still win). The honest signal is the naming confusion: most headline "quantum government" wins are classical AI. Realistic operational value: gated on fault tolerance, late 2020s at the earliest; today it is essentially zero.

SOURCES

- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11727351/> (QUBO anti-fraud community detection)
- <https://arxiv.org/pdf/2208.01203> (unsupervised quantum ML for fraud)
- https://www.ey.com/en_us/insights/tax/how-quantum-computing-will-improve-tax-administration-and-compliance
- <https://www.ciat.org/quantum-computer-impacts-on-tax-administration-at-first-glance/?lang=en>
- <https://thenextweb.com/news/quantexa-hmrc-ai-tax-fraud-sovereignty> (the AI-not-quantum contrast)

Quantum in Healthcare Imaging & Diagnostics · I-healthimaging

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from I-pharma drug-discovery; this is the nearer-term quantum-*sensing* health story)

THE PITCH

Split from I-pharma (molecular simulation), this node is the quantum-**sensing** side of healthcare — and it is far closer to the clinic than quantum computing. The flagship is **OPM-MEG**: optically-pumped magnetometers (a quantum sensor) replacing cryogenic SQUIDs for magnetoencephalography, giving wearable, room-temperature brain imaging. Adjacent threads: NV-diamond and hyperpolarization for enhanced MRI/NMR contrast, quantum-enhanced (sub-shot-noise) optical imaging, and diagnostic ML.

REAL ACTIVITY (NAMED, DATED)

- **Cerca Magnetics** (UK, Nottingham spinout) — commercial **wearable OPM-MEG** helmet systems; a real product shipped to research sites. As of 2025 it is sold as a **research system with no medical/regulatory approval** in any jurisdiction — the honest ceiling on the claim.
- **384-channel OPM-MEG** (arXiv 2509.03107, 2025) — toward whole-head high-channel-count systems; the technical frontier.
- **EU Horizon OPMMEG project** (CORDIS 101099379) — pushing OPM-MEG toward routine clinical practice; epilepsy pre-surgical workup is the most advanced application, with mild-TBI use growing.
- **NIHR Innovation Observatory report** (Jan 2025) — surveys quantum sensing in healthcare; notes only two sensing technologies are beyond **TRL 6** (entering clinical testing), i.e. still early.

KEY GRADED CLAIMS

- [T2/T3] OPM-MEG matches SQUID sensitivity without cryogenics, wearable — peer-reviewed OPM literature + PMC epilepsy studies (demonstrated capability)
- [T3] 384-channel OPM-MEG system — arXiv 2509.03107 (preprint, prototype)

- [T4] Cerca commercial OPM-MEG — vendor product (real, research-only; no regulatory approval as of 2025)
- [T5] Quantum-healthcare-sensing market forecasts — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** OPM-MEG works — wearable, cryogen-free brain magnetometry with SQUID-class sensitivity, running in research labs today. This is a real quantum sensor in health use.
- **Promise:** routine clinical MEG (epilepsy, TBI), NV/hyperpolarized MRI contrast — advancing through clinical testing and regulatory pathways.
- **Hype:** "quantum computers diagnose disease" — this node is sensing, not compute; and OPM-MEG is not yet cleared for diagnosis.

HONEST ASSESSMENT

This is one of the strongest "quantum is real in health now" stories, and the thing that is real is a *sensor*. OPM-MEG is a shipped quantum-sensing product with genuine clinical promise (epilepsy pre-surgical mapping), held back by regulatory approval and magnetically-shielded-room requirements rather than physics. NV-diamond and hyperpolarization for MRI are earlier. The whole node needs no fault-tolerant computer, which is exactly why it is nearer-term than **I-pharma** drug discovery. Realistic clinical adoption: **this decade** for MEG in specialist centers, contingent on regulatory clearance. Keep it separate from quantum-computing hype.

SOURCES

- NIHR IO "Emerging applications of quantum sensing technology in healthcare" (Jan 2025): https://io.nihr.ac.uk/wp-content/uploads/2025/03/NIHR-IO-Quantum-Sensing-Technology-Report_Jan-2025.pdf
- 384-channel OPM-MEG: <https://arxiv.org/pdf/2509.03107>
- Cerca Magnetix OPM-MEG: <https://www.cercamagnetix.com/>
- OPM epilepsy presurgical workup: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10896867/>

Quantum in Insurance & Risk · I-insurance

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Insurance is finance's cousin (*I-finance*): Monte Carlo-heavy catastrophe modeling, capital/reserve optimization under reinsurance, actuarial pricing, tail-risk and dependency modeling, and fraud detection. Quantum is pitched for amplitude-estimation speedups on Monte Carlo (*S-qmc*) and better modeling of extreme-event correlations (tail dependence) that classical Gaussian copulas capture poorly.

REAL ACTIVITY (NAMED, DATED)

- **Allstate** — among the first major US insurers to join the **Chicago Quantum Exchange** (2025), partnering with quantum researchers on complex risk modeling.
- **Method / academic work** — quantum-AI copulas for catastrophe and reinsurance capital modeling: *Quantum computational insurance and actuarial science* (Science China Information Sciences, 2025 review); *Quantum Computing in Insurance Capital Modelling under Reinsurance Contracts* (AppliedMath, 2024); World Quantum Summit 2025 sessions on quantum-AI copulas.
- Reinsurers (Swiss Re, Munich Re) and actuarial bodies (IFoA) are watching and running small experiments; no named production deployment has surfaced.

KEY GRADED CLAIMS

- [T3] Quantum methods for non-life/life/reinsurance capital modeling — Science China Info Sci 2025, AppliedMath (peer-reviewed numerical demonstrations on small instances)
- [T4] Allstate/Chicago Quantum Exchange risk-modeling exploration — membership announcement (early)
- [T5] Quantum-AI copulas improve tail-dependence/catastrophe modeling at scale — summit + review framing (forecast)

- [T5] Insurance slice of quantum-value forecasts — analyst; tracks finance, lags in adoption

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** small numerical demonstrations of quantum capital/copula methods exist in the literature — no advantage over classical actuarial compute.
- **Promise:** amplitude-estimation Monte Carlo for cat modeling and tail dependence — the same finance bet, one adoption step behind.
- **Hype:** any claim of quantum catastrophe modeling in production (none exists).

HONEST ASSESSMENT

Insurance is one of the *least* mature L4 verticals — mostly academic papers and consortium memberships, few named corporate pilots, no deployments. The theoretical hook (quadratic Monte Carlo speedup via amplitude estimation) is the same one finance has chased for years and never realized in production, because the speedup needs fault-tolerant depth that erases the near-term advantage. Catastrophe modeling is genuinely hard and valuable, so the motivation is real, and nothing here beats classical actuarial compute today. This vertical is almost entirely promise: realistic value tracks finance's **late 2020s–2030s** timeline and lags it in adoption.

SOURCES

- The Actuary Weekly "Quantum Computing in Catastrophe Modeling"
- Science China Information Sciences "Quantum computational insurance and actuarial science": <https://link.springer.com/article/10.1007/s11432-024-4411-8>
- AppliedMath "Quantum Computing in Insurance Capital Modelling under Reinsurance Contracts": <https://doi.org/10.3390/appliedmath3040040>
- World Quantum Summit 2025 "Insurance Risk Modelling with Quantum-AI Copulas"

Quantum in Intelligence & Cryptanalysis · I-intelligence

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from I-aerospace defense and I-cyber defense; the offensive SIGINT/cryptanalysis angle is distinct)

THE PITCH

Distinct from defense hardware (I-aerospace) and defensive PQC (I-cyber), this node is the **offensive signals-intelligence** dimension: a cryptographically relevant quantum computer (CRQC) running Shor (S-shor) would let a state actor decrypt intercepted RSA/ECC traffic. The strategic behavior this *already* drives is **harvest-now-decrypt-later (HNDL)** — bulk collection and storage of today's encrypted traffic for future decryption. Secondary: quantum-enhanced pattern-analysis over massive intercept datasets (the same QML caveats as I-ai).

REAL ACTIVITY (NAMED, DATED)

- **Intelligence-community warnings** — NSA, CISA, and NIST have jointly warned that adversaries **may already be harvesting** encrypted data with long-term value to decrypt once a CRQC exists (the HNDL doctrine). Nation-state SIGINT programs (US, China, Russia) operate the bulk-collection infrastructure that makes HNDL feasible.
- **NSA CNSA 2.0** — mandates quantum-resistant algorithms for new national-security systems by **Jan 2027**, full transition by 2035 — an intelligence-driven timeline (defend own secrets against adversary CRQC).
- **Resource-estimate shift (2025–26)** — the RSA-2048 qubit estimate fell from ~20M (2019) toward **<1M, possibly ~100k** on newer architectures — which compresses the intelligence risk horizon without a CRQC existing yet.
- **No CRQC exists.** No public evidence any agency can run Shor at cryptographic scale. Adversary capability is classified — the honest overhang.

KEY GRADED CLAIMS

- [T1] Shor breaks RSA/ECC given a CRQC — established theory (S-shor); the CRQC itself is [T6]

- [T3] Bulk HNDL collection is technically feasible and warned-of by NSA/CISA/NIST — official advisories (established as doctrine; specific programs classified)
- [T3] RSA-2048 resource estimate reduced toward <1M qubits — 2025–26 preprints (revised estimate, not a demonstration)
- [T6] Any agency currently possesses cryptanalytic quantum capability — speculative/classified (no public evidence)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** the doctrine and the math. HNDL is a real, warned-of collection strategy; Shor's threat is certain given a CRQC.
- **Promise (or threat):** a future CRQC that decrypts harvested traffic — timing genuinely unknown (2030–2040+, 0-scaling).
- **Hype:** claims that a nation "already breaks encryption with quantum," stated with false certainty from classified inference.

HONEST ASSESSMENT

Intelligence is the vertical where quantum's impact is most certain in *kind* and most uncertain in *timing*, wrapped in classification. Shor's threat to public-key crypto is established; HNDL collection is real doctrine, which is precisely why CNSA 2.0's deadlines exist. What no one outside classified spaces knows is whether — or how close — any adversary is to a CRQC, and the recent qubit-count reductions tighten the risk window without resolving it. The rational response is the same as I-cyber : migrate to PQC now, because HNDL makes the risk retroactive. Treat any "they already have it" claim as unverifiable. Offensive quantum SIGINT-analytics (QML over intercepts) inherits I-ai 's dequantization caveats.

SOURCES

- "Harvest now, decrypt later": https://en.wikipedia.org/wiki/Harvest_now,_decrypt_later
- The Quantum Insider "Harvest Now, Decrypt Later — Why Should You Care?" (May 2026)
- NSA CNSA 2.0 (quantum-resistant mandate, 2027/2035)
- MDPI "Harvest-Now, Decrypt-Later: A Temporal Cybersecurity Risk in the Quantum Transition"

Quantum in Logistics & Supply Chain · I-logistics

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Routing, scheduling, bin-packing, and supply-chain optimization are combinatorial (often NP-hard), and vendors pitch QAOA/annealing for vehicle routing, traffic flow, cargo loading, and network design. This is the most "demo-friendly" industry because small optimization instances run on today's hardware and produce photogenic results. It is also where the **quantum-inspired vs quantum-hardware** conflation is most misleading in the press.

REAL ACTIVITY (NAMED, DATED)

- **Volkswagen** — early real-world **traffic-routing pilot in Lisbon** (2019 WebSummit), near-real-time bus routes via a quantum hybrid solver over live traffic/passenger data. A landmark demo, small fleet.
- **Airbus + IonQ** — cargo/aircraft loading via the **MAL-VQA** algorithm on trapped-ion hardware (arXiv 2504.01567, 2025). Airbus + BMW ran a **2024 Quantum Computing Challenge** including a sustainable-manufacturing logistics task.
- **Deutsche Bahn (Netz AG) + Cambridge Quantum** — **train rescheduling** with the F-VQE algorithm on realistic delayed timetables (part of Digitale Schiene Deutschland). DLR also pursuing quantum rail scheduling.
- **DHL** — quantum-inspired route-optimization work; IBM collaboration factoring customs, weather, fuel.
- **IonQ freight/logistics pilots** (2024–25) — container/freight optimization; IonQ has leaned into logistics as a commercial narrative.

KEY GRADED CLAIMS

- [T3] Aircraft cargo loading via MAL-VQA on IonQ hardware — arXiv 2504.01567 (demonstrated at small scale)
- [T3] Train rescheduling via F-VQE — Cambridge Quantum/DB Netz (demonstrated, realistic-instance study)
- [T4] VW Lisbon traffic routing — press pilot (hybrid solver, small fleet)

- [T5] Multi-billion logistics-optimization TAM projections — analyst forecasts (speculative; classical solvers already dominate)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** the workflow *runs* on today's hardware for small instances — a feasibility fact, not an advantage claim.
- **Promise:** vehicle routing, rail rescheduling, cargo loading at production scale — awaits fault tolerance and better encodings.
- **Hype:** "quantum optimizes global supply chains today" — most near-term wins are quantum-*inspired* (classical) methods relabeled.

HONEST ASSESSMENT

Logistics produces the most demos and the least proven advantage. Every result is a small instance where classical solvers (Gurobi, CPLEX, OR-Tools) and quantum-*inspired* methods still win at production scale. The value shown is that the workflow runs, and the genuine near-term wins come from quantum-inspired algorithms that borrow quantum structure while running on classical hardware — honest progress that is not a quantum computer. Genuine routing advantage awaits fault tolerance; realistic: **end of decade**.

SOURCES

- Q-CTRL Airbus/BMW logistics: <https://q-ctrl.com/blog/exploring-the-future-of-quantum-powered-logistics-with-airbus-and-bmw-group>
- Cambridge Quantum/DB train scheduling: <https://www.hpcwire.com/off-the-wire/cambridge-quantum-and-deutsche-bahn-netz-ag-leverage-latest-quantum-algorithms/>
- Aircraft loading: <https://arxiv.org/html/2504.01567v1>
- DHL "The quantum leap in logistics"; IonQ cargo optimization

Quantum in Manufacturing & Industrials · I-manufacturing

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Manufacturing quantum pitches cluster on **combinatorial optimization**: production sequencing, job-shop scheduling, factory-floor layout, paint-shop switching, supply/quality allocation, and quantum-enhanced digital twins that simulate whole production lines to find bottlenecks. Because these are annealing-friendly optimization problems, manufacturing has produced some of the earliest claimed *production* deployments in the whole field.

REAL ACTIVITY (NAMED, DATED)

- **Ford Otosan + D-Wave** (2024) — deployed a **hybrid-quantum application in production** for Ford Transit manufacturing sequencing; reported **~50% reduction in scheduling time** at one site (annealing-based). See [I-auto](#). The single most-cited "quantum in production" case.
- **D-Wave** (Qubits 2025 user conference, March 2025) — announced expanded commercial optimization offering and customer growth across manufacturing/scheduling; D-Wave reported its first meaningful optimization *revenue* — real dollars, on annealing whose advantage-vs-classical is contested.
- **Digital-twin case studies** — a "global automotive company" reported ~20% production-time reduction using quantum-enhanced digital-twin scenario simulation (case-study framing, unnamed).
- **WEF "How manufacturing is harnessing quantum technologies"** (Oct 2025) — frames process optimization, quality, and materials as the leading near-term manufacturing targets.

KEY GRADED CLAIMS

- [T3/T4] Ford Otosan production sequencing ~50% faster — D-Wave, 2024 (demonstrated, in production; annealing, advantage-vs-classical contested)
- [T4] ~20% digital-twin production-time reduction — unnamed automotive case study (claimed)
- [T4] D-Wave optimization revenue growth — vendor report, Mar 2025 (real revenue, contested advantage)

- [T5] Broad manufacturing process-optimization TAM — WEF/analyst framing (forecast)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** an annealing-based scheduling app runs in a live plant (Ford Otosan) and customers pay for optimization services.
- **Promise:** gate-model optimization and quantum digital twins beating classical HPC — no demonstrated advantage yet.
- **Hype:** "quantum computer beats classical solvers on scheduling" (contested); fully quantum digital twins.

HONEST ASSESSMENT

Manufacturing shares automotive's honest split. Annealing and quantum-inspired optimization deliver real workflow value today (Ford Otosan is a legitimate production case), and whether a *quantum* processor beats a good classical solver on the same scheduling problem is contested — D-Wave's own advantage claims live under **0-advantage**, and classical and quantum-inspired methods frequently match it. Digital twins remain mostly classical HPC with a quantum-optimization add-on. The near-term story is "quantum-flavored optimization services that sometimes help." Gate-model manufacturing advantage: **end of decade**. This is one of the more grounded L4 verticals precisely because it under-claims relative to finance/pharma.

SOURCES

- D-Wave customer growth / Ford Otosan: <https://www.dwavequantum.com/company/newsroom/press-release/d-wave-highlights-quantum-optimization-customer-growth-and-introduces-expanded-offering-to-accelerate-adoption-and-deployment/>
- WEF "How manufacturing is harnessing quantum technologies" (Oct 2025)
- Augmented Qubit "Definitive Report on Quantum-Enhanced Digital Twins"

Quantum in Media & Entertainment · I-media

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

The thinnest but genuinely pitched industry node: procedural content generation (PCG) for games/worlds seeded by true quantum randomness (A-qrng), quantum-assisted rendering and physics simulation, content-recommendation optimization, and ad/creative optimization. The one place quantum offers something classical cannot fake is **certified true randomness** as a generative resource; the rest is optimization/ML reheated for a creative audience. (Sports analytics/betting, sometimes grouped here, is a marginal recommendation/optimization case with no quantum-specific hook.)

REAL ACTIVITY (NAMED, DATED)

- **James Wootton / Qiskit** — the reference thread: "Procedural Generation Using Quantum Computation" (Qiskit blog + IEEE CG&A 2024, *Quantum Wave Function Collapse for PCG*, arXiv 2312.13853). Demonstrates quantum-seeded wave-function-collapse PCG on real hardware/simulators. Research and demos, not shipped games.
- **arXiv 2508.09683** (2025) — *Procedural Generation and Games at the Dawn of Fault-Tolerant Quantum Computing* frames the near-term realistic role as hybrid quantum-classical subroutines injecting variability, not standalone quantum rendering.
- **QRNG in market** — ID Quantique, Quantinuum, and cloud QRNG feeds ship true randomness usable for generative seeds; a component, not a media product.
- **Telefónica Servicios Audiovisuales** and similar trade pieces sketch 1–3 year horizons for recommendation optimization and 3–5 years for rendering/simulation. Vendor/advisory forecasts.

KEY GRADED CLAIMS

- [T3] Quantum wave-function-collapse PCG runs on quantum hardware/simulators — Wootton, IEEE CG&A 2024 / arXiv 2312.13853 (demonstrated; research artifact, no advantage)

- [T4] Quantum recommendation/ad optimization near-term — vendor framing (claimed)
- [T6] Quantum rendering/simulation productized within 3–5 years — trade-press forecast (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** QRNG-seeded generative art/PCG runs in toy form — a novelty resource, no performance advantage (classical PRNGs are indistinguishable for entertainment).
- **Promise:** hybrid quantum-classical PCG subroutines for variability.
- **Hype:** quantum rendering, quantum recommendation engines, quantum-accelerated content pipelines — no deployed pilots.

HONEST ASSESSMENT

This is a curiosity node. The QRNG-as-creativity angle is real and shipped in toy form, and it is a novelty resource rather than a performance advantage — classical PRNGs are indistinguishable for entertainment purposes. Rendering, recommendation, and ad optimization are the usual optimization/ML pitches with no media-specific advantage and no deployed pilots. No studio or platform runs quantum hardware in production. Keep this node for completeness; its honest weight is low and mostly T3-demo plus T6-forecast.

SOURCES

- <https://medium.com/qiskit/introducing-procedural-generation-using-quantum-computation-956e67603d95>
- <https://arxiv.org/pdf/2312.13853> (Quantum Wave Function Collapse for PCG)
- <https://arxiv.org/html/2508.09683v1> (PCG at the dawn of FTQC)
- <https://www.telefonicaserviciosaudiovisuales.com/en/audiovisual-divulging-articles/quantum-computing-and-its-impact-on-the-media-and-entertainment-industry/>

Quantum in Mining & Mineral Exploration · I-mining

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from I-extractive ; mining's story is sensing-led and distinct from oil & gas reservoir compute)

THE PITCH

Mining's quantum story is led by **sensing**, not computing — the opposite balance from oil & gas (I-extractive). Cold-atom **gravimeters** and NV-diamond **magnetometers** (A-gravimetry / A-magneto / A-nvsensing) detect ore bodies and density anomalies from the air, sharpening exploration for critical minerals (copper, lithium, nickel, cobalt, rare earths) needed for the energy transition. Secondary compute threads: mine-plan/haulage optimization and mineral-processing chemistry (froth flotation, leaching catalysts).

REAL ACTIVITY (NAMED, DATED)

- **SBQuantum + Silicon Microgravity — QUAMINEX** (announced March 2024) — a drone-mounted system fusing a **diamond quantum vector magnetometer** with a MEMS gravimeter to map underground deposits in 3D. Funded by Canada (CAD \$500k) and the UK (£414k). SBQuantum's magnetometer is miniaturized to a handheld <1 lb, <4 W device; claims ≥30% better resolution than the current magnetic-mapping standard.
- **SBQuantum** (2025) — also pursuing defense/aerospace magnetic navigation; a June 2025 Metal Tech News update tracks continued exploration-tech progress.
- **Rio Tinto** and other majors — using advanced geophysical exploration platforms; quantum sensors are entering the toolchain as a nearer-term option than quantum compute.
- **Mine-optimization compute** — QUBO haulage/scheduling and processing-chemistry pilots exist at the same small-instance, no-advantage stage as I-logistics / I-chem .

KEY GRADED CLAIMS

- [T3/T4] Drone-based diamond-magnetometer + gravimeter mineral mapping, ≥30% resolution gain — SBQuantum/Silicon Microgravity, 2024 (demonstrated dev program; resolution claim vendor-stated)

- [T4] Handheld <1 lb diamond quantum magnetometer — SBQuantum (vendor product spec)
- [T3] QUBO mine-plan/processing optimization — method papers (toy scale, no advantage)
- [T5] Critical-minerals quantum-exploration market — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** diamond quantum magnetometers and cold-atom gravimeters are real, deployable field instruments — mining's genuine near-term quantum tech.
- **Promise:** drone-fused magneto+gravity 3D exploration (QUAMINEX) at production scale, quantum processing-chemistry.
- **Hype:** "quantum computers optimize the mine" — the compute side is the same small-instance no-advantage story as logistics.

HONEST ASSESSMENT

Mining is a sensing-led vertical, and that is what makes it credible. Diamond quantum magnetometers (`A-nvsensing`) and cold-atom gravimeters are real instruments with a plausible short horizon for critical-minerals exploration — a strategically important use given lithium/copper/rare-earth demand. The QUAMINEX program is dated, government-funded, and concrete, though the resolution numbers are vendor-stated and the systems are still commercializing (niche, expensive). Quantum *computing* for mine planning and mineral-processing chemistry is at the small-instance no-advantage stage. Split the story: sensing has a credible near path; compute is a **2030s** bet.

SOURCES

- SBQuantum + Silicon Microgravity QUAMINEX: <https://thequantuminsider.com/2024/03/01/sbquantum-and-silicon-microgravity-partner-to-accelerate-mining-exploration-using-quantum-sensing-with-funding-from-uk-canada/>
- SBQuantum diamond magnetometry: <https://magneticmag.com/sbquantum-pioneers-quantum-magnetometry-for-defense-aerospace-minerals-exploration/>
- Canadian Mining Journal "Quantum technologies in mining exploration"
- Metal Tech News "Quantum leap in mineral exploration tech" (June 2025)

Quantum in Nuclear & Fusion · I-nuclear

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from I-energy ; fusion/nuclear is deep enough to stand alone)

THE PITCH

Three threads. (1) **Fusion plasma** — magnetized-plasma dynamics in tokamaks/stellarators is a quantum many-body and stiff-PDE problem; quantum simulation and QAOA-for-stability-optimization are pitched to accelerate confinement modeling and coil design. (2) **Nuclear structure & reactions** — lattice/nuclear-physics Hamiltonians are a native quantum-simulation target (the same S-qsim logic as chemistry, at the nuclear scale). (3) **Fission engineering** — reactor core optimization, fuel-loading, and materials-under-irradiation modeling as QUBO/chemistry problems.

REAL ACTIVITY (NAMED, DATED)

- **Frontiers in Physics** (March 2025) — *The role of quantum computing in advancing plasma physics simulations for fusion energy and high-energy physics* — a substantive review; QAOA is applied to optimize tokamak plasma-stability criteria, Hamiltonian-simulation to plasma dynamics.
- **PPPL (Princeton Plasma Physics Lab)** — 2025 Kaul Foundation Prize to a team optimizing 3D magnetic fields to control tokamak edge instabilities; the broader lab explores quantum-computing methods for plasma. HPC-classical today, quantum-curious.
- **Nuclear-structure quantum simulation** — VQE/quantum-simulation demos of light-nucleus (deuteron, ^4He -scale) binding energies on superconducting/trapped-ion hardware (Oak Ridge and others, ongoing since ~2018) — small but genuine physics results.
- **Fusion startups** (Commonwealth Fusion, TAE, Tokamak Energy) and **ENN** (proton-boron) run heavy classical HPC plasma simulation; quantum is on watch-lists, not in the loop.

KEY GRADED CLAIMS

- [T2/T3] VQE/quantum-simulation of light-nucleus binding energies on real hardware — nuclear-physics literature (demonstrated at small scale; landmark physics, not engineering advantage)
- [T3] QAOA for tokamak plasma-stability optimization / Hamiltonian-sim for plasma — Frontiers 2025 review (method framing + small demos)
- [T4] Quantum-accelerated fusion reactor design — vendor/lab aspiration (roadmap)
- [T5] Fusion/nuclear quantum TAM — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** quantum computers have simulated small nuclei (deuteron-scale) — real nuclear-physics results, no engineering payoff.
- **Promise:** plasma-stability optimization and full plasma dynamics — strong theoretical fit, no advantage over classical HPC yet.
- **Hype:** "quantum computing will crack fusion" — fusion's bottleneck is engineering and confinement, and its modeling is elite classical HPC.

HONEST ASSESSMENT

Nuclear/fusion has the best *physics* pedigree of any young L4 node — simulating nuclear and plasma Hamiltonians is exactly what quantum computers are for, and the deuteron/light-nucleus results are genuine science. It also has the widest gap to engineering value: fusion plant design runs on world-class classical supercomputing (M3D-C1, JOEKE, NIMROD), and quantum methods are at the review-and-small-demo stage. Plasma-stability QAOA is a promising formulation without demonstrated advantage. Realistic engineering value: **2030s+**, gated on fault tolerance. Near-term, the honest role is accelerating specific nuclear/plasma sub-Hamiltonians as a research tool.

SOURCES

- Frontiers in Physics "Role of quantum computing in advancing plasma physics simulations for fusion energy": <https://www.frontiersin.org/journals/physics/articles/10.3389/fphy.2025.1551209/full>
- PPPL 3D-field / Kaul Prize: interestingengineering.com/energy/3d-magnetic-field-breakthrough-nuclear-fusion

- Nuclear-structure VQE (deuteron) — Oak Ridge quantum-simulation literature
- AZoQuantum "Tokamak Reactors and the Key to Nuclear Fusion"

Quantum in Pharma & Healthcare · I-pharma

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Drug discovery is chemistry, and chemistry is quantum — simulating molecular ground/excited states, protein folding, and binding affinities is the canonical Feynman-era pitch (see `S-qsim`). Quantum promises to model molecules classical methods approximate poorly, shrinking discovery cycles. Pharma is a top-three enterprise adopter. This node covers **drug discovery / molecular simulation**; healthcare *imaging and diagnostics* (a nearer-term quantum-sensing story) is split into `I-healthimaging`.

REAL ACTIVITY (NAMED, DATED)

- **Cleveland Clinic + IBM** — "Discovery Accelerator," first on-site private-sector quantum computer (IBM System One at Lerner Research Institute, live 2023). 50+ joint projects; Heron chip upgrade + HPC coupling Dec 2025. **Algorithmiq + Cleveland Clinic + IBM won the \$2M Wellcome Leap Q4Bio prize** (~April 2026) for simulating photodynamic-therapy processes on up to 100 qubits — the most concrete pharma-quantum result to date.
- **IBM "quantum-centric supramolecular interactions"** (arXiv 2410.09209, Oct 2024) — accurate simulation of supramolecular systems on Heron; a refereeable capability demo, still small-molecule scale.
- **Boehringer Ingelheim** — formal quantum program since 2021, partnered **Google Quantum AI** for molecular-dynamics work.
- **AstraZeneca + IonQ** (via AWS/NVIDIA, published 2025) — reported **~20x speedup** on a drug-discovery workflow subroutine; vendor-framed.
- **Qubit Pharmaceuticals + Pasqal** — drug-discovery tasks on neutral-atom hardware. **Menten AI + D-Wave** — peptide design via hybrid annealing. **Roche pRED + QC Ware; Moderna + IBM** (mRNA structure).
- **Quantinuum Helios** (48 logical qubits, late 2025) — approaching the scale where quantum chemistry begins to *challenge* classical on specific systems (~20 active orbitals), per 2026 pharma reviews.

KEY GRADED CLAIMS

- [T2/T3] Photodynamic-therapy ground/excited-state simulation on up to 100 qubits (Q4Bio) — Algorithmiq/Cleveland Clinic/IBM (competitively judged; demonstrates a *path*, not advantage over DFT/coupled-cluster)
- [T3] Quantum-centric supramolecular simulation — IBM, arXiv 2410.09209 (preprint capability demo)
- [T4] AstraZeneca/IonQ 20x workflow speedup — vendor+pharma announcement (narrow subroutine)
- [T4] Boehringer/Google, Qubit/Pasqal, Menten/D-Wave pilots — partnership PRs (exploratory)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** quantum hardware can now simulate ~20-orbital molecules accurately — matching what classical methods already do well.
- **Promise:** photodynamic-therapy chemistry, peptide design, binding affinity — real refereeable demos at 100-qubit scale, still short of design-grade accuracy on hard targets (FeMoco, cytochrome P450).
- **Hype:** "quantum discovered a drug," end-to-end "20x faster pipelines," near-term clinical impact.

HONEST ASSESSMENT

No approved drug has been discovered by a quantum computer as of mid-2026, and none is close — a fact 2026 pharma reviews state plainly (no quantum computer yet simulates a drug-relevant molecule better than classical chemistry on the same system). The Q4Bio work is the strongest signal: real chemistry circuits at 100-qubit scale, competitively judged. The "20x" claims are narrow subroutines. Pharmaceutically useful accuracy on hard active sites needs fault tolerance (millions of gates). Realistic operational impact: **late 2020s into the 2030s**. Pharma keeps investing because the eventual payoff is enormous.

SOURCES

- IBM Quantum healthcare: <https://newsroom.ibm.com/2026-04-16-how-ibm-quantum-is-enabling-healthcare-and-biology-research>
- Supramolecular interactions: <https://arxiv.org/pdf/2410.09209>
- Quantum-machine-assisted drug discovery, npj Drug Discovery: <https://www.nature.com/articles/s44386-025-00033-2>

- Quantum Computing Watch List for Pharma R&D Leaders 2026 (Sakara Digital)
- Cleveland Clinic Discovery Accelerator: <https://my.clevelandclinic.org/research/computational-life-sciences/discovery-accelerator>

Quantum in Retail · I-retail

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Retail's pitch is demand forecasting, dynamic pricing, assortment and shelf-space optimization, inventory/replenishment, and recommendation. All reduce to optimization (QUBO/QAOA over combinatorial assortment and stocking) or ML (quantum-enhanced forecasting, quantum neural networks). The claimed payoff is fewer stockouts and less overstock across huge SKU catalogs — a combinatorial space classical solvers already handle well, which is the tension.

REAL ACTIVITY (NAMED, DATED)

- **Strangeworks / Quantagonia** (Aug 2025) — Strangeworks acquired Quantagonia to fold quantum-enhanced demand forecasting and decision-making into its platform. Vendor consolidation, not a named retailer deployment.
- **Method research** — *Exploring Quantum Neural Networks for Demand Forecasting* (Entropy/MDPI, 2025, vol 27:490): QNN forecasting on benchmark datasets, no advantage over classical shown.
- **Analyst-heavy** — most visible activity is market-report noise ("Quantum-Enhanced Demand Forecasting" TAM projected ~\$2B (2025) → ~\$2.64B (2026), ~32% CAGR). T5 forecasts, not pilots.
- Named marquee retail deployments (Walmart/Amazon/Kroger running a QPU in the loop) do not exist in the public record; retail quantum work sits inside logistics (**I-logistics**) and finance-style optimization pilots.

KEY GRADED CLAIMS

- [T3] Quantum neural networks tested for demand forecasting on benchmark data — MDPI Entropy 2025 (demonstrated on toy data; no advantage vs classical)
- [T4] Quantum-enhanced forecasting productized (Strangeworks/Quantagonia) — vendor consolidation announcement (claimed)

- [T5] "Quantum-enhanced demand forecasting" multi-billion-dollar market — GII/Research&Markets forecasts (speculative; double-counted with logistics/finance, inflation-unadjusted)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** nothing operational — QNN forecasting demos on benchmark data, matched by classical ML.
- **Promise:** assortment/pricing optimization at catalog scale — high bar because classical MILP is strong and cheap.
- **Hype:** the TAM reports themselves — this node is where market-sizing noise outweighs any real pilot.

HONEST ASSESSMENT

Retail is one of the thinner industry pitches. Demand forecasting and assortment are problems where classical ML and modern MILP solvers are strong and cheap, so the quantum bar for advantage is high and unmet. The visible "activity" is dominated by market-sizing reports and one vendor acquisition — a T5-heavy corner with essentially no retailer pilots. No retailer runs quantum hardware operationally. Realistic value is far off and, even then, likely marginal versus classical optimization. Grade the TAM numbers hardest here.

SOURCES

- <https://www.giiresearch.com/report/tbrc1982975-quantum-enhanced-demand-forecasting-global-market.html>
- <https://www.researchandmarkets.com/reports/6177364/quantum-enhanced-demand-forecasting-market-report>
- <https://www.mdpi.com/1099-4300/27/5/490> (QNN demand forecasting)
- <https://www.grandviewresearch.com/industry-analysis/quantum-computing-market> (retail segment CAGR)

Quantum in Semiconductors & EDA · I-semiconductor

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — genuinely new, non-duplicative)

THE PITCH

Two directions, both real. (1) **Quantum FOR chips** — quantum computing applied to semiconductor design: electronic-structure simulation of new device materials (high-k dielectrics, 2D channels, photoresists), computational lithography, and QUBO placement/routing in EDA. (2) **Chips FOR quantum** — the semiconductor industry is quantum's manufacturing base: foundries fabricate transmon and spin-qubit chips, and a "fabless quantum chip" model is emerging (design house + foundry). This node covers the *application* direction; the fab/supply-chain physics lives in **H-fab / H-silicon**, cryo-CMOS in **H-cryocmos**.

REAL ACTIVITY (NAMED, DATED)

- **NVIDIA + TSMC** (GTC 2025) — the honest near-term truth is *classical* acceleration: **cuLitho** (20–50% better cost/cycle-time on computational lithography) and **cuEST** (~50x faster electronic-structure chemistry for materials). This is GPU HPC, and it is the bar quantum must clear in this vertical.
- **Q-EDA research stack** — EDA-Q, KQCircuits, QPDK, GDSII-to-wafer flows (arXiv 2606.17956, "Fabless Quantum Chip Design and Commercial Production," 2026) demonstrate automated topology design, layout, DRC/LVS/DFM, and mask-data prep — for designing *quantum* chips, using classical EDA methods.
- **Materials simulation pilots** — Mitsubishi Chemical + Xanadu (semiconductor materials), IBM/Azure Quantum Elements chemistry platforms pitched to fabs and materials houses.
- **Infineon, imec, TSMC** — running spin/superconducting qubit process R&D that doubles as advanced-node process learning.

KEY GRADED CLAIMS

- [T4] Classical GPU acceleration of lithography/materials (cuLitho/cuEST) — NVIDIA/TSMC, 2025 (vendor; not quantum, sets the bar)

- [T3] Automated quantum-chip EDA flow (topology→mask) — arXiv 2606.17956 (preprint; classical EDA for quantum chips)
- [T4] Quantum electronic-structure simulation for device materials — vendor pilots (exploratory)
- [T5] Semiconductor-design quantum TAM — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** classical GPU HPC (cuLitho/cuEST) accelerates real fab workflows now — this is the competition, not quantum.
- **Promise:** quantum simulation of next-node device materials and QUBO place-and-route — exploratory, fault-tolerance-gated.
- **Hype:** "quantum computers design chips today."

HONEST ASSESSMENT

The semiconductor industry is central to quantum as its *fab base* — that part is real and load-bearing (**H-fab** , **H-silicon** , **H-cryocmos**). Quantum computing *applied to* chip design is early: materials-simulation pilots inherit **I-chem** 's fault-tolerance wall, and EDA placement/routing is QUBO optimization that classical solvers dominate. The most concrete 2025–26 wins in "AI/compute for semiconductors" are GPU-classical (NVIDIA cuLitho/cuEST), which is a useful reality check on where the value actually flows. Realistic quantum-for-design value: **2030s**, gated on fault tolerance.

SOURCES

- NVIDIA + TSMC bring AI into fabs (cuLitho/cuEST): <https://nvidianews.nvidia.com/news/nvidia-and-tsmc-bring-ai-into-fabs-to-advance-semiconductor-design-and-manufacturing>
- "Fabless Quantum Chip Design and Commercial Production": <https://arxiv.org/pdf/2606.17956>
- DCD "Quantum of promise: How to build a quantum chip"
- Mitsubishi Chemical/Xanadu semiconductor materials (IBM case studies)

Quantum in Space & Earth Observation · I-space

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Space is a natural home for quantum *sensing and communication* more than computing: satellite-based QKD (A-satqkd) for global secure comms, ultra-stable atomic clocks (A-clocks) for PNT and constellation timing, quantum magnetometers/gravimeters and atom interferometers for GPS-free spacecraft navigation and Earth observation, plus optimization for orbit/constellation/mission planning. The harsh, GPS-independent environment rewards quantum's absolute-reference advantages.

REAL ACTIVITY (NAMED, DATED)

- **ESA ACES** (Atomic Clock Ensemble in Space) — installed on the ISS **April 2025**; the most accurate clock ensemble ever flown, targeting fractional stability $\sim 1 \times 10^{-16}$, comparing on-board cesium (PHARAO) + H-maser clocks to ground stations — a step toward relativistic geodesy and SI-second work (A-timedist).
- **Boeing Q4S** — space-qualified quantum-networking payload passed environmental qualification and entanglement-swapping ground tests; orbital demo planned. Boeing/Space Force flew "the highest-performing quantum inertial sensor in space" on X-37B (**Aug 2025**, see I-aerospace).
- **University of Vienna** (2025) — launched a compact **photonic quantum computer into orbit** (~ 10 W, ~ 3 L) — a demonstration that quantum hardware survives launch/space, not a useful computation.
- **China Micius** — the landmark satellite-QKD platform (A-satqkd), with a follow-on constellation program.
- **NASA CAL** (Cold Atom Lab, ISS) — BEC/atom-interferometry science platform underpinning future space gravimetry.

KEY GRADED CLAIMS

- [T2/T3] ACES $\sim 1 \times 10^{-16}$ optical-clock-class stability in orbit — ESA, 2025 (demonstrated hardware milestone)

- [T3] Space-qualified quantum-networking payload (entanglement swapping) — Boeing Q4S (demonstrated ground test; orbital pending)
- [T3] Photonic quantum computer operates in orbit — U. Vienna 2025 (demonstrated survival, not useful compute)
- [T5] Space-quantum market forecasts — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** ACES clocks, quantum inertial sensors, and satellite QKD are flying and returning data now — mostly A-layer sensing/timing/comms.
- **Promise:** operational satellite quantum networking (Q4S), space gravimetry for Earth observation.
- **Hype:** quantum computers doing orbit/constellation optimization in space — classical HPC handles it fine.

HONEST ASSESSMENT

Space quantum is real where it is sensing/timing/comms and aspirational where it is computing. ACES, quantum inertial sensors, and satellite QKD fly and return data now — meaningful, near-term, and mostly A-layer tech rather than L4 compute. Orbit/constellation optimization by quantum computers is speculative and well-served by classical HPC. The Vienna orbital "quantum computer" is a survivability demo. Split cleanly: quantum sensors and clocks are arriving in space this decade; quantum computing in/for space is a long-horizon bet. As with aerospace, the near-term space win is a sensor.

SOURCES

- ESA "Europe's quantum decade extends into space": https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Europe_s_quantum_decade_extends_into_space
- Quantum sensing for NASA science missions (PMC): <https://pmc.ncbi.nlm.nih.gov/articles/PMC12095411/>
- VisionSpace "Quantum Horizons: Redefining Satellite Operations in 2025"
- aero-defence.tech Boeing Q4S payload integration

Quantum in Telecom & Networking · I-telecom

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth

THE PITCH

Telecom's quantum story is mostly about being the *substrate* for quantum-secure communication — deploying QKD (A-qkd) and post-quantum crypto (A-pqc) across metro and backbone networks, plus longer-term ambitions to build the quantum internet (A-qinternet). Secondary threads: network-optimization and radio-resource scheduling. Carriers monetize "quantum-safe-as-a-service." Two tracks run at very different speeds: PQC is software crypto shipping now; QKD is hardware, deployed but niche.

REAL ACTIVITY (NAMED, DATED)

- **BT** — deployed the **UK's first commercial quantum-secured metro network** in London (2022, with Toshiba; customers included HSBC, EY); trialing long-distance and satellite QKD.
- **SK Telecom** — launched a **hybrid QKD-PQC** encryption solution (Oct 2024) for 5G backbone/VPN/data centers with Nokia and ID Quantique at Equinix Seoul; partnered IonQ (Feb 2025) to embed trapped-ion compute into AI platforms.
- **Verizon** — PQC migration focus; participating in NIST PQC standardization; early QKD experiments.
- **NTT** — Quantum Datacenter Alliance member (2025); long-standing quantum research arm; IOWN all-photonics network program.
- **Telefónica, Orange, KT** — national QKD testbeds (EuroQCI-aligned in Europe).
- Industry-wide: by end-June 2025, ~29 operators had made ~75 quantum-related statements/deployments.

KEY GRADED CLAIMS

- [T3/T4] Commercial quantum-secured metro network live with enterprise customers — BT/Toshiba (demonstrated deployment)
- [T3/T4] Hybrid QKD-PQC service in production data centers — SK Telecom/Nokia/ID Quantique, Oct 2024 (demonstrated)

- [T2] PQC (ML-KEM/ML-DSA) is the NIST-standardized path most carriers actually adopt — see [A-pqc](#) (established standards)
- [T5] "Quantum-safe telecom" TAM projections — forecast

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** PQC migration (classical software crypto) is real, mandated, and underway across carriers now.
- **Promise:** commercial QKD links (BT, SKT) — genuine but small, expensive, distance-limited footprints.
- **Hype:** "the quantum internet is here," QKD as a general-purpose replacement for PQC (security agencies disagree).

HONEST ASSESSMENT

Telecom is a two-track reality. PQC migration is real and happening — software crypto, no quantum hardware, where carriers spend seriously. QKD is deployed and niche — expensive, distance-limited, point-to-point, and NSA/UK NCSC prefer PQC over QKD for general use. The BT/SKT deployments are genuine but small commercial footprints. The quantum internet is a research decade away. Telecom's honest near-term quantum value is defensive crypto, mostly the classical-software PQC kind.

SOURCES

- Lightreading "2025 in review: Telecom gets entangled with quantum"
- Bain "How Telecom Carriers Are Preparing for Quantum"
- postquantum.com telecom use cases
- The Quantum Insider "Telecom at the Edge of Scale" (Oct 2025)

Quantum in Weather & Climate Modeling · I-weather

Layer: L4 Industries · **Chapter:** §05 · **Status:** depth **Added:** 2026-07-08 (cycle 3 random-walk — split from I-climate , which is chemistry/carbon-capture)

THE PITCH

Distinct from I-climate (quantum *chemistry* for carbon capture/fertilizer), this node is about **numerical weather prediction (NWP) and climate dynamics** — the fluid-dynamics side. Weather/climate models solve Navier-Stokes and radiative-transfer PDEs on enormous grids; the pitch is quantum linear-solvers (HHL, S-hhl) for the linear-algebra core, quantum algorithms for turbulent/CFD flow, and QML for pattern-finding in satellite/sensor data (extreme-event and precipitation nowcasting).

REAL ACTIVITY (NAMED, DATED)

- **Open Quantum Institute (OQI, at CERN, pilot phase 2024–2026)** — runs a **"Weather and Climate Forecasting" project** whose core is a *quantum fluid-dynamics* solver to improve forecast reliability. Multi-institution, pre-commercial, the most concrete organized effort.
- **arXiv 2502.10488** (Feb 2025) — *Opportunities and challenges of quantum computing for climate modelling* — a serious survey mapping HHL/QLSA and Hamiltonian-simulation methods onto dynamical cores, with honest caveats on data loading (S-qram) and readout.
- **arXiv 2509.01422** (2025) — *Exploring Quantum Machine Learning for Weather Forecasting* — QNNs for wind-speed prediction on benchmark data; potential-outperformance framing, small scale.
- **BAMS 2023** — *Quantum Computers for Weather and Climate Prediction: The Good, the Bad, and the Noisy* — the reference sober review from the meteorology community.

KEY GRADED CLAIMS

- [T3] Quantum linear-solver / Hamiltonian-simulation methods mapped onto NWP dynamical cores — arXiv 2502.10488 (survey; no end-to-end advantage)
- [T3] QNN wind-speed/precipitation nowcasting on benchmark data — arXiv 2509.01422 (toy-scale demonstration)

- [T4] OQI quantum fluid-dynamics forecasting pilot — CERN/OQI (exploratory, pre-commercial)
- [T5] Quantum-weather market projections — analyst (speculative)

PROVEN TODAY VS PROMISE VS HYPE

- **Proven:** nothing operational — surveys and toy-scale QNN nowcasting demos, matched by classical ML.
- **Promise:** quantum linear-solvers for the PDE core and quantum CFD — real theoretical hooks, blocked by data-loading and readout bottlenecks.
- **Hype:** "quantum will fix weather forecasting" — today's NWP is world-class classical HPC that quantum cannot yet touch.

HONEST ASSESSMENT

Weather/climate modeling is one of the harder places to get a quantum win: the HHL-style speedup for linear systems is real on paper but dies on the input/output problem (loading a petabyte-scale state and reading out a full field), which the field's own 2025 surveys flag plainly. NWP is a mature, heavily-optimized classical-HPC discipline (ECMWF, NOAA), so the bar is very high. The credible near-term thread is QML *nowcasting* on sensor/satellite data, which is small and classically matchable. The OQI pilot is the honest bright spot: organized, dated, and explicitly pre-commercial. Realistic value: **2030s+**, gated on fault tolerance and a qRAM-class data-loading solution.

SOURCES

- OQI Weather and Climate Forecasting: <https://open-quantum-institute.cern/project/weather-and-climate-forecasting/>
- Opportunities and challenges of quantum computing for climate modelling: <https://arxiv.org/pdf/2502.10488>
- Exploring Quantum ML for Weather Forecasting: <https://arxiv.org/html/2509.01422v1>
- BAMS "Quantum Computers for Weather and Climate Prediction: The Good, the Bad, and the Noisy" (2023)

§06 · Ecosystem & Geopolitics

National programs, money, standards, talent, patents, export controls — and the numbers, graded for inflation.

China's parallel software stack — fork & soft-power risk · E-china-stack

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk, field-flagged)

SUMMARY

China runs a **full parallel quantum-software stack** that mirrors the Qiskit/Cirq/PennyLane ecosystem (E-oss) end to end, so a Chinese developer, chip, and cloud can operate without touching any US-governed toolchain. The anchors: **Origin Quantum** (Hefei) with **QPanda** (programming framework), **Origin Pilot** (a quantum operating system, open-sourced Feb 2026), **ChemiQ** (chemistry), and **Tianji** (measurement-and-control, v4.0 supporting 500+ qubits, May 2025); **Baidu's Paddle Quantum** (quantum-ML on the PaddlePaddle DL framework); and **Huawei's HiQ** (simulator + algorithm library on Huawei Cloud). This is the software counterpart to the hardware story in E-china. The strategic point is a **fork risk**: whoever's framework developers learn first shapes which hardware they later reach for, and export controls (E-export) make convergence on one global toolchain less likely each year. Origin is notable as arguably the only company anywhere building the *entire* stack — processor, cryogenics, control system (Tianji), OS (Origin Pilot), framework (QPanda), and cloud — in-house.

THE LANDSCAPE (GRADED)

- [T2] **Origin QPanda** is China's leading open-source quantum programming framework; Origin operates the Wukong superconducting machine on a public cloud — company/press (established)
- [T3] **Origin Pilot** (quantum OS for multi-task scheduling / resource management) was **open-sourced in February 2026** — a deliberate soft-power move to seed adoption around Chinese hardware — postquantum.com (reported)
- [T3] **Tianji 4.0** measurement-and-control system supports 500+ qubits and reduces PhD-level bring-up to standard engineering workflows (May 2025) — Origin (claimed; vendor)
- [T2] **Baidu Paddle Quantum** — open-source QML toolkit on PaddlePaddle; Baidu built the 36-qubit Qianshi processor (2023) before

donating its quantum lab to BAAI (2024) — Baidu docs / press

(established). The software outlived the hardware program.

- [T2] **Huawei HiQ** — cloud quantum simulator + algorithm library integrated with Huawei Cloud — Huawei (established)

KEY GRADED CLAIMS

- [T3] The Chinese stack is a **genuine fork**, not a veneer over Western tools — QPanda/Paddle Quantum/HiQ have independent APIs and do not depend on Qiskit/Cirq (reported)
- [T3] **Soft-power play**: open-sourcing Origin Pilot/QPanda mirrors IBM's Qiskit funnel strategy — free tooling as a channel to Chinese cloud + hardware, aimed at Belt-and-Road / non-aligned markets (analysis)
- [T4] Interop with OpenQASM/QIR (E-oss) exists partially, but a determined bifurcation along the export-control line would strand cross-stack portability (claimed)

CONFLICTS / OPEN QUESTIONS

- **Does the world converge or fork?** OpenQASM/QIR portability (E-oss) vs export-control-driven decoupling (E-export). Resolution: whether Chinese and Western clouds keep honoring each other's IRs through 2027–28.
- Corporate retreat paradox: Baidu/Alibaba exited *hardware* (E-china, E-patents) yet their *software* (Paddle Quantum) persists — soft power can outlast the lab that spawned it.
- Adoption outside China is unmeasured — GitHub stars and cloud-usage numbers for QPanda/HiQ are not independently audited.

SOURCES

- <https://postquantum.com/quantum-computing/china-quantum-os-origin-pilot/>
- <https://postquantum.com/quantum-computing-companies/origin-quantum/>
- <https://thequantuminsider.com/2026/05/15/10-plus-companies-leading-the-quantum-technologies-race-in-china/>
- <https://entangledfuture.com/guides/quantum-computing-in-china/>
- <https://quantumzeitgeist.com/china-quantum-computing-companies/>

National programs — China · E-china

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

China runs the largest state-directed quantum program in the world, centered on **USTC (Hefei)**, the CAS system, and the **National Laboratory for Quantum Information Sciences** (Hefei; construction reported in the ~\$1B+ range). Signature results: the **Micius** QKD satellite (2016), the **Jiuzhang** photonic-advantage series (now Jiuzhang 4.0, 3,050 photons), and the **Zuchongzhi** superconducting line (Zuchongzhi 3.0, 105 qubits, 2025). The famous "\$15B commitment" is the least verifiable headline number in the field — cited for years with no primary source, with insiders putting real spending at a fraction of it. The **15th Five-Year Plan** (adopted early 2026) elevated quantum to the top-ranked future industry, ahead of AI and semiconductors. (China's parallel *software* stack — Origin/QPanda, Baidu Paddle Quantum, Huawei HiQ — is broken out as its own node, [E-china-stack](#).)

THE NUMBERS (GRADED, WITH CAVEATS)

- [T5] "\$15B+" (often \$15.3B) — the perennial headline commitment. No primary Chinese budget document supports it; postquantum.com traces it to circular citation. A USTC physicist estimated actual money at "maybe 25 percent of that" (~\$3.75–5B). Treat as an upper bound on rhetoric, not a budget line.
- [T5] >\$1B — reported investment in the Hefei National Laboratory for Quantum Information Sciences (CSIS / secondary reporting). Construction-cost figures and research budgets get conflated.
- [T5] ~\$17.5B — estimated National Guidance Venture Fund backing quantum under the 15th FYP (postquantum.com, 2026). The fund spans multiple deep-tech sectors; quantum's slice is unknown — clear double-counting risk.
- [T5] \$15B in 2024 alone vs ~\$7B combined US public+private — Asia Times framing (May 2026); methodology opaque, likely folds provincial + venture-guidance money.

KEY GRADED CLAIMS

- [T2] Micius demonstrated space-to-ground QKD and entanglement distribution — Science/Nature (established). Mar 2025: the **Jinan-1** microsatellite sent a quantum-encrypted image ~13,000 km China→South Africa (USTC) — the longest intercontinental QKD link reported (T3).
- [T3] Jiuzhang 4.0 claims sampling advantage $\sim 10^{54}$ over classical — USTC (claimed; boson-sampling advantage claims have historically eroded under classical spoofing — see S-bench)
- [T2] Quantum named a frontier priority in the 14th FYP; top-ranked future industry in the 15th FYP — plan texts (established)
- [T4] Program secrecy: military-adjacent work (sensing, comms) is unpublished, so open-source estimates undercount capability and possibly overcount civilian spend (claimed)

CONFLICTS / OPEN QUESTIONS

- **C-china-spend:** \$15B headline vs ~25% insider estimate. Resolution: an audited budget line, which will likely never surface.
- How much of Jiuzhang/Zuchongzhi advantage survives classical counterattack (see O-advantage)?
- **Corporate retreat vs state ascent:** Baidu **donated** its quantum lab to BAAI and **Alibaba shuttered** its DAMO quantum effort (2023–24, equipment to Zhejiang University) — state labs and Origin Quantum now carry nearly all momentum. This is the inverse of the US private-capital surge (see E-vc).
- China is building domestic dilution-fridge, laser, and cryo supply to route around export controls (see E-export, E-supplychain).

SOURCES

- <https://postquantum.com/china-quantum-ambition/china-quantum-investment-unknowable/>
- <https://asiatimes.com/2026/05/us-china-escalate-quantum-race-with-rival-investment-drives/>
- <https://www.uscc.gov/research/vying-quantum-supremacy-us-china-competition-quantum-technologies>
- <https://postquantum.com/china-quantum-ambition/china-15th-five-year-plan-quantum/>

- <https://www.csis.org/analysis/understanding-chinas-quest-quantum-advancement>
- <https://www.nature.com/nature-index/news/ten-best-countries-for-quantum-physics-research>

Defense funding channel — DARPA QBI & procurement · E-defense

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Defense is a distinct funding channel from the civilian national program (E-us), and its centerpiece is **DARPA's Quantum Benchmarking Initiative (QBI)** — a reframing of the earlier US2QC program whose stated question is whether an **industrially-useful quantum computer is buildable by 2033** ("computational value exceeds its cost"). QBI's design move is unusual: DARPA **does not fund development directly**; it runs an interagency verification-and-validation team to separate hype from reality across every architecture, evaluating ~20 companies since mid-2024. This makes it as much a **national due-diligence instrument** as a grant program — a defense-run reality check the whole field cites. The June 2026 EOs added a second defense-adjacent thread by renaming DoD as the "Department of War" in tasking language and expanding the Quantum Counterintelligence Protection Team.

THE MECHANISM & NUMBERS (GRADED)

- [T2] QBI operates in three stages — A (concept), B (yearlong R&D plan + risk-reduction prototypes), C (government V&V build feasibility) — darpa.mil (established)
- [T2] Funding ladder: ~\$1M (Stage A concept) → up to ~\$15M (Stage B roadmap) → up to ~\$300M (Stage C build/demo) — DARPA (established)
- [T2] As of **6 Nov 2025**, **11 companies advanced to Stage B**; ~20 evaluated across modalities since mid-2024 — DARPA / Quantum Insider (established)
- [T3] Named performers include Quantinuum (Stage B), with Microsoft and PsiQuantum in the final US2QC phase and Google joining Stage A (Sep 2025) — company/DARPA releases (claimed/reported)

KEY GRADED CLAIMS

- [T2] QBI's premise is **verification, not development funding** — its output is independent feasibility assessment, deliberately vendor-neutral — DARPA (established)

- [T4] Defense **procurement of operational quantum systems** — essentially none yet; the channel today funds evaluation and prototypes, not fielded machines (claimed/absent). The nearest-term fielded quantum tech for defense is sensing/PNT (A-sensing, A-gravimetry), not compute.
- [T5] "Industrially useful by 2033" is DARPA's framing question, not a prediction — a benchmark date, not a roadmap (forecast)

CONFLICTS / OPEN QUESTIONS

- **QBI is a filter, not a bet** — it explicitly is "not a competition," so being cut from a stage signals DARPA's V&V team found the concept unconvincing: a rare public *negative* signal in a hype-heavy field. How much weight the market gives those cuts is open, but it is one of the few credible independent screens (contrast the equity-picking turn in E-us/E-sovereign).
- The broader defense channel (NSA/DoD codebreaking interest — the original HNDL motive behind PQC, see A-pqc — plus service-lab sensing/PNT) is **opaque and classified**; public numbers understate real defense spend.
- **Double-counting risk:** QBI dollars and IARPA/service-lab programs get folded into "US quantum spending" tallies (E-us) — keep the defense channel separate.

SOURCES

- <https://www.darpa.mil/research/programs/quantum-benchmarking-initiative>
- <https://www.darpa.mil/research/programs/quantum-benchmarking-initiative/stage-b-selection>
- <https://thequantuminsider.com/2025/11/07/darpa-advances-quantum-computing-initiative/>
- <https://www.quantum.gov/darpa-announces-participants-for-the-initial-stage-of-the-quantum-benchmarking-initiative/>
- <https://www.darpa.mil/news/2025/companies-targeting-quantum-computers>

National programs — European Union · E-eu

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The EU's base layer is the **Quantum Technologies Flagship** (2018, €1B over ten years). On top of it sit **EuroQCI** (a quantum-secure communication network across all 27 member states plus ESA's **Eagle-1** satellite), **EuroHPC** quantum-computer procurements co-located with supercomputers, and — since **2 July 2025** — the **Quantum Europe Strategy** (COM(2025) 363), a five-pillar plan to make Europe a "quantum industrial powerhouse" by 2030. A binding **EU Quantum Act** is scheduled for adoption in 2026. Europe's strength is public research and startup density; its acknowledged weakness is private capital and scale-up, which the Act is meant to address. The 2025–26 story is EuroHPC turning procurement pledges into **six machines actually inaugurated on European soil** (detailed in E-sovereign).

THE NUMBERS (GRADED, WITH CAVEATS)

- [T5] €1B / 10 yr — Quantum Flagship (digital-strategy.ec.europa.eu). Frequently *added* to member-state programs in "Europe spends €X" claims — the same euros counted at EU and national level.
- [T5] ~€11B+ — commonly cited combined EU + member-state public commitment (Qureca tallies). Sums multi-year pledges of different vintages/inflation bases; order-of-magnitude only.
- [T5] Member-state headline programs (nationally administered; see E-others for Germany): Germany ~€5B total (€2B stimulus 2020 + €3B action plan 2023), France ~€1.8B national strategy (2021), Netherlands ~€615M Quantum Delta NL.
- [T5] Quantum Act 2026 — "potentially billions" in industrialization incentives; nothing appropriated yet (twobirds.com / EC call for evidence). Likely structured as a Joint Undertaking modeled on EuroHPC.

KEY GRADED CLAIMS

- [T2] All 27 member states signed the EuroQCI Declaration (since June 2019) — EC (established)

- [T2] Quantum Europe Strategy adopted 2 July 2025, five pillars: R&I, infrastructures, ecosystem, space & dual-use, skills — COM(2025) 363 (established)
- [T2] **EuroHPC JU procured six quantum computers** now inaugurating across DE, CZ, FR, IT, PL, ES (2025–26) — multiple platforms (superconducting, trapped-ion, photonic, neutral-atom, annealer) integrated with Tier-0 supercomputers — eurohpc-ju.europa.eu (established; see E-sovereign for the roster)
- [T3] Quantum Act proposal expected 2026 — EC signaling (roadmap)
- [T4] "Europe leads in quantum science, lags in commercialization" — EC's own framing; self-serving for a funding push but consistent with VC data (see E-vc) (claimed)

CONFLICTS / OPEN QUESTIONS

- Will the Quantum Act carry **new money** or restructure existing Horizon Europe funds?
- EuroQCI deployment status per member state is uneven and poorly reported; Eagle-1's launch slipped repeatedly.
- Flagship phase-2 evaluation: what did the first €1B actually produce beyond papers and pilots?
- Europe's scale-up gap is structural — its best startups (IQM, Pasqal, Quandela, Alice&Bob) raise less than US peers and some list via US SPACs (IQM → Nasdaq, 2026; see E-mna).

SOURCES

- <https://digital-strategy.ec.europa.eu/en/policies/quantum>
- https://qt.eu/media/pdf/Quantum_Europe_Strategy_July_2025.pdf
- <https://digital-strategy.ec.europa.eu/en/policies/european-quantum-communication-infrastructure-euroqci>
- https://www.eurohpc-ju.europa.eu/eurohpc-quantum-computers/our-quantum-computers_en
- <https://postquantum.com/industry-news/eu-quantum-act/>
- <https://www.quireca.com/quantum-initiatives-worldwide/>

Quantum export controls · E-export

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

On **6 September 2024** the US Commerce Department's **Bureau of Industry and Security (BIS)** issued an interim final rule adding export controls on quantum computing items — full quantum computers, related equipment, components, materials, software, and development/maintenance technology — bundled with parallel controls on advanced semiconductors and additive manufacturing (new **ECCNs 3A901, 3D901, 3E901** and related). The defining feature is that it is **plurilateral**: BIS built an Export Administration Regulations framework with a **license exception (IEC — Implemented Export Controls)** authorizing exports to partner countries that adopt **harmonized** controls, so allies coordinating with the US face fewer barriers than non-aligned states. Allied measures landed in lockstep — the **Netherlands** issued its own controls **7 September 2024**, with the **UK, France, Spain, Australia, and Japan** adopting similar rules. Controls affecting exports to certain allies took effect **5 November 2024**; the rest applied from 6 September 2024. This is the first time quantum was made a first-class multilateral export-control category rather than folded into dual-use catch-alls.

THE MECHANISM (GRADED)

- [T2] BIS interim final rule, 6 Sep 2024, controls quantum computers + enabling equipment/materials/software/technology — bis.gov / Federal Register (established)
- [T2] Plurilateral framework + IEC license exception rewards allies with harmonized controls — Covington / Mayer Brown / Cleary analyses (established)
- [T2] Netherlands announced coordinated controls 7 Sep 2024; UK/France/Spain/Australia/Japan aligned — Orrick / APS News (established)
- [T2] **"Deemed export" reach**: sharing controlled quantum tech with foreign nationals *inside* the US is itself a controlled export — UChicago/ LBNL guidance (established); directly ties this node to E-immigration and E-talent.

KEY GRADED CLAIMS

- [T2] The rule is explicitly **coordination-first** — the license exception makes allied harmonization the point, not a byproduct (established)
- [T3] BIS simultaneously **expanded CFIUS mandatory-filing scope** to cover quantum-relevant inbound investment — Cleary (claimed/analysis); the June 2026 Quantum EO reinforced the inbound-investment-screening posture.
- [T4] Real-world chilling effect on academic collaboration and hiring of foreign nationals — reported concern, not yet measured (contested)

CONFLICTS / OPEN QUESTIONS

- **Effectiveness is unproven:** controls slow but do not stop a determined state, and China is documented building domestic supply to route around them (see E-supplychain, China self-sufficiency).
- **Scope ambiguity** — "quantum computer" thresholds and dual-use materials (cryogenics, lasers) are hard to police; enforcement precedent is thin, and the "logical/physical qubit" line is undefined in standards (see E-standards).
- **Bloc-cohesion risk:** does the plurilateral bloc hold when a member's national champion is disadvantaged? The IonQ acquisition of UK-based **Oxford Ionics** (E-mna) and Australia's bet on US-domiciled PsiQuantum (E-others) both test whether "allied" firms are treated as domestic — echoes of the ASML/Netherlands semiconductor tensions.

SOURCES

- <https://www.bis.gov/press-release/departments-commerce-implements-controls-quantum-computing-other-advanced-technologies-alongside>
- <https://www.aps.org/apsnews/2024/10/export-quantum-computer>
- <https://www.orrick.com/en/Insights/2024/09/US-Joins-Allies-to-Impose-Export-Controls-on-Quantum-Computing-and-Other-Emerging-Technologies>
- <https://www.cov.com/en/news-and-insights/insights/2024/09/us-implements-plurilateral-export-controls-framework-and-additional-controls>

- <https://www.clearytradewatch.com/2024/09/commerce-imposes-export-controls-on-quantum-computing-and-other-advanced-technologies-expands-scope-of-cfius-mandatory-filing-requirement/>

Workforce immigration & clearance policy · E-immigration

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

The workforce *gap* is E-talent; this node is the **policy layer that decides who is allowed to fill it** — immigration/visa rules and security-clearance requirements, which pull in opposite directions. Quantum is one of the few deep-tech fields where domestic supply clearly cannot meet demand, so employers rely on **international hires**; yet the same technology is export-controlled (E-export "deemed exports") and clearance-gated for defense work (E-defense), which **shrinks the eligible pool**. The result is a structural tension: the US and allies need foreign quantum talent to build the workforce, and simultaneously restrict foreign nationals' access to the controlled technology those jobs involve.

THE LANDSCAPE (GRADED)

- [T5] **50% of advanced-STEM-degree holders in the US defense industrial base are foreign-born**; 82% of DIB firms report difficulty finding qualified STEM workers — IFP / CSIS (reported)
- [T3] Quantum roles are sponsored across **national labs (Argonne, ORNL, LBNL), universities, and IBM/Google/Microsoft/AWS**, plus contractors (Booz Allen, Leidos) — though clearance requirements vary by project and exclude many non-citizens — F1Jobs / migratemate (reported)
- [T2] The **22 June 2026 Quantum EO** directs a **domestic** workforce push (apprenticeships, credentials) and **expands the Quantum Counterintelligence Protection Team** — a tightening, not a loosening, of the foreign-access posture — whitehouse.gov (established)

THE TWO-SIDED TENSION (GRADED)

- [T2] **Deemed-export rule (E-export)**: sharing controlled quantum technology with a foreign national *inside* the US is legally an export — so hiring a non-citizen into a controlled-tech role can itself require a license — BIS / university guidance (established)

- [T3] **Clearance bottleneck:** defense quantum work (E-defense) requires clearances that most non-citizens cannot hold, and clearance backlogs further slow staffing — RAND / reporting (reported)
- [T3] **Global competition for talent:** other states (Canada, Australia, EU, Gulf) court the same PhDs with faster visa tracks — a talent-attraction race layered on the funding race — Robinson Immigration / CSIS (reported)

KEY GRADED CLAIMS

- [T4] US immigration policy volatility (H-1B cost/reform debates, 2025–26) adds hiring uncertainty precisely where the shortage is most acute — CSIS/press (claimed)
- [T5] The clearance + export-control filter is **largely absent from the quantum-shortage literature** (E-talent), which counts open roles without netting out who is *eligible* to fill them (analyst gap)

CONFLICTS / OPEN QUESTIONS

- **Self-defeating loop?** Restricting foreign access protects IP (E-export, E-patents) but starves the very workforce the national strategies say is the binding constraint (E-talent). No government has published how it resolves the tradeoff.
- Does the June 2026 EO's domestic-apprenticeship push produce hybrid technicians fast enough to reduce reliance on foreign PhDs, or is it a decade too slow (O-talent-attrition)?
- Measurement gap: no dataset separates "open quantum roles" from "roles a security-cleared citizen could fill" — the true domestic shortfall is unquantified.

SOURCES

- <https://ifp.org/stem-immigration-is-critical-to-american-national-security/>
- <https://www.csis.org/analysis/practical-h-1b-reforms-serve-us-economic-interests>
- <https://www.f1jobs.io/resources/blog/quantum-computing-jobs-visa-sponsorship>
- <https://www.whitehouse.gov/presidential-actions/2026/06/ushering-in-the-next-frontier-of-quantum-innovation/>

- <https://robinsonimmigration.com/resources/global-competition/>

Insurance & liability for quantum risk · E-insurance

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

This node is about **insuring against the quantum threat** — cyber underwriters, D&O liability, and regulators pricing the risk that current cryptography breaks — which is the mirror image of I-insurance (using quantum computing *for* actuarial modeling). The 2026 industry consensus is a split: the **long-run** exposure is real and potentially systemic (a cryptographically-relevant quantum computer could invalidate the confidentiality assumption under most cyber policies at once), but **near-term claims are considered unlikely**, so insurers are folding **PQC-readiness into underwriting questionnaires** rather than repricing books today. The annual cyber-renewal cycle is itself the industry's stated hedge — it lets insurers adjust before quantum becomes a material exposure.

THE LANDSCAPE (GRADED)

- [T3] **Lloyd's** analysis: the largest potential insurance impact comes from the **cybersecurity risk of cryptographically-relevant quantum computers**; underwriters describe it as a "long-term issue" managed via annual renewals — Lloyd's / Finadium (reported)
- [T3] **Cyber insurers see little near-term claims threat** (Jul 2026) — HNDL was a central conference theme, but insurance lawyers argued it is unlikely to become a major near-term claims source — The Insurer (reported)
- [T3] Insurers are **integrating PQC-readiness evaluations into underwriting** to keep portfolios profitable and to nudge clients to begin migration (E-pqcmarket) — Gallagher / Insurance Business (reported)

THE EXPOSURE CATEGORIES (GRADED)

- [T3] **Confidentiality/data risk** — customer, claims, health, and underwriting data stored today could be decrypted later (HNDL); a *silent aggregation* risk because one crypto break hits many policies simultaneously — Hogan Lovells / PLUS (reported)

- [T3] **Operational & contractual risk** — long-term technology arrangements (outsourcing, PKI, long-dated bonds/records) that assume today's crypto — Lloyd's (reported)
- [T3] **D&O / governance liability** — as regulators (see E-us June 2026 PQC EO deadlines) increasingly *expect* firms to plan for quantum, boards that fail to act face governance-failure exposure — PLUS "Quantum Risk Series" (reported)

KEY GRADED CLAIMS

- [T4] No dedicated "quantum insurance" product line exists yet; quantum risk is being **absorbed into existing cyber/D&O lines** with questionnaire changes, not new policies (reported/emerging)
- [T5] "Q-Day could reshape cyber claims" — insurance-industry framing (PLUS); a scenario, not a priced event (speculative)

CONFLICTS / OPEN QUESTIONS

- **Systemic-correlation problem:** most cyber risk is idiosyncratic, but a crypto break is a **single correlated event across an entire book** — classic uninsurable/catastrophe structure. How (or whether) reinsurance handles this is unresolved.
- Tension with E-pqcmarket: insurers say near-term claims are unlikely, while PQC vendors sell urgency — the truth depends on *data lifetime* vs *Q-day* timing (O-scaling).
- Liability allocation: if crypto breaks, who is at fault — the software vendor, the enterprise that didn't migrate by the EO deadline, or the standards body? No case law exists.

SOURCES

- <https://www.hoganlovells.com/en/publications/emerging-risks-in-2026-preparing-for-the-legal-and-regulatory-implications-of-quantum>
- <https://finadium.com/lloyds-impacts-of-quantum-computing-on-insurance/>
- <https://www.theinsurer.com/ti/news/cyber-insurers-see-little-near-term-quantum-computing-claims-threat-2026-07-08/>
- <https://plusweb.org/news/quantum-risk-series-q-day-is-coming-how-quantum-computing-could-reshape-cyber-insurance-claims/>

- <https://www.ajg.com/news-and-insights/the-emerging-threat-of-quantum-computing/>

Academic-lab leaderboard · E-labs

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

Where E-patents ranks corporate IP and E-talent counts the workforce, this node maps the **academic research base** — the university labs that produce the physics, the people, and the spinouts. It matters geopolitically because national programs (E-us..E-others) are downstream of a small set of anchor institutions, and because the leaderboard reveals a **US/China/Europe tripole** with distinct modality specializations. Rankings are bibliometric (Nature Index Share, publication/citation counts) and therefore favor *output volume over breakthrough* — read them the way E-patents reads filing counts: informative about mass, silent about which lab produced the load-bearing result.

THE LEADERBOARD (GRADED, WITH CAVEATS)

- [T3] **USTC (Hefei)** ranks **first in quantum-physics output** in the Nature Index (2024–25 window) — home of Micius, Jiuzhang, Zuchongzhi, and the 2025 Jinan-1 13,000 km QKD link (E-china) — Nature Index (reported; output-weighted)
- [T3] **US anchors:** MIT, Harvard (Lukin — neutral-atom simulation/ computing, spun out QuEra), Caltech, Chicago (Pritzker/PME + Argonne/ Fermilab), Maryland/JQI (ion trapping — IonQ lineage), Princeton, Berkeley/LBNL — Nature Index / TQI (reported)
- [T3] **Chinese Academy of Sciences** system + Tsinghua, plus **USTC**, dominate the China cluster — bibliometric studies (reported)
- [T3] **Europe:** TU Delft / **QuTech** (superconducting + spin + networking; Intel/Microsoft ties), ETH Zurich, Oxford (ion traps → Oxford Ionics), Sydney/UNSW (silicon spin — Diraq, SQC), Waterloo/IQC (Canada), Innsbruck (ions), Paris (CNRS, neutral atoms → Pasqal) — Nature Index / TQI (reported)

KEY GRADED CLAIMS

- [T3] The field's output grew **exponentially since ~2015**, led by the US, China, and Europe, with MIT, CAS, CNRS, and Waterloo cited as

bibliometric leaders — EPJ Quantum Technology mapping study (demonstrated)

- [T3] **Labs map to modalities and to spinouts:** Maryland/Innsbruck→ions/IonQ-Quantinuum, Harvard-MIT→neutral atoms/QuEra, UNSW→silicon/Diraq, Delft→spin+networking, Sydney→control (Q-CTRL). The academic map *is* the industry map one generation earlier (analysis)
- [T4] Rankings favor volume; a single below-threshold-QEC result (Google Willow, a corporate lab) can outweigh a year of a university's papers — bibliometrics miss this (claimed)

CONFLICTS / OPEN QUESTIONS

- **Output vs breakthrough:** USTC tops output counts, but the field's landmark QEC results came from *corporate* labs (Google, IBM, Quantinuum) — academic leaderboards understate the industrial research frontier (E-patents trade-secret shadow).
- **Brain-drain direction:** do top academic labs feed *domestic* champions or foreign ones (E-immigration, E-mna)? Oxford→IonQ (US) and UNSW→global shows leakage across borders.
- No neutral, methodologically stable "quantum lab ranking" exists — Nature Index, QS, and TQI lists diverge; treat any single ordering as one lens.

SOURCES

- <https://www.nature.com/nature-index/news/ten-best-countries-for-quantum-physics-research>
- <https://link.springer.com/article/10.1140/epjqt/s40507-026-00464-4>
- <https://thequantuminsider.com/2025/03/26/top-universities-for-quantum-research-2025/>
- <https://thequantuminsider.com/2026/05/22/top-20-quantum-computing-masters-ph-d-programs/>
- <https://quantumcomputingreport.com/universities/>

Market forecasts — read with gloves on · E-market

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Every number in this card is **T5 by construction**. The forecast spread is enormous and the biggest inflation trick is **unit-switching**: "economic value created" (trillions) gets quoted as if it were vendor revenue (tens of billions), a 20–50x difference. Current reality check: **actual sector revenue was ~\$1B in 2025** (McKinsey's own figure), which the same report projects to as much as \$4.4B by 2028. McKinsey's 2026 Monitor declared a "commercial tipping point" and revised its own market numbers sharply upward year-over-year — which says as much about forecast elasticity as about the market. The honest reading is to track each house's **revision history**, not its point estimate.

THE NUMBERS (GRADED, WITH CAVEATS)

- [T5] **McKinsey (2026 Monitor)**: quantum-technology market **\$60–100B by 2035**, of which quantum computing **\$43–71B** — revised UP from ~\$28B in the prior edition. A >2x self-revision in 12 months is the caveat.
- [T5] **McKinsey**: up to **\$2.7T "economic value"** by 2035 (prior editions said \$1.3–2.7T). Value-created ≠ revenue — do not mix with the line above; most press does.
- [T5] **McKinsey**: present quantum-computing revenue **~\$1B (2025) → up to \$4.4B (2028)**; >300 companies (Airbus, JPMorgan, Boehringer) now engaged with vendors. Motivated forecaster caveat: McKinsey sells quantum-readiness engagements.
- [T5] **BCG**: \$450–850B economic value by 2040, supporting a \$90–170B hardware+software vendor market. Different horizon (2040 vs 2035) makes it incomparable to McKinsey's headline despite constant press comparison.
- [T5] **Gartner** keeps quantum computing on the long-horizon end of its hype cycle (10+ years to plateau); **IDC** has repeatedly cut near-term revenue forecasts — the revision trail beats any single number.
- [T5] Normalized spread: 2035 vendor-market estimates ~\$40–100B; near-term (2027–28) estimates cluster in single-digit billions.

KEY GRADED CLAIMS

- [T5] "Commercial tipping point in 2026" — McKinsey framing; provenance-flagged (claimed)
- [T5] Analyst TAMs assume fault-tolerant machines arrive on vendor roadmap schedules — the single load-bearing assumption; every year of FTQC slip pushes the hockey stick right (contested)
- [T2] Present-day sector revenue ~\$1–1.5B/yr, dominated by hardware R&D contracts, cloud access, and government programs — company filings + McKinsey (established)

CONFLICTS / OPEN QUESTIONS

- **C-market-size:** McKinsey \$43–71B by 2035 vs BCG's flatter path to 2040. Resolution: audited vendor revenue through 2030, and whether any commercial advantage case (see O-killerapp) closes.
- No analyst house publishes its **miss rate** on prior quantum forecasts; a backtest of 2019–2022 editions would likely be brutal.
- **Double-counting:** "quantum market" tallies often fold in **PQC migration** spending — a classical cybersecurity market (\$1–15B depending on definition, see E-pqcmarket) that exists regardless of quantum hardware progress.

SOURCES

- <https://www.mckinsey.com/capabilities/mckinsey-technology/our-insights/mckinsey-quantum-technology-monitor-2026-a-commercial-tipping-point>
- <https://postquantum.com/industry-news/mckinsey-quantum-monitor-2026/>
- <https://www.bcg.com/publications/2024/long-term-forecast-for-quantum-computing-still-looks-bright>
- <https://www.grandviewresearch.com/industry-analysis/quantum-computing-market>
- <https://aimultiple.com/quantum-computing-stats>

Metrology standards & SI-redefinition governance · E-metrology-gov

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

Quantum has a **second, quieter standards front** from the PQC/QKD one (E-standards): the governance of the **International System of Units (SI)** itself, run by **BIPM / CIPM / CGPM** and the consultative committees (CCTF for time, CCU for units). The 2019 SI redefinition already anchored the kilogram, ampere, kelvin, and mole to fundamental constants (Planck, elementary charge, Boltzmann, Avogadro), making national metrology institutes able to **realize units from quantum physics** rather than artifacts. The live 2025–26 governance question is the **redefinition of the second** onto an **optical atomic-clock** transition — a decision headed for **CGPM 2026**. This is geopolitically loaded because whoever's clocks and realizations anchor the new second holds metrological soft power, and the outcome underpins timing, PNT (A-clocks, A-timedist), and finance/telecom infrastructure.

THE GOVERNANCE TRACK (GRADED)

- [T2] **BIPM operates under CIPM supervision**; the CGPM (member states) is the decision body; the **CCTF** drafts the second-redefinition roadmap — bipm.org (established)
- [T2] **2019 SI redefinition** fixed the kilogram to the **Planck constant** (realized via the **Kibble balance** or the X-ray-crystal-density/Avogadro route), the ampere to elementary charge, kelvin to Boltzmann — a quantum-constant-based system (established)
- [T3] **Roadmap toward redefining the second**: several optical frequency standards already beat caesium fountains by ~2 orders of magnitude in systematic uncertainty; the CCTF needed a draft ready for its **Sep 2025 meeting** to present at **CGPM 2026** — IOP Metrologia roadmap (demonstrated)
- [T3] Multiple NMIs (PTB, NIST, INRIM, NPL, SYRTE, KRISS) are cross-evaluating candidate optical transitions; **>80–90% clock uptime**

demonstrations are the readiness signal — Nature Index / Metrologia (reported)

KEY GRADED CLAIMS

- [T2] The realization of quantum-based units is **decentralized by design** — any NMI can realize the kilogram/second independently, which is itself a governance principle (no single artifact, no single owner) — BIPM (established)
- [T3] Optical-clock redefinition of the second is likely "**within the current decade**" and is the flagship deliverable of the 2025 International Year of Quantum Science & Technology metrology agenda — BIPM (reported)
- [T4] Metrological leadership is soft power: the NMIs whose clocks define the ensemble shape traceability chains the whole economy inherits (analysis)

CONFLICTS / OPEN QUESTIONS

- **Which optical transition wins?** Competing candidates (Sr, Yb, Al⁺, Yb⁺ E3, etc.) across rival NMIs — a standards choice with national prestige attached; resolution at/after CGPM 2026.
- **Realization inequality:** not every member state can build an optical clock or Kibble balance, so "decentralized realization" is decentralized only among wealthy NMIs — a metrology-access gap.
- Overlap with A-timedist: redefinition needs clock **comparison** networks (optical fiber / free-space time transfer) to disseminate the new second — the governance decision is downstream of that hardware.

SOURCES

- <https://www.bipm.org/en/faq-redefinition-second>
- <https://iopscience.iop.org/article/10.1088/1681-7575/ad17d2>
- <https://iopscience.iop.org/article/10.1088/1681-7575/ab0013>
- <https://www.bipm.org/en/-/2025-01-20-year-of-quantum-science-and-technology>
- <https://www.patsnap.com/resources/blog/articles/optical-atomic-clock-technology-landscape-2026/>

The consolidation wave — M&A and public listings 2025–26 · E-mna

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node**
(cycle 2 random-walk)

SUMMARY

2025–26 is the field's **consolidation phase**, distinct from primary VC rounds (E-vc). Two forces run in parallel: **roll-ups** — public hardware companies buying components, foundries, and rivals to build full stacks — and an **exit wave** of SPAC mergers and IPO filings that let 2021-era private companies reach public markets. IonQ ran the most aggressive acquisition spree in the sector's history; D-Wave, Xanadu, IQM, and Quantinuum each made defining 2026 moves. The strategic reading: the sector is maturing toward a **small number of vertically integrated players**, which concentrates IP (E-patents) and interacts with export controls when the target is foreign (E-export).

THE DEALS (GRADED, WITH CAVEATS)

- [T5] **IonQ 2025 spree:** acquired **Oxford Ionics (~\$1.08B, the year's largest deal)**, Lightsynq, Qubitekk, Capella Space, and Vector Atomic; took a **majority stake in ID Quantique** — McKinsey/Crunchbase (reported; some values are stock-heavy)
- [T5] **IonQ → SkyWater Technology, ~\$1.8B (announced Jan 2026, expected close 2026)** — buying a US semiconductor foundry with government/defense contracts; a vertical-integration + supply-security play (E-supplychain) — Crunchbase
- [T5] **D-Wave → Quantum Circuits Inc., \$550M** (stock + cash, early 2026) — annealing leader buying a gate-model developer — Crunchbase
- [T5] **Xanadu** went public via SPAC on Nasdaq + TSX (~\$3.1B enterprise value cited) — photonic; H1-2026 exit — Crunchbase/PitchBook
- [T5] **IQM → Real Asset Acquisition Corp. SPAC** — first publicly traded *European* quantum-computing company on Nasdaq; ~\$233.5M injected incl. a \$145.5M PIPE — press
- [T5] **Quantinuum** filed for a **Nasdaq IPO in 2026** after its ~\$839M round at \$10B pre-money (E-vc) — Crunchbase News

- [T5] H1-2026 exits totaled ~**\$5.7B across four listings** (Xanadu, Inflektion, Quantum Circuits, Horizon Quantum) — ~15x the prior three years' exit value combined (PitchBook)

KEY GRADED CLAIMS

- [T5] M&A + exit activity "accelerated markedly in 2025" and signals market maturation/consolidation — McKinsey 2026 Monitor (analyst framing)
- [T3] Vertical integration is the explicit thesis — hardware firms buying foundries (SkyWater), photonic interconnect (Lightsynq), sensing/space (Capella, Vector Atomic) to own the stack (reported)

CONFLICTS / OPEN QUESTIONS

- **Is consolidation strength or distress?** Roll-ups can signal maturation *or* that standalone economics don't work at ~\$1B total sector revenue (E-market). Resolution: post-merger revenue at the acquirers through 2027.
- **Sovereignty friction:** IonQ (US) absorbing UK-based Oxford Ionics and Swiss ID Quantique puts allied national champions under US ownership + export-control reach (E-export, E-uk, E-others).
- Stock-heavy deal values inflate headline M&A totals when acquirer shares are themselves richly valued (E-vc) — a circularity worth flagging.

SOURCES

- <https://entangledfuture.com/acquisitions/>
- <https://news.crunchbase.com/venture/quantum-computing-startup-investment-data-quantinuum-ipo/>
- <https://www.mckinsey.com/capabilities/mckinsey-technology/our-insights/mckinsey-quantum-technology-monitor-2026-a-commercial-tipping-point>
- <https://entangledfuture.com/guides/quantum-computing-spac-ipo-guide-2026/>
- <https://thequantuminsider.com/2025/09/23/top-quantum-computing-companies/>

Open-source ecosystem & soft power · E-oss

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The quantum software layer is dominated by **vendor-backed open-source SDKs**, and whoever's framework a developer learns first shapes which hardware they reach for later — soft power and lock-in dressed as free tooling. The big three are **Qiskit** (IBM), **Cirq** (Google), and **PennyLane** (Xanadu), joined by tket/pytket (Quantinuum), Braket SDK (AWS), Q#/QDK (Microsoft), and D-Wave's Ocean. All are permissively licensed and free, but each is a funnel toward its sponsor's cloud and hardware. The counter-force is interop standards — **OpenQASM** (IBM-originated assembly, now OpenQASM 3) and **QIR** (Microsoft-led LLVM-based intermediate representation under the **QIR Alliance / Linux Foundation**) — which keep code portable across backends and blunt pure lock-in. The open geopolitical question is whether the world converges on this Western toolchain or **forks** along the export-control line into China's parallel stack (broken out as **E-china-stack**).

THE LANDSCAPE (GRADED)

- [T2] Qiskit, Cirq, PennyLane are the leading full-stack SDKs, all corporate-maintained (IBM/Google/Xanadu) yet open and free — multiple 2025 surveys (established)
- [T2] PennyLane is hardware-agnostic by design (device plugins target Qiskit/Cirq/Forest/Braket) — Xanadu docs (established)
- [T2] OpenQASM + QIR provide cross-backend portability; a Cirq circuit can be lowered to OpenQASM and run via Qiskit on IBM hardware — 2025 ecosystem writeups (established)
- [T3] The ecosystem is partly siloed — specialized QEC/chemistry tools don't interoperate cleanly; cohesion is an open community effort — surveys (reported)
- [T3] NVIDIA **CUDA-Q** is emerging as a de facto hybrid GPU-QPU orchestration layer, giving Nvidia soft-power leverage over the classical side of every quantum workflow — Nvidia (reported)

KEY GRADED CLAIMS

- [T3] Framework choice functions as soft-power lock-in — learning Qiskit funnels toward IBM Quantum cloud; governance-by-defaults, not contract lock-in (analysis/claimed)
- [T2] Governance is **corporate, not neutral-foundation**, for most SDKs — Qiskit/Cirq/PennyLane steered by their sponsors; the **QIR Alliance (Linux Foundation)** is the main neutral-governance exception (established)
- [T3] A 2026 formal-verification study (arXiv 2604.06712) found security vulnerabilities across ~45 open-source quantum simulators — software-supply-chain risk (preprint)

CONFLICTS / OPEN QUESTIONS

- Is portability (OpenQASM/QIR) real enough to defeat lock-in, or do deep tooling layers (transpilers, error mitigation, pulse control) **re-lock** users to one vendor in practice?
- No dominant *neutral* foundation governs the top SDKs — unlike Linux/ Apache — so "open source" here is closer to open-source-as-marketing than a community-governed commons. The one durable neutral layer is the IR (QIR), not the SDKs.
- **Fork risk (see E-china-stack)**: China's Origin QPanda / Baidu Paddle Quantum / Huawei HiQ are a full parallel stack; export controls (E-export) make convergence less likely each year.

SOURCES

- <https://www.opensourceforu.com/2025/11/the-top-open-source-quantum-computing-frameworks/>
- <https://postquantum.com/quantum-computing/quantum-programming/>
- <https://www.ibm.com/quantum/ecosystem>
- <https://arxiv.org/html/2604.06712v2>
- <https://medium.com/@adnanmasood/the-quantum-platforms-briefing-day-5-open-source-ecosystem-51109e78df70>

National programs — the rest of the pack · E-others

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Beyond the US/China/EU/UK quartet, roughly a dozen countries run serious national quantum programs. Common pattern: a multi-year headline pledge (T5, usually mixing new money with relabeled research budgets), one flagship national lab or mission, and a bet on a niche — India on indigenous full-stack hardware, Japan on industrialization, Australia on photonics via PsiQuantum, Canada on a deep startup bench, Israel on defense-adjacent depth, South Korea on late-but-large catch-up, Germany on error-corrected machines within the EU frame. The near-universal caveat: **pledge ≠ disbursement**, and almost none of these countries publish the gap.

THE NUMBERS (GRADED, WITH CAVEATS — ALL T5, ALL MULTI-YEAR PLEDGES, CURRENCIES AS ANNOUNCED)

- [T5] **India** — National Quantum Mission, ₹6,003.65 crore (~\$730M), 2023–2031 (dst.gov.in). Four thematic hubs (computing at IISc Bengaluru, comms, sensing/metrology, materials/devices). 17 supported startups; QpiAI shipped a 25-qubit "Indus" (Apr 2025) and a 64-qubit "Kaveri" chip (Nov 2025) — both T4 vendor claims. Still imports key lasers/cryogenics (see E-supplychain).
- [T5] **Japan** — 2025 declared "first year of quantum industrialization"; ¥1.05T (~\$7.4B) headline covers semiconductors **and** quantum (severe double-counting if quoted whole); ~\$900M is quantum-specific. Goal: 10M quantum users by 2030. RIKEN (which fielded Japan's first homegrown superconducting machine, 2023) + the Moonshot R&D program underpin the base.
- [T5] **Australia** — ~AU\$2.3B cumulative, dominated by the ~AU\$940M federal+Queensland investment in **PsiQuantum's Brisbane** machine (2024) — a bet on one (US-headquartered) company, politically contested and audited.
- [T5] **Canada** — National Quantum Strategy C\$360M (2021) + C\$334M/5yr defence (2025 budget) + C\$92M Quantum Champions Phase 1 (Dec 2025) + C\$50M DISH defence R&D (Feb 2026). Deep startup bench

(Xanadu, D-Wave origins, Photonic Inc., Nord Quantique) — though Xanadu went public via US SPAC in 2026 (see E-mna).

- [T5] **Israel** — NIS 1.2B (~\$380M) six-year program; small headline, high density (Quantum Machines, Classiq, Quantum Source), deep defense overlap that hides real totals.
- [T5] **South Korea** — ~₩3T (~\$2.3B) toward 2035 quantum goals under the 2023 Quantum Science & Technology Act; late entrant, chaebol-backed (SK Telecom in QKD, Samsung/SK in supply chain).
- [T5] **Germany** — ~€5B total: €2B (2020 stimulus) + €3B Action Plan (2023) targeting a **universal error-corrected** computer; hosts EuroHPC's Euro-Q-Exa (IQM, 54→150 qubits, LRZ Munich). Counted again inside EU tallies — see E-eu.

KEY GRADED CLAIMS

- [T4] QpiAI "India's first full-stack quantum computer" — vendor claim, no independent benchmark (claimed)
- [T3] Germany's strategy centers error-corrected systems over NISQ demos — government strategy documents (roadmap)
- [T3] Japan/RIKEN operates the first non-US/China domestically built superconducting quantum computer at national-lab scale (demonstrated)

CONFLICTS / OPEN QUESTIONS

- Pledge-vs-disbursement gap is unmeasured for nearly every country here.
- Australia's single-vendor concentration (PsiQuantum) has no public plan B, and the vendor is US-domiciled — a sovereignty wrinkle.
- Still missing / candidates for a future split: **Singapore** (CQT, early mover), **Netherlands** (QuTech/Delft, ASML-adjacent supply leverage), **Taiwan**, **Denmark** (Novo-backed), **Russia** (opaque, sanctioned).

SOURCES

- <https://dst.gov.in/national-quantum-mission-nqm>
- <https://thequantuminsider.com/2026/03/26/leading-quantum-computing-countries/>
- <https://www.qureca.com/quantum-initiatives-worldwide/>

- https://www.oecd.org/en/publications/an-overview-of-national-strategies-and-policies-for-quantum-technologies_5e55e7ab-en/full-report/component-4.html
- <https://thequantuminsider.com/2023/05/03/germany-announces-3-billion-euro-action-plan-for-a-universal-quantum-computer/>
- <https://www.quantumworldcongress.com/news-and-updates/germany-outlines-quantum-strategy-focused-on-error-corrected-systems-applied-research-and-international-collaboration>

Patents & the IP landscape · E-patents

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The patent race splits cleanly on a **volume vs quality** axis. China files the most quantum patents by a wide margin (~60% of global *filings* in 2024); the US dominates **International Patent Families** — patents protected in multiple jurisdictions, the proxy for inventions someone expects to monetize globally (~48% US vs ~11% China). Specialization mirrors national strategy: China owns QKD/communications IP (~80% of families in that niche); the US leads computing hardware and algorithms, driven by IBM, Google, and Microsoft. The load-bearing lesson: **which country "leads" depends entirely on the denominator you pick**, and headlines swap weekly using the same underlying data.

THE NUMBERS (GRADED, WITH CAVEATS)

- [T5] ~60% China vs ~19% US — share of global quantum patent *filings*, 2024 (patentailab.com). Raw counts are inflated by Chinese subsidy programs that reward filing volume; many are domestic-only and never examined abroad.
- [T5] ~48% US vs ~11% China — share of **International Patent Families** (multi-jurisdiction). The better quality proxy, still imperfect: family-counting methods differ across analyses.
- [T5] ~80% — China's share of QKD-related patent families (patentailab.com); consistent with its QKD infrastructure build (see E-standards, A-qkd).
- [T5] **IBM holds the largest single corporate quantum-computing portfolio** — repeated across landscape studies; exact counts vary by taxonomy (CPC class **G06N 10/00** vs keyword searches disagree by 2–3x).

KEY GRADED CLAIMS

- [T3] **Filing-subsidy distortion:** Chinese volume partly reflects state per-filing incentives — patent-policy literature (demonstrated)
- [T2] Baidu **donated** its quantum lab to BAAI and Alibaba **shuttered** its quantum effort (2023–24), transferring equipment to Zhejiang University —

the corporate patent engines went quiet while state/Origin filings continued (established)

- [T6] Whether quantum patents filed 2015–2025 matter commercially before they expire (20-year term vs FTQC timelines) — the quiet joke of the whole race (speculative)

CONFLICTS / OPEN QUESTIONS

- **C-patent-lead:** "China leads quantum IP" vs "US leads quantum IP" — both are printed weekly from the same data; the answer depends on filings-vs-families methodology. Resolution: none needed — always state the denominator.
- **Trade-secret shadow:** error-correction decoders (see S-decoders), control calibration, and fab recipes are increasingly kept as **secrets**, so patent landscapes systematically undercount the real IP frontier. This is growing as the field industrializes.
- **M&A concentrates IP:** IonQ's 2025 acquisition spree (Oxford Ionics, Lightsynq, Qubitekk, ID Quantique stake — see E-mna) folds independent patent estates into single US owners, interacting with export controls (E-export).

SOURCES

- <https://patentailab.com/global-quantum-patent-race-2026/>
- <https://entangledfuture.com/guides/quantum-patent-strategy/>
- <https://www.globallegalinsights.com/practice-areas/quantum-computing-laws-and-regulations/china/>
- <https://sagaciousresearch.com/blog/future-of-quantum-computing>
- <https://www.uscc.gov/research/vying-quantum-supremacy-us-china-competition-quantum-technologies>

The quantum-safe migration market · E-pqcmarket

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

Post-quantum cryptography migration is an **economic market in its own right**, separate from the crypto science (A-pqc, technical) and from the quantum-computing vendor market (E-market). It is driven not by quantum hardware progress but by **"harvest now, decrypt later" (HNDL)** — adversaries archiving encrypted traffic today to decrypt once a cryptographically-relevant quantum computer exists — and by hard regulatory deadlines. It is the one quantum-adjacent market with **real, mandated near-term revenue**, because it sells classical software/services (crypto discovery, crypto-agility, PKI replacement) that must ship regardless of whether FTQC ever arrives. This is exactly why it inflates "quantum market" tallies (E-market double-counting flag).

THE NUMBERS (GRADED — ALL T5 FORECASTS, WIDE SPREAD BY DEFINITION)

- [T5] **Narrow "PQC market"**: ~\$0.5–0.8B in 2025 → ~\$1.1B in 2026 (marketsandmarkets / imarc-style). Counts PQC products specifically.
- [T5] **"PQC migration market"**: ~\$1.9B (2025) → ~\$12.4B (2035), CAGR ~20.6% (Future Market Insights). Adds migration services.
- [T5] **"PQC migration solutions"**: ~\$15.0B (2025) → ~\$86.5B (2034), CAGR ~22% (MarketIntel). Broadest definition — folds in consulting + managed services; the outlier high end.
- [T5] The **NIST FIPS 203/204/205 finalization (Aug 2024)** is estimated to trigger ~\$4.2B in net-new federal + regulated-industry spending in **2025–2026 alone** (industry analysis).
- [T5] Nasdaq/press framing: PQC migration is now a **"trillion-dollar imperative"** across the global installed base — motivated-vendor rhetoric; treat as TAM-of-everything, not revenue.

THE REGULATORY ENGINE (GRADED)

- [T2] NIST finalized ML-KEM/ML-DSA/SLH-DSA Aug 2024; HQC backup Mar 2025 (E-standards) — the standards that make procurement possible (established)

- [T2] **US 22 June 2026 PQC EO**: OMB deadlines — high-value-asset **key establishment by 31 Dec 2030, digital signatures by 31 Dec 2031**; Commerce migration pilot by 31 Dec 2027 (E-us) — whitehouse.gov (established)
- [T5] NSA **CNSA 2.0** targets ~2033 for full NSS transition (E-standards) — policy deadline (slips possible)

KEY GRADED CLAIMS

- [T3] **HNDL is the demand driver**, assessed by US intelligence to involve China/Russia/DPRK archiving encrypted data now — Cloudflare/Gopher/agency framing (reported)
- [T5] This market exists **independent of quantum-hardware timelines** — its risk is set by *data lifetime* vs *Q-day*, so long-lived secrets justify migrating today even under optimistic FTQC-slip scenarios (analyst consensus)

CONFLICTS / OPEN QUESTIONS

- **Definitional 30x spread** (\$0.5B vs \$15B for "2025") makes any single headline meaningless — always name the definition (product vs migration vs solutions/services).
- Does mandated federal migration crowd in private-sector spend, or does the private base wait until a public break (contrast the insurance view, E-insurance, that near-term claims are unlikely)?
- Overlap: is PQC spend "quantum" spend at all? It is classical cryptography — folding it into E-market inflates the sector.

SOURCES

- <https://www.futuremarketinsights.com/reports/post-quantum-cryptography-pqc-migration-market>
- <https://marketintelo.com/report/post-quantum-cryptography-pqc-migration-solutions-market>
- <https://www.nasdaq.com/press-release/post-quantum-cryptography-migration-now-trillion-dollar-imperative-2026-02-19>
- <https://blog.cloudflare.com/post-quantum-eo-2026/>

- <https://www.marketsandmarkets.com/Market-Reports/post-quantum-cryptography-market-126986626.html>

Sovereign compute & national champions · E-sovereign

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth · **New node** (cycle 2 random-walk)

SUMMARY

A distinct 2025–26 pattern cutting across the national-program cards: states are no longer just *funding* quantum research, they are **acquiring sovereign compute** — government-owned or government-hosted machines on national soil, coupled to national supercomputers, plus **equity stakes** in national-champion firms. The thesis is that quantum, like advanced semiconductors, is strategic infrastructure that a country must **own and physically host**, not merely rent from a foreign cloud (E-oss soft-power concern). The clearest instance is **EuroHPC** turning procurement pledges into six machines inaugurated across Europe; the sharpest US instance is the **~\$2B Commerce package reportedly taking equity** in nine firms (E-us).

THE MOVES (GRADED)

- [T2] **EuroHPC JU procured six quantum computers**, each hosted at a European supercomputing centre and integrated with a Tier-0 machine for hybrid HPC-QC — a deliberate **digital-sovereignty** program — eurohpc-ju.europa.eu (established):
 - **Germany (Munich, LRZ):** Euro-Q-Exa — IQM superconducting, **54 qubits**, inaugurated **Feb 2026**, upgrade to **150 qubits** planned end-2026
 - **France (Bruyères-le-Châtel, CEA/GENCI):** **Lucy** — Quandela **12-qubit photonic**, **80% EU-sourced components**, coupled to Joliot-Curie / Alice Recoque exascale (inaugurated Apr 2026)
 - **Poland (Poznań):** PIAST-Q — trapped-ion (Jun 2025); **Czechia (Ostrava):** VLQ (Oct 2025); **Spain (Barcelona, BSC):** EuroQCS-Spain — analog annealer (May 2026); **Italy (Bologna):** SOL — neutral-atom simulator
- [T2] **EuroHPC Quantum Grand Challenge** call opened Oct 2025 to drive utilization of the fleet — eurohpc-ju.europa.eu (established)
- [T4] **US ~\$2B Commerce awards to nine firms, ~half to IBM, reportedly with government equity** (May 2026) — a sovereign-stake model new to US tech policy (E-us) — CNN/Forbes (claimed)

KEY GRADED CLAIMS

- [T3] **On-prem hosting is the sovereignty move** — owning the physical machine (not cloud access) is what "sovereign compute" means; Lucy's 80%-EU-component sourcing makes the supply-sovereignty point explicit (E-supplychain) (reported)
- [T3] Multi-platform hedging: EuroHPC deliberately bought **five different modalities** (superconducting, ion, photonic, annealer, neutral-atom) rather than betting on one — a portfolio approach to technological uncertainty (reported)
- [T4] The equity-stake turn (US) shifts government from **grant-maker to shareholder**, aligning fiscal upside with national champions but risking picking-winners distortion (E-us, E-vc) (claimed)

CONFLICTS / OPEN QUESTIONS

- **Does hosting a 12–54 qubit NISQ machine confer real sovereignty**, or is it symbolic until fault tolerance? The machines inaugurated are small; sovereignty of a pre-utility technology is a bet on the trajectory (O-scaling).
- **Champion capture:** Australia's PsiQuantum (US-domiciled) and IonQ's absorption of Oxford Ionics (E-mna) show "national champion" and "foreign-owned" can collide under export controls (E-export).
- Utilization risk: sovereign machines can sit idle without a user pipeline — hence EuroHPC's Grand Challenge call.

SOURCES

- https://www.eurohpc-ju.europa.eu/eurohpc-quantum-computers/our-quantum-computers_en
- https://www.eurohpc-ju.europa.eu/inauguration-euro-q-exa-expanding-european-quantum-computing-infrastructure-2026-02-12_en
- https://www.eurohpc-ju.europa.eu/eurohpc-ju-inaugurates-quantum-computer-france-strengthening-europes-sovereignty-2026-04-14_en
- https://www.eurohpc-ju.europa.eu/eurohpc-ju-opens-quantum-grand-challenge-call-2025-10-14_en
- <https://www.cnn.com/2026/05/21/business/ibm-quantum-computing-firms-grants>

Standards bodies · E-standards

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Standards are where quantum stops being physics and becomes procurement. The mature front is **post-quantum cryptography**: NIST finalized **FIPS 203 (ML-KEM)**, **FIPS 204 (ML-DSA)**, and **FIPS 205 (SLH-DSA)** in **August 2024**, selected **HQC** as a backup KEM in **March 2025**, and US EO 14144 (Jan 2025), reinforced by the **22 June 2026 PQC EO** (deadlines 2030/2031, see E-us, E-pqcmarket), pushed federal migration. Everything else — terminology, benchmarking, QKD certification — runs years behind, spread across ISO/IEC, ETSI, ITU, IEEE, and the new **ISO/IEC JTC 3** committee dedicated to quantum technologies. Standards racing is itself geopolitical: China dominates **QKD** standardization the way it dominates QKD patents (see E-patents), while the SI-metrology standards war is its own node (E-metrology-gov).

THE NUMBERS (GRADED, WITH CAVEATS)

- [T2] 3 finalized NIST PQC standards (FIPS 203/204/205, Aug 2024) + HQC selected Mar 2025 — csrc.nist.gov. Published, testable standards — the hardest artifact in this layer.
- [T5] Migration deadlines: NSA **CNSA 2.0** targets ~2033 for full national-security-system transition; the June 2026 EO adds OMB high-value-asset deadlines (key establishment 2030, signatures 2031). Deadlines are policy, so they slip.

KEY GRADED CLAIMS

- [T2] ISO/IEC JTC 1/WG 14 (renamed "Quantum Information Technology") published ISO/IEC 4879 terminology — jtc1info.org (established)
- [T2] ISO/IEC 23837-1/-2 define security requirements + test/evaluation methods for QKD systems — ISO/IEC SC 27 WG 3 (established)
- [T2] **ISO/IEC JTC 3** on quantum technologies stood up (kick-off spring 2024) — the first top-level ISO committee dedicated to quantum, a governance milestone — CEN/CENELEC / NPL TQE19 (established)

- [T3] **ETSI ISG-QKD** maintains the GS QKD series — the oldest quantum standards effort, running since 2008 (established as activity; deployment adoption thin)
- [T3] ITU-T ran **FG-QIT4N** producing terminology for quantum networks/ QKD; SG13/SG17 carry follow-on work (demonstrated)
- [T4] IEEE efforts (P7130 terminology, P7131 benchmarking) remain drafts with limited industry pull (claimed)
- [T5] **NSA's position that QKD is unsuitable for national-security systems** (favoring PQC) is a live standards-war fault line against China's QKD-heavy strategy (contested)

CONFLICTS / OPEN QUESTIONS

- **C-qkd-vs-pqc:** US agencies favor PQC over QKD; China builds QKD backbones. Resolution: a decade of deployment economics and any PQC break.
- **Benchmarking vacuum:** what counts as a "logical qubit" or a "QuOp" remains unstandardized while procurement targets (E-uk ProQure) already cite them — buyers are writing contracts against undefined units.
- Terminology fragmentation: QED-C, CEN/CENELEC JTC 22, ISO WG14, and JTC 3 all issue overlapping vocabularies.

SOURCES

- <https://csrc.nist.gov/projects/post-quantum-cryptography>
- <https://jtc1info.org/sd-2-history/jtc1-working-groups/wg-14/>
- <https://www.nist.gov/document/worldwide-standardization-activity-quantum-key-distribution>
- <https://eprintspublications.npl.co.uk/9298/1/TQE19.pdf>
- https://www.cenelec.eu/media/CEN-CENELEC/AreasOfWork/CEN-CENELEC_Topics/Quantum%20technologies/Documentation%20and%20Materials/fgqt_q06_standardizationroadmapquantumtechnologies_release1-1.pdf
- <https://quantumconsortium.org/publication/quantum-terminology-standards/>

Quantum supply-chain chokepoints · E-supplychain

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

This is the **geopolitics/supply** angle on the physical inputs a quantum computer needs — distinct from **H-fab** (the technical fabrication node). A short list of components has no elastic supply and few qualified vendors, turning each into a strategic chokepoint: **dilution refrigerators** (superconducting/spin), **helium-3** (their coldest stage), **quantum-grade UV/visible lasers** (trapped-ion, neutral-atom), **²⁸Si enrichment** (spin qubits), **CVD quantum-grade diamond** (NV), **cryo-CMOS foundry access**, **high-density cryogenic connectors/coax**, and **SNSPD** uniformity (photonic/QKD). The NATO Transatlantic Quantum Community study reportedly found **>90% of high-purity material processing for quantum happens outside NATO territory** — the strategic worry that motivates both export controls (E-export) and reshoring. The chokepoint map is **modality-dependent**: no single supply strategy covers all qubit types.

THE CHOKEPOINTS (GRADED)

- [T3] **Dilution fridges** — effective build rate ~1 system/day industry-wide (Bluefors, Oxford Instruments dominant) becomes binding when builders re-spin hardware every 12–18 months; the single most critical bottleneck for superconducting/spin qubits — War on the Rocks / PostQuantum 2025 analyses (demonstrated/reported)
- [T3] **Helium-3** — not naturally abundant; primary source is **tritium decay from nuclear-weapons stockpiles** (a national-security-controlled feedstock); ~\$2,000–15,000/L (per-mass more valuable than gold); structural deficit ~20,000–40,000 L/yr — PostQuantum cryogenics guide (reported estimates)
- [T2] **Kiutra** (Germany) raised €13M in 2025 for helium-3-free **solid-state magnetic (ADR) cooling**, explicitly a supply-resilience play — company/press (established funding fact)
- [T3] Lasers are a real trapped-ion/neutral-atom constraint, but the broad laser market is huge (\$100Bs); the chokepoint is *quantum-grade* specificity (linewidth, stability, specific wavelengths), not volume — PostQuantum (reported)

- [T2] **NIST launched a ~\$20M quantum manufacturing center** targeting the cryostat/cryogenic bottleneck (mid-2026) — a direct policy response to the fridge chokepoint — Tech Times / NIST (established)

KEY GRADED CLAIMS

- [T3] **>90% of high-purity quantum material processing occurs outside NATO territory** — NATO Transatlantic Quantum Community study, via 2025 supply-chain analyses (claimed; primary study not directly read)
- [T3] China is building domestic dilution-fridge/laser capacity partly in response to export controls — PostQuantum China self-sufficiency reporting (contested/reported)
- [T3] Rare inputs beyond He-3: ²⁸Si enrichment capacity is thin and partly ex-NATO; CVD electronic-grade diamond is near-monopoly (Element Six) — supply-chain reports (reported)

CONFLICTS / OPEN QUESTIONS

- Helium-3 deficit figures vary widely and fold in **non-quantum demand** (neutron detection, medical MRI, security) — treat the L/yr numbers as order-of-magnitude.
- Whether alternative cooling (Kiutra-style ADR, or higher-temp qubits) relieves the fridge/He-3 chokepoint before it binds hard.
- Reshoring economics: a \$20M NIST center is small against the capital intensity of the bottleneck — is national supply resilience even affordable without industrial-policy money (see E-us \$2B Commerce package, E-sovereign)?

SOURCES

- <https://warontherocks.com/2025/10/the-supply-chain-chokepoints-in-quantum/>
- <https://postquantum.com/building-quantum-computers/quantum-cryogenic-infrastructure-helium3/>
- <https://www.techtimes.com/articles/319456/20260701/nists-20m-quantum-manufacturing-center-targets-cryostat-bottleneck.htm>
- <https://www.futuremarketsinc.com/the-global-quantum-computing-supply-chain-2026-2036-materials-components-and-enabling-hardware-across-qubit-modalities/>

- <https://postquantum.com/china-quantum-ambition/china-supply-chain-self-sufficiency/>

Talent & workforce · E-talent

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The binding constraint is **hybrid practitioners** — people who can run a dilution fridge, write control software, or translate a chemistry problem into circuits. PhD physicists exist; production engineers, cryo technicians, and quantum-literate domain scientists are scarce. The commonly cited ratio is **one qualified candidate per three open roles**. Every national strategy (E-us through E-others) now carries a workforce pillar, and the university pipeline is growing but still thin at the applied end — most programs mint researchers while industry increasingly hires engineers. The immigration and security-clearance dimension of this — which reshapes *who is eligible* to fill the gap — is broken out as **E-immigration**.

THE NUMBERS (GRADED, WITH CAVEATS)

- [T5] ~1 qualified candidate per 3 open roles — widely repeated (WEF/industry surveys); the survey base is small and self-reported by employers with hiring incentives.
- [T5] ~30,000 — rough estimate of the global quantum workforce (industry reports); definitional soup (does a PQC migration engineer count?).
- [T5] US quantum-skill job postings roughly tripled since 2018; ~180% growth 2020–2024 — MIT Quantum Index Report 2025. Posting counts overstate headcount (reposts, evergreen reqs).
- [T5] **176 university quantum programs worldwide; only 29 offer a graduate degree specifically in quantum computing.** Germany leads master's programs (12), UK 10, US 9 (Technical.ly / program censuses).
- [T5] 82% of US defense-industrial-base companies report difficulty finding qualified STEM workers; **50% of advanced-STEM-degree holders in the DIB are foreign-born** (IFP/CSIS) — the tie to immigration policy.

KEY GRADED CLAIMS

- [T3] The gap is concentrated in **hybrid roles** — technicians, control engineers, research software engineers — not PhDs — RAND workforce study (demonstrated)

- [T5] Salary inflation for senior quantum engineers (\$200k–500k+ at US firms) reflects the shortage — press/recruiter data (claimed)
- [T4] Vendor education pushes (IBM Qiskit ecosystem, QuEra/others' programs) double as funnel marketing; completion-to-employment rates unpublished (claimed)
- [T2] The **22 June 2026 EO** directs a domestic-workforce push via expanded apprenticeships and credentials — whitehouse.gov (established policy direction)
- [T6] Whether AI-assisted development compresses the needed workforce (fewer, more leveraged engineers) is open and rarely modeled (speculative)

CONFLICTS / OPEN QUESTIONS

- **C-talent-size:** "crippling shortage" narrative vs the 2022–23 period when quantum startups ran layoffs. Both were true at once — shortage in niche roles, surplus in generalists. Resolution: longitudinal hiring data, which nobody publishes.
- Security-clearance requirements shrink the eligible pool further for defense quantum work — mostly undiscussed in the shortage literature (see E-immigration).
- Does the field need quantum-native curricula or physics + software retraining? Program design diverges by country.

SOURCES

- <https://technical.ly/workforce/quantum-workforce-shortage-guest-post/>
- <https://www.weforum.org/stories/2025/11/upskilling-quantum-talent/>
- https://www.rand.org/pubs/research_reports/RRA3889-1.html
- <https://ifp.org/stem-immigration-is-critical-to-american-national-security/>
- <https://www.forbes.com/sites/bernardmarr/2025/05/02/quantum-talent-wars-the-hidden-crisis-threatening-techs-next-frontier/>

National programs — United Kingdom · E-uk

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The UK has run a continuous national program since **2014** (the National Quantum Technologies Programme, ~£1B over its first decade) — the earliest sustained national quantum bet among Western states. The **March 2023 National Quantum Strategy** committed **£2.5B over ten years** toward being a "leading quantum-enabled economy by 2033," structured around **five quantum missions** (announced Nov 2023) with the **National Quantum Computing Centre (NQCC, Harwell)** as the compute hub. In 2026 the government layered a **~£2B scaling-and-procurement package** whose centerpiece is **ProQuire** — using state procurement as the demand signal (an advanced market commitment to buy large-scale machines) with explicit QuOps milestones. The UK's distinctive move is demand-side pull; most rivals push supply-side grants.

THE NUMBERS (GRADED, WITH CAVEATS)

- [T5] £2.5B / 10 yr — National Quantum Strategy (gov.uk, Mar 2023). A pledge across research, innovation, skills; annual appropriations arrive via spending reviews and can lag the headline.
- [T5] ~£1B — cumulative NQTP investment 2014–2023 (uknqt.ukri.org). Sometimes added to the £2.5B in "UK has spent £3.5B" claims even though the strategy partly *continues* the same funding streams — double-counting risk.
- [T5] £2B (~\$2.67B) — 2026 scaling + procurement commitment: ~£1B new R&D over four years + ~£1B advanced market commitment to procure a billion-quantum-operation platform by 2031 (Quantum Computing Report / NQCC).
- [T5] **QuOps milestones (NQCC):** 1 MQuOp by 2030, 1 GQuOp by 2033, 1 TQuOp by 2035. Targets set by policy; feasibility is a T6 question and depends on the still-unstandardized "QuOp" unit (see E-standards).

KEY GRADED CLAIMS

- [T2] NQCC operational at Harwell; testbed program hosts multiple modalities (superconducting, ion, photonic, neutral-atom) — NQCC (established)
- [T3] **ProQure** launched late March 2026, inviting prototype submissions for evaluation-led procurement — NQCC/QCR (roadmap)
- [T4] "Billion quantum operations by 2031" procurement target — government/vendor co-produced timeline; no machine today is close (roadmap, contested feasibility)
- [T3] Five missions span compute, networks, PNT/navigation, and healthcare imaging (brain scanners via OPM magnetometry — see A-magneto) — gov.uk (established framing)

CONFLICTS / OPEN QUESTIONS

- Is the £2B package **inside or on top of** the £2.5B strategy envelope? Reporting is ambiguous — flag until a budget document settles it.
- **Post-Brexit exclusion** from parts of Horizon Europe quantum calls (the UK re-associated to Horizon in 2024 but on limited terms): how much does it cost the ecosystem?
- Can procurement pull (ProQure) work when no vendor can yet deliver the spec — or does it just subsidize R&D under a purchasing label?
- UK champions (Oxford Ionics, Quantum Motion, ORCA, Riverlane) are strong but small; Oxford Ionics was **acquired by IonQ** (2025, ~\$1.08B) — a sovereignty question given US export-control leverage (see E-mna, E-export).

SOURCES

- <https://www.nqcc.ac.uk/about-us/national-quantum-strategy/>
- https://assets.publishing.service.gov.uk/media/6411a602e90e0776996a4ade/national_quantum_strategy.pdf
- <https://quantumcomputingreport.com/uk-government-commits-2-billion-2-67-billion-usd-to-national-quantum-scaling-and-procurement/>
- <https://www.nqcc.ac.uk/blogs/the-uk-has-thrown-down-the-gauntlet-on-quantum/>
- <https://uknqt.ukri.org/our-programme/>

National programs — United States · E-us

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

The US program is built on the **National Quantum Initiative (NQI) Act of 2018** (Public Law 115-368), which stood up the National Quantum Coordination Office in OSTP, tasked NIST, NSF, and DOE with QIS Research Centers, and authorized ~\$1.2B over its first five years. The authorization **lapsed 30 September 2023**; the field then ran on annual appropriations for over two years. Two 2026 events redefined the posture: a **reauthorization bill (S.3597, Young/Cantwell, Jan 2026)** extending NQI to 2034, and — decisively — **two executive orders signed 22 June 2026** that pivoted federal policy from grant-making toward industrial policy and a hard national build target.

THE NUMBERS & MECHANISMS (GRADED, WITH CAVEATS)

- [T5] ~\$1.2B — original NQI five-year authorization (2019–2023) (quantum.gov). *Authorized ≠ appropriated*; annual QIS budgets ran ~\$900M–1B/yr per the NQI budget supplements, which fold in pre-existing agency programs (double-counting risk).
- [T5] ~\$1.8B / 5 yr — House NQI Reauthorization figure; Senate S.3597 (Jan 2026) authorizes NIST ~\$85M/yr FY26–30, up to three new NIST quantum centers at \$18M/yr each, \$25M for NASA QIST (congress.gov). Authorization only until appropriated.
- [T5] \$625M — DOE next-phase funding for the five National QIS Research Centers (energy.gov). Same headline as the 2020 tranche; verify renewal vs recycled 2020 announcement.
- [T5] ~\$2B — Commerce Dept. awards to nine quantum firms (May 2026), ~half to IBM, reportedly taking equity stakes (CNN/Forbes). Terms not fully public; blurs public/private tallies (see E-vc, E-sovereign).
- [T2] **EO "Ushering in the Next Frontier of Quantum Innovation," 22 Jun 2026** — launches **QC-ADDS** (Quantum Computer for Application Development and Discovery Science), a DOE-led effort (launched 23 Jun 2026) to deploy a discovery-capable quantum computer **by 2028**; directs Commerce/War/Energy/NASA capability + workforce plans; expands the Quantum Counterintelligence Protection Team (whitehouse.gov).

- [T2] **Companion PQC EO "Securing the Nation Against Advanced Cryptographic Attacks," 22 Jun 2026** — OMB deadlines for high-value-asset PQC: **key establishment by 31 Dec 2030, digital signatures by 31 Dec 2031**; Commerce PQC-migration pilot due 31 Dec 2027 (see A-pqc, E-pqcmarket).

KEY GRADED CLAIMS

- [T2] NQI authorization lapsed Sept 2023; programs continued via appropriations — congress.gov / AIP FYI (established)
- [T3] S.3597 introduced Jan 2026; extends NQI to Dec 2034 — congress.gov (claimed; passage status stale-checks fast)
- [T4] Equity-stake structure of the \$2B Commerce package — press reporting on unreleased terms (claimed)
- [T4] "Quantum computer by 2028" (QC-ADDs) is a DOE program target, not a peer-reviewed roadmap — treat as policy ambition (roadmap)

CONFLICTS / OPEN QUESTIONS

- Has S.3597 passed either chamber, and does the June 2026 EO framework moot it? Recheck congress.gov.
- How much of the \$2B Commerce package is new money vs repackaged CHIPS/DOE funds?
- NQI supplement totals mix new-start and legacy programs — the "US spends \$X/yr on quantum" number is soft.
- The 2026 EOs shift the model from vendor-neutral grants (cf. DARPA QBI, E-defense) to picking champions with equity — a strategic turn the NQI Act's authors did not design for (see E-sovereign).

SOURCES

- <https://www.whitehouse.gov/presidential-actions/2026/06/ushering-in-the-next-frontier-of-quantum-innovation/>
- <https://www.whitehouse.gov/presidential-actions/2026/06/securing-the-nation-against-advanced-cryptographic-attacks/>
- <https://www.mayerbrown.com/en/insights/publications/2026/06/president-trump-signs-two-executive-orders-on-quantum-computing-and-accelerated-post-quantum-cryptography-migration>
- <https://www.congress.gov/bill/119th-congress/senate-bill/3597/text>

- <https://www.energy.gov/articles/energy-department-announces-625-million-advance-next-phase-national-quantum-information>
- <https://www.cnn.com/2026/05/21/business/ibm-quantum-computing-firms-grants>
- <https://www.quantum.gov/wp-content/uploads/2024/12/NQI-Annual-Report-FY2025.pdf>

Private investment & VC · E-vc

Layer: L5 Ecosystem & geopolitics · **Chapter:** §06 · **Status:** depth

SUMMARY

Private quantum capital ran through three phases: the **2021 SPAC wave** (IonQ, Rigetti, D-Wave, Arqit went public early, most then traded far below listing), a **2022–23 trough**, and a **2024–26 boom** in which generalist mega-capital (BlackRock, Nvidia, Temasek, sovereign funds) replaced specialist quantum VCs. The two most-cited 2025 totals **disagree by 3x because they count different things**: PitchBook's ~\$3.9B is pure quantum-*computing* VC; McKinsey's ~\$12.6B (6.3x over 2024) counts *all* quantum-technology startup investment including sensing/comms **and** M&A (see E-mna). Either way, government's share of investment fell from ~a third in 2024 to ~3% in 2025 as private capital took over. In H1 2026 the action moved to **exits** (see E-mna).

THE NUMBERS (GRADED, WITH CAVEATS)

- [T5] ~\$3.9B across 127 deals — 2025 global quantum-**computing** VC record (PitchBook). Narrow definition.
- [T5] ~\$12.6B — 2025 quantum-**technology** startup investment, 6.3x over 2024, ~90% to computing (McKinsey 2026 Monitor). Broad definition + includes acquisition value. **Do not add to or compare directly with PitchBook's \$3.9B** — different denominators.
- [T5] \$1B at ~\$6B pre / ~\$7B post — PsiQuantum Series E, BlackRock-led, Sept 2025 (press variously reports \$6B pre and \$7B valuation — flag the discrepancy).
- [T5] ~\$838.9M at \$10B pre-money — Quantinuum Series B, Nov 2025; Nasdaq IPO filed 2026 (Crunchbase News).
- [T5] Government share of quantum investment fell ~**33% (2024) → ~3% (2025)** as private funds/capital markets took over (McKinsey).
- [T5] Top capital deployers: BlackRock (~\$1.7B), Nvidia (~\$1.6B), Baillie Gifford, Temasek (Quantum Insider, Jun 2026).

KEY GRADED CLAIMS

- [T2] 2021 SPAC cohort (IonQ, Rigetti, D-Wave) went public years before product-market fit; all traded >75% below peak during 2022–23 — market record (established)
- [T5] Venture-growth share of deal value jumped from ~1% to ~27.5%; sovereign funds and crossover investors now lead the largest rounds (PitchBook) (demonstrated, as data)
- [T3] Crunchbase (mid-2026): startup *investment* actually **slowed** in 2026 even as public markets held strong — the money rotated from private rounds to IPOs/M&A (demonstrated)
- [T6] Whether 2024–26 valuations survive contact with revenue (sector revenue ~\$1B/yr, see E-market) is the open bet (speculative)

CONFLICTS / OPEN QUESTIONS

- **C-vc-bubble:** "capital cycle rationally pricing 2030s FTQC" vs "SPAC-wave redux with bigger names." Resolution: 2027–28 revenue growth at IonQ/Quantinuum/D-Wave versus guidance.
- Double-counting: government equity stakes (E-us \$2B Commerce package; see E-sovereign) now blur public/private tallies.
- Keyword pollution: defense-drone maker "Quantum Systems" (\$1.2B raise, Jul 2026) is **unrelated** to quantum computing but contaminates naive funding scrapes.
- IonQ +700% trailing-year and peers trade at revenue multiples with no historical analog outside dot-com — a valuation the ~\$1B sector revenue cannot yet explain.

SOURCES

- <https://pitchbook.com/news/articles/inside-quantum-computings-record-setting-investment-run>
- <https://www.mckinsey.com/capabilities/mckinsey-technology/our-insights/mckinsey-quantum-technology-monitor-2026-a-commercial-tipping-point>
- <https://news.crunchbase.com/venture/quantum-computing-startup-investment-data-quantinuum-ipo/>
- <https://fortune.com/2026/06/29/blackrock-nvidia-temasek-betting-billions-quantum-computing/>

- <https://thequantuminsider.com/2026/06/26/top-quantum-computing-investors-in-2026/>

§07 · History

The full arc, 1900 → today — from Planck's quantum to below-threshold error correction.

The "second quantum revolution" framing (2003) · T-2ndrev

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

By the early 2000s the field needed a way to explain what was new. The first quantum revolution (1900–1935, see T-old, T-formalism) discovered the *rules* — quantization, superposition, entanglement — and used them passively to *understand* nature: why atoms are stable, why metals conduct, why the sun shines. It gave the 20th century the transistor, the laser, and the atomic clock, all of which exploit quantum mechanics in bulk without controlling individual quantum states. The **second quantum revolution** is the shift from understanding the rules to *engineering with them* — deliberately preparing, controlling, and measuring individual quantum systems (single atoms, photons, spins, superconducting circuits) and harnessing superposition and entanglement as active resources. The phrase was crystallized by Jonathan Dowling and Gerard Milburn in a 2003 Royal Society paper (arXiv 2002) that opened flatly: "We are currently in the midst of a second quantum revolution." It became the standard framing for the whole quantum-technology enterprise — computing, communication, sensing, simulation — and now anchors funding programs (the EU Quantum Flagship explicitly invokes it), popular explanations, and strategy documents worldwide.

WHAT THE FRAMING CAPTURES

The distinction is real, not just rhetorical. The first revolution's technologies (transistor, laser, LED, MRI, GPS clocks) rely on quantum mechanics governing the collective behavior of many particles; you never address a single electron. The second revolution's technologies require *single-quantum control*: trap one ion and watch it, entangle two photons on demand, keep a superconducting qubit coherent long enough to run a gate. That capability is exactly what the 2012 Nobel to Haroche and Wineland recognized — "measuring and manipulating individual quantum systems" — and it is the enabling condition for everything in this manual's hardware and stack layers (see H-, S-). Entanglement moves from a philosophical puzzle (the first revolution's leftover, see T-epr) to an engineering resource.

MILESTONE TIMELINE

- 1980s–1990s — Single-system control becomes real: single trapped ions (Wineland, Dehmelt), cavity QED (Haroche), the first superconducting and ion qubits (see T-early) — the technical precondition for the framing — [T1] historical record
- 2002 (Jun) — "Quantum Technology: The Second Quantum Revolution" posted to arXiv — Jonathan P. Dowling & Gerard J. Milburn — [T1] arXiv:quant-ph/0206091
- 2003 (Aug) — Published in *Philosophical Transactions of the Royal Society A* — the citation that fixes the term — [T1] Phil. Trans. R. Soc. A 361, 1655 (2003)
- 2012 — Nobel Prize to Haroche & Wineland "for ground-breaking experimental methods that enable measuring and manipulation of individual quantum systems" — the revolution's core capability, honored (see T-nobel) — [T1] nobelprize.org
- 2016– — The term is adopted into policy: the EU Quantum Flagship (€1B) and subsequent national strategies frame themselves as advancing the second quantum revolution (see E-eu, E-us) — [T5] program documents
- 2025 (UN) — "International Year of Quantum Science and Technology," marking 100 years since matrix mechanics — the framing goes fully mainstream — [T1] UN/UNESCO designation

WHY IT BELONGS ON THE TIMELINE

Naming an era changes how it is funded and pursued — the same way "NISQ" (see T-nisq) organized the noisy-hardware research program. "Second quantum revolution" gave governments and grant agencies a clean story: the first revolution paid off enormously (transistor, laser), so the second is worth a national bet. It also sets an honest expectation bar. If the first revolution's payoff was the transistor, the second revolution's advocates are implicitly promising something comparably transformative — which is precisely the promise the skeptics (see T-winter, O-hype) say has not yet been delivered.

KEY GRADED CLAIMS

- claim: The term "second quantum revolution" was crystallized by Dowling & Milburn (2002/2003) · tier: T1 · source: arXiv:quant-ph/0206091; Phil.

Trans. R. Soc. A 361, 1655 (2003) · status: established (the underlying idea has older roots, e.g. Alain Aspect's usage, but this is the canonical citation)

- claim: The distinction (passive understanding vs. active single-system engineering) is technically substantive, not just branding · tier: T1 · status: established — single-quantum control is a real experimental capability, recognized by the 2012 Nobel
- claim: The second revolution has delivered a transistor-scale economic payoff · tier: T6 · status: speculative/contested — the central open question (see O-killerapp, T-winter)

SOURCES

- Dowling & Milburn, "Quantum Technology: The Second Quantum Revolution," Phil. Trans. R. Soc. A 361, 1655 (2003), arXiv:quant-ph/0206091
- Nobel Prize in Physics 2012 (Haroche & Wineland); EU Quantum Flagship documentation; UN International Year of Quantum 2025
- **Go deeper:** T-nisq (a later era-naming); T-nobel; E-eu, E-us (how the framing drives policy); T-winter (the counter-narrative)

Loophole-free Bell tests (2015) · T-belltests

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

For fifty years after Bell's 1964 theorem, every experiment that violated a Bell inequality left a door open. A determined local-hidden-variable believer could always invoke one of two escape hatches. The **detection loophole**: real detectors miss most particles, so maybe the detected sub-sample is unrepresentative and a "fair sampling" assumption is smuggling in the result. The **locality (communication) loophole**: if the two measurement stations are close enough, or fix their settings early enough, a subluminal signal could in principle coordinate them. Every experiment before 2015 closed one loophole while leaving the other open — Aspect (1982) closed locality with fast-switched analyzers but used inefficient photon detectors; later ion experiments closed detection but sat centimeters apart. The believer could always retreat to the unclosed hatch. In 2015 three groups, on three independent platforms, closed *both loopholes simultaneously* for the first time, finally ruling out local realism with no remaining escape. This is the experimental capstone of the foundations program (see T-foundations) and the work most directly recognized by the 2022 Nobel to Aspect, Clauser, and Zeilinger (see F-bell, T-nobel).

THE TECHNICAL BREAKTHROUGH

Closing both loopholes at once requires a brutal combination: detectors efficient enough ($> \sim 2/3$, well above the fair-sampling threshold) *and* stations far enough apart that no light-speed signal can traverse the gap during a measurement. Delft solved it with an **event-ready entanglement-swapping scheme** — two distant NV-center electron spins are entangled by having their emitted photons interfere at a midpoint, so only genuinely entangled, heralded events are counted, sidestepping the detection loophole entirely while the 1.3 km separation guarantees locality. The photonic groups (Vienna, NIST) instead used near-unity-efficiency superconducting nanowire single-photon detectors (SNSPDs — see H-detect) with fast quantum-random setting choices over tens of meters.

MILESTONE TIMELINE

- 2015 (Aug submitted, Oct published) — First loophole-free test: two NV-center electron spins in diamond 1.3 km apart, event-ready entanglement swapping, 245 trials, $S = 2.42 \pm 0.20$ — Hensen, Bernien et al. (Hanson group, TU Delft) — closes detection + locality together — [T2] Hensen et al., Nature 526, 682 (2015), doi:10.1038/nature15759
- 2015 (Nov, same PRL issue) — Photon experiment with high-efficiency SNSPDs, violation at ~ 11.5 standard deviations — Giustina, Zeilinger et al. (Vienna) — independent confirmation, photonic platform — [T2] PRL 115, 250401 (2015)
- 2015 (Nov, same PRL issue) — Independent photon experiment, $\sim 7\sigma$ — Shalm, Nam et al. (NIST Boulder) — third platform, same conclusion — [T2] PRL 115, 250402 (2015)
- 2016–2017 — Ion (Rosenfeld/Weinfurter, 398 m) and further photon tests tighten statistics — [T2]
- 2018 (May) — Cosmic Bell test uses light from stars (and, in the BIG Bell Test, $\sim 100,000$ human volunteers generating bits) to choose measurement settings, pushing the settings' causal origin back centuries/light-years and closing the **freedom-of-choice** loophole — Handsteiner et al.; BIG Bell Test Collaboration — [T2] PRL 118, 060401 (2017); Nature 557, 212 (2018)

THE HUMAN CONTEXT

The three 2015 papers appeared within months of each other, a near-simultaneous finish after a fifty-year race — the same pattern of independent convergence that marked the birth of the formalism in 1925–26. Delft's event-ready scheme was slower and yielded far fewer events (245 trials over weeks) but was conceptually airtight; the photonic experiments were statistically overwhelming but relied on detector efficiency. Together they left the local-realist with only philosophically desperate positions.

KEY GRADED CLAIMS

- claim: Local hidden-variable theories are experimentally excluded with detection and locality loopholes closed simultaneously · tier: T2 (three independent 2015 experiments; now textbook) · status: established — Hensen (Delft), Giustina (Vienna), Shalm (NIST), three platforms

- claim: The residual "superdeterminism" loophole cannot be closed by any experiment · tier: T1 (logical) · status: established — it denies the free choice of settings and is unfalsifiable in principle; noted, not a live scientific dispute

SOURCES

- Hensen et al., Nature 526, 682 (2015), doi:10.1038/nature15759
- Giustina et al., PRL 115, 250401 (2015); Shalm et al., PRL 115, 250402 (2015)
- Handsteiner et al. (cosmic Bell), PRL 118, 060401 (2017); The BIG Bell Test Collaboration, Nature 557, 212 (2018)
- **Go deeper:** F-bell (the inequality and its meaning); T-foundations (the 1935→1982 lead-up); H-detect (the SNSPDs that made it possible); T-nobel (2022)

Birth of quantum computing (1980–1994) · T-birth

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The idea arrived from several directions at once. Yuri Manin (1980, in a Russian book) and Paul Benioff (1980) asked whether computation itself could be quantum-mechanical — Benioff building an explicit quantum Turing machine to show computation is compatible with unitary evolution. Richard Feynman flipped the question in a now-legendary May 1981 keynote at the first MIT Physics of Computation conference: classical computers choke on simulating quantum systems because the state space grows exponentially, so build a quantum machine to simulate quantum physics natively. "Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical." David Deutsch made it rigorous in 1985 — a universal quantum computer and the first quantum algorithm — motivated partly by wanting to test the many-worlds interpretation. Bennett and Brassard, building on Wiesner's dormant conjugate-coding idea (see T-qinfobirth), showed quantum states could carry provably secure keys (BB84, 1984). The era ends with a detonation: Peter Shor's 1994 algorithm factored integers in polynomial time, threatening RSA and the entire public-key infrastructure. Overnight, quantum computing went from a physicists' curiosity to a national-security priority. Every funding line that exists today traces back to that one result.

MILESTONE TIMELINE

- 1980 — Quantum computation floated in *Computable and Uncomputable* — Yuri Manin — earliest published suggestion (in Russian, so little-noticed in the West for years) — [T1] Manin, Sovetskoye Radio (1980)
- 1980 — Quantum-mechanical model of a Turing machine — Paul Benioff (Argonne) — showed computation is compatible with reversible quantum evolution — [T1] Benioff, J. Stat. Phys. 22, 563 (1980)
- 1981 (6–8 May) — "Simulating Physics with Computers" keynote, MIT Physics of Computation conference — Richard Feynman — the field's founding motivation: simulate quantum with quantum — [T1] published Feynman, Int. J. Theor. Phys. 21, 467 (1982)

- 1982 — No-cloning theorem: an unknown quantum state cannot be copied — Wootters & Zurek; Dieks — the security root of quantum crypto (see F-nocloning) — [T1] Nature 299, 802 (1982)
- 1984 — BB84 quantum key distribution protocol — Charles Bennett (IBM) & Gilles Brassard (Montréal) — first quantum information technology; still deployed today — [T1] Proc. IEEE Int. Conf. on Computers, Systems and Signal Processing, Bangalore, 175 (1984)
- 1985 — Universal quantum computer; the quantum Turing machine; the first quantum algorithm (Deutsch's problem) — David Deutsch (Oxford) — the theoretical foundation of the whole field — [T1] Deutsch, Proc. R. Soc. A 400, 97 (1985)
- 1992 — Deutsch–Jozsa algorithm, first provable exponential quantum–classical separation in the exact query model — David Deutsch & Richard Jozsa — proof the speedups could exist — [T1] Proc. R. Soc. A 439, 553 (1992)
- 1993 — Bernstein–Vazirani and the first steps toward complexity-theoretic separation; quantum teleportation protocol — Bernstein & Vazirani (STOC 1993); Bennett, Brassard, Crépeau, Jozsa, Peres, Wootters — entanglement as a communication resource — [T1] PRL 70, 1895 (1993)
- 1994 (20 Nov) — Polynomial-time factoring and discrete-log algorithm — Peter Shor (AT&T Bell Labs) — broke public-key crypto in principle; ignited the field and its funding — [T1] Shor, Proc. 35th FOCS, 124 (1994); SIAM J. Comput. 26, 1484 (1997)
- 1994 — Simon's problem: the exponential separation that directly inspired Shor — Daniel Simon — the algorithmic template (period-finding) Shor generalized — [T1] Proc. 35th FOCS, 116 (1994)

THE HUMAN CONTEXT

Feynman's talk is often called the birth of the field, but he was pitching a simulator, not a general-purpose computer, and he was skeptical the machine could be built. Deutsch's universal model came from a physicist worried about interpretation, not computation. Shor's algorithm reportedly grew out of Simon's problem, which Shor heard about at a talk; he generalized period-finding to factoring within weeks. The result was so consequential that within two years the field went from a few dozen people to a global program — and cryptographers began the migration that is still underway 30 years later (see A-pqc).

KEY GRADED CLAIMS

- claim: Shor's algorithm factors N in polynomial time on an ideal fault-tolerant quantum computer · tier: T1 · source: Shor 1994; SIAM J. Comput. 26, 1484 (1997) · status: established (mathematically proven; large-scale execution still awaits hardware — see S-shor, O-scaling)
- claim: Feynman's simulation pitch is the application most likely to pay off first · tier: T6 · provenance: widely held field belief, unproven at scale · status: speculative — see S-qsim, O-killerapp
- claim: The no-cloning theorem (1982) makes BB84 security information-theoretic, not computational · tier: T1 · status: established — see F-nocloning, A-qkd

SOURCES

- Manin (1980); Benioff, J. Stat. Phys. 22 (1980); Feynman, IJTP 21 (1982); Wootters–Zurek, Nature 299 (1982); Bennett–Brassard, BB84 (1984); Deutsch, Proc. R. Soc. A 400 (1985); Shor, FOCS (1994); Simon, FOCS (1994)
- **Go deeper:** T-qinfobirth (Wiesner→Holevo→Bennett-Brassard lineage); S-shor; F-nocloning; T-early (turning theory into hardware)

Early experiments (1995–2010) · T-early

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

After Shor, the question stopped being "is it useful?" and became "can you actually build one?" This era answered "in principle, yes" — while quietly revealing how hard the engineering would be. Cirac and Zoller published a realistic trapped-ion two-qubit gate in May 1995, and Wineland's NIST group demonstrated a working quantum logic gate on a single beryllium ion within months. The deepest result was theoretical: Shor (1995) and Steane (1996) proved that quantum errors could be corrected *without measuring* — *and thereby destroying* — *the encoded data*, removing the single strongest objection to the whole enterprise. The threshold theorem (Aharonov–Ben-Or, Knill–Laflamme–Zurek, Kitaev, 1996–98) then showed that below a critical error rate, arbitrarily long computations are possible. Grover added a second headline algorithm. NMR machines ran the first real algorithms — including Shor's algorithm factoring 15 — before hitting a hard scalability wall. DiVincenzo wrote down the five-item checklist any real platform must satisfy, still the field's scorecard. And superconducting circuits went from the first coherent charge qubit (NEC 1999) through circuit QED (Yale 2004) to the transmon (Yale 2007) — the noise-tolerant design that would dominate the next two decades.

MILESTONE TIMELINE

- 1995 (May) — Trapped-ion CNOT gate proposal using a shared vibrational mode as the bus — Ignacio Cirac & Peter Zoller (Innsbruck) — first realistic hardware blueprint for any platform — [T1] Cirac & Zoller, PRL 74, 4091 (1995)
- 1995 (Dec) — First quantum logic gate demonstrated on a single ${}^9\text{Be}^+$ ion — Monroe, Meekhof, King, Itano, Wineland (NIST) — theory becomes experiment in seven months — [T1] Monroe et al., PRL 75, 4714 (1995)
- 1995–96 — Quantum error-correcting codes: the 9-qubit Shor code, then the 7-qubit Steane (CSS) code — Peter Shor; Andrew Steane — errors are correctable; fault tolerance is possible — [T1] Shor, Phys. Rev. A 52, R2493 (1995); Steane, PRL 77, 793 (1996)

- 1996 — Threshold theorem: below a critical physical error rate, arbitrarily long computation is possible — Aharonov–Ben-Or; Knill–Laflamme–Zurek; Kitaev — the theoretical license for fault tolerance (see S-qec) — [T1] STOC 1997; Science 279, 342 (1998)
- 1996 — Unstructured-search algorithm, quadratic speedup $O(\sqrt{N})$ — Lov Grover (Bell Labs) — the second canonical algorithm — [T1] Grover, Proc. 28th STOC, 212 (1996)
- 1997–98 — First NMR quantum algorithms (Deutsch–Jozsa, Grover on 2 qubits) — Chuang, Gershenfeld, Jones et al. — first algorithms run on any hardware — [T2] Nature 393, 143 (1998); Chuang et al., PRL 80, 3408 (1998)
- 1999 — First coherent superconducting (charge) qubit; Rabi oscillations in a Cooper-pair box — Nakamura, Pashkin, Tsai (NEC) — solid-state qubits arrive — [T2] Nakamura et al., Nature 398, 786 (1999)
- 2000 — DiVincenzo criteria: five requirements (plus two for communication) any quantum computer must meet — David DiVincenzo (IBM) — the checklist every new modality is still graded against — [T1] DiVincenzo, Fortschr. Phys. 48, 771 (2000)
- 2001 — Shor's algorithm factors $15 = 3 \times 5$ on 7 NMR qubits — Vandersypen, Steffen, Chuang et al. (IBM Almaden/Stanford) — proof-of-principle for the algorithm that started the field — [T2] Nature 414, 883 (2001)
- 2004 — Circuit QED: a superconducting qubit strongly coupled to a single microwave photon in a resonator — Wallraff, Schoelkopf, Girvin et al. (Yale) — the readout-and-coupling architecture of all superconducting QCs — [T2] Nature 431, 162 (2004)
- 2007 — Transmon qubit: charge-noise-insensitive by design — Koch, Schoelkopf, Devoret et al. (Yale) — the workhorse used by IBM, Google, Rigetti to this day — [T2] Koch et al., Phys. Rev. A 76, 042319 (2007)

THE HUMAN CONTEXT

NMR's early lead was real and then evaporated — Braunstein and colleagues showed in 1999 that room-temperature liquid-state NMR states are so mixed they are arguably never entangled, and the signal decays exponentially with qubit count, so the "factoring 15" demo could not have scaled. Two of the 2007 transmon co-authors, Devoret and Martinis, would later share the 2025 Nobel

Prize for the macroscopic-quantum-tunneling experiments (1984–85) that made superconducting qubits conceivable in the first place (see T-nobel). This era is where the modern hardware map was drawn: ions and superconductors as the two front-runners, error correction as the organizing goal.

KEY GRADED CLAIMS

- claim: Quantum error correction works — errors can be detected and reversed without measuring the logical data · tier: T1 · source: Shor (1995), Steane (1996) · status: established (theory); experimentally realized below threshold only in 2024 (see T-ecera)
- claim: Liquid-state NMR does not scale (signal decays exponentially; states arguably unentangled) · tier: T1 · source: Braunstein et al., PRL 83, 1054 (1999) · status: established — why NMR's early lead evaporated
- claim: A working platform must satisfy the DiVincenzo criteria (scalable qubits, initialization, universal gates, long coherence, measurement) · tier: T1 · status: established as the field's standard scorecard

SOURCES

- Cirac–Zoller, PRL 74 (1995); Monroe et al., PRL 75 (1995); Shor (1995); Steane (1996); Grover, STOC (1996); DiVincenzo, Fortschr. Phys. 48 (2000); Vandersypen et al., Nature 414 (2001); Wallraff et al., Nature 431 (2004); Koch et al., PRA 76 (2007)
- **Go deeper:** S-qec (codes and threshold); H-ion, H-supercon (the two platforms this era chose); T-nobel (Devoret/Martinis 2025); T-race (industrial scale-up)

The error-correction era (2020–now) · T-ecera

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The scoreboard metric changed. Through the 2010s the field measured itself in physical qubit counts and one-off "supremacy" stunts; from about 2023 it measures itself in *logical* qubits and error suppression. Two results reset expectations: QuEra/Harvard's 48-logical-qubit neutral-atom demo (late 2023) showed logical qubits could be operated at scale, and Google's Willow (Dec 2024) achieved the first *below-threshold* surface code — the regime where making the code bigger makes the logical error rate exponentially *smaller*, which is the threshold theorem finally realized in hardware rather than proven on paper. Fault tolerance became an engineering trajectory instead of a hope, and vendors began publishing logical-qubit roadmaps with dates attached. That is exactly where the grading discipline earns its keep: this era mixes peer-reviewed landmarks (T2) with contested vendor claims (T4), sometimes inside the same press release, and by 2026 nearly every "advantage" headline draws a classical counterattack within months. The 2025 Nobel Prize — to Clarke, Devoret, and Martinis for the 1984–85 experiments that first showed a macroscopic circuit obeys quantum mechanics — was the establishment's formal blessing of the hardware lineage that made all of this possible (see T-nobel).

MILESTONE TIMELINE

- 2020 (Dec) — Jiuzhang photonic boson-sampling advantage claim — USTC (Pan group) — second beyond-classical claim; non-programmable, so no algorithm runs on it — [T2, contested] Zhong et al., Science 370, 1460 (2020)
- 2021 (Nov) — Eagle, 127 qubits, first processor over 100 — IBM — physical scaling continues — [T4] IBM announcement
- 2023 (Feb) — A distance-5 surface code logical qubit outperforms distance-3 for the first time — Google Quantum AI — first hardware sign QEC scaling works, though not yet below threshold — [T2] Nature 614, 676 (2023)
- 2023 (Jun) — "Utility" experiment: 127-qubit Eagle + zero-noise extrapolation beats brute-force classical on a 2D Ising spin model — IBM

- [T2 experiment, contested] Kim et al., Nature 618, 500 (2023); matched by tensor-network methods within weeks
- 2023 (Oct) — 1,180-qubit neutral-atom array announced — Atom Computing — first 1,000+ physical qubits — [T4] company announcement
- 2023 (Dec) — 48 logical qubits, error-corrected operations, on 280 neutral atoms — Bluvstein et al. (Harvard/QuEra/MIT) — logical qubits at scale, first credible demonstration — [T2] Nature 626, 58 (2024)
- 2024 (Apr) — 12 logical qubits on trapped-ion H2 at $\sim 800\times$ fewer errors than physical, with active syndrome extraction — Microsoft + Quantinuum — logical-over-physical advantage demonstrated — [T3→T2] arXiv:2404.02280
- 2024 (9 Dec) — Willow, 105 qubits: below-threshold surface code, logical error rate $\sim$ halves per code-distance step ($d=3\rightarrow 5\rightarrow 7$), $\Lambda \approx 2.14$ — Google Quantum AI — the threshold theorem realized in hardware — [T2] Nature 638, 920 (2025); the companion RCS claim "under 5 min vs. 10^{25} classical years" is [T4, contested]
- 2025 (Feb) — Majorana 1 "first topological qubit" — Microsoft — [T4, contested] Nature's referees noted the published data "does not, on its own, determine whether Majorana zero modes are present"; strong skepticism (APS Physics, Science News, Aaronson)
- 2025 (Jun) — Large-scale FTQC roadmap: qLDPC codes; "Starling" targeting ~ 200 logical qubits / 100M gates by 2029 — IBM — [T4, roadmap]
- 2025 (Oct 7) — Nobel Prize in Physics to John Clarke, Michel Devoret, John Martinis for macroscopic quantum tunneling and energy quantization in Josephson-junction circuits (experiments 1984–85) — the superconducting-qubit lineage honored (see T-nobel) — [T1] nobelprize.org (2025)
- 2025 (Oct) — "Quantum Echoes" out-of-time-order-correlator experiment claimed as a first *verifiable* quantum advantage — Google (Willow) — [T3/T4, contested by default per SCHEMA rule 2]
- 2025 (5 Nov) — Helios launched: 98 $^{137}\text{Ba}^+$ ion qubits, 99.921% two-qubit fidelity, up to 48 logical qubits via color codes (2:1 encoding) — Quantinuum — [T4] company launch; named customers include JPMorganChase, Amgen, BMW, SoftBank

- 2025 (12 Nov) — Nighthawk (120 qubits, higher connectivity) and Loon (qLDPC-connectivity test chip) delivered; real-time decoder progress — IBM — [T4] IBM newsroom
- 2026 (Jan) — 96 logical qubits via a high-rate $[[16,6,4]]$ code on 448 neutral atoms ($\sim 4.7:1$ encoding), with below-threshold error suppression across all logical qubits — QuEra — the logical-qubit record roughly doubles in ~ 13 months — [T2] peer-reviewed, Nature s41586-025-09848-5 (doi:10.1038/s41586-025-09848-5, Jan 2026) — primary-source verified
- 2026 (Jan) — Quantinuum files confidentially for a US IPO ($\sim \$20B$ valuation reported) — [T5] press reports
- 2026 (Jun) — First fully-distributed GHZ (tripartite) state across a three-node network of remote trapped-ion memories linked by photonic interconnects; fidelity 0.841–0.881, Mermin violation with the detection loophole closed — Duke Quantum Center + IonQ (Monroe group) — a step toward modular/networked QC (see H-intercon) — [T3] preprint, Goetting et al., arXiv:2606.17173 (2026) — primary-source verified
- 2026 (Jun) — Microsoft reports Majorana 2 progress (lead-based superconducting stack, larger topological gap) — [T4, contested] company claim; extends the disputed 2025 result, independent confirmation still absent. On 24 Jun 2026 Henry Legg (St Andrews) published a formal Nature critique (d41586-026-01788-y) citing flawed tune-up routines, software errors in the analysis code, and omitted transport measurements; Microsoft filed a formal Nature reply (Nayak: "we stand by our results and our roadmap")

THE HONEST READ (MID-2026)

Below-threshold QEC is real and peer-reviewed (Willow). Logical-qubit counts are climbing fast ($48 \rightarrow 96$ in about a year) but the headline numbers span the full grading range: some are Nature papers, some are launch-day press. The topological-qubit claim remains the era's sharpest credibility test — two years of Microsoft announcements without independent confirmation. And every "verifiable advantage" so far has drawn a classical response, so per SCHEMA rule 2 they are contested by default. The trajectory is genuine; the specific superlatives are provisional.

KEY GRADED CLAIMS

- claim: Below-threshold QEC is demonstrated in hardware (error suppression grows with code distance, $\Lambda \approx 2.14$) · tier: T2 · source: Nature 638, 920 (2025) · provenance: Google, but peer-reviewed and independently discussed · status: demonstrated
- claim: A topological (Majorana) qubit exists · tier: T4 · provenance: Microsoft (sells Azure Quantum) · status: contested — no independent confirmation as of mid-2026
- claim: Logical-qubit counts are doubling on a ~roughly annual cadence (48→96) · tier: T2 · source: QuEra, Nature s41586-025-09848-5 (2026) · status: demonstrated for the specific 2024→2026 neutral-atom results (both peer-reviewed); not a guaranteed trend
- claim: Utility-scale / verifiable-advantage results resist classical counterattack · tier: T3 · status: contested — every claim so far has drawn a classical response within months

SOURCES

- Bluvstein et al., Nature 626 (2024); Google Quantum AI, Nature 638, 920 (2025); Kim et al., Nature 618 (2023); Zhong et al., Science 370 (2020); Microsoft/Quantinuum arXiv:2404.02280; nobelprize.org Physics 2025; quantinuum.com Helios launch (2025-11-05); newsroom.ibm.com Nighthawk/Loon (2025-11-12); ibm.com/quantum/blog/large-scale-ftqc
- QuEra 96-logical (Jan 2026): primary source confirmed — Nature s41586-025-09848-5 (doi:10.1038/s41586-025-09848-5), peer-reviewed, [[16,6,4]] high-rate code on 448 atoms → promoted to T2. Duke/IonQ networking (Jun 2026): primary source confirmed — Goetting et al., arXiv:2606.17173 (preprint → T3)
- **Go deeper:** S-logical, S-qec, S-decoders; T-nobel (2025 prize); O-scaling, O-hype (the roadmap-vs-reality gap); T-winter (the funding-downturn risk)

The EPR paradox & Bohr's reply (1935) · T-epr

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

In May 1935 Einstein, Boris Podolsky, and Nathan Rosen published four pages in *Physical Review* under a title posed as a question: "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" Their answer was no. Using a pair of particles that had interacted and separated, they argued that measuring one instantly fixes a property of the other; since no signal could travel between them (locality) and the outcome is definite (realism), that property must have been *real all along* — an "element of reality" the wavefunction fails to include. Either quantum mechanics is incomplete, or nature is nonlocal. Bohr replied in the same journal five months later with a subtle, much-debated argument that the two particles form a single indivisible system and that EPR's criterion of reality is ambiguous when applied to complementary quantities. Schrödinger, energized by the paper, coined "entanglement" (*Verschränkung*) to name exactly the correlation EPR had isolated, and wrote his cat into the record the same year. For thirty years the exchange looked like unresolvable philosophy. Then Bell showed EPR's own premises — locality plus realism — make a prediction that quantum mechanics violates, and the argument became an experiment (see T-foundations, T-belltests).

WHY IT EARNS ITS OWN CARD

EPR is usually folded into the 1925–35 formalism, but it deserves standing as a milestone in its own right because it is the single most consequential *question* in the field's history. Every later development in quantum foundations and much of quantum information traces to it: Bell's theorem is a direct answer to EPR; the loophole-free tests of 2015 are the experimental verdict; entanglement — named in response to EPR — is the resource behind teleportation, dense coding, device-independent cryptography, and the speedups of quantum computing. Einstein intended the paper as a critique; it became the field's most fertile seed. The word "paradox" is a misnomer: there is no logical contradiction, only a forced choice between locality and completeness that nature (via Bell) later made for us.

MILESTONE TIMELINE

- 1935 (15 May) — "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" — Einstein, Podolsky, Rosen (Institute for Advanced Study, Princeton) — poses the locality-vs-completeness dilemma; the "EPR criterion of reality" — [T1] Phys. Rev. 47, 777 (1935)
- 1935 (Jun–Aug) — Schrödinger's correspondence with Einstein; the three-part "Die gegenwärtige Situation in der Quantenmechanik" coins *Verschränkung* (entanglement) and presents the cat — Erwin Schrödinger — names the phenomenon EPR isolated — [T1] Naturwissenschaften 23, 807/823/844 (1935)
- 1935 (15 Oct) — Bohr's reply, same title as EPR — Niels Bohr — argues the system is indivisible and EPR's reality criterion is ambiguous under complementarity — [T1] Phys. Rev. 48, 696 (1935)
- 1951 — Bohm recasts EPR in discrete spin- $\frac{1}{2}$ variables (the version everyone now teaches) — David Bohm — makes EPR concrete and testable-in-principle — [T1] Bohm, *Quantum Theory* (1951)
- 1964 — Bell converts EPR's premises into a testable inequality — John S. Bell — turns the paradox into a measurement (see T-foundations) — [T1] Physics 1, 195 (1964)

THE HUMAN CONTEXT

Einstein reportedly disliked the final EPR text — Podolsky drafted it, and Einstein felt the essential point (the tension between locality and completeness) was buried under formalism. He restated it more sharply in later letters, especially to Schrödinger, using the phrase *spukhafte Fernwirkung* — "spooky action at a distance." Podolsky, a Soviet-born physicist, is a footnote to most accounts but is the reason the paper exists in the form it does. Bohr, according to Léon Rosenfeld, dropped everything to respond and spent weeks agonizing over the wording — a sign that EPR had landed a real blow, even if the eventual verdict went against Einstein.

KEY GRADED CLAIMS

- claim: EPR is a valid argument, not a fallacy: *if* locality and realism both hold, QM is incomplete · tier: T1 · status: established as logic; the conditional's premises were later shown false by experiment

- claim: Nature is nonlocal in the EPR/Bell sense — one of locality or realism must go · tier: T1 · source: Bell tests, loophole-free 2015 (see T-belltests) · status: established
- claim: Entanglement, the phenomenon EPR isolated, is a physical resource · tier: T1 · status: established — it underlies teleportation, QKD, and quantum computing (see F-entangle, F-teleport)

SOURCES

- Einstein, Podolsky, Rosen, Phys. Rev. 47, 777 (1935); Bohr, Phys. Rev. 48, 696 (1935); Schrödinger, Naturwissenschaften 23 (1935) and Proc. Camb. Phil. Soc. 31, 555 (1935)
- Bell, Physics 1, 195 (1964); Gilder, *The Age of Entanglement* (2008)
- **Go deeper:** T-solvay (the debate EPR grew out of); T-foundations (Bell's answer); T-belltests (the verdict); F-entangle; F-interp

The formalism (1925–1935) · T-formalism

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

In one astonishing decade the ad-hoc patches of the old quantum theory were replaced by a complete mathematical framework — twice, independently, from opposite instincts. Heisenberg, 23 and recovering from hay fever on the island of Helgoland in June 1925, built matrix mechanics by refusing to talk about anything unobservable (electron orbits) and keeping only measurable transition amplitudes. Schrödinger, 38 and unusually old for a revolution, took de Broglie's matter waves literally and wrote down a wave equation over the Christmas holidays of 1925–26. The two formalisms looked nothing alike and their authors disliked each other's approach — Heisenberg called wave mechanics "disgusting," Schrödinger found matrix mechanics "repellent" — yet within months Schrödinger, Pauli, and later von Neumann proved them mathematically equivalent. Born supplied the interpretation that made the whole thing physics: $|\psi|^2$ is a probability, a claim so uncomfortable that Einstein spent the rest of his life objecting to it. Dirac, the quietest of them, unified the formalism, married it to special relativity, and predicted antimatter from the mathematics alone. The decade closed with the framework's own architects flagging what it left unresolved: EPR named the spooky correlations later called entanglement, and Schrödinger's cat dramatized the unsolved measurement problem. Both puzzles became the raw material of the foundations era and, eventually, of quantum information itself.

MILESTONE TIMELINE

- 1925 (Jul) — Matrix mechanics (the Umdeutung / "reinterpretation" paper), drafted on Helgoland — Werner Heisenberg, formalized by Born & Jordan and then the Dreimännerarbeit (three-man paper) with Born, Heisenberg, Jordan — first complete quantum formalism — [T1] Heisenberg, Zeitschrift für Physik 33, 879 (1925)
- 1926 (Jan–Jun) — Wave equation for matter; hydrogen solved analytically in the first paper — Erwin Schrödinger (four-part series) — the workhorse equation of all quantum science — [T1] Schrödinger, Annalen der Physik 79–81 (1926)

- 1926 (Jun) — Born rule: $|\psi|^2$ is a probability density — Max Born (added in proof, in a footnote) — probability enters physics at the fundamental level; the interpretive keystone — [T1] Born, *Zeitschrift für Physik* 37, 863 (1926)
- 1927 (Feb–Mar) — Uncertainty principle: $\Delta x \Delta p \geq \hbar/2$ — Heisenberg — conjugate observables cannot be jointly sharp — [T1] Heisenberg, *Zeitschrift für Physik* 43, 172 (1927)
- 1927 — Electron diffraction observed independently — Davisson & Germer (Bell Labs, nickel crystal); G.P. Thomson (Aberdeen, thin films) — de Broglie's matter waves confirmed; father J.J. Thomson had shown the electron was a particle, son showed it was a wave — [T1] *Phys. Rev.* 30, 705 (1927)
- 1927 (Oct 24–29) — Fifth Solvay Conference, Brussels — the Bohr–Einstein debates begin (see T-solvay for the full card) — the interpretation argument that never ended — [T1] historical record; Bacciagaluppi & Valentini (2009)
- 1928 — Relativistic electron equation; antimatter implied by the negative-energy solutions — Paul Dirac — QM meets special relativity; the positron was found by Anderson in 1932 — [T1] Dirac, *Proc. R. Soc. A* 117, 610 (1928)
- 1932 — Rigorous Hilbert-space axiomatization and measurement theory; the density operator — John von Neumann — the mathematical foundation still in use, including a (later-critiqued) no-hidden-variables argument — [T1] von Neumann, *Mathematische Grundlagen der Quantenmechanik*
- 1935 (May) — EPR paper: if QM is complete, reality is nonlocal — Einstein, Podolsky, Rosen; Bohr's reply followed in October — defined the question Bell made testable 29 years later (see T-epr) — [T1] *Phys. Rev.* 47, 777 (1935)
- 1935 — "Entanglement" (Verschränkung) named; the cat thought experiment published — Erwin Schrödinger, in dialogue with Einstein — the resource quantum computing runs on gets its name — [T1] *Naturwissenschaften* 23, 807/823/844 (1935); *Proc. Camb. Phil. Soc.* 31, 555 (1935)

THE HUMAN CONTEXT

The speed is the story: from Heisenberg's Helgoland notebook to von Neumann's axioms was seven years. Most of it happened at Göttingen,

Copenhagen, and Munich, among people in their twenties. Born, whose interpretation is arguably the deepest single idea of the decade, was not included in Heisenberg and Jordan's 1932 Nobel and had to wait until 1954. Dirac derived a new particle from an equation before anyone had seen one — the modern template for theory leading experiment.

KEY GRADED CLAIMS

- claim: Matrix and wave mechanics are mathematically equivalent formulations of the same theory · tier: T1 · source: Schrödinger (1926); von Neumann (1932) · status: established
- claim: The Born rule (probabilities as squared amplitudes) is a postulate, not yet derived from the other axioms · tier: T1 · status: established — deriving it is an open program (see F-measure, O-foundations)
- claim: EPR exposed a genuine tension between QM and local realism · tier: T1 (as a historical argument) · status: contested-then-resolved — Bell tests ruled for QM (see T-foundations, T-belltests)

SOURCES

- Primary papers above; van der Waerden, *Sources of Quantum Mechanics* (1967); Jammer, *The Conceptual Development of Quantum Mechanics* (1966); Mehra & Rechenberg, *The Historical Development of Quantum Theory* (6 vols)
- **Go deeper:** T-solvay (1927 debates); T-epr (the 1935 exchange as its own milestone); F-uncertainty; F-interp; T-nobel (Heisenberg 1932; Schrödinger & Dirac 1933)

Foundations era (1935–1980) · T-foundations

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

For three decades the EPR question sat in philosophical exile while physicists built QED and the bomb — asking what the wavefunction "really" meant was, as John Clauser later put it, a good way to end a career. Bohm reopened it in 1952 with a working hidden-variable theory (pilot waves) that the orthodoxy had declared impossible, citing von Neumann's 1932 no-go proof; Bohm's counterexample showed the proof assumed too much. That provocation reached John Stewart Bell, a particle physicist at CERN who did foundations "on the side." In 1964 Bell proved that *any* local hidden-variable theory obeys an inequality that quantum mechanics violates — turning a metaphysical dispute into a laboratory measurement. CHSH (1969) recast the inequality into a form real detectors could test. Freedman and Clauser ran the first experiment in 1972 on a shoestring at Berkeley; Aspect's 1981–82 experiments in Orsay added time-varying analyzers that switched settings while photons were in flight, closing the locality loophole of the day. Quantum mechanics won every round. The 2022 Nobel Prize to Clauser, Aspect, and Zeilinger certified this quiet, career-risky arc as the origin of quantum information science: entanglement is real, physical, and usable.

MILESTONE TIMELINE

- 1948–1950 — QED renormalized; the mainstream of physics moves to fields, leaving foundations to a marginalized few — Feynman, Schwinger, Tomonaga, Dyson — the intellectual climate the foundations workers had to fight — [T1] historical record (see T-nobel, 1965)
- 1952 — Pilot-wave (hidden-variable) reformulation of QM — David Bohm — proved a deterministic completion is logically possible, refuting the "von Neumann forbids it" folklore and provoking Bell — [T1] Bohm, Phys. Rev. 85, 166 & 180 (1952)
- 1957 — Relative-state ("many-worlds") interpretation — Hugh Everett III (Princeton PhD under Wheeler) — collapse-free QM; ignored for a decade, now mainstream — [T1] Everett, Rev. Mod. Phys. 29, 454 (1957)

- 1964 — Bell's theorem: local hidden variables bound correlations; QM violates the bound — John S. Bell, on sabbatical from CERN — made metaphysics falsifiable; the pivot of the whole era — [T1] Bell, *Physics* 1, 195 (1964)
- 1966 — Kochen–Specker / Bell's review of no-hidden-variable proofs — Bell (*Rev. Mod. Phys.* 38, 447), Kochen & Specker (1967) — contextuality as a separate no-go (see F-contextuality) — [T1]
- 1969 — CHSH inequality, an experimentally testable form usable with real (inefficient) detectors — Clauser, Horne, Shimony, Holt — the version every Bell test since has used — [T1] *PRL* 23, 880 (1969)
- 1970 — Decoherence program begins: environment-induced superselection — H. Dieter Zeh — why the classical world emerges from the quantum; later the enemy of every qubit — [T1] Zeh, *Found. Phys.* 1, 69 (1970)
- 1972 — First experimental Bell test (calcium cascade photons); QM violates CHSH — Stuart Freedman & John Clauser (Berkeley) — first data against local realism, on borrowed equipment — [T1] Freedman & Clauser, *PRL* 28, 938 (1972)
- 1973 — Holevo bound: a qubit carries at most one classical bit of accessible information — Alexander Holevo (Moscow) — a founding theorem of quantum information theory (see T-qinfobirth) — [T1] Holevo, *Probl. Peredachi Inf.* 9, 3 (1973)
- 1981–82 — Bell tests with locality-motivated timing, including analyzers switched while photons are in flight — Alain Aspect, Dalibard, Roger (Orsay) — closed the most glaring locality loophole of the day; $S = 2.697 \pm 0.015$ — [T1] *PRL* 47, 460 (1981); 49, 91 & 1804 (1982)
- 2022 — Nobel Prize in Physics to Aspect, Clauser, Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science" — the era's retroactive coronation — [T1] [nobelprize.org](https://www.nobelprize.org) (2022)

THE HUMAN CONTEXT

Bell called quantum foundations his "hobby" and published his theorem in an obscure, short-lived journal (*Physics*) that paid authors — partly so he could keep his real job doing accelerator physics. Clauser was warned the Bell test would sink his career; he ran it anyway and got the "wrong" answer (QM was

right, disappointing his own realist hopes). Everett left academia for defense work after his interpretation was dismissed. The people who built the conceptual foundation of a trillion-dollar field mostly did it against professional incentives.

KEY GRADED CLAIMS

- claim: Nature violates Bell inequalities; local hidden-variable theories are ruled out · tier: T1 · source: Aspect (1982); loophole-free tests 2015 sealed it (see T-belltests) · status: established
- claim: Bohm's 1952 theory shows von Neumann's no-hidden-variables proof was not decisive · tier: T1 · status: established (the proof's flawed assumption was later pinpointed by Bell 1966)
- claim: The measurement problem itself remains unresolved · tier: T1 (as consensus about openness) · status: contested — see O-foundations, F-interp

SOURCES

- Bell, *Physics* 1, 195 (1964); CHSH, *PRL* 23, 880 (1969); Freedman–Clauser, *PRL* 28, 938 (1972); Aspect et al. (1981–82); Zeh, *Found. Phys.* 1, 69 (1970); NobelPrize.org 2022 scientific background
- Bell, *Speakable and Unsayable in Quantum Mechanics* (1987) — the collected essays; Gilder, *The Age of Entanglement* (2008)
- **Go deeper:** T-belltests (the 2015 loophole-free capstone); T-epr (the question Bell answered); F-bell; F-decoher; T-nobel

The NISQ era (Preskill 2018) · T-nisq

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

By 2017 the field needed a word for the machines it was actually building. They were no longer the handful-of-qubit demos of the 2000s, and they were nowhere near the fault-tolerant, error-corrected computers of theory. They were processors with roughly 50–100 physical qubits, imperfect gates, and no error correction — powerful enough to be hard to simulate classically, too noisy to run a deep useful circuit. John Preskill supplied the name in a December 2017 keynote (arXiv 2 January 2018, published in *Quantum* 6 August 2018): **NISQ** — **Noisy Intermediate-Scale Quantum**. "Intermediate-scale" for a qubit count beyond brute-force classical simulation yet far short of utility; "noisy" because uncorrected gate errors cap achievable circuit depth. The coinage stuck instantly and organized an entire research program — variational algorithms (VQE, QAOA), error mitigation (ZNE, PEC), and near-term "advantage" claims all live under it (see S-nisq, S-variational, O-advantage).

WHY THE WORD MATTERED

Preskill's essay did two things at once, and both were acts of expectation-management. It gave hardware groups a realistic near-term target instead of a fault-tolerant fantasy that was still decades off — you could publish NISQ results now. And it sober-set expectations: his often-quoted line is that NISQ technology "will not change the world by itself" and should be regarded "as a step toward more powerful quantum technologies we will develop in the future." He explicitly named the gap between the NISQ present and the fault-tolerant future that the whole field is now trying to cross (see T-ecera, S-logical, O-scaling). The term became load-bearing in three registers at once: it frames vendor roadmaps, it frames funding pitches, and it frames the honest-skeptic critique — because if the NISQ era ends without a single unambiguous practical advantage, that is itself the verdict.

THE VERDICT, EIGHT YEARS ON

The uncomfortable fact by mid-2026 is that no NISQ algorithm has produced a clear, durable practical advantage. Variational methods hit barren plateaus and optimization walls; every "utility" or "advantage" demo on a noisy device has

drawn a classical counterattack within months (see T-race, T-ecera, O-advantage). The field's own answer has been to shift focus from NISQ to early fault tolerance — which is why "post-NISQ" and "the era of logical qubits" now appear in the same roadmaps that once sold NISQ. Preskill framed the practical-advantage question as open in 2018; the honest reading is that it is still open, and increasingly the bet is that the payoff waits on error correction, not on cleverer noisy circuits.

KEY GRADED CLAIMS

- claim: The 2018-onward regime of ~50–100 noisy, uncorrected qubits is the defining near-term phase, and it is named NISQ · tier: T2 · status: established (as terminology) — Preskill, *Quantum* 2, 79 (2018), arXiv:1801.00862
- claim: NISQ devices will yield near-term commercial advantage · tier: T4 · status: contested — Preskill himself framed it as open; eight years on, no unambiguous practical NISQ advantage has survived classical counterattack (see O-advantage, O-killerapp)
- claim: The field is transitioning from NISQ to early fault tolerance · tier: T3 · status: demonstrated in direction (below-threshold QEC, logical-qubit demos — see T-ecera), not yet in delivered applications

SOURCES

- Preskill, "Quantum Computing in the NISQ era and beyond," *Quantum* 2, 79 (2018). arXiv:1801.00862, doi:10.22331/q-2018-08-06-79
- Preskill, Q2B keynote, Dec 2017 — the origin of the term
- **Go deeper:** S-nisq, S-variational (the algorithms of the era); O-advantage (the classical-counterattack pattern); T-ecera (the shift to logical qubits)

Nobel Prizes as the quantum spine (1918→2025) · T-nobel

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The Nobel Prize in Physics is an imperfect but useful backbone for the whole history of quantum science: each prize marks the moment the establishment ratified a result as settled. Reading the prizes in order traces the field's two revolutions. The early prizes (1918–1933) canonized the *discovery of the rules* — Planck's quantum, Einstein's photon, Bohr's atom, the matrix and wave mechanics. A middle cluster (1954–1972) honored the *deep theory and the materials physics* that would later become hardware — Born's probability rule, QED, BCS superconductivity, the Josephson effect. The recent prizes (1997–2025) reward the *engineering of single quantum systems* and the *foundations made experimental* — laser cooling, Bose–Einstein condensation, individual-quantum control, the Bell-test verdict, and finally macroscopic quantum circuits, the direct ancestors of the superconducting qubit. Two centuries-apart bookends define the span: Planck in 1918 for the constant that started it, and Clarke, Devoret, and Martinis in 2025 for showing an electrical circuit big enough to hold in your hand still obeys quantum mechanics — the physics every superconducting quantum computer runs on.

THE SPINE (PHYSICS PRIZES CENTRAL TO QUANTUM)

- 1918 — Max Planck — energy quanta (the constant h) — first revolution begins — [T1]
- 1921 (awarded 1922) — Albert Einstein — the photoelectric effect (explicitly, not relativity) — light is quantized — [T1]
- 1922 — Niels Bohr — the structure of atoms and their radiation — [T1]
- 1929 — Louis de Broglie — the wave nature of electrons — [T1]
- 1932 (announced 1933) — Werner Heisenberg — the creation of quantum mechanics — [T1]
- 1933 — Erwin Schrödinger & Paul Dirac — new productive forms of atomic theory — [T1]
- 1945 — Wolfgang Pauli — the exclusion principle — [T1]
- 1954 — Max Born — the statistical interpretation of the wavefunction (Born rule), a 28-year wait; shared with Walther Bothe — [T1]

- 1956 — Shockley, Bardeen, Brattain — the transistor, the first quantum revolution's defining technology — [T1]
- 1965 — Sin-Itiro Tomonaga, Julian Schwinger, Richard Feynman — quantum electrodynamics (QED) — [T1]
- 1972 — John Bardeen, Leon Cooper, Robert Schrieffer — BCS theory of superconductivity (Bardeen's second physics Nobel) — the physics under every superconducting qubit — [T1]
- 1973 — Brian Josephson (shared) — the Josephson effect, tunneling supercurrent across a junction — the nonlinear element of the transmon (see H-supercon) — [T1]
- 1997 — Chu, Cohen-Tannoudji, Phillips — laser cooling and trapping of atoms — enables ion/neutral-atom qubits — [T1]
- 2001 — Cornell, Ketterle, Wieman — Bose–Einstein condensation — ultracold quantum matter — [T1]
- 2012 — Serge Haroche & David Wineland — measuring and manipulating *individual* quantum systems — the second revolution's core capability (see T-2ndrev) — [T1]
- 2022 — Alain Aspect, John Clauser, Anton Zeilinger — Bell-inequality violation with entangled photons, "pioneering quantum information science" (see T-foundations, T-belltests) — [T1]
- 2023 — Agostini, Krausz, L'Huillier — attosecond light pulses (quantum-adjacent; probes electron dynamics) — [T1]
- 2025 — John Clarke, Michel Devoret, John Martinis — macroscopic quantum tunneling and energy quantization in Josephson-junction circuits (experiments 1984–85) — the superconducting-qubit lineage (see T-early, T-ecera) — [T1] nobelprize.org (7 Oct 2025)

WHAT THE SPINE REVEALS – AND HIDES

Three patterns stand out. **The long lag:** Born waited 28 years, Aspect/Clauser/Zeilinger about 40, Clarke/Devoret/Martinis about 40 — the prize ratifies, it does not predict. **The three-person cap distorts credit:** teams and unmentioned collaborators vanish (Rosalind Franklin's chemistry analogue haunts physics too — Freeman Dyson, who made QED calculable, was excluded in 1965). **The hardware bias toward the past:** the 2025 prize honored circuit experiments from 1984, and no gate-model quantum *computer* has yet produced a Nobel-worthy verified advantage, which is itself an honest signal about where the field

really is versus where the press releases say it is (see T-ecera, O-hype). The spine is a reliable record of what is *established*; by construction it lags the frontier by decades.

KEY GRADED CLAIMS

- claim: The 2025 Physics Nobel went to Clarke, Devoret, Martinis for macroscopic quantum tunneling in superconducting circuits · tier: T1 · source: nobelprize.org (2025) · status: established
- claim: The 2022 Physics Nobel recognized Bell-test experiments as the origin of quantum information science · tier: T1 · source: nobelprize.org (2022) scientific background · status: established
- claim: No quantum *computer* result has yet earned a Nobel; the prizes track established physics, lagging the hardware frontier by decades · tier: T1 (observation) · status: established

SOURCES

- NobelPrize.org prize records and scientific-background documents, 1918–2025 (especially 2012, 2022, 2025)
- Physics World, C&EN, Nature Physics coverage of the 2025 prize (Clarke/Devoret/Martinis)
- **Go deeper:** every era card in §07 cross-references its prize; T-2ndrev (the 2012 capability); T-ecera (the 2025 lineage); O-hype (prizes as a lag indicator)

Old quantum theory (1900–1925) · T-old

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

Classical physics broke on two experimental fronts — blackbody radiation and the photoelectric effect — and the patches applied between 1900 and 1925 became the "old quantum theory": Newtonian mechanics and Maxwell's electromagnetism with ad-hoc quantization rules bolted on. It was a quarter-century of brilliant guessing without a governing equation. Max Planck, a conservative 42-year-old thermodynamicist who disliked his own result, quantized energy exchange to fit the blackbody spectrum and later called it "an act of desperation." Einstein — then a patent clerk in Bern — took the quantum literally as a particle of light in his 1905 *annus mirabilis*, a step so radical that Planck himself resisted it for a decade. Bohr quantized the atom's orbits and, in a single stroke, computed the hydrogen spectrum's Rydberg constant from first principles. De Broglie closed the loop in his 1924 doctoral thesis by giving *matter* a wavelength, an idea his examiners found so strange they forwarded it to Einstein for a verdict (he approved). By 1925 the patchwork was visibly failing — it could not handle helium's two electrons or the anomalous Zeeman effect, and the Bohr–Sommerfeld quantization rules had no principled justification. That failure is exactly what forced the full formalism of the next era.

MILESTONE TIMELINE

- 1900 (14 Dec) — Quantum hypothesis: energy exchanged in units $E = h\nu$, fitting the blackbody spectrum — Max Planck, presented to the German Physical Society — introduced h , the constant the whole field is named after; Planck saw it as a mathematical device, not a physical quantum — [T1] Planck, *Verhandlungen der DPG* 2, 237 (1900)
- 1905 — Photoelectric effect explained via light quanta (later "photons," a term coined by Gilbert Lewis in 1926) — Albert Einstein — light itself is quantized; the specific work later cited for the 1921 Nobel — [T1] Einstein, *Annalen der Physik* 17, 132 (1905)
- 1913 — Quantized atomic orbits explaining the hydrogen spectrum (the Bohr trilogy) — Niels Bohr, then 27, working from Rutherford's Manchester

lab — first quantum model of matter's structure; derived the Rydberg constant — [T1] Bohr, *Philosophical Magazine* 26, 1 (1913)

- 1916–1919 — Sommerfeld's elliptical orbits and relativistic fine-structure; the Bohr–Sommerfeld quantization $\oint p \, dq = nh$ — Arnold Sommerfeld — the high-water mark of old quantum theory, and the model that would soon break — [T1] Sommerfeld, *Atombau und Spektrallinien* (1919)
- 1922 — Space quantization of angular momentum observed — Otto Stern & Walther Gerlach — direct evidence quantization is real; later read as the prototype of a projective qubit measurement — [T1] *Zeitschrift für Physik* 9, 349 (1922)
- 1923 — X-ray scattering off electrons shows photon momentum $p = h/\lambda$ — Arthur Compton — sealed the particle nature of light against still-widespread skepticism — [T1] *Phys. Rev.* 21, 483 (1923)
- 1924 — Bose–Einstein statistics for indistinguishable particles — Satyendra Nath Bose & Einstein — the first quantum statistics, later the basis of BEC and bosonic modes — [T1] Bose, *Zeitschrift für Physik* 26, 178 (1924)
- 1924 — Matter waves: $\lambda = h/p$ for any particle — Louis de Broglie (doctoral thesis, Sorbonne) — the conceptual bridge to wave mechanics, vindicated by electron diffraction three years later — [T1] de Broglie, *Recherches sur la théorie des quanta* (1924)
- 1925 (Jan) — Exclusion principle: no two electrons share a quantum state — Wolfgang Pauli — why atoms have shell structure; the seed of fermionic statistics — [T1] *Zeitschrift für Physik* 31, 765 (1925)

THE HUMAN CONTEXT

This was physics done by very young people under an old guard's skeptical eye. Bohr was 27, Heisenberg would soon be 23, Pauli 25. Planck, the reluctant revolutionary, spent years trying to reconcile his quantum with classical continuity. Einstein's light-quantum was so heterodox that when Planck and others nominated him for the Prussian Academy in 1913, they added a note excusing his "occasional" overreach on the light-quantum. The field's founders did not believe their own equations meant what they said — which is precisely why the 1925–27 formalism landed like a shock.

KEY GRADED CLAIMS

- claim: Energy quantization was forced by experiment (the ultraviolet failure of the classical blackbody law), not chosen for elegance — Planck's own "act of desperation" · tier: T1 · source: Planck 1900; Kuhn, *Black-Body Theory* (1978) · provenance: historical record · status: established
- claim: De Broglie's matter waves were confirmed by electron diffraction (Davisson–Germer 1927; G.P. Thomson 1927, covered in T-formalism) · tier: T1 · status: established
- claim: The "old quantum theory" was a genuine research program, not merely a set of mistakes — it produced correct spectra while lacking a governing equation · tier: T1 · status: established

SOURCES

- Planck (1900), *Verhandlungen der DPG* 2, 237; Einstein (1905), *Annalen der Physik* 17, 132; Bohr (1913), *Phil. Mag.* 26, 1; de Broglie thesis (1924)
- Kuhn, *Black-Body Theory and the Quantum Discontinuity* (1978); Pais, *Subtle is the Lord* (1982); Jammer, *The Conceptual Development of Quantum Mechanics* (1966) — secondary T1 histories
- **Go deeper:** T-formalism (the equation that replaced the patchwork); T-nobel (Planck 1918, Einstein 1921, Bohr 1922); F-statistics (Bose/Fermi–Dirac lineage)

Birth of quantum information theory (1970–1984) · T-qinfobirth

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

Before there was quantum *computing*, there was quantum *information* — the recognition that a quantum state is a carrier of information governed by its own rules, distinct from Shannon's classical theory. The idea's patient zero is Stephen Wiesner, a Columbia graduate student who around 1970 wrote a manuscript called "Conjugate Coding" describing how to encode two messages in conjugate observables (like the linear and circular polarization of light) so that a receiver can read one or the other but never both — and, as a bonus, how to make banknotes that cannot be counterfeited because an unknown quantum state cannot be copied. It was so far ahead of its time that it was rejected (reportedly by IEEE Transactions on Information Theory) and sat unpublished for thirteen years, circulating only as a curiosity among a few physicists. One of the few who took it seriously was Charles Bennett, Wiesner's friend; another was Gilles Brassard. In 1984 they turned Wiesner's conjugate coding into BB84, the first quantum key distribution protocol — and quantum information had its first practical technology. In parallel, Alexander Holevo (Moscow, 1973) proved the fundamental limit on how much classical information a quantum state can carry, and Wootters, Zurek, and Dieks (1982) proved the no-cloning theorem that makes the whole edifice secure. By 1984 the conceptual toolkit — qubits as information carriers, no-cloning as a security primitive, channel capacities, conjugate coding — was in place, waiting for Deutsch and Shor to add computation (see T-birth).

MILESTONE TIMELINE

- ~1970 — "Conjugate Coding" manuscript: quantum money and multiplexed messages via conjugate observables — Stephen Wiesner — the founding idea of quantum cryptography, rejected and unpublished for over a decade — [T1] Wiesner, eventually in SIGACT News 15(1), 78 (1983)
- 1973 — Holevo bound: a qubit conveys at most one bit of accessible classical information; the general capacity limit for quantum channels — Alexander Holevo — a founding theorem of quantum information theory

(later completed by the HSW theorem, 1996–98) — [T1] Holevo, *Probl. Peredachi Inf.* 9(3), 3 (1973)

- 1975–1976 — Early quantum detection and estimation theory; distinguishability limits — Helstrom; Holevo — the quantum analogue of statistical inference — [T1] Helstrom, *Quantum Detection and Estimation Theory* (1976)
- 1982 — No-cloning theorem: an unknown quantum state cannot be perfectly copied — Wootters & Zurek (*Nature*); Dieks (*Phys. Lett. A*) — the security root of quantum cryptography; makes eavesdropping detectable (see F-nocloning) — [T1] *Nature* 299, 802 (1982)
- 1983 — Wiesner's "Conjugate Coding" finally appears in print — SIGACT News — thirteen years late — [T1] SIGACT News 15(1), 78–88 (1983)
- 1984 — BB84 quantum key distribution: Wiesner's conjugate coding turned into a working key-exchange protocol — Charles Bennett & Gilles Brassard — quantum information's first technology; still deployed (see A-qkd) — [T1] *Proc. IEEE Int. Conf. Computers, Systems and Signal Processing*, Bangalore, 175 (1984)
- 1991 — Entanglement-based QKD (E91), grounding security in Bell-inequality violation — Artur Ekert — links quantum crypto to the foundations program (see T-foundations) — [T1] Ekert, *PRL* 67, 661 (1991)

WHY IT EARNS ITS OWN CARD

The birth-of-computing card (T-birth) starts the clock at 1980 with Manin, Benioff, and Feynman, but that framing hides an older lineage: the information-theoretic roots that predate the computational ones. Wiesner's 1970 insight — that conjugate observables plus no-cloning give you security for free — is arguably the first genuinely quantum *application* ever conceived, and it arrived before anyone was thinking about computation at all. Giving it a dedicated card corrects the "Feynman started everything" simplification and makes visible the crypto→computing sequence: the field learned to *protect* information with quantum mechanics before it learned to *process* it.

THE HUMAN CONTEXT

Wiesner (1942–2021) never held a conventional physics career commensurate with his ideas; Scott Aaronson's memorial notes that he spent years as a construction laborer in Israel while the field he seeded became a global

enterprise. Bennett has said BB84 was essentially Wiesner's idea with a use case attached. The pattern — a foundational insight, rejected and dormant for a decade, then activated by someone who happened to know the originator — recurs throughout quantum history (Everett's many-worlds, Bell's theorem in an obscure journal).

KEY GRADED CLAIMS

- claim: Quantum key distribution (BB84) derives its security from the no-cloning theorem and measurement disturbance, making it information-theoretic rather than computational · tier: T1 · status: established — see A-qkd, F-nocloning
- claim: The Holevo bound caps accessible classical information in a quantum state at one bit per qubit · tier: T1 · status: established — a founding result of quantum Shannon theory
- claim: Wiesner's ~1970 conjugate coding is the earliest conceived quantum information application · tier: T1 (historical) · status: established, though "earliest" is a claim about priority and framing

SOURCES

- Wiesner, "Conjugate Coding," SIGACT News 15(1), 78 (1983, written ~1970); Holevo, Probl. Peredachi Inf. 9 (1973); Wootters & Zurek, Nature 299 (1982); Bennett & Brassard, BB84 (1984); Ekert, PRL 67 (1991)
- Aaronson, "Stephen Wiesner (1942–2021)," scottaaronson.blog; Wikipedia "Quantum money," "Holevo's theorem"
- **Go deeper:** F-qinfo (the modern theory); F-nocloning; A-qkd; T-birth (where computation joins); T-nobel

The engineering race (2011–2019) · T-race

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

Quantum computing left the physics lab and became an industrial race. D-Wave sold the first commercial machine in 2011 — an annealer, not a gate-model computer — kicking off a decade-long argument about whether it delivered any quantum speedup at all. IBM put a real processor on the open cloud in 2016 and normalized the practice of publishing qubit counts as press events. Google hired John Martinis's UCSB group in 2014 and pointed it at a single target: beat a classical supercomputer at *something*. Governments answered with billion-dollar national programs (EU Flagship 2016, US NQI Act 2018 — see E-eu, E-us). The era peaked in October 2019 with Google's Sycamore "quantum supremacy" claim — a 53-qubit chip sampling random circuits in 200 seconds against an estimated 10,000 classical years. IBM contested the estimate within days (arguing a well-configured Summit supercomputer could do it in 2.5 days), and over the next three years classical algorithms shrank the gap to hours. That claim-and-counterattack cycle — a headline number, a rapid classical response, a milestone that survives while its superlative does not — became the template governing every advantage announcement since (see SCHEMA rule 2, O-advantage).

MILESTONE TIMELINE

- 2011 (May) — First commercial quantum computer sold: D-Wave One (128-qubit annealer) to Lockheed Martin — D-Wave — a market exists; the underlying Nature paper shows quantum annealing in the hardware, but generic speedup stays contested for years — [T4 sale → T2 physics] D-Wave; Johnson et al., Nature 473, 194 (2011)
- 2013 — Google, NASA, and USRA buy a D-Wave Two for the Quantum AI Lab at Ames — big tech enters, betting on annealing — [T4] press record
- 2014 — Martinis group (UCSB) joins Google; superconducting fidelity crosses the surface-code threshold in the lab — Google — the start of the supremacy program — [T2] Barends et al., Nature 508, 500 (2014)

- 2014 — No generic quantum speedup found in D-Wave benchmarks — Rønnow, Troyer et al. — the defining skeptical result on annealing — [T2] Rønnow et al., *Science* 345, 420 (2014)
- 2016 (May) — IBM Quantum Experience: a 5-qubit processor free on the public cloud — IBM — anyone can run a circuit; the cloud-QC business model is born (see S-cloud) — [T2] IBM + subsequent literature
- 2016 — EU announces the €1B Quantum Flagship — European Commission — governments commit at scale (see E-eu) — [T5] EC announcement
- 2017–2018 — 50-qubit IBM prototype; 72-qubit Bristlecone (Google); 49-qubit Tangle Lake (Intel) — the qubit-count arms race in full swing — [T4] vendor announcements
- 2018 (21 Dec) — US National Quantum Initiative Act signed into law (\$1.2B over 5 years) — US Congress — quantum becomes a statutory national priority (see E-us) — [T2] Public Law 115-368
- 2019 (Jan) — IBM Q System One, the first integrated, rack-and-chandelier commercial quantum system — IBM — productization and industrial design begin — [T4] IBM announcement
- 2019 (23 Oct) — "Quantum supremacy" on the 53-qubit Sycamore chip: random circuit sampling in ~200 s vs. an estimated 10,000 classical years — Google (Martinis et al.) — first claimed beyond-classical computation — [T2 experiment] Arute et al., *Nature* 574, 505 (2019)

THE HUMAN CONTEXT

The supremacy paper leaked early — a draft appeared on a NASA server in September 2019 before the embargo — and IBM published its rebuttal the same week the *Nature* paper ran, a public spat between the two biggest players that set the tone for the field's credibility wars. Martinis left Google in 2020 after a reorganization, a reminder that the era's central figures were people, not logos. D-Wave's long speedup argument never resolved cleanly; the honest summary is that annealing found niche uses without ever demonstrating the general exponential speedup its early marketing implied.

KEY GRADED CLAIMS

- claim: Sycamore performed a sampling task infeasible for any classical computer in 2019 · tier: T2 (experiment) · status: contested — IBM's 2.5-

day Summit estimate (Oct 2019); tensor-network methods later reproduced the sampling in hours-to-days (Pan, Chen, Zhang, PRL 129, 090502, 2022). The demo stands; the "10,000 years" figure does not.

- claim: D-Wave annealers deliver a generic quantum speedup on practical problems · tier: T4 · status: contested — Rønnow et al., Science 345, 420 (2014) found no generic speedup; niche advantages remain debated
- claim: Publishing qubit counts as milestones tracks progress · tier: T4 · status: contested — the field itself later shifted the scoreboard to logical qubits and error rates (see T-ecera)

SOURCES

- Johnson et al., Nature 473 (2011); Rønnow et al., Science 345 (2014); Barends et al., Nature 508 (2014); Arute et al., Nature 574 (2019); IBM "On quantum supremacy" blog (Oct 2019); Pan et al., PRL 129, 090502 (2022)
- **Go deeper:** S-bench (how "advantage" is measured); T-ecera (the pivot to logical qubits); E-us, E-eu (the national programs); O-advantage

The 1927 Solvay Conference & the Bohr–Einstein debates · T-solvay

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The Fifth Solvay Conference, held in Brussels 24–29 October 1927 on the theme "Electrons and Photons," is the most famous meeting in the history of physics. Twenty-nine physicists attended; seventeen were or would become Nobel laureates. The group portrait — Einstein front-center, Curie the only woman, Bohr, Planck, Dirac, Heisenberg, Schrödinger, de Broglie, Born, Pauli arrayed around them — is often called "the most intelligent photo ever taken." It marked the moment the new quantum mechanics, barely two years old, was presented as a finished theory, and the moment its interpretation split the field. Bohr and Heisenberg's Copenhagen view — that the wavefunction is a complete description and that a quantum system has no definite properties until measured — was challenged head-on by Einstein, who could not accept that physics had become irreducibly probabilistic. Their exchange did not settle anything in 1927; it opened the interpretation argument that runs to this day and that eventually became testable through Bell's theorem and the loophole-free experiments (see T-foundations, T-belltests, F-interp).

WHAT ACTUALLY HAPPENED

The popular story compresses a week of hard argument into one quip. Einstein's real attack came through a series of ingenious thought experiments — variants of the double slit with movable diaphragms and recoiling screens — each designed to show that one could, in principle, beat the uncertainty relation and pin down both a particle's path and its interference pattern. Bohr answered each over breakfast and dinner, showing that the very apparatus needed to measure the particle's momentum recoil would blur its position by exactly enough to preserve the uncertainty bound. Einstein conceded the internal consistency of quantum mechanics but never its completeness. The famous lines — Einstein's "God does not play dice" and Bohr's "Einstein, stop telling God what to do" — capture the temperature of the debate, though both are paraphrases, not verbatim transcripts of a single moment.

MILESTONE TIMELINE

- 1927 (24–29 Oct) — Fifth Solvay Conference, "Electrons and Photons," Brussels; Bohr, Born, Heisenberg present the new mechanics; Einstein and Schrödinger dissent — the public debut of the interpretation debate — [T1] proceedings; Bacciagaluppi & Valentini, *Quantum Theory at the Crossroads* (2009), a full reconstruction from the original French/German records
- 1930 (Oct) — Sixth Solvay Conference: Einstein's "photon box" (light-box weighed to defeat the energy–time uncertainty relation); Bohr rebuts using Einstein's *own* general relativity (gravitational time dilation) — the debate's second round — [T1] historical record
- 1935 — Unable to defeat consistency, Einstein shifts ground to *completeness* and *locality*: the EPR paper (see T-epr) — the debate's most productive turn — [T1] Phys. Rev. 47, 777 (1935)

WHY IT BELONGS ON THE TIMELINE

Solvay 1927 is where quantum mechanics stopped being an equation and started being a worldview problem. The Copenhagen interpretation became the working default not because it won an argument but because it was operationally sufficient and its rivals were dismissed. Einstein's refusal looks, in hindsight, less like stubbornness and more like the first statement of the questions — locality, realism, completeness — that the second half of the 20th century turned into experiments and the 21st turned into technology. Entanglement, the thing Einstein was circling at Solvay, is the resource quantum computers run on.

KEY GRADED CLAIMS

- claim: The Solvay debates concerned the *interpretation* and *completeness* of QM, not its predictive accuracy, which was never in dispute · tier: T1 · source: Bacciagaluppi & Valentini (2009) · status: established
- claim: Bohr successfully defended the uncertainty principle against each of Einstein's thought experiments · tier: T1 (as the historical consensus) · status: established; the deeper completeness question Einstein raised was resolved only much later, and against him (Bell tests)
- claim: "God does not play dice" / "stop telling God what to do" are paraphrases, not a verbatim single exchange · tier: T1 · status: established

SOURCES

- Bacciagaluppi & Valentini, *Quantum Theory at the Crossroads: Reconsidering the 1927 Solvay Conference* (Cambridge, 2009), arXiv:quant-ph/0609184
- Bohr, "Discussion with Einstein on Epistemological Problems in Atomic Physics" (1949); Wikipedia "Bohr–Einstein debates"
- **Go deeper:** T-epr (the 1935 sequel); T-formalism (the theory being debated); F-interp; F-uncertainty

The "quantum winter" question (AI-winter analogy) · T-winter

Layer: L6 History · **Chapter:** §07 · **Status:** depth

THE ARC

The term "quantum winter" borrows directly from the "AI winters" — the two multi-year collapses of funding and interest (roughly 1974–1980 and 1987–1993) that followed cycles of over-promising in artificial intelligence. A quantum winter would be the same pattern: a period of stalled progress and, more sharply, a withdrawal of capital and attention when the field's grand promises fail to materialize on the advertised schedule. It is not (yet) a historical event — it is a live risk and a recurring debate. The bull case says quantum has never been advancing faster (below-threshold QEC, logical-qubit records — see T-ecera). The bear case, argued by longtime skeptics like Mikhail Dyakonov and by cautious investors, is that quantum advantage has so far appeared only in contrived, useless tasks; that no application pays for itself; and that a downturn could drain talent and funding before fault-tolerant machines arrive (see O-talent-attribution, O-killerapp). Including this card is a matter of intellectual honesty: the manual grades vendor roadmaps as marketing until peer-reviewed (SCHEMA rule 1), and a field that runs on hype is a field exposed to a winter.

THE EVIDENCE FOR THE RISK

- **Funding volatility.** Private quantum investment reportedly fell sharply in 2023 (widely cited as ~50% globally, ~80% in the US year-over-year) before recovering — a swing consistent with hype-cycle dynamics, not steady industrial build-out. Figures vary by source and are T5.
- **AI cannibalization.** From 2023 the generative-AI boom pulled capital, talent, and executive attention toward a technology with immediate revenue, leaving quantum — still pre-revenue for computing — competing for a shrinking share of "frontier tech" budgets.
- **The advantage gap.** Every beyond-classical claim (Sycamore 2019, Jiuzhang 2020, IBM utility 2023, Google Quantum Echoes 2025) has drawn a classical counterattack (see T-race, T-ecera, O-advantage). A decade of NISQ produced no durable practical advantage (see T-nisq). If the gap persists, patience — and money — can run out.

- **Public-market fragility.** Several pure-play quantum stocks are richly valued relative to near-zero revenue; a single high-profile stumble or accounting controversy could reprice the sector and chill private funding (the "QCI saga" cited in the postquantum.com "quantum winter warning" is the archetype).

THE EVIDENCE AGAINST

- **Genuine technical acceleration.** Below-threshold QEC (Willow, Nature 2025) and logical-qubit counts doubling (48→96 in ~13 months) are peer-reviewed or near-peer-reviewed, not press-release physics (see T-ecera).
- **Deep-pocketed patrons.** National programs (US NQI, EU Flagship, China's multi-billion commitment — see E-us, E-eu, E-china) provide funding floors decoupled from VC sentiment; governments do not exit on a bad quarter.
- **The 2025 Nobel** (Clarke/Devoret/Martinis) and the UN International Year of Quantum reinforce institutional legitimacy at exactly the moment a winter narrative might take hold.

MILESTONE / DISCOURSE TIMELINE

- 2019 — Early "quantum winter is not coming" essays push back on premature doom (Ryan Shaffer and others) — [T5/opinion]
- 2022 (Jan) — Mainstream "what if quantum faces a winter?" coverage (Slate) — the analogy enters public discourse — [T5]
- 2023 — Reported ~50% global / ~80% US drop in private quantum funding; simultaneous generative-AI boom — the sharpest data point for the bear case — [T5]
- 2024 — "No quantum winter in 2024" counter-narratives cite rapid hardware advances — [T5]
- 2025 — "Quantum winter warning" analyses tie the risk explicitly to overhype and specific public-company controversies — [T5]

KEY GRADED CLAIMS

- claim: A "quantum winter" analogous to the AI winters is a plausible risk, driven by an advantage gap plus hype-cycle funding · tier: T5/T6 · provenance: analysts, skeptics, some investors · status: contested — a scenario, not a fact

- claim: Quantum advantage has so far been shown only in contrived tasks with no practical utility · tier: T2 (about the record to date) · status: established as of mid-2026 — every practical-advantage claim has been contested (see O-advantage)
- claim: National-program funding insulates the field from a purely VC-driven downturn · tier: T5 · status: contested — floors exist, but budgets can still be cut

SOURCES

- postquantum.com "Quantum Winter Warning"; Slate (2022) "What if quantum computing faces a winter?"; Cointelegraph (2024) "no quantum winter"; Infosys "Quantum Winter: Reality or Myth?"
- Dyakonov, *Will We Ever Have a Quantum Computer?* (2020) — the standing skeptical case
- AI-winter history: standard accounts of the 1974–80 and 1987–93 downturns
- **Go deeper:** O-hype, O-killerapp, O-advantage, O-talent-attribution; E-vc (the capital cycle); T-nisq (the unmet-promise precedent)

§08 · The Honest Frontier

What's unsettled — fault-tolerant scaling, hype vs reality, the killer-app question, the measurement problem.

Quantum Advantage in Practice — Which Speedups Survive • O-advantage

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

"Quantum advantage" is a moving target: every claim faces a classical counterattack, and the record so far favors the attackers. Google's 2019 random-circuit-sampling supremacy ("10,000 years") was matched within ~3 years by tensor-network methods on GPUs. The sharp question splits in two: (1) is there any *useful* task with a speedup that survives sustained classical attack, and (2) how much of the theoretical speedup catalog was ever real once loading, readout, and overhead are counted? The 2025–26 record is a live back-and-forth on both spin-glass simulation and OTOC-based "quantum echoes."

WHERE THE DISAGREEMENT IS

- **Advantage-is-real camp.** Google's **Quantum Echoes** (Oct 2025, 105-qubit Willow) claims the first *verifiable* quantum advantage — an OTOC-based algorithm run **13,000× faster** than the best known classical method on a top supercomputer, with results reproducible and cross-checkable on another quantum device [T4→T3; vendor-led, published, young]. A March 2026 preprint, "Tensor Networks with Belief Propagation Cannot Feasibly Simulate Google's Quantum Echoes Experiment" (arXiv:2604.15427), argues the leading classical attack *fails* here — an early sign the claim may hold [T3]. D-Wave's March 2025 *Science* paper claimed beyond-classical spin-glass dynamics on 5,000+ qubits [T2 paper, contested claim], and in May 2026 D-Wave **rebutted** the Flatiron follow-up, holding that the classical framework fails to scale across the most complex topologies and measurements [T4]. Shor's and quantum-simulation speedups remain provably superpolynomial given a fault-tolerant machine [T1/T2 theory].
- **Attrition camp. Ewin Tang's dequantization** (2018→) erased the exponential speedup of quantum recommendation systems and much of the QML linear-algebra family — classical sampling matches them up to polynomial factors [T2]. HHL's fine print (state preparation, condition number, readout) hollows out most quantum-linear-algebra applications [T2]. On D-Wave, the Flatiron Institute (Joseph Tindall et al., belief-

propagation tensor networks) published a peer-reviewed *Science* result (May 2026) computing large-scale annealing dynamics classically, and EPFL's Dries Sels ran comparable instances on a laptop; critics showed the original comparison used weak classical baselines [T2/T3]. No neutral third party has adjudicated the spin-glass dispute as of mid-2026: D-Wave's May-2026 rebuttal holds that the belief-propagation method never attempted the hardest 3D lattice geometries, the largest 3D simulations, the low-precision ensembles where correlations grow fastest, or the full-state / fourth-order observables of the original *Science* paper — and those specific instances remain unmatched classically, so the claim stays contested rather than overturned. A 2025 result gives quasi-polynomial classical sampling of noisy circuits under approximate Markovianity, squeezing noisy-sampling advantage generally [T3, arXiv:2510.06324]. Grover's quadratic speedup is widely expected to be consumed by QEC overhead [T2/T3, "beyond quadratic speedups"].

WHAT WOULD RESOLVE IT

A verifiable advantage claim that stands **~3+ years** against motivated classical algorithm hunters. Quantum Echoes is the live test case: if belief-propagation and other tensor-network attacks keep failing on it through 2027–28, it becomes the first durable useful-ish advantage; if a classical method matches it (as happened to 2019 supremacy), the pattern of attrition holds. On the theory side, dequantization of the remaining chemistry/simulation candidates would collapse the practical case to fault-tolerant Shor and quantum simulation alone.

SOURCES

- Google Research, "A verifiable quantum advantage" (Quantum Echoes, Oct 2025) [T4/T3]
- arXiv:2604.15427 — belief-propagation tensor networks cannot feasibly simulate Quantum Echoes (2026) [T3]
- D-Wave *Science* (Mar 2025) vs Flatiron *Science* (adx2728, May 2026) + D-Wave rebuttal — quantumcomputingreport.com [T2/T3/T4, contested]
- Tang, dequantization line — ewintang.com; Nature Rev. Physics s42254-022-00511-w [T2]

- [arXiv:2510.06324](#) (classical sampling of noisy circuits); [arXiv:2508.05720](#) ("The vast world of quantum advantage") [T3]

The Benchmarking Standardization Crisis · O-benchmark-standard

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

There is no agreed way to say one quantum computer is better than another. Vendors report different, self-favoring metrics — IBM's **Quantum Volume** and CLOPS (speed), IonQ/QED-C's **Algorithmic Qubits (#AQ)**, raw physical-qubit count, gate fidelities, "logical qubits" under vendor-chosen definitions — and each metric can be gamed or is sensitive to different subsystems. Unlike classical computing, which converged on SPEC, LINPACK, and MLPerf, quantum has no common yardstick, which makes cross-vendor comparison, procurement, and even honest progress-tracking hard. The sharp question: can the field standardize application-oriented, spoof-resistant benchmarks before the metric zoo lets marketing outrun physics? This is the measurement layer under O-hype and O-advantage — if you can't benchmark honestly, you can't grade the claims.

WHERE THE DISAGREEMENT IS

- **Standardization-is-arriving camp.** The **QED-C** (Quantum Economic Development Consortium) ships an open-source **application-oriented benchmark suite** (15+ applications: Grover, Monte Carlo, Shor period-finding, VQE, etc.) using volumetric benchmarks, and runs a Standards & Performance Metrics committee convening vendors and labs; its 2026 plan puts standards work front and center [T3/T4]. Quantum Volume is a genuine full-system metric sensitive to qubit number, fidelity, and connectivity together. New suites (QuSquare, QPack, the QuantumBenchmarkZoo aggregation) push toward quality-oriented, reproducible measurement. ISO/IEC JTC1, IEEE, and ETSI have quantum working groups building toward formal standards.
- **Metrics-are-a-mess camp.** "There is not yet a common understanding of standardized metrics in quantum computing" [T3, direct]. Vendor metrics are chosen to flatter the vendor's architecture — #AQ favors high-fidelity low-qubit ion machines, raw counts favor superconducting, "logical qubits" often means memory not a full gate set. Cross-entropy benchmarking,

used for supremacy claims, is not a proof and has been spoofed classically (O-advantage, O-verification). Application benchmarks depend heavily on compilation and error-mitigation choices, so the same hardware scores differently under different software, and results are rarely independently reproduced. Quantum Volume saturates and stops discriminating at the top of the field. The absence of a neutral, funded, adversarial benchmarking body (a "quantum NIST-benchmark") is a structural gap.

WHAT WOULD RESOLVE IT

Broad adoption of a small set of **application-oriented, independently reproduced, spoof-resistant** benchmarks — reported by an independent body, not the vendor — that procurement and peer review actually use, the way MLPerf disciplined ML hardware claims. Concretely: a standard that reports *logical* performance with a defined universal gate set at stated code distance, plus an end-to-end application score run by a third party. Until then, the manual treats every single-metric vendor headline as T4 and asks which subsystem the number really measures.

SOURCES

- QED-C, Standards & Performance Metrics committee + application-oriented benchmark suite — quantumconsortium.org/tac/standards [T3/T4]
- SRI/QED-C, QC-App-Oriented-Benchmarks (open source, 15+ applications) [T3]
- IBM Quantum Volume / CLOPS; IonQ + QED-C Algorithmic Qubits (#AQ) [T4, vendor metrics]
- arXiv:2512.19665 — "QuSquare: Scalable Quality-Oriented Benchmark Suite" (2025) [T3]
- QuantumBenchmarkZoo benchmarking-initiatives aggregation [T3]

The Classical-Simulation Frontier — Where the Boundary Actually Is · O-classical-sim

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Every quantum-advantage claim implicitly draws a line: on *this* side, classical computers can reproduce the result; on *that* side, they can't. The line moves. It is set by the best classical algorithm anyone has written, and clever people keep writing better ones — so a claim that looked beyond-classical in year N is routinely pulled back inside the boundary in year $N+2$. The sharp question is not "can a quantum computer beat brute-force state-vector simulation" (it can, at ~ 50 qubits) but "where does the boundary sit against the *best* classical method — tensor networks, belief propagation, neural quantum states, Clifford-plus-few-T sparsification — for the specific structured problems quantum machines actually run?" Getting this boundary right is what separates real advantage (O-advantage) from a benchmark artifact.

WHERE THE DISAGREEMENT IS

- **Boundary-is-close-and-moving camp.** Classical simulation has repeatedly caught up: Google's 2019 "10,000 years" fell to days on tensor-network/GPU methods within ~ 3 years; IBM's 2023 utility experiment was matched within days by 2D tensor networks (gPEPS, Tindall et al.) and sparse Pauli methods (Sandia, Caltech). The Flatiron Institute's belief-propagation tensor networks reproduced D-Wave's flagship spin-glass dynamics (*Science*, May 2026). Neural quantum states and matrix-product-state methods keep extending reach. The lesson: most "advantage" demos sit *near* the boundary, on structured problems (low entanglement, shallow depth, geometric locality) that classical heuristics exploit, and headline speedups often used weak classical baselines [T2/T3].
- **Boundary-is-being-crossed-for-real camp.** Not every problem yields to classical heuristics. The March 2026 result "Tensor Networks with Belief Propagation Cannot Feasibly Simulate Google's Quantum Echoes" (arXiv:2604.15427) argues the leading classical attack *fails* on that OTOC problem — the first advantage claim in a while that resists the usual counterattack [T3]. Volume-law-entangled, deep, non-geometrically-local

circuits are believed hard for all known classical methods; fault-tolerant Shor and genuine quantum simulation live provably beyond the boundary given a working FTQC machine [T1/T2 theory]. The camp argues the boundary, while it moves, is *converging* on a stable frontier that quantum machines are now starting to clear.

WHAT WOULD RESOLVE IT

For any specific claim: a sustained failure of *all* classical methods (tensor networks with belief propagation, neural quantum states, Clifford perturbation theory, MPS/PEPS, quantum Monte Carlo) to match it over ~3 years of motivated attack. Quantum Echoes is the current live probe — if it holds against classical attack through 2027–28 it marks a real, stable boundary crossing; if a classical method matches it, the "near-the-boundary" reading wins again. Field-wide, a rigorous complexity-theoretic characterization of *which structured problems* admit efficient classical simulation would convert this from an empirical arms race into settled theory.

SOURCES

- Tindall et al., "Efficient tensor network simulation of IBM's largest quantum processors," arXiv:2309.15642 / PRX Quantum 5, 010308 [T2]
- Flatiron/BU classical annealing-dynamics algorithm — *Science* adx2728 (May 2026) [T2]
- arXiv:2604.15427 — belief-propagation tensor networks cannot feasibly simulate Quantum Echoes (2026) [T3]
- arXiv:2510.06324 — classical sampling of noisy circuits under approximate Markovianity (2025) [T3]
- arXiv:2508.05720 — "The vast world of quantum advantage" [T3]

The Cryptographically-Relevant Quantum Computer (CRQC) Timeline · O-crqc-timeline

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

A **cryptographically-relevant quantum computer** is one that can run Shor's algorithm at scale — factoring RSA-2048 or solving the discrete logarithm behind ECC, breaking the public-key crypto that secures nearly all internet traffic, finance, and government data. This is a *specific* fault-tolerant milestone, distinct from the general "useful FTQC" question (O-scaling): it needs thousands of logical qubits sustained through $\sim 10^9$ operations. It carries its own urgency because of **harvest-now-decrypt-later (HNDL)**: adversaries can record encrypted traffic *today* and decrypt it once a CRQC exists, so any secret that must stay confidential for a decade is already at risk. The sharp question: when does the CRQC arrive, and is PQC migration (A-pqc) outrunning it? This is where **Mosca's inequality** bites — if (migration time) + (secrecy shelf-life) > (time to CRQC), you are already exposed.

WHERE THE DISAGREEMENT IS

- **Sooner / resources-are-falling camp.** Craig Gidney (Google, arXiv:2505.15917, 2025) cut the RSA-2048 estimate $\sim 20\times$, to **<1M noisy physical qubits at ~1 week runtime** — down from $\sim 20M$ (2019) — via approximate residue arithmetic, yoked surface codes, and magic-state cultivation [T3]. Expert-elicitation surveys (Global Risk Institute) put meaningful probability on a CRQC within a decade: roughly **22.7% of surveyed cryptographers expect RSA-2048 to fall by 2030, and ~50% by 2035**; point estimates cluster around **2030 $\pm$ 3 years** [T5]. NIST has already standardized PQC (ML-KEM/Kyber, ML-DSA/Dilithium) precisely because the threat is treated as near enough to act on now [T2, standards].
- **Later / hardware-isn't-close camp.** A $<1M$ -qubit *estimate* is not a $<1M$ -qubit *machine* — the largest devices are $\sim 1,000$ – $6,000$ physical qubits, so the gap is 2–3 orders of magnitude in count while *holding* the fidelity Shor needs, which nobody has shown at scale (O-scaling, O-materials). Broad expert consensus places the *heaviest* weight in the mid-2030s, and Gil Kalai's minority position says a CRQC never arrives at all (O-hype). Recent

toy Shor demonstrations factored only tiny numbers and do not extrapolate. The energy cost of the run is itself nontrivial (O-energy). Skeptics stress that the resource *estimates* keep improving on paper faster than the *hardware* improves in the lab.

WHAT WOULD RESOLVE IT

A quantum computer factoring a cryptographically meaningful RSA modulus (even RSA-1024, then 2048) with fault tolerance — or credible interim milestones: factoring a 100+-bit semiprime with Shor (not a special-form or annealing shortcut), then scaling. On the defensive side, completion of PQC migration across critical infrastructure before any CRQC appears would neutralize the threat regardless of the exact date. The dials: physical-qubit count *at Shor-relevant fidelity*, published Shor demonstrations on growing moduli, and the pace of NIST-PQC deployment versus HNDL exposure windows.

SOURCES

- Gidney, "How to factor 2048 bit RSA integers with less than a million noisy qubits," arXiv:2505.15917 (2025) [T3]
- Global Risk Institute Quantum Threat Timeline expert survey (2030 $\pm$ 3; ~50% by 2035) [T5]
- NIST FIPS 203/204/205 (ML-KEM, ML-DSA, SLH-DSA) post-quantum standards (2024) [T2]
- quantumzeitgeist.com, "Cryptographically Relevant Quantum Computer: Complete 2026 Guide"; postquantum.com HNDL + RSA-2048 energy analyses [T3/T5]
- Mosca, "Cybersecurity in an era with quantum computers" (Mosca's inequality) [T3]

Real-Time QEC Decoding Throughput · O-decoder

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Error correction only works if a classical co-processor decodes each round of syndrome measurements and tells the quantum hardware what correction to apply *before the next round arrives*. Superconducting qubits emit a syndrome roughly every microsecond, so the decoder must sustain about **1 MHz per logical qubit** at sub-microsecond latency — for thousands of logical qubits at once. If decoding lags, syndromes pile up (the "backlog problem," Terhal 2015): the effective clock speed collapses exponentially in the backlog. That makes the classical decoder a first-class bottleneck on the road to fault tolerance. A machine can have perfect qubits and still be useless if it can't decode fast enough — and the raw syndrome-data rate off a large cryostat runs to terabytes per second, which is itself an open cryo-electronics problem (see H-cryocmos, O-interconnect-loss).

WHERE THE DISAGREEMENT IS

- **Tractable-engineering camp.** Riverlane has built dedicated decoders — the Local Clustering Decoder targets $<1\ \mu\text{s}$ per round, and a Dec 2025 *Nature Communications* result reports hitting a **MegaQuOp** (10^6 reliable operations) with $\sim 4\times$ fewer qubits via better decoding [T2]. Riverlane's roadmap frames the path from MegaQuOp (late 2020s) to **TeraQuOp** (10^{12} operations, early 2030s) as an engineering scaling problem, and flags that commodity PCIe/GPU interfaces bottleneck beyond ~ 300 physical qubits, motivating purpose-built interconnects (QECi, $<400\ \text{ns}$ round-trip) [T3/T4]. Google has run real-time decoding on Willow's distance-5/7 codes keeping pace with the syndrome stream; FPGA and ASIC decoders (union-find, sliding-window) are demonstrated at small distance [T2/T3].
- **Unproven-at-scale camp.** The accuracy/latency/bandwidth *trilemma* is unsolved at scale: high-accuracy decoders (full minimum-weight matching, neural decoders) are slow; fast decoders are less accurate, and lower accuracy silently raises the logical error rate. Nobody has demonstrated real-time decoding across the **thousands-of-logical-qubit** regime a useful computation needs, and qLDPC codes (which IBM now bets on) are *harder*

to decode fast than surface codes because their checks are non-local. Transversal-gate architectures need correlated decoding across logical qubits, which is more expensive still (arXiv:2505.13587). Off-chip syndrome bandwidth at million-qubit scale has no demonstrated solution [T3].

WHAT WOULD RESOLVE IT

A machine running many logical qubits through millions of error-corrected cycles with decoding kept ahead of the syndrome stream, at demonstrated MHz throughput and **constant, non-growing backlog**, reproduced independently — ideally on a qLDPC code, not just surface codes. Riverlane's TeraQuOp milestone hitting spec, or IBM Kookaburra/Starling shipping with an integrated real-time decoder that holds pace, would settle the direction. The negative signal: decoder latency that grows with qubit count and forces the quantum clock to slow down.

SOURCES

- Riverlane, "QEC Technology Roadmap" (2026) + Local Clustering Decoder papers — riverlane.com [T4]
- Riverlane et al., *Nature Communications* (Dec 2025), MegaQuOp with ~4× fewer qubits [T2]
- Terhal, "Quantum error correction for quantum memories," RMP 87, 307 (2015) — backlog problem [T2]
- arXiv:2605.30765 — real-time QEC system stack (architecture/algorithms/engineering, 2026) [T3]
- arXiv:2505.13587 — fast correlated decoding of transversal logical algorithms [T3]

Energy Consumption — Quantum vs Classical · O-energy

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

A quantum computer's headline runtime is only half of a fair comparison; the other half is *energy*. A dilution refrigerator draws roughly **10 kW continuously**, dominated by the mechanical compressor on the 3 K stage, and a full quantum system with control electronics runs on the order of **~18 kW** — a fixed cost paid whether the machine is computing or idle. A classical supercomputer like Frontier or Summit draws megawatts. The sharp question: for the specific tasks quantum machines win on, does the quantum system deliver the answer at *lower total energy* than the classical alternative — and does that "energy advantage" arrive earlier or later than the runtime advantage? This is a distinct axis from speed, and it becomes a scaling argument as classical data-center power grows 20–40% per year.

WHERE THE DISAGREEMENT IS

- **Quantum-is-energy-efficient camp.** For sampling-type tasks, Google's supremacy analysis found the classical simulation on Summit would consume **~7 orders of magnitude more energy** than the quantum processor's run [T3]. A quantum system at ~18 kW is orders of magnitude below a supercomputer; when *idle* the gap is even larger (~432 kWh/day vs a supercomputer's draw) [T5]. Recent theory identifies an **"energetic advantage before computational advantage"**: in boson sampling and in superconducting cat-qubit computation, the quantum machine can win on energy at a problem size *smaller* than where it wins on wall-clock time (arXiv:2601.08068; arXiv:2605.19854, 2026) [T3]. As classical compute's energy demand climbs 20–40%/year, quantum looks comparatively sustainable at scale.
- **Cryo-overhead-dominates camp.** The cryogenic and classical-control overhead is *fixed and large*, so for any small or short computation the quantum machine's energy-per-useful-answer is terrible — you pay 18 kW to run a circuit a laptop finishes for watts. Full-system energy accounting (arXiv:2605.09580, 2026) stresses that control electronics, the classical co-processor (decoders — O-decoder), and the dilution plant must all be

counted, and at *fault-tolerant scale* the millions of physical qubits plus their real-time decoding could push total draw far above today's single-fridge figure. The RSA-2048 energy cost, even at ~1M qubits over a week, is nontrivial. The favorable comparisons are for cherry-picked sampling problems with no practical use (O-advantage); no useful workload has yet demonstrated a *net* energy win.

WHAT WOULD RESOLVE IT

A full-system, end-to-end energy measurement — fridge + control + classical co-processor — for a *useful* computation, compared against the best classical method's energy for the same answer done in good faith. A published joules-per-useful-result that undercuts classical would establish a real energy advantage; full-stack accounting showing cryo + decoding overhead swamps the benefit at fault-tolerant scale would deflate it. The theory prediction to test: does the energetic-advantage crossover really precede the computational-advantage crossover on a problem someone cares about?

SOURCES

- arXiv:2605.09580 — "Estimating the Energy Consumption of Quantum Computing from a Full System Aspect" (2026) [T3]
- arXiv:2601.08068 — "Quantum Energetic Advantage before Computational Advantage in Boson Sampling" (2026) [T3]
- arXiv:2605.19854 — energetic advantage in superconducting cat-qubit computation (2026) [T3]
- Google supremacy energy comparison (Summit, ~7 orders of magnitude) [T3]
- postquantum.com, "The Enormous Energy Cost of Breaking RSA-2048"; Springer, "Energy Efficiency for Quantum Computers" [T3/T5]

Interpretational Openness — The Measurement Problem, Still Unsolved · O-foundations

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

A century after the formalism was written down (1925 → 2025 centenary), physics still lacks an agreed answer to its central question: what happens during measurement, and what the quantum state *is*. The dual evolution of quantum states — unitary and reversible between measurements, discontinuous and probabilistic at them — remains the field's oldest open problem. Quantum computing sharpens it: every working quantum computer is an experiment in maintaining coherent superposition at ever-larger scales, probing exactly the regime where interpretations differ, and a large-scale fault-tolerant machine would be the most macroscopic superposition ever engineered. The sharp question is whether the problem is *empirically* open (collapse models make testable predictions) or *permanently* open (the surviving no-collapse interpretations agree on all observations).

WHERE THE DISAGREEMENT IS

- **Resolvable / already-resolved camp.** Everettians argue unitarity-all-the-way-down dissolves the problem; David Deutsch famously framed a scalable quantum computer as evidence for many-worlds — "where was the computation performed?" [T3/philosophical]. Objective-collapse theorists (GRW, CSL, Diósi–Penrose) make the strongest claim: collapse is *physical and testable*, and their parameter space is being steadily squeezed — underground experiments (Gran Sasso, using the spontaneous X-ray emission collapse models predict) and space-based data have excluded large regions of Diósi–Penrose and CSL parameter space [T2]. Decoherence theory rigorously explains the *appearance* of collapse [T1], which some read as most of the answer.
- **Stays-open camp.** A 2025 *Nature* centenary survey of 1,100+ physicists found **no consensus** — Copenhagen leading at ~36%, epistemic/QBist views next, then many-worlds, with pilot-wave, spontaneous-collapse, and relational holding steady minorities; respondents split even on whether the question is physics or philosophy [T3/survey]. Recent reviews

(arXiv:2502.19278) conclude the measurement problem is *clarified but unsolved* [T3]. Decoherence explains suppressed interference and leaves the single-outcome ("why this result") question untouched — a point near-universal among foundations specialists [T2/T3].

WHAT WOULD RESOLVE IT

An experimental deviation from unitarity — detection of the spontaneous radiation or heating that collapse models predict — would end the debate in favor of objective collapse. Continued null results, plus verified coherence at ever-more-macroscopic scales (large-mass matter-wave interferometry pushing toward 10^6 – 10^9 amu, and fault-tolerant quantum computers themselves), keep squeezing collapse models toward irrelevance *without* adjudicating between Copenhagen, QBism, and many-worlds. Distinguishing those three may require no experiment at all — which is precisely why the problem may be permanent rather than merely hard. A working million-qubit FTQC machine would be a data point for the no-collapse camps and against any collapse model whose threshold it exceeds.

SOURCES

- arXiv:2502.19278 — "The Quantum Measurement Problem: A Review of Recent Trends" (2025) [T3]
- Nature centenary survey (2025): physicists disagree on what QM says about reality [T3/survey]
- Diósi–Penrose / CSL experimental constraints — Gran Sasso spontaneous-X-ray tests [T2]
- CERN Courier, "The measurement problem, measured"; MDPI *Quantum Reports* 7(2):28 [T3]
- Deutsch, *The Fabric of Reality* (many-worlds / quantum computing argument) [T3/philosophical]

Hype vs Reality — Timelines, Quantum-Washing, Valuations · O-hype

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

The industry's market capitalization runs decades ahead of its revenue. As of May 2026, IonQ, Rigetti, and D-Wave traded at price-to-sales ratios of roughly **109, 836, and 791**; IonQ carried a **\$21B+ market cap on ~\$187M trailing revenue and a \$711M EBITDA loss**. Total industry revenue sits near **\$1.4B** (QED-C 2026, computing segment). Every vendor roadmap promises early fault tolerance by 2029–2030 [all T4/roadmap]. The open question is which part of the enthusiasm is early-stage conviction and which part is a bubble. This manual grades every vendor timeline T4/T6 on principle and the valuation debate T5.

WHERE THE DISAGREEMENT IS

- **Substance camp.** IBM, Google, Quantinuum, and Microsoft roadmaps independently converge on 2029–2030 for early fault tolerance [T4]; real technical milestones keep landing — below-threshold QEC confirmed by four teams (Google Willow, Quantinuum, Harvard/QuEra, USTC) [T2], the 96-logical-qubit QuEra record (Jan 2026), Quantum Echoes (Oct 2025). National programs and hyperscalers keep funding. Scott Aaronson (March 2026) said 2025 "met or exceeded my expectations on hardware" and updated toward taking 2028–29 pronouncements more seriously [T3]. Analysts project tens of billions in value creation by the 2030s [T5, BCG/McKinsey TAM — flagged for optimism].
- **Bubble camp.** **Sabine Hossenfelder** predicted the hype would crash and argues claimed advantages in logistics and finance dissolve on contact with real-world constraints [T5]. **Gil Kalai** holds the strongest position: FTQC is impossible because noise correlations kill the threshold theorem — a published minority view; Aaronson (2026) counters that Kalai "keeps saying [the 2019] experiment was fallacious, but meanwhile a dozen other experiments by other companies have reached the same conclusion, so he's fighting a losing battle," adding that Kalai's "path has been getting narrower and narrower" [T3/contested]. Kalai's direct reply (gilkalai.wordpress.com, 10 Mar 2026) concedes he "starts with quantum

computation being impossible as [an] axiom" but disputes the interpretation, holding that the observed exponential error decay "does not contradict my conjectures regarding correlated errors" and that his predicted correlations show up at "hundreds... and thousands of gates," not millions [T3/contested]. **Jensen Huang** (Jan 2025): useful machines 15–30 years away; quantum indices fell double digits on the remark, recovering when he walked it back at GTC Quantum Day [T5]. Insiders at IonQ/Rigetti/D-Wave were net sellers of ~\$931M of stock over five years — a signal analysts flag [T5]. "Quantum-washing" — rebranding classical or hybrid products as quantum — is tracked as a distinct risk; Quantum Computing Inc.'s Q1 2026 "9,000% revenue surge" to \$3.6M came largely from acquisitions and peripheral products [T5]. The 2026 stress test, per multiple analysts: "Cool story. Where's the revenue?" [T5]

WHAT WOULD RESOLVE IT

Revenue and reproduction. A commercial customer paying for a quantum computation because it beats the classical alternative on cost would end the debate one way; a funding winter after missed 2029–2030 roadmap dates would end it the other. Kalai's in-principle claim dies the day a large machine sustains fault tolerance; it strengthens with every correlated-noise surprise at scale. Distinguish the *technical* roadmap question (O-scaling) from the *valuation* question — a company can be overpriced while the physics is on track.

SOURCES

- Kalai, "Scott Aaronson's View of my View About Quantum Computing," gilkalai.wordpress.com/2026/03/10/ (Mar 2026); Aaronson, "More on whether useful quantum computing is 'imminent'," scottaaronson.blog/?p=9425 (2026) — the Mar-2026 exchange, both sides [T3/contested]
- Hossenfelder public commentary + UK parliamentary written evidence 121097 [T5/T3]
- The Motley Fool / Nasdaq / 24-7 Wall St., IonQ-Rigetti-D-Wave valuation + insider-selling coverage (2026) [T5]
- QED-C, "State of the Global Quantum Industry 2026" (market \$1.9B, computing \$1.4B → \$3B by 2028) [T5]

Chip-to-Chip Entanglement Fidelity · O-interconnect-loss

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

No single chip will hold the millions of qubits a useful computer needs, so every serious roadmap is **modular**: many smaller processors linked by quantum interconnects. The catch is that gates *across* a link are slower and lower-fidelity than gates on a chip. Long-range links (fridge-to-fridge, via microwave-to-optical transduction and photons) are lossy and probabilistic; even short on-substrate couplers add error. The sharp question is whether inter-module entanglement can be made good enough — high enough fidelity, high enough rate — that error correction still works *across* module boundaries, or whether the interconnect becomes the dominant error source that caps the whole machine (see H-intercon, H-transduce, F-teleport, A-qmemory-hw).

WHERE THE DISAGREEMENT IS

- **Modular-is-inevitable camp.** IBM's 2024 Flamingo demonstrated two Heron R2 chips linked over ~1 m couplers, and the entire roadmap (Cockatoo 2027 establishes module-to-module entanglement via I-couplers; Starling 2028–29 does magic-state injection across modules) is built on chip-to-chip and fridge-to-fridge links. Trapped-ion and neutral-atom platforms shuttle qubits or photons between zones with high fidelity; Quantinuum uses photonic links between QCCD units and reports high-fidelity entanglement across them. Codes tolerant of noisy inter-module links (lattice surgery across boundaries, qLDPC blocks connected by comparatively few long-range checks) exist on paper, and proponents argue distributed/networked QC (see A-qinternet) makes modularity a feature that also enables scaling past a single cryostat [T3/T4].
- **Interconnect-caps-the-machine camp.** IBM's own Flamingo link reported roughly **3.5% error** across the coupler in 2024 — about two orders of magnitude worse than the best on-chip two-qubit gates [T4]. Photon-loss-limited remote-entanglement rates are slow relative to qubit coherence, and if inter-module error correction demands many rounds of noisy entanglement purification, the overhead could swamp the benefit. Microwave-to-optical transduction (the fridge-to-fridge path) still sits at low

conversion efficiency with added noise, an unsolved component problem (H-transduce). Nobody has yet shown *fault-tolerant* logical operations spanning modules at scale [T3].

WHAT WOULD RESOLVE IT

A logical qubit whose stabilizer measurements *straddle two modules*, kept below threshold across the interconnect, sustained over many rounds — reproduced independently. IBM Cockatoo (2027) demonstrating below-threshold cross-module operation, or a trapped-ion/atom system running a logical gate across a photonic link at on-chip-comparable fidelity, would settle the direction. The negative signal: cross-link error rates that stay stuck near percent-level while on-chip gates keep improving, forcing architectures back toward monolithic chips they can't actually build.

SOURCES

- IBM Quantum roadmap + Flamingo (Heron R2 link demo, 2024; ~3.5% coupler error) — ibm.com/quantum/blog [T4]
- Monroe et al., "Large-scale modular quantum-computer architecture with atomic memory and photonic interconnects," PRA 89, 022317 (2014) [T2]
- Benchmarking cavity-mediated interconnects, arXiv:2407.15651 (2024) [T3]
- Quantinuum QCCD photonic-link entanglement reports (2025–26) [T3/T4]

The Killer-App Question — What Pays For the Machine · O-killerapp

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Every platform technology needed one application valuable enough to fund the buildout. For quantum computing the candidates are chemistry/materials simulation, optimization, and codebreaking — and each has a problem. Chemistry needs hundreds-to-thousands of error-corrected logical qubits that don't exist yet. Optimization speedups are mostly quadratic and likely to be eaten by error-correction overhead. Shor's algorithm destroys value (breaking RSA) once, after which the world migrates to post-quantum cryptography and that market evaporates. The sharp question: is there any application whose paying demand arrives *before* investor patience runs out? This node is the technical-feasibility twin of O-roi-business (who pays) and shares the chemistry resource numbers with O-crqc-timeline's cousin problem.

WHERE THE DISAGREEMENT IS

- **Chemistry-is-the-app camp.** The Aspuru-Guzik-line argument that "chemistry is quantum computing's killer app" [T2 theory], with Quantinuum and pharma partners (Boehringer, Roche) building toward scalable chemistry-on-quantum workflows [T4]. Commonly cited commercial timeline: quantum molecular simulation contributing to drug discovery in **5–10 years**, contingent on hundreds of logical qubits [T5]. A June 2025 perspective maps applications reachable with **25–100 logical qubits** (arXiv:2506.19337), a nearer target than the flagship FeMoco problem [T3]. Quantum sensing already produces revenue today and de-risks the supply chain [T3].
- **Feasibility-skeptic camp.** The canonical FeMoco (nitrogen-fixation cofactor) target keeps needing more, not fewer, resources at the physical level: modern estimates put it at **~1,137 logical qubits and ~3.4×10⁸ logical T-gates**, and "first useful QPE chemistry" is often pegged near **~1,500 logical qubits** — far beyond 2026 hardware [T3]. Google once estimated ~4M physical qubits, ~4 days for FeMoco [T3]. 2025–26 feasibility studies find quantum computers face bigger hurdles in quantum

chemistry than advertised [T3; New Scientist: "chemistry may not be the killer application"]. Optimization/logistics/finance value is the softest: Hossenfelder's critique that real operations are constrained by everything except raw compute [T5], plus Babbush et al.'s "focus beyond quadratic speedups" showing quadratic advantages evaporate under QEC overhead [T2/T3]. Scott Aaronson's standing honest answer: the killer app may be **simulating quantum physics itself**, a market whose dollar size nobody knows [T3].

WHAT WOULD RESOLVE IT

One paying, repeated commercial workload where the quantum result is cheaper or better than the best classical alternative — audited, with the classical baseline done in good faith (density-matrix renormalization, coupled cluster, tensor networks all tried honestly). A pharma company reordering compute budget toward QPU time, or a materials firm shipping a product whose design used a quantum simulation no classical method could do, would settle it. Ten more years of pilot projects that never convert would settle it the other way. The nearest real test: whether the 25–100-logical-qubit chemistry targets produce a *decision-changing* answer once that hardware arrives.

SOURCES

- Aspuru-Guzik et al., "Chemistry is quantum computing's killer app" line — C&EN (2017) [T2]
- arXiv:2506.19337 — "A Perspective on Quantum Computing Applications...using 25–100 Logical Qubits" (2025) [T3]
- FeMoco resource estimates ($\sim 1,137$ logical qubits / 3.4×10^8 T-gates; Google $\sim 4\text{M}$ physical) [T3]
- New Scientist / Ground News chemistry-feasibility skepticism (2025–26) [T3/T5]
- Babbush et al., "Focus beyond quadratic speedups for error-corrected quantum advantage" [T2]

Decoherence & Materials Defects — The TLS Bottleneck · O-materials

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

The dominant coherence killer in superconducting qubits is a materials problem: **two-level systems (TLS)** — microscopic defects in amorphous oxides, interfaces, and Josephson-junction tunnel barriers — that couple to qubits, drift in frequency, and cause both decoherence and readout failures. Coherence times have climbed from nanoseconds (1999) to the millisecond range, yet TLS make individual qubits *unpredictable*: a qubit fine today can be lossy tomorrow as a defect diffuses onto resonance. The sharp question is whether materials science suppresses TLS density fast enough for million-qubit processors, or whether the defect burden scales with the system so that some fraction of a large chip is always out of spec. This is where the threshold theorem's iid-noise assumption meets a physical process that may not obey it.

WHERE THE DISAGREEMENT IS

- **Suppressible-engineering camp.** Fabrication research is finding handles. 2026 high-throughput junction studies correlate TLS density with Al electrode thickness and grain size, demonstrating a **two-thirds reduction in TLS density** through fabrication changes alone [T3, arXiv:2602.11469]. Site-specific TLS frequency tuning in qubit arrays lets operators steer defects off resonance [T3, arXiv:2503.04702], and ML-accelerated TLS characterization speeds the search [T2/T3, Adv. Quantum Technol. 2026]. The camp's analogy: semiconductor yield problems looked fatal in the 1960s and became a controlled engineering discipline. Tantalum and titanium-nitride films, improved surface treatments, and encapsulation have each bought coherence-time gains.
- **Scaling-wall camp.** TLS-induced dropouts **scale with system size** — on a 1,000+ qubit chip some fraction of qubits is always out of spec, and QEC decoders assume error rates that TLS spectral diffusion silently violates during a run [T3, arXiv:2605.02755]. Correlated error *bursts* — quasiparticle poisoning and cosmic-ray/ionizing-radiation impacts that knock out many qubits at once — add a second materials-linked failure

mode that surface codes handle badly [T2]. Gil Kalai's deeper claim generalizes this: noise in engineered many-body systems is inherently correlated, so the iid assumption underlying the threshold theorem never holds at scale [T3/contested]. Even friendly reviews call TLS "a critical bottleneck for scalable quantum processors" [T3, arXiv:2602.04831].

WHAT WOULD RESOLVE IT

A junction process with TLS density low enough — and stable enough — that a 10,000-qubit chip holds every qubit within QEC spec across weeks of continuous operation, with published defect statistics showing the out-of-spec fraction does *not* grow with chip area. Alternatively, a demonstration that decoders plus real-time TLS-tuning absorb realistic defect drift and cosmic-ray bursts at scale (leakage-aware decoding, gap engineering against quasiparticles). Either result converts the materials bottleneck from open problem to engineering roadmap. The negative signal: persistent out-of-spec fractions that scale with system size, exactly Kalai's predicted regime.

SOURCES

- arXiv:2602.11469 — structural control of TLS density in Josephson junctions (2/3 reduction) [T3]
- arXiv:2605.02755 — readout failures from junction TLS defects, dropouts scaling with size [T3]
- arXiv:2503.04702 — scalable site-specific TLS frequency tuning [T3]
- arXiv:2602.04831 — review of superconducting qubit large-scale integration [T3]
- Science Advances abc5055 — TLS from trapped quasiparticles; cosmic-ray correlated-error literature [T2]

Error-Correction Overhead — The Physical:Logical Ratio · O-overhead

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Error correction is a tax: every logical qubit is paid for in physical qubits, and every non-Clifford (T) gate is paid for in magic states. Standard surface codes at useful code distance imply roughly **1,000 physical qubits per logical qubit**, and magic-state distillation factories have historically consumed a large fraction of the total footprint. The sharp question is how far new code families and distillation tricks compress that tax — and, critically, whether the headline low ratios from recent demos survive at *algorithm-relevant code distance and gate depth*. A 4.7:1 ratio at distance 4 on a hundred-qubit demo and a 1,000:1 ratio at distance 25 for a billion-gate algorithm are both true; the manual must say which regime a number lives in.

WHERE THE DISAGREEMENT IS

- **Optimist camp.** qLDPC codes cut overhead an estimated **~10–20×** **versus surface codes** [T3], and IBM's public Starling accounting claims up to **90% overhead reduction** with its bivariate-bicycle qLDPC codes [T4]. QuEra demonstrated **96 logical qubits from 448 physical atoms** (~4.7:1) with a high-rate $[[16,6,4]]$ code, running error-corrected gates across all 96 at once, below threshold (Nature, Jan 2026) [T2/T3]. Quantinuum reported **94 logical qubits at ~2:1** on H-series with logical error below 10^{-4} (March 2026), and Microsoft+Quantinuum logged an **~800×** **error-rate improvement** (Nature, June 2026) [T2/T4]. GKP bosonic codes reach effectively 1:1 with the logical qubit outliving its physical parts [T2/T3]. Gidney's magic-state **cultivation** (2024–25) collapses distillation cost enough to help drive the 20× drop in RSA-2048 estimates; Litinski's line already argued "magic state distillation is cheaper than you think" [T3].
- **Skeptic camp.** The flattering ratios come from **low-distance, high-rate codes on small systems** — a $[[16,6,4]]$ code has distance 4, far from the distance ~25 needed to hit 10^{-12} logical error for a billion-gate algorithm. At those target error rates, realistic architectures still project **hundreds-to-1,000:1 all-in** once routing, ancillas, and T-factories are counted [T3].

qLDPC's long-range connectivity is natural for neutral atoms and ions and remains awkward for fixed-lattice superconducting chips, which is why IBM needs new c-/l-couplers to use it at all [T3]. Vendor "logical qubit" counts use vendor-chosen definitions and often report memory rather than a full logical *gate set* including T-gates [T4 — grade hard].

WHAT WOULD RESOLVE IT

An end-to-end fault-tolerant algorithm — thousands of logical operations including T-gates, with the full physical budget published — at a measured all-in ratio. If a qLDPC machine sustains **<100:1 at distance ≥ 11** with a working magic-state supply, the overhead pessimists lose. If every deep-circuit demo quietly reverts to ~1,000:1 surface-code accounting once T-gates and routing enter, they win. The tell is whether the low ratios are reported *with* a universal gate set at algorithm-relevant distance, or only for memory/Clifford operation.

SOURCES

- QuEra, 96 logical qubits from 448 atoms, $[[16,6,4]]$ qLDPC, below threshold — Nature (Jan 2026) [T2/T3]
- Quantinuum 94 logical qubits (Mar 2026); Microsoft+Quantinuum 800× improvement, Nature (Jun 2026) [T2/T4]
- Gidney, arXiv:2505.15917 (yoked surface codes, magic-state cultivation) [T3]
- IBM Starling qLDPC overhead accounting — ibm.com/quantum/blog/large-scale-ftqc [T4]
- postquantum.com qLDPC overview; arXiv:2410.07327 (low-overhead magic states via code switching); arXiv:2606.20263 (Vine Codes, planar qLDPC) [T3]

The Business-Model & ROI Question — Who Actually Pays · O-roi-business

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

O-killerapp asks whether a quantum computer *can* do something useful. This node asks the separate question: even granting a technical use case, does the *unit economics* work — who pays, at what price, for what margin, on what horizon? A technically real speedup can still be a bad business if the QPU-hour costs more than the classical alternative it beats, if the customer would rather wait for the cloud price to fall, or if the value accrues to a one-time result rather than a recurring service. The sharp question: is there a repeatable commercial workload where a customer chooses quantum on cost/performance grounds *and* the provider earns a margin — before the capital that funds the buildout runs out (O-talent-attribution, O-hype)?

WHERE THE DISAGREEMENT IS

- **Revenue-is-forming camp.** Quantum-as-a-service is monetizing now: cloud access (Amazon Braket pay-as-you-go, Azure Quantum, IBM), hardware-as-a-service, on-prem sovereign installs, and post-quantum-security consulting. D-Wave has a real commercial book — a \$20M Florida Atlantic system sale and a ~\$10M QCaaS deal with a Fortune 100 buyer in 2026 — and its survey found businesses expecting up to **20× ROI** on quantum-optimization projects, with a claimed **\$51.5B** aggregate potential impact [T5, vendor survey — grade hard]. IonQ guided to **\$225–245M revenue for 2026** (from ~\$130M in 2025). The global quantum-computing market is put near \$1.4B (2025), projected to ~\$3B by 2028 (QED-C) [T5].
- **Economics-don't-close-yet camp.** Most reported "quantum revenue" comes from **acquisitions, government contracts, and peripheral hardware/consulting**, not customers paying for quantum computations that beat classical. Quantum Computing Inc.'s Q1 2026 "9,000% revenue surge" to \$3.6M was largely M&A and peripheral products [T5]. Classical IT investments return in 12–36 months; quantum ROI horizons often exceed a decade, which strains any normal capital-budgeting case [T5]. Today's machines remain "too fragile for broad, repeatable economic advantage

across mainstream enterprise workloads" [T5]. Annealing/hybrid workflows generate *some* real revenue, but whether they beat a good classical solver on the customer's actual problem is exactly the contested O-advantage question. Extreme valuations (P/S of 100–800+) price in a business that doesn't exist yet.

WHAT WOULD RESOLVE IT

An audited, repeated commercial workload where a customer pays for QPU time *because* it is cheaper or better than the best classical alternative, and the provider books it at a margin — not a pilot, not a grant, not a bundled-services deal. A published cost-per-useful-answer that undercuts classical would settle it positively; a decade of pilots that never convert to recurring paid production, plus valuations compressing toward revenue, would settle it negatively. The dial to watch: the fraction of vendor revenue that is *recurring quantum-compute consumption* versus one-time hardware sales, grants, and consulting.

SOURCES

- QED-C, "State of the Global Quantum Industry 2026" (market \$1.9B; computing \$1.4B → \$3B by 2028) [T5]
- D-Wave, "Quantum Means Business" ROI survey (up to 20×; \$51.5B potential) [T5, vendor]
- IonQ 2026 revenue guidance (\$225–245M); D-Wave 2026 commercial deals [T5]
- Cryptobriefing, "Quantum Computing revenue surges 9,000%...the fine print" (Q1 2026) [T5]
- The Strategy Institute / entangledfuture.com quantum-ROI analyses [T5]

Room-Temperature & Photonic Scaling Paths · O-roomtemp

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Superconducting and spin qubits live at millikelvin temperatures inside dilution refrigerators — a cost, wiring, and I/O ceiling that gets brutal at a million qubits. Photonic qubits run gates at room temperature, ship through ordinary fiber, and can be fabbed in semiconductor foundries; NV-diamond systems also operate warm. If either path scales, the economics change category: data-center racks instead of cryostats, networking for free, CMOS-foundry manufacturing. The sharp question is whether photonics' *own* tax — photon loss, probabilistic entangling gates, and the cryogenic single-photon detectors it still needs — comes in cheaper than the dilution-refrigerator tax it removes. "Room temperature" is a claim about *where the hard part moves*, not about eliminating it.

WHERE THE DISAGREEMENT IS

- **Path-changes-the-economics camp. Xanadu's Aurora** (Jan 2025, *Nature*): a modular, networked photonic machine — 4 server racks, 35 photonic chips, 13 km of fiber, 12 physical qubits — presented as scalable in principle to thousands of racks and millions of qubits; Xanadu targets ~1,000 logical qubits by 2029 at ~100:1 [T2 for the demo; T4/T6 for the extrapolation]. **PsiQuantum** partners with GlobalFoundries to fab million-qubit photonic systems, with utility-scale sites announced in Brisbane and Chicago and a multi-hundred-million-dollar raise [T4]. Fusion-based-QEC papers argue photonic loss thresholds are reachable [T3]. On the warm-hardware side, Quantum Brilliance's NV-diamond modules run fully cryo-free.
- **Extrapolation-is-huge camp.** "Room temperature" carries an asterisk — the superconducting-nanowire single-photon detectors (SNSPDs) at the heart of every photonic scheme are cryogenic, so the cryoplant shrinks rather than disappears [T3, CACM]. Photon loss plays the role decoherence plays elsewhere, and current per-component losses sit orders of magnitude above fusion thresholds; **Aurora's 12 physical qubits against a million-qubit roadmap is the largest extrapolation ratio in**

the industry [T3/T6]. PsiQuantum has published no full-system demo at any qubit count, making its million-qubit target the field's biggest single T4 claim. NV-diamond is genuinely cryo-free but sits far behind on qubit count and two-qubit gate fidelity [T3]. Room-temperature *superconductivity* — which would remove the fridge for the transmon path entirely — remains unconfirmed after the retracted LK-99 and Ranga Dias episodes [T3].

WHAT WOULD RESOLVE IT

A photonic machine demonstrating error-corrected logical qubits with **published end-to-end loss budgets** — even ~10 logical qubits would validate the architecture and move it from T4 to T2. For the economics claim: a cost-per-logical-qubit-hour comparison against a superconducting system once both exist at comparable capability. PsiQuantum's first full-stack public benchmark is the single most information-rich pending event on this node; Aurora scaling from 12 to hundreds of physical qubits with loss held constant would be the second.

SOURCES

- Xanadu Aurora — *Nature* s41586-024-08406-9 (2025) [T2]; Xanadu roadmap [T4]
- PsiQuantum / GlobalFoundries partnership + Brisbane/Chicago site announcements [T4]
- CACM, "Making Qubits with Photons" (SNSPD-cryogenics caveat) [T3]
- Springer J. Supercomputing photonic FTQC review (2025); Tom's Hardware photonics roadmap [T3/T5]

Fault-Tolerant Scaling — How Far Off, Really · O-scaling

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

A cryptographically or chemically useful quantum computer needs thousands of logical qubits, which under current architectures means hundreds of thousands to millions of physical qubits running billions of gates. As of mid-2026 the largest devices sit at roughly 1,000–6,000 physical qubits; the best *verified* logical-qubit demonstrations count 96 (QuEra, Nature Jan 2026). The distance between "dozens of logical qubits, a few thousand operations" and "thousands of logical qubits, a billion gates" is the whole game. The estimates on the table differ by an order of magnitude in both qubit count and calendar time, and this manual grades every roadmap date T4/T6 on principle.

WHERE THE DISAGREEMENT IS

- **Optimist camp.** IBM published a modular, milestone-by-milestone path in June 2025: **Loon** (2025, qLDPC architectural elements + c-couplers linking distant on-chip qubits), **Kookaburra** (2026, the first fault-tolerant module integrating logic + memory), **Cockatoo** (2027, entanglement between modules via I-couplers), and **Starling** (2028–29): ~200 logical qubits, 100 million gates, "large-scale fault tolerance." **Blue Jay** (2033) targets 2,000+ logical qubits and a billion gates. IBM's move to qLDPC claims up to a **90% reduction in physical-qubit overhead** versus surface codes [all T4/roadmap]. Google targets a useful error-corrected machine on a similar 2029–2030 horizon [T4]. Craig Gidney (Google, arXiv:2505.15917, 2025) cut the RSA-2048 estimate ~20×, from ~20M physical qubits (2019) to **<1M noisy qubits at ~1 week runtime** [T3]. Scott Aaronson (March 2026) called 2025 a year that "met or exceeded my expectations on hardware, with multiple platforms now boasting >99.9% fidelity two-qubit gates," and said he had updated "in favor of taking more seriously the aggressive pronouncements...about where they could be in 2028 or 2029" [T3, expert commentary].
- **Skeptic camp.** Jensen Huang (Nvidia, Jan 2025) put "very useful" quantum computers **15–30 years out** (~20 central), a remark that cratered quantum stocks before he softened it at GTC's Quantum Day [T5]. Gil Kalai

argues fault-tolerant scaling is impossible in principle: correlated noise defeats the threshold theorem before you reach a million qubits, and he holds the correlations are observable in straightforward simulations of the noisy circuits that build highly entangled states [T3/contested]. The field has demonstrated well under 1% of the physical-qubit count Gidney's optimized RSA estimate needs, at anything like the required fidelity — closing five orders of magnitude in qubit count while *holding* fidelity is a claim, not a datum. Aaronson keeps a "little voice...that asks 'but what if Gil Kalai turned out to be right'," while adding that in 2026 it "can barely be heard," and that Kalai's "path has been getting narrower and narrower" [T3]. Kalai's March-2026 reply (gilkalai.wordpress.com, 10 Mar 2026) grants that he "starts with quantum computation being impossible as [an] axiom" but insists the measured exponential error decay "does not contradict my conjectures regarding correlated errors," which he expects to surface at "hundreds... and thousands of gates" [T3/contested].

WHAT WOULD RESOLVE IT

A single machine running hundreds of logical qubits through millions of error-corrected gates on a problem with a checkable answer — IBM Kookaburra (2026) and then Starling (2028–29) hitting spec, verified independently, would settle the direction. Each vendor deadline that holds or slips recalibrates the distribution. Kalai's in-principle claim dies the day any large machine sustains fault tolerance; it strengthens with every correlated-noise surprise at scale. Watch: does fidelity hold as qubit count grows an order of magnitude, or degrade?

SOURCES

- IBM, "IBM lays out clear path to fault-tolerant quantum computing" — ibm.com/quantum/blog/large-scale-ftqc (Jun 2025) [T4]
- Gidney, "How to factor 2048 bit RSA integers with less than a million noisy qubits," [arXiv:2505.15917](https://arxiv.org/abs/2505.15917) (2025) [T3]
- Aaronson, "More on whether useful quantum computing is 'imminent'," scottaaronson.blog/?p=9425 (2026) [T3, expert]; Kalai reply, gilkalai.wordpress.com/2026/03/10/ (Mar 2026) [T3, contested]
- Google Quantum AI roadmap — quantumai.google/roadmap [T4]

- IEEE Spectrum, "IBM's Starling: Quantum Computing's Future"; DataCenterDynamics roadmap coverage (2025) [T4/T5]

Funding-Winter Risk & Talent Attrition · O-talent-attrition

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Fault-tolerant timelines run to 2029–2033 and beyond (O-scaling), but venture funding, public patience, and skilled people operate on much shorter cycles. The open risk: a "quantum winter" — a stretch of disappointing milestones and cheap capital drying up — could drain talent and shutter startups *before* the long roadmaps mature, the way the AI winters of the 1970s and late 1980s set that field back years. Quantum is unusually exposed: capital-intensive, pre-revenue, and its "advantage" claims keep getting matched classically (O-advantage, O-hype). Human capital is thin and slow to grow — quantum PhDs take years, and a few big players absorb most of the trained pool, so a shock disperses expertise fast and refills it slowly.

WHERE THE DISAGREEMENT IS

- **Winter-is-a-real-risk camp.** Q1 2026 quantum-startup funding fell to ~\$240M, down 51% quarter-over-quarter and ~68% year-over-year (Crunchbase), and funding concentrates hard — in 2024 two firms (PsiQuantum, Quantinuum) took ~half of all VC; in Q3 2025 three firms took >80%. That concentration means a downturn strands the long tail of startups and disperses their teams into AI and quant finance, where they rarely return. A widely cited workforce gap — roughly one qualified candidate per three open quantum roles, and ~250k professionals needed by 2030 (McKinsey) — means attrition can't be easily backfilled. Insider stock sales (~\$931M across IonQ/Rigetti/D-Wave over five years) suggest even operators are hedging [T5].
- **No-winter camp.** Government programs (US NQI, EU Flagship, UK, China, others — E-us/E-eu/E-china/E-others) provide counter-cyclical funding that private downturns don't touch, and public-market quantum names held up into 2026 even as private rounds cooled. Aggregate 2025 equity funding still roughly doubled year-over-year (~\$3.8B by September). The real technical milestones landing in 2025–26 (four teams below threshold, QuEra's 96 logical qubits, Quantum Echoes) argue the field is *earning* its capital, not just burning it. Optimists hold that a base of

committed national money plus a handful of well-capitalized leaders carries the field through any private-capital lull [T5].

WHAT WOULD RESOLVE IT

A 2–3 year stretch that goes one of two ways: (a) a genuinely useful, revenue-generating application appears (killing winter risk — O-killerapp/O-roi-business), or (b) sustained funding collapse plus visible startup closures and a net outflow of researchers into other fields. The observable dials are quarterly VC totals, headcount at the pure-plays, PhD-pipeline enrollment, and whether national programs *sustain* commitments through a private downturn or trim them in sympathy. The 2026 funding drop is the first real stress test; whether it is a blip or a trend is the live question.

SOURCES

- Crunchbase, "Quantum Computing Startup Investment Slows in 2026" (Q1 2026: ~\$240M, –51% QoQ) [T5]
- McKinsey, "Quantum Technology Monitor" / talent-gap analyses (2024–2025, ~250k needed by 2030) [T5]
- The Quantum Insider, funding-concentration data (2024–2025) [T5]
- The Motley Fool / Nasdaq, IonQ-Rigetti-D-Wave insider-selling coverage (2026) [T5]

The "Quantum Utility" Definitional Debate · O-utility-definition

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Words do heavy lifting in this field, and "utility," "advantage," and "supremacy" are not interchangeable — yet marketing blurs them constantly. IBM's June 2023 *Nature* paper, "Evidence for the utility of quantum computing before fault tolerance," coined **quantum utility**: a noisy 127-qubit processor producing *reliable* results at a scale beyond brute-force classical simulation, using error mitigation. The claim was carefully *not* "advantage" (beating all classical methods) — utility meant "useful for scientific exploration, competitive with existing classical approximations." The sharp question: is "utility" a meaningful, defensible category, or a rhetorical hedge that lets vendors claim a win while conceding the harder "advantage" bar? Getting the vocabulary honest is prerequisite to grading everything else in this chapter.

WHERE THE DISAGREEMENT IS

- **Utility-is-a-real-milestone camp.** IBM and allies argue "utility" names a genuine transition: quantum processors now operate at a scale where they can *contribute to research*, run circuits classical brute force can't, and serve as a scientific tool even without asymptotic advantage. The 2023 kicked-Ising experiment on Eagle was, at minimum, a demonstration that error-mitigated NISQ machines can produce trustworthy expectation values at 127 qubits. IBM later reframed toward "the dawn of quantum advantage" as hardware improved [T4]. Proponents note the honest hedge is a *feature* — the field distinguishing "useful now" from "provably beyond classical later" is exactly the neutrality this manual asks for.
- **Utility-is-a-goalpost-move camp.** The 2023 utility claim was **classically simulated within days** — 2D tensor networks (gPEPS, Tindall et al.) and sparse Pauli-dynamics methods (Sandia, Caltech) reproduced or beat the quantum results on laptops-to-clusters, showing the specific problem was *not* beyond classical reach [T2]. Critics argue "utility before fault tolerance" became a movable goalpost: when "supremacy" claims got matched, the bar quietly relaxed to "utility," a term with no agreed operational definition

and no independent adjudicator (see O-benchmark-standard). If a result is reproduced classically within a week, calling it "utility" risks quantum-washing (O-hype). The word invites exactly the ambiguity that lets a T4 press release read like a T2 result.

WHAT WOULD RESOLVE IT

A precise, community-adopted operational definition — a taxonomy that fixes "utility" (useful/competitive, may be classically reproducible), "advantage" (beats the best classical method on a task people want), and "supremacy" (beats all classical methods on *any* task, useful or not), with independent adjudication of which bar a given demo clears. Failing a standard, the manual applies its own: "utility" claims are T4 until an independent party confirms the classical baseline was strong and the result competitive; a claim that is classically reproduced within weeks is graded a *classical-simulation-frontier* data point (O-classical-sim), not an advantage.

SOURCES

- IBM et al., "Evidence for the utility of quantum computing before fault tolerance," *Nature* 618, 500 (2023) [T2 paper / T4 framing]
- Tindall et al., "Efficient tensor network simulation of IBM's Eagle kicked Ising experiment," *PRX Quantum* 5, 010308 (2024) [T2]
- IBM, "The dawn of quantum advantage" — ibm.com/quantum/blog/quantum-advantage-era [T4]
- arXiv:2306.14887 (Eagle kicked-Ising tensor network); sparse Pauli-dynamics classical rebuttals (Sandia/Caltech) [T2/T3]

Verifying a Quantum Computation Too Large to Check · O-verification

Layer: L7 Frontier & open · **Chapter:** §08 · **Status:** depth

THE OPEN QUESTION

Once a quantum computer solves a problem no classical machine can redo, how do you know the answer is *right*? For factoring you multiply the factors back — an easy classical check. For quantum simulation, sampling, or optimization there may be no efficient classical way to verify the output, and the whole point is that you can't re-run it classically. This is the verification problem: can a *limited* verifier (classical, or a small trusted quantum device) certify the result of an untrusted, powerful quantum computer? It matters commercially — why pay a cloud QPU if you can't trust its output (S-cloud) — and scientifically: an unverifiable "advantage" is a weak claim (O-advantage, S-bench). Quantum Echoes' selling point is precisely that it is *cross-checkable on another quantum device*, a weaker but real form of verification.

WHERE THE DISAGREEMENT IS

- **Solvable-in-principle camp.** Urmila Mahadev (2018, FOCS Best Paper) proved a *classical* verifier can verify an arbitrary quantum computation using a single quantum prover, assuming the device can't break post-quantum cryptography (Learning With Errors) [T2]. This is a landmark: interactive classical verification of all of BQP under a standard computational assumption. Related "tests of quantumness" and certified-randomness protocols (Brakerski–Christiano–Mahadev–Vazirani–Vidick) have run on real hardware — Quantinuum's 56-qubit H2 generated *certified* random bits verified classically (Nature 2025, arXiv:2503.20498), the first delivered verification-flavored application. Blind/verifiable observable-estimation protocols advanced in 2025–26 (arXiv:2510.08548). Self-testing, cross-entropy benchmarking, and classical spot-checks give partial verification today [T2/T3].
- **Impractical-where-it-matters camp.** Mahadev's protocol is far from usable — it needs a fault-tolerant device well beyond today's, deep circuits, and many interactive rounds. Cross-entropy benchmarking (the metric behind supremacy claims) is *not a proof* and has itself been spoofed by

classical spoofing algorithms. For the near-term applications people actually want — chemistry ground-state energies — there is no efficient rigorous verifier; you fall back to trusting variational convergence heuristics or loose classical bounds [T3/contested]. Certified randomness verifies *randomness*, not a chosen useful computation, so the gap between "we can certify random bits" and "we can certify a drug-discovery answer" is wide open.

WHAT WOULD RESOLVE IT

A demonstrated, resource-realistic protocol that verifies a *beyond-classical, useful* computation on a real device with a classical (or exponentially weaker) verifier — going past certified randomness to a chosen task. Failing full Mahadev-style verification, a widely accepted intermediate (device-independent self-testing at useful scale, or robust quantum-checks-quantum cross-validation of an advantage claim) would move the node forward. The near-term test: whether Quantum Echoes-style quantum-verifies-quantum cross-checks become a trusted standard, or whether the community holds that only classical-verifier protocols count.

SOURCES

- Mahadev, "Classical Verification of Quantum Computations," FOCS 2018, arXiv:1804.01082 [T2]
- Brakerski, Christiano, Mahadev, Vazirani, Vidick, "Certifiable Randomness from a Single Quantum Device," JACM (2021) [T2]
- Certified randomness on Quantinuum H2, Nature 645 (2025), arXiv:2503.20498 [T2]
- arXiv:2510.08548 — "Verifiable blind observable estimation" (2025) [T3]
- Gheorghiu, Kapourniotis, Kashefi, "Verification of quantum computation: an overview," Theory Comput. Syst. (2019) [T3]

Appendix · How every claim is graded

Evidence Schema — how every quantum claim is graded

Inherited from the health manual's neutrality mechanism, adapted for a field where the loudest claims come from vendors with something to sell.

Every claim carries

- **claim** : the statement, in plain language
- **tier** : evidence strength (below)
- **source** : primary citation (arXiv ID, DOI, journal, or named press release)
- **provenance** : who is making the claim and what they gain from it
- **status** : **established** · **demonstrated** · **claimed** · **contested** · **roadmap**

Tiers (strongest → weakest)

Tier	Meaning	Example
T1 Established physics	Textbook, reproduced for decades	Superposition, Bell violation
T2 Peer-reviewed result	Published, refereed, ideally reproduced	Google Willow below-threshold QEC (Nature 2024)
T3 Preprint / conference	arXiv or talk, not yet refereed	Most fresh hardware milestones
T4 Vendor claim	Company announcement, not independently verified	Roadmap qubit counts, "advantage" PRs
T5 Analyst / forecast	Market projection, punditry	BCG/McKinsey TAM numbers
T6 Speculative	Plausible but unproven	Room-temp qubit timelines, killer-app dates

Rules

1. A vendor announcing its own benchmark is **T4 until independently reproduced** — never promote to T2 on the strength of a blog post.

2. "Quantum advantage / supremacy" claims are **contested by default** and must carry the classical-counterattack status (was it later matched classically?).
3. National-program dollar figures are **T5** and flagged for double-counting.
4. Conflicts are first-class objects — see `evidence/CONFLICTS.md` . When two credible sources disagree (e.g. "FTQC by 2029" vs "not before 2040"), keep both.
5. Timeline / roadmap nodes are **T4/T6** and say so in the manual text.

Conflict object format

```
id: C-<slug>
topic: <one line>
positions:
  - who: <source>  claim: <...>  tier: T?
  - who: <source>  claim: <...>  tier: T?
what_would_resolve_it: <the experiment/event that settles it>
```

Appendix · Conflict register

Conflicts — first-class disagreement objects

Per `SCHEMA.md` rule 4: when two credible sources disagree, keep both. This file is the registry. Cards link here; the manual never silently picks a side.

id: C-ftqc-timeline

topic: When (if ever) does large-scale fault-tolerant quantum computing arrive

positions:

- who: IBM (modular roadmap, Jun 2025) claim: Loon 2025 → Kookaburra 2026 (final)
- who: Google Quantum AI (roadmap) claim: useful error-corrected machine ~2028
- who: Scott Aaronson (Mar 2026) claim: 2025 met/exceeded hardware expectations
- who: Jensen Huang (Nvidia, Jan 2025) claim: very useful QC is 15-30 years out
- who: Gil Kalai claim: FTQC impossible in principle (correct)
- who: Manifold prediction markets (2026) claim: steady engineering gains, no leaps

what_would_resolve_it: IBM Kookaburra (2026) then Starling (2028-29) hitting spec

id: C-overhead-ratio

topic: The realistic all-in physical:logical qubit ratio for useful algorithms

positions:

- who: QuEra / Quantinuum / qLDPC papers claim: ratios of ~2:1 to ~13:1 demonstrated
- who: surface-code resource analyses claim: at target logical error $1e-9$, ~1000:1

what_would_resolve_it: an end-to-end fault-tolerant algorithm (incl. T-gates) at target error

id: C-tls-scaling

topic: Whether materials defects (TLS) are an engineering nuisance or a scaling wall

positions:

- who: fabrication groups (arXiv:2602.11469 etc.) claim: TLS density is controlled
- who: TLS-dropout studies (arXiv:2605.02755) + Kalai claim: dropouts and correlated errors

what_would_resolve_it: a 10k-qubit chip holding all qubits within QEC spec for wall

id: C-advantage-survival

topic: Whether any claimed quantum speedup survives classical counterattack positions:

- who: Google (Quantum Echoes, Oct 2025) claim: first verifiable quantum advantage
- who: arXiv:2604.15427 (2026) claim: belief-propagation tensor networks
- who: D-Wave (Science, Mar 2025) claim: beyond-classical spin-glass simulation
- who: Flatiron Institute (Tindall et al., Science adx2728, May 2026) + Sels (2026)
- who: D-Wave rebuttal (May 2026) claim: the classical framework fails
- who: third-party adjudication status (mid-2026) claim: NONE – no neutral body
- who: Ewin Tang (dequantization program) claim: quantum recommendation systems
- who: arXiv:2510.06324 claim: noisy circuit sampling classically

what_would_resolve_it: a verifiable advantage claim standing ~3+ years against multiple

id: C-hype-valuation

topic: Whether 2024-26 quantum valuations reflect substance or a bubble positions:

- who: vendors + analysts (BCG/McKinsey) claim: convergent 2029-30 roadmaps +
- who: Hossenfelder claim: hype will crash; real-world use cases
- who: Kalai claim: the entire FTQC premise fails
- who: market analysts (2025-26) claim: 150x-3,000x sales multiples are

what_would_resolve_it: revenue – a paying customer choosing quantum on cost/performance

id: C-photonic-scaling

topic: Whether photonics' room-temperature path actually changes the economics positions:

- who: Xanadu (Aurora, Nature 2025) claim: modular photonic architecture scaling
- who: PsiQuantum + GlobalFoundries claim: million-qubit photonic systems manufacturing
- who: photonics skeptics (CACM etc.) claim: SNSPD detectors stay cryogenic, photonic

what_would_resolve_it: any photonic machine demonstrating error-corrected logical qubits

id: C-killer-app

topic: Whether a near-term application exists that pays for the machine

positions:

- who: chemistry camp (Aspuru-Guzik line, Quantinuum, pharma partners) claim:
- who: feasibility skeptics (2025-26 studies, New Scientist) claim:
- who: optimization skeptics (Hossenfelder; quadratic-speedup analyses) claim:
- who: Aaronson claim:

what_would_resolve_it: one audited, repeated commercial workload where quantum b

id: C-measurement-problem

topic: Whether the measurement problem is solved, dissolvable, or permanently op

positions:

- who: Everettians (Deutsch et al.) claim: unitarity all the way down; sca
- who: objective-collapse camp (GRW/DP) claim: collapse is physical and testab
- who: Nature centenary survey (2025) claim: no consensus – Copenhagen ~36%
- who: foundations reviews (arXiv:2502.19278) claim: clarified by decoherence,

what_would_resolve_it: detection of collapse-model radiation/heating (ends it fo

id: C-psi-ontology

topic: Whether the quantum state is a real physical property (ψ -ontic) or an obs

positions:

- who: Pusey, Barrett & Rudolph (Nat. Phys. 8, 475, 2012; arXiv:1111.3328) cl
- who: PBR experimental tests (Ringbauer et al., Nat. Phys. 11, 249, 2015; Nig
- who: ψ -epistemic defenders (Leifer review 2014; Spekkens toy model; Harrigan
- who: instrumentalists / QBists claim: the ontic-vs-

what_would_resolve_it: a theorem removing the preparation-independence assumption

id: C-discord-resource

topic: Whether quantum discord (correlation beyond entanglement) is an operation

positions:

- who: DQC1 / mixed-state-power camp (Knill-Laflamme; Datta, Shaji, Caves, PRL
- who: discord-consumption protocols (quantum state merging, encoding – Madhok
- who: skeptics claim: nonzero discord

what_would_resolve_it: an audited task where discord (with zero entanglement) is

id: C-majorana-existence

topic: Whether Microsoft's topological hardware actually hosts Majorana zero modes
positions:

- who: Microsoft (Majorana 1, Nature, Feb 2025) claim: InAs/Al topoisolator
- who: Henry Legg critique (formal Nature comment, d41586-026-01788-y, Jun 24 2025)
- who: prior retraction (2021, Delft/Microsoft) claim: earlier quantized-Majorana
- who: Microsoft response (formal Nature reply, 2026) claim: defends the topology
- who: independent reproduction status (mid-2026) claim: NONE – no lab has independently

what_would_resolve_it: a demonstrated topological qubit with fusion-rule / braiding

id: C-crqc-timeline

topic: When a cryptographically-relevant quantum computer (Shor at RSA-2048 scale)
positions:

- who: Gidney (Google, arXiv:2505.15917, 2025) claim: RSA-2048 in <1M noisy physical
- who: Global Risk Institute expert survey claim: ~22.7% of cryptographers
- who: NIST (2024 PQC standards) claim: threat is near enough to
- who: hardware skeptics / Kalai claim: estimates improve on previous

what_would_resolve_it: a Shor factorization of a growing genuine semiprime (RSA-2048)

id: C-quantum-utility

topic: Whether "quantum utility" is a meaningful milestone or a rhetorical hedge
positions:

- who: IBM (Nature 618:500, 2023) claim: 127-qubit error-mitigated
- who: Tindall et al. (PRX Quantum 5, 010308, 2024) + Sandia/Caltech claim: the
- who: skeptics (0-hype) claim: "utility" became a moving

what_would_resolve_it: a community-adopted taxonomy fixing utility (useful/comparable)

id: C-energy-advantage

topic: Whether quantum computing delivers a net energy advantage over classical
positions:

- who: energetic-advantage theory (arXiv:2601.08068, 2605.19854, 2026) claim: quantum
- who: Google supremacy energy analysis claim: classical simulation of

- who: full-system accounting (arXiv:2605.09580, 2026) claim: fixed cryo (~10
what_would_resolve_it: a full-stack (fridge + control + classical co-processor) ;

id: C-benchmark-metrics

topic: Whether quantum computing can standardize honest cross-vendor benchmarks

positions:

- who: QED-C + standards bodies claim: application-oriented o
- who: metrics-are-a-mess camp claim: no common yardstick ex

what_would_resolve_it: broad adoption of application-oriented, independently-rep

Quantum Operating Manual · Bucket Foundation · 2026-07-08. Not investment advice. Every timeline attributed to a vendor is marketing until a peer reviews it.